Proximal Origin — A Tragicomedy of our Times (Part 1)

This story is in 3 parts: Summary, Part 1, Part 2

Gilles Demaneuf 177 min read · Aug 4, 2024

The Accidental Heroes

. . .

Warning:

This piece is rather long by necessity. I chose not to introduce simplifications that could end up misrepresenting the events in their totality and complexity. Instead, please read the Summary first, then jump through the various Insight sections for some major points, or you may view this 6-minute podcast-style briefing.

Notes on dates and times:

Throughout this piece, the dates and times are most often provided in a convenient time zone; typically UK time or Eastern Standard Time (NY time), as indicated. Particular efforts have been made to understand the actual time zones of each email, before conversion to the chosen time zone [see footnote ❖].

. . .

Table of Content:

Part1:
1. CONTEXT: The fight for the sequence publication (Zhang, Holmes, Farrar, WHO)
2. CONTEXT: Questions about human transmission (Holmes, Farrar, WHO)

3. Concerns about the sequence (unspecified US scientists, MI5, Farrar, Holmes)
4. The Farrar call
5. Farrar and the ‘core-group’ compare notes and estimates
6. Farrar shares the passaging hypothesis with the ‘Bethesda boys’
7. In Parallel: Trying to get the WHO to pick up the issue
8. CONTEXT: US parallel effort: Track II via NASEM
9. Writing a confidential report while waiting for the WHO
10. Drafts are shared with Fauci and Collins (4 Feb)
11. WHO agrees to get involved (5–7 Feb)
12. ‘Pangomania’ and a fateful decision (7 Feb)
13. Sharing the decision with the call group and some pushback (8 Feb)

Part 2:

14. CONTEXT: Pressure mounts as NASEM, US Gov and WHO all urgently need access to China
15. Farrar has to keep pushing for publication (8–11 Feb)
16. Working towards publication (10–16 Feb)
17. CONTEXT: Main-street media turn ugly just as the full WHO mission arrives in China (16–17 Feb)
18. Publication of manuscript on virological.org and submission to Nature (15–17 Feb)
19. In parallel: Daszak’s pre-emptive strike (6–18 Feb)
20. The pangolin claims implode (18 Feb-20 Feb)
21. Rejection by Nature (20 Feb)
22. RmYN02 and its consequences (21–25 Feb)
23. End-Game: Publication in Nature Medicine (26 Feb-17 Mar)
24. CONTEXT: White House vs. China Wolf-Warriors (6-17 Mar)
25. The Nature Medicine version
26. Post publication doubts
27. Egomaniac strikes back (Jul 2020)
28. CONTEXT: A long feud and some funding woes

footnotes
Appendix A: The evolution of a scenario
Appendix B: The evolution of a name
Appendix C: Plausible research-related accident scenarios
Appendix D: Virus Genesis Scenarios
Appendix E: Evolutionary vs. Outbreak Origins in P.O.

1. CONTEXT: The fight for the sequence publication (Zhang, Holmes, Farrar, WHO)

To understand the Proximal Origin story, we need to go back to the first interactions between Jeremy Farrar, Eddie Holmes, and the WHO, in January 2020, as part of their efforts to try to wriggle data out from China, such as the virus sequence or reliable epidemiological data.

1.1 Sequencing of SARS-CoV-2 (5 Jan)

a. Surveillance hospital

For many years, the **Wuhan Central Hospital** had been a Sentinel Hospital for detection of infectious diseases in Central China. Practically, this meant that the Wuhan Central Hospital would regularly sample patients with pneumonia symptoms, especially if unexplained, and would send the samples to the Wuhan CDC. The CDC would then split the samples before sending one split to **Zhang Yongzhen** of the Shanghai Public Health Clinical Center, for further analysis.

Zhang's team would use some shotgun sequencing to figure out what pathogen were present in the samples. At that stage **Eddie Holmes** from Sydney University, a long-time collaborator of Zhang and a well-known virologist, would help with the analysis.

The choice of the top-rated Wuhan Central hospital allowed for sampling in the large Wuhan population that may be referred to it. Crucially, it was also meant to allow for viral surveillance of people from the nearby Wuhan Huanan market, one of the three main markets in Wuhan, for which there was no systematic sampling of animals, their human handlers and store handlers. That market viral surveillance was instead meant to be — hopefully — a by-catch of the Wuhan Central Hospital.

Talking about a trip he did to Wuhan (during which he took a photo of a raccoon dog in a cage at the market), Holmes later explained:

> *When I visited the market in 2014, a variety of live wildlife was for sale including raccoon dogs and muskrats. At the time I suggested to my Chinese colleagues that we sample these market animals for viruses. Instead, they set up a virological surveillance study at the nearby Wuhan Central Hospital, which later cared for many of the earliest COVID patients.*
>
> *source: Eddie Holmes, quoted in "The COVID lab leak theory is dead. Here's how we know the virus came from a Wuhan market", The Conversation, Aug 2022.*

b. Striking gold

That viral surveillance involving the Wuhan Central Hospital, the Wuhan CDC, Zhang, and his friend Holmes, was a fairly well-oiled mechanic that was funded under some key national research program, the '863' program, as the 'Survey of major epidemic virus sources'. At the end of December 2019, that routine surveillance did strike gold when Zhang's lab identified a new coronavirus in a patient sample:

> *On the afternoon of December 30, the samples were taken by a chief physician of the Wuhan Center for Disease Control and Prevention. On Jan. 2, another researcher at the Wuhan CDC wrapped the samples heavily in dry ice, iron boxes and foam boxes and sent them, along with other animal specimens, by rail express to Shanghai. On Jan. 3, Professor Zhang Yongzhen's team at the Shanghai Public Health Clinical Center received the samples. [..]*
>
> *In the early morning of January 5, Zhang Yongzhen's research team detected a novel SARS-like coronavirus from the sample and obtained the whole genome sequence of the virus through high-throughput sequencing; the evolutionary tree drawn from the sequencing data also confirmed that the novel coronavirus in Wuhan had never been seen before in history. The Shanghai Public Health Center immediately reported to the Shanghai Municipal Health and Wellness Commission and the National Health and Wellness Commission and other authorities that day, alerting them that the new virus was* ***homologous with SARS and should be transmitted via the respiratory tract, and suggesting appropriate disease control and epidemic prevention measures in public places.***
>
> *source: 'New coronavirus gene sequencing traceability: when the alarm was sounded', Caixin, 26 Feb 2020*

1.2 Holmes, Farrar, and Rambaut join hands to release the sequence (7 Jan–10 Jan)

a. Something is not right

After identifying that new SARS-like coronavirus in a patient sample on Jan 5, Holmes and Zhang quickly realised that the Chinese government was unwilling to confirm the finding as the likely pathogen behind the recently disclosed cluster of infections.

First, Holmes was sent images of Little Mountain Dog's blog describing how, many days earlier, Vision Medicals in **Shenzhen** had identified a virus with a high homology to SARS on 26 Dec, and produced a new full sequence on 27 Dec, immediately alerting the CDC and the Chinese Academy of Medical Sciences in Beijing (which received the sequence) and insisting on the risk of human-to-human transmission. [footnote 1]

Then, Holmes was told by Zhang that his lab was not authorised to release the virus sequence.

Q Why -- and I'm now asking you to speculate -- but would there be a reason to hold off on the sequence between the 27th of December and January 10th?

Mr. Rowley. By who?

Mr. Benzine. By China.

Dr. Andersen. Yeah. I think -- well, the thing -- again, so this is not specific to China.

Now, I'll say Eddie and I, for example, disagree on this, because I think Eddie

---

believes that because they had it on the 27th of December, they should have released it on the 27th of December.

b. Holmes and Zhang try to put pressure via some article in Nature (7–9 Jan)
Hoping to find a way to convince China to release the sequence, which they had been holding on since 27 Dec, Zhang and Holmes prepared a paper. On 7 Jan, Holmes sent the paper for review to Clare Thomas at Nature's office in London [source: Spike]. The paper described the new coronavirus identified by Zhang, and its position in a phylogenic tree, without explicitly quoting the sequence. Holmes is co-author and responsible for the key virological analysis.

A publication would be expected for a newly found virus of importance, as soon as the pathogen has been correctly identified, and it would be expected to include the sequence. But nothing prevented Holmes and Chang from still writing an analysis of the new sequence and from sending the manuscript for review.

The calculation was that, faced with the present review of the manuscript in a top scientific publication, by some top coronavirus experts, including Baric, the Chinese authorities were expected to release the sequence officially, in time for the paper publication. If they didn't, they would lose face, as details about the virus (and possibly the sequence itself) would have been already shared with the reviewers, and thus knowledge about the new virus would have already spread through the scientific community — as it should. [footnote 6]

On 9 Jan, Chinese authorities confirmed that the new virus was a SARS-like coronavirus, but they still did not release the sequence, which Chinese authorities had had for about two weeks by that time. As the day passed, the delay became more and more an issue, for Zhang, Holmes, and others in the know, given the similarity of the virus to SARS-1.

Insight: Baric was a reviewer of that paper

Baric's congressional interview leaves little doubt that he was one of the three reviewers of that essential paper. Which makes perfect sense, since the top experts should be chosen for the purpose.

> *A: I also had received emails from other people that it was a coronavirus on January 5th. And by the 6th or so, I also knew it was a coronavirus because I was asked to review a paper*
> *Q: For the January 6th paper that you reviewed, do you recall if it had the sequence of the virus?*
> *A: It did. When it was first sent, it did not. All three reviewers immediately asked for the sequence.*
> *Q: You mentioned earlier that the sequence of the virus was not initially provided. Do you recall when you got access to the sequence?*
> *A: Within about 12 hours from requesting it from the journal.*
>
> *source: Baric-TI-Transcript.pdf, extracts from p.30–32*

c. Holmes and Farrar get in touch (9–10 Jan)

Starting on Thursday evening, 9 Jan 2020 (UK), Holmes and Farrar had a series of calls [footnote 7]. Holmes explained to Farrar that the Chinese authorities had had access to a sequence of the virus since the 27 Dec 2020, and that his friend Zhang had also independently sequenced the virus from another patient's sample, and that he and Zhang were trying to get a paper through that could force the sequence publication.

By around 9 pm London time on 9 Jan, Farrar and Holmes decided to increase the stakes: Holmes and Zhang would force the release of the sequence by giving the Chinese authorities a 24-hour ultimatum, after which they would publish an analysis and release the genome themselves if the China CDC (which had sent the samples to Zhang) had not been able to do so already.

d. The WHO was already frustrated with China's lack of collaboration

During the second week of January, the WHO got increasingly frustrated with the lack of information shared by China, but the WHO still praised China in public statements to avoid being totally locked out. Tapes of WHO meetings at the time, obtained by Associated Press, perfectly summarised the situation:

> *"We're going on very minimal information," said American epidemiologist **Maria Van Kerkhove**, now WHO's technical lead for COVID-19, in one internal meeting. "It's clearly not enough for you to do proper planning."*
>
> *In the second week of January, WHO's chief of emergencies, Dr. **Michael Ryan**, told colleagues it was time to "shift gears" and apply more pressure on China, fearing a repeat of the outbreak of Severe Acute Respiratory Syndrome that started in China in 2002 and killed nearly 800 people worldwide.*
>
> *"This is exactly the same scenario, endlessly trying to get updates from China about what was going on," he said. "WHO barely got out of that one with its neck intact given the issues that arose around transparency in southern China." [..] "We need to see the data…..It's absolutely important at this point." [..]*
>
> *Dr. Tom Grein, chief of WHO's acute events management team, said the agency looked "doubly, incredibly stupid." Van Kerkhove, the American expert, acknowledged WHO was "already late" in announcing the new virus and told colleagues that it was critical to push China.*
>
> *Ryan, WHO's chief of emergencies, was also upset at the dearth of information. "The fact is, we're two to three weeks into an event, we don't have a laboratory diagnosis, we don't have an age, sex or geographic distribution, we don't have an epi curve".*
>
> *source: AP article*

e. The WHO (Van Kerkhove) contacts Farrar who then adds pressure with a tweet (10 Jan)

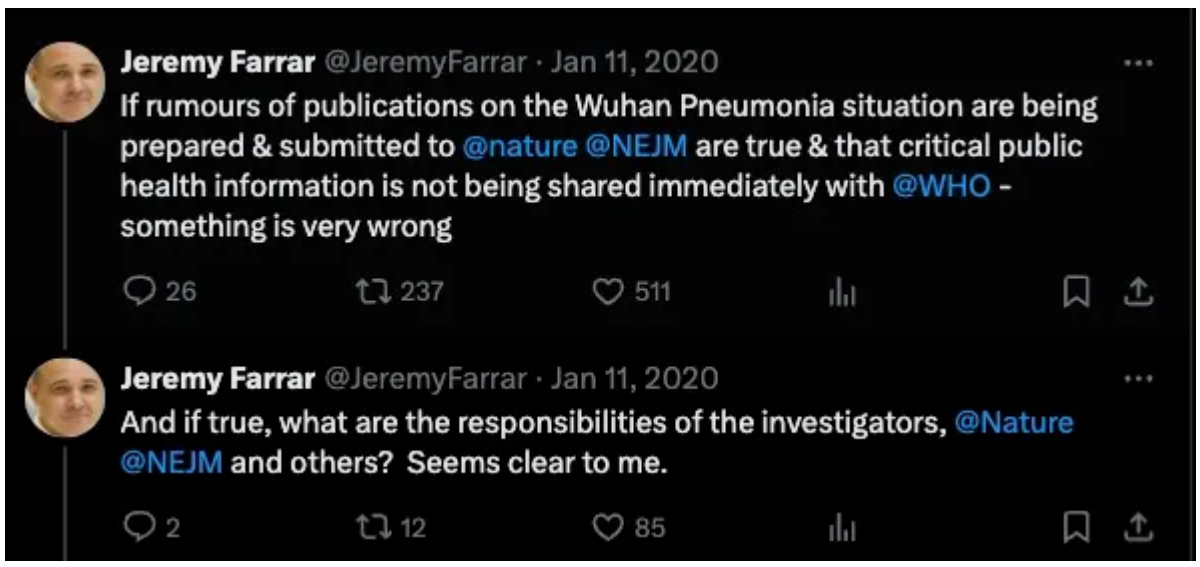
On Friday 10 Jan 2020, Maria Van Kerkhove of the WHO Health Emergencies Program, told Farrar that the WHO had independently learnt that two papers about the current outbreak were in preparation at Nature and NEJM. Most likely, Van Kerkhove would have been told that the teams behind the two papers had each sequenced the virus, and that Holmes was one of the co-authors of the Nature paper [footnote 8].

It is not clear whether Van Kerkhove contacted Farrar first after hearing of the Nature and NEJM papers in review, or if he contacted her after Holmes reached out to him. What seems clear is that:

1. The WHO very much needed to get that sequence, short of which it would look rather powerless in this crisis, but was at the same time unwilling to directly confront China.
2. Farrar being at the Wellcome Trust, so at arm-length from the WHO, and Holmes, who was the only Westerner of the two papers in review, were effectively in the best position to force some release of the sequence for the WHO.

3. Hence the conjunction of Van Kerkhove for the WHO, Farrar for Wellcome, and Holmes for his insider links to China, however it happened over the 9–10 Jan 2020 [footnote 7], made sense.

Following that call with Van Kerkhove, Farrar tweeted at about 4 pm on 10 Jan. The tweet articulated the need for China to share information with the WHO, with a not very subtle hint at the fact that Nature and NEJM may have to release the information themselves if it was not forthcoming. Since Holmes and Zhang had already sent a 24-hours ultimatum to the Chinese authorities, these authorities should have been able to see the reading on the wall.

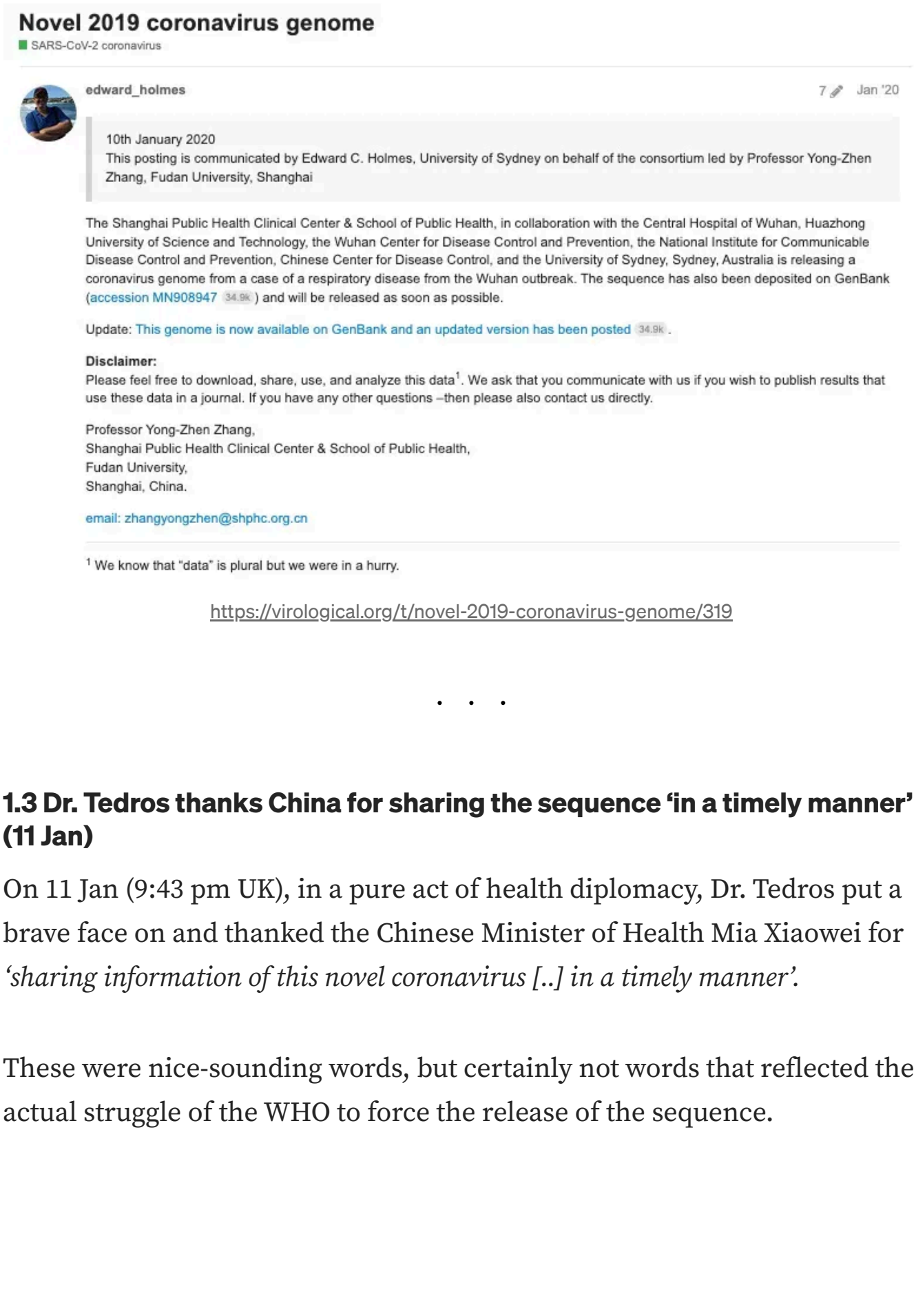


10 Jan 2020, afternoon (UK Time), date in tweet is NZ time

f. Publication of the sequence on virological.org with the help of Rambaut (11 Jan)

The Chinese CDC was unfortunately not in a position to take any decision, being purely a consulting body with no administrative of its own. As the deadline neared, Holmes then tuned to a trusted contact, **Andrew Rambaut**, to be ready to quickly publish the sequence on virological.org after the necessary quality checks (QCs). As per the ultimatum, a short message about the virus and its sequence was posted on virological.org at 1.05 am (GMT) on 11 Jan, where it was introduced by a quick notes from Zhang Yong-Zhen. The message included a link to the sequence stored in GenBank.

The Nature paper by Holmes and Zhang would be published online by Nature on 3 Feb 2020, after being accepted on 28 Jan 2020 (with some minor updates by then).


Novel 2019 coronavirus genome
SARS-CoV-2 coronavirus
edward_holmes
7 Jan '20
10th January 2020
This posting is communicated by Edward C. Holmes, University of Sydney on behalf of the consortium led by Professor Yong-Zhen Zhang, Fudan University, Shanghai
The Shanghai Public Health Clinical Center & School of Public Health, in collaboration with the Central Hospital of Wuhan, Huazhong University of Science and Technology, the Wuhan Center for Disease Control and Prevention, the National Institute for Communicable Disease Control and Prevention, Chinese Center for Disease Control, and the University of Sydney, Sydney, Australia is releasing a coronavirus genome from a case of a respiratory disease from the Wuhan outbreak. The sequence has also been deposited on GenBank (accession MN908947 34.9k) and will be released as soon as possible.
Update: This genome is now available on GenBank and an updated version has been posted 34.9k .
Disclaimer:
Please feel free to download, share, use, and analyze this data[1]. We ask that you communicate with us if you wish to publish results that use these data in a journal. If you have any other questions –then please also contact us directly.
Professor Yong-Zhen Zhang,
Shanghai Public Health Clinical Center & School of Public Health,
Fudan University,
Shanghai, China.
email: zhangyongzhen@shphc.org.cn
[1] We know that "data" is plural but we were in a hurry.

https://virological.org/t/novel-2019-coronavirus-genome/319

. . .

1.3 Dr. Tedros thanks China for sharing the sequence 'in a timely manner' (11 Jan)

On 11 Jan (9:43 pm UK), in a pure act of health diplomacy, Dr. Tedros put a brave face on and thanked the Chinese Minister of Health Mia Xiaowei for *'sharing information of this novel coronavirus [..] in a timely manner'.*

These were nice-sounding words, but certainly not words that reflected the actual struggle of the WHO to force the release of the sequence.

Tedros' tweet, 11 Jan, 9:43 pm UK (time and date in the tweet are NZ)

2. CONTEXT: Questions about human transmission (Holmes, Farrar, WHO)

2.1 The absence of data

While Zhang and Holmes managed to force the publication of the virus sequence on 11 Jan, the WHO was facing another very urgent problem: the complete absence of any new case being reported over the first two weeks of January.

Hardy a single new case was reported by the Chinese CDC from 3 Jan to 15 Jan, as if the closing of the market (on 1st Jan) had been enough to stop transmission — in line with the Chinese official line that all cases were likely due to animal contacts and not to human-to-human transmissions. This would have left the WHO rather doubtful and unimpressed. But the WHO still had to put a brave and conciliant face on, to try to engage China into properly sharing data [footnote 41].

2.2 Farrar redirects some key concerns to the WHO (19 Jan)

On 16 Jan, Thijs Kuiken, an adviser to the Dutch government on zoonotic diseases, received a key Lancet manuscript for review. That paper described clear cases of human-to-human transmission, that was still denied by China at the time. The paper also mentioned possible asymptomatic shedding, which would be a very concerning feature if correct, as this could mean asymptomatic transmission (in contrast to SARS), and much more difficulty in controlling the outbreak. Kuiken sent his review back to Lancet on 18 Jan, and became worried by the fact that neither the Lancet nor the authors were willing or able to make these key finding immediately public.

Kuiken next contacted Farrar (with whom he had worked before) on 18 Jan to discuss what the best solution would be to force the disclosure of these essential elements. They agreed that they should contact Maria Van Kerkhove at the WHO and tell her about the main findings in that paper under review. Accordingly, after letting Lancet know of his plans, Kuiken sent a 2-page summary of the manuscript main findings to Farrar and to the WHO on 19 Jan [footnote 9]. China conceded human-to-human transmission the following day (20 Jan).

2.3 First WHO visit to China (20–21 Jan)

On 20–21 Jan, the WHO sent a small team to Wuhan to evaluate the information first-hand and observe the situation. The team visited (i) the Wuhan Tianhe Airport on arrival, where they saw the temperature checking process there, (ii) the Zhongnan Hospital, where they learnt about the fever clinic, screening, isolation and admission, and last (iii) the BSL-3 laboratory of the Hubei provincial CDC. The five-people team was led by Gauden Galea (WHO Representative in China) and Olowokure Babatunde (WHO Regional Emergency Director for Western Pacific, based in Manilla).

Unbeknown to the WHO at the time, this little advertised first WHO mission would be the best access that the WHO would have to Wuhan for a year, until the WHO mission in early 2021.

WHO team at Tianhe (Wuhan) airport, source: https://archive.ph/oALoU

2.4 WHO meeting in Geneva to review the situation (22–23 Jan)

After China started admitting human-to-human transmission on 20 Jan, and while the small WHO team was evaluating the situation in Wuhan (20–21 Jan), the WHO organised a meeting in Geneva for 22–23 Jan, to decide if the outbreak should be declared a Public Health Emergency of International Concern (PHEIC) [footnote 69].

On the first day of the meeting (22 Jan), despite the recent disclosure of human-to-human transmission, no decision could be reached. Later that day, news came in that Wuhan would move into lockdown.

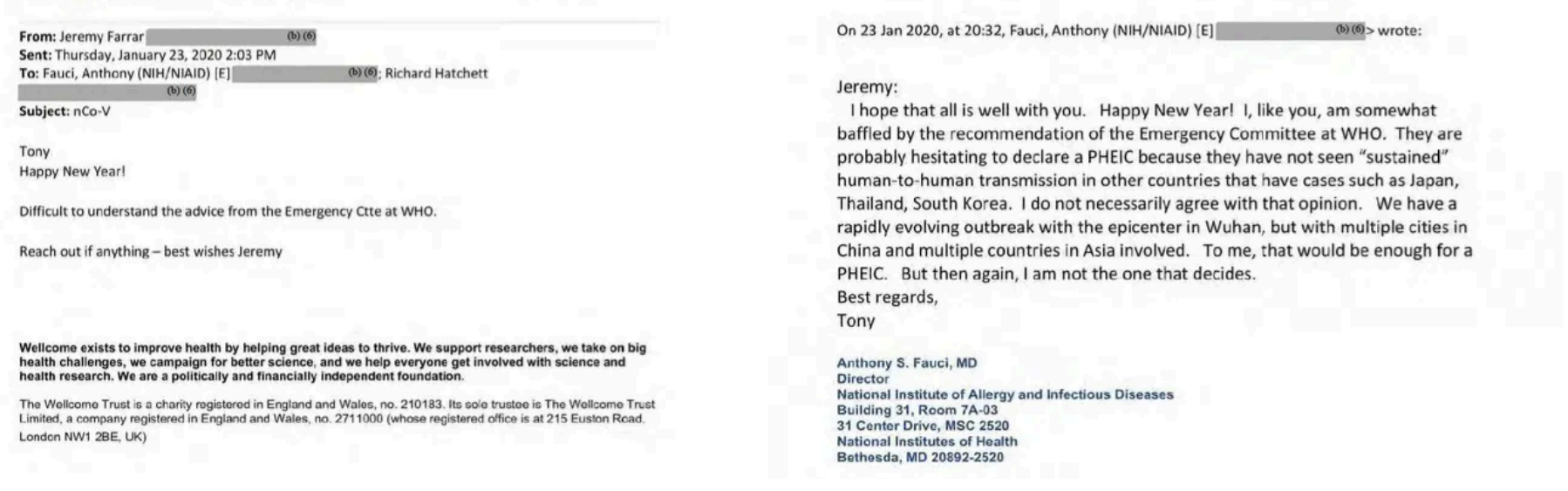
That lockdown decision followed immediately after the busting of the charade that there was no human-to-human transmission (as tentatively supported by the supposedly total absence of any new cases after the closure of the market), and the simultaneous disclosure that they may be asymptomatic transmissions making the types of controls used with SARS-1 ineffectual. Beijing basically totally flip-flopped once the truth was out, after bursting through the multiple layers of local government obstructions nurtured by Beijing reward systems, monolithic decision taking, and emphasis on saving face.

2.5 No PHEIC declared — Farrar contacts Fauci (23 Jan)

On the second day of the meeting in Geneva (23 Jan), the WHO decided again not to declare a PHEIC, apparently assuaged by the lockdown decision. Behind the scene, Beijing also let the WHO know that declaring a PHEIC would be considered a vote of no-confidence, and would thus endanger any further cooperation.

On the same day, Farrar reached out to Fauci (Director of NIAID), to show some support to him while expressing his perplexity about the decision not to declare a PHEIC (‘Difficult to understand the advice for the Emergency

Ctte at WHO'). Fauci concurred, especially as some first cases were by now appearing abroad (Japan, Thailand, South Korea, HK, Singapore). This was likely the first time Farrar got in touch with Fauci that year, given the greetings in the emails they exchanged ('Happy New Year!').



From: Jeremy Farrar (b) (6)
Sent: Thursday, January 23, 2020 2:03 PM
To: Fauci, Anthony (NIH/NIAID) [E] (b) (6); Richard Hatchett (b) (6)
Subject: nCo-V

Tony
Happy New Year!

Difficult to understand the advice from the Emergency Ctte at WHO.

Reach out if anything – best wishes Jeremy

Wellcome exists to improve health by helping great ideas to thrive. We support researchers, we take on big health challenges, we campaign for better science, and we help everyone get involved with science and health research. We are a politically and financially independent foundation.

The Wellcome Trust is a charity registered in England and Wales, no. 210183. Its sole trustee is The Wellcome Trust Limited, a company registered in England and Wales, no. 2711000 (whose registered office is at 215 Euston Road. London NW1 2BE, UK)

On 23 Jan 2020, at 20:32, Fauci, Anthony (NIH/NIAID) [E] (b) (6) > wrote:

Jeremy:
I hope that all is well with you. Happy New Year! I, like you, am somewhat baffled by the recommendation of the Emergency Committee at WHO. They are probably hesitating to declare a PHEIC because they have not seen "sustained" human-to-human transmission in other countries that have cases such as Japan, Thailand, South Korea. I do not necessarily agree with that opinion. We have a rapidly evolving outbreak with the epicenter in Wuhan, but with multiple cities in China and multiple countries in Asia involved. To me, that would be enough for a PHEIC. But then again, I am not the one that decides.
Best regards,
Tony

Anthony S. Fauci, MD
Director
National Institute of Allergy and Infectious Diseases
Building 31, Room 7A-03
31 Center Drive, MSC 2520
National Institutes of Health
Bethesda, MD 20892-2520

farrar-fauci-comms.pdf, p. 174, 23 Jan 2020, and p. 173, 24 Jan 2020

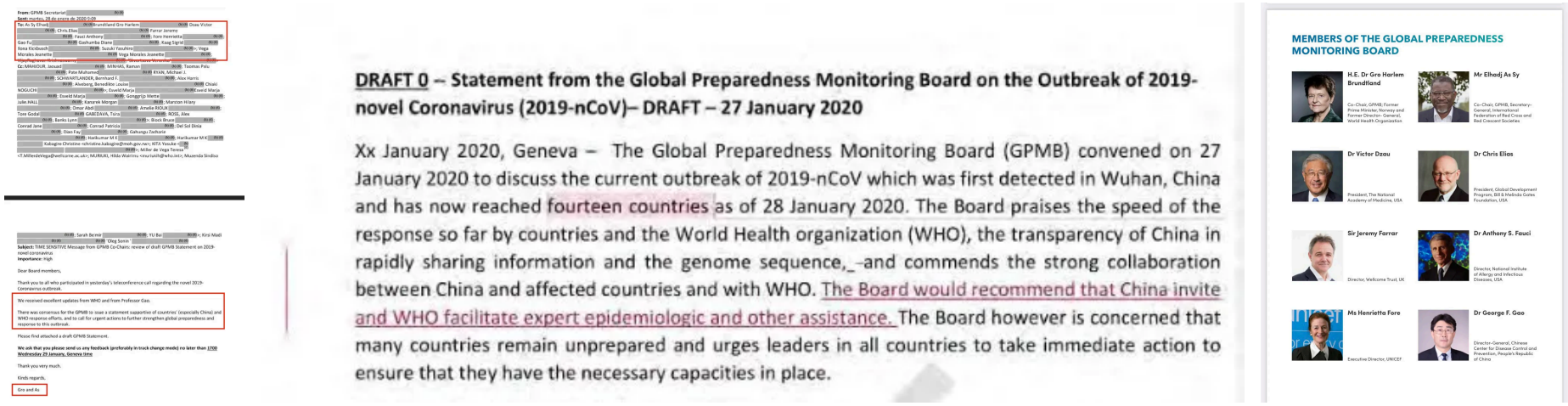
2.6 Global Preparedness Monitoring Board steps in (27–28 Jan)

On 27 Jan, the Global Preparedness Monitoring Board (GPMB, a joint arm of the WHO and the World Bank), convened online to discuss the outbreak. Farrar and Fauci were board member of the GPMB, as well as George Gao, (Chinese CDC director at the time) and Victor Dzau (the president of the US National Academy of Medicine).

On 28 Jan, an email was sent by the two co-chairs to the online attendants to summarise the decision. One of the co-chairs happened to be Gro Harlem Brundtland, the former WHO DG who had been very firm with China during SARS-1 and helped expose their cover-up, forcing Beijing to action. The email explained that *'There was consensus for the GPMB to issue a statement supportive of countries' (especially China) and WHO response efforts [..]'.*

Such a draft statement was circulated with the email. It included some praises for the *'transparency of China in rapidly sharing information and the genome sequence'* as well as the *'strong collaboration between China [..] and the WHO'.* One version of the draft also included the recommendation that *'China invite[s] and WHO facilitate[s] expert epidemiologic and other assistance'*.

In other words, the GPMB, lead by SARS-1 cover-up busting Brundtland, was still happy to praise China at that stage, and ignore all the obstructions that the WHO experts had been fuming on, to secure access for the WHO.



DRAFT 0 – Statement from the Global Preparedness Monitoring Board on the Outbreak of 2019-novel Coronavirus (2019-nCoV)– DRAFT – 27 January 2020

Xx January 2020, Geneva – The Global Preparedness Monitoring Board (GPMB) convened on 27 January 2020 to discuss the current outbreak of 2019-nCoV which was first detected in Wuhan, China and has now reached fourteen countries as of 28 January 2020. The Board praises the speed of the response so far by countries and the World Health organization (WHO), the transparency of China in rapidly sharing information and the genome sequence, and commends the strong collaboration between China and affected countries and with WHO. The Board would recommend that China invite and WHO facilitate expert epidemiologic and other assistance. The Board however is concerned that many countries remain unprepared and urges leaders in all countries to take immediate action to ensure that they have the necessary capacities in place.

MEMBERS OF THE GLOBAL PREPAREDNESS MONITORING BOARD

H.E. Dr Gro Harlem Brundtland — Co-Chair, GPMB; Former Prime Minister, Norway and former Director- General, World Health Organization

Mr Elhadj As Sy — Co-Chair, GPMB; Secretary-General, International Federation of Red Cross and Red Crescent Societies

Dr Victor Dzau — President, The National Academy of Medicine, USA

Dr Chris Elias — President, Global Development Program, Bill & Melinda Gates Foundation, USA

Sir Jeremy Farrar — Director, Wellcome Trust, UK

Dr Anthony S. Fauci — Director, National Institute of Allergy and Infectious Diseases, USA

Ms Henrietta Fore — Executive Director, UNICEF

Dr George F. Gao — Director-General, Chinese Center for Disease Control and Prevention, People's Republic of China

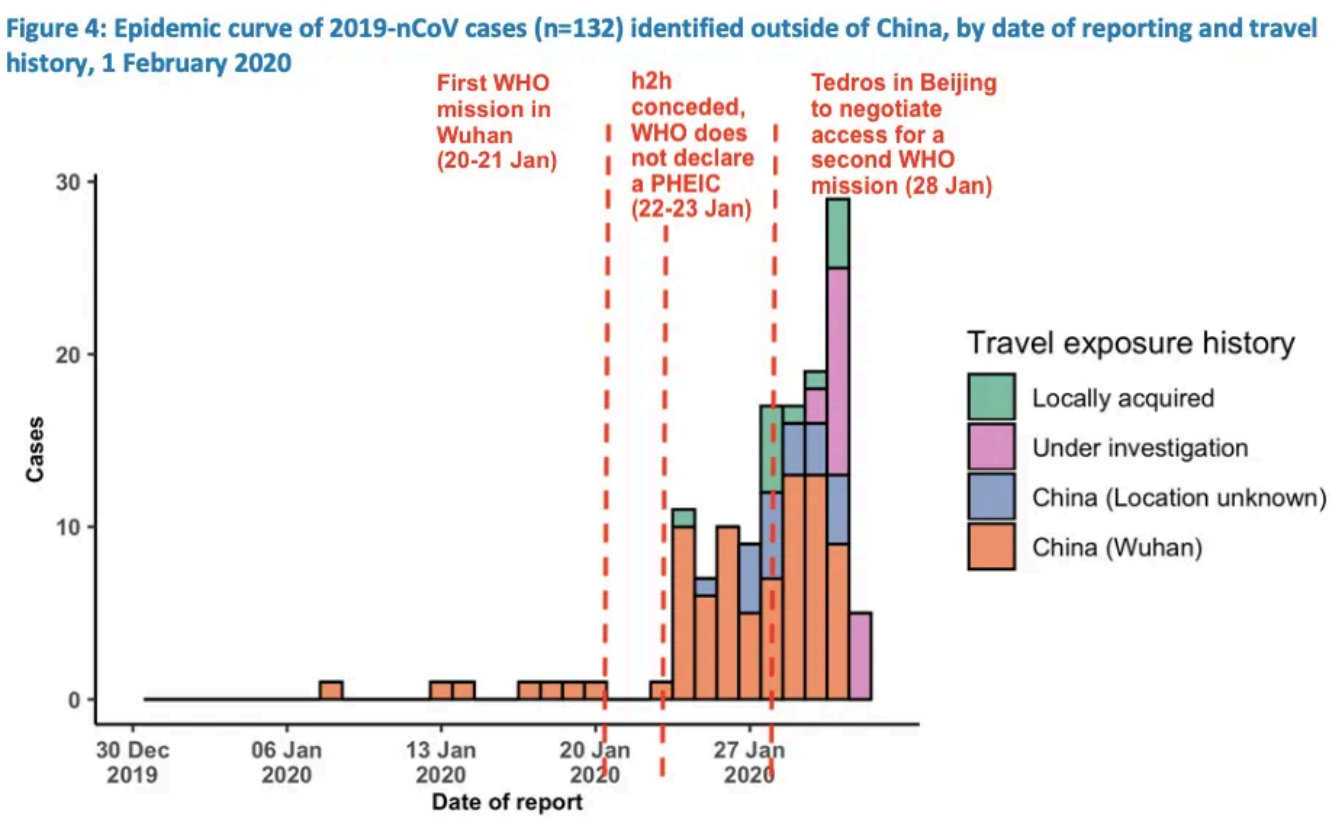
farrar-fauci-comms.pdf, p. 169–170 and 166, 28 Jan 2020, and WHO/World Bank GPMB Annual Report 2019, p.32

2.7 Dr. Tedros is in Beijing to try to negotiate WHO access (end Jan)

a. Reported cases start exploding (23–28 Jan)

The first WHO visit in Wuhan on 20–21 Jan had some fairly limited access to Wuhan and had not been anywhere close to the market. Immediately after, China put pressure on the WHO not to declare a PHEIC on 23 Jan.

Unfortunately, the WHO decision not to declare a PHEIC immediately started looking bad as cases mounted in China, raising serious doubts that the outbreak could be controlled. While 571 cases in China had been reported to the WHO up to 23 Jan (included, by 10 am CET), another 1,414 were reported to the WHO in the following 3 days. On 28 Jan 2020 only, close to 1,800 new cases would be reported in China. With the delay between infection and reporting and the risk of asymptomatic infections, the real number of infected people was clearly in the 10,000s already by 26 Jan. The situation did not look good at all, and was now unlikely to be limited to China, as cases were also climbing abroad:



source: WHO Situation report of 1 Feb 2020, with annotations

Having been wrong-footed, Dr. Tedros knew that his credibility was now at stake if he could not quickly arrange for a team of WHO experts to visit China and especially Wuhan, in a physical demonstration of the WHO ability

to gather true data at the source. A visit that — contrary to the first WHO visit of 20–21 Jan, would be strongly advertised and avidly followed, as all the projectors were now on the WHO and Wuhan.

b. Tedros, Ryan, and Briand fly to Beijing (28–29 Jan)

On Tuesday 28 Jan, Dr. Tedros, seconded by Michael Ryan, and accompanied by Dr. Sylvie Champaloux Briand (French citizen, director of Global Infectious Hazard Preparedness at WHO), met Xi Jinping in Beijing to negotiate such access. The official WHO press-release includes the usual praises for the supposed commitment and transparency of China:

> *'We appreciate the seriousness with which China is taking this outbreak, especially the commitment from top leadership, and the transparency they have demonstrated, including sharing data and genetic sequence of the virus''.*

The Chinese media reported more excerpts from Dr. Tedros' praises on 28 Jan:

> *China's measures are not only protecting its people but also protecting the people of the world. It is admirable that the Chinese government has shown solid political resolve and taken timely and effective measures in dealing with the epidemic.*
> *The high speed and massive scale of China's response are rarely seen in the world. It shows the advantages of China's system. The experience of China is worth learning for other countries. I believe the measures China has taken will effectively control and finally prevail over the epidemic.*

c. Tedros extracts a vague promise of access

Possibly as a result of that kowtowing, the Chinese side agreed that the WHO should 's*end international experts to visit China to visit China as soon as possible to work with Chinese counterparts*'. As the NY Times later put it:

> *The agreement was critical for Dr. Tedros, who the previous week had decided against declaring an international emergency after convening a committee to*

> *advise him.*
>
> *What was not publicly known, though, was that the committee's Jan. 23 decision followed intense lobbying, notably by China, according to diplomats and health officials. Committee members are international experts largely insulated from influence. But in Geneva, China's ambassador made it clear that his country would* ***view an emergency declaration as a vote of no confidence****.*
>
> *China also presented data to the committee, portraying a situation under relative control.*

d. Xi Jin-ping is rather clear

While the WHO made the most of the supposed promise of access negotiated by Tedros during his talk with Xi Jinping, the reality of what had been agreed was quite clear from the Chinese government's reporting of the talks.

The official reporting noted that '*the World Health Organization plays an important role in coordinating global health affairs, and China attaches great importance to cooperation with the World Health Organization*', and '*China stands ready to work with the World Health Organization and the international community to jointly safeguard public health security in the region and globally*'. Crucially, the official Chinese report added that '**the *WHO have gone to Wuhan for an on-the-spot visit*'**, referring to the 20–21 Jan WHO visit in Wuhan.

What this means is that Xi Jin-ping saw no need for the WHO to go back to Wuhan. First, that had already been done with the first WHO mission, and secondly, **the next mission was supposed to be about the WHO learning from China how to fight the epidemic, which could as well be done away from Wuhan, by talking to officials in Beijing and elsewhere.**

e. Dr. Briand stays a few more days to get some answers

Dr. Sylvie Champaloux Briand (French) stayed a few more days in Beijing for talks with Chinese officials, and to try to get answers to practical epidemiological questions. She may have also started discussions on a list of Western experts to be allowed in China at the time. She was back in Geneva by the 5 Feb. As reported by the WSJ, describing Briand's experience, it took until mid-February to make arrangements and get the WHO team there.

In fact, that supposed access to China for the February 2020 mission was just a saving exercise for both the WHO and China: as we shall see later ['Key Insight: Saving Face' under 18.15], the WHO had to settle for a very short and limited visit of Wuhan, conceded only well into the February trip, while the rest of the time the team was kept about as far away from Wuhan as possible without making it look too absurd.

2.8 Tedros praises China further back in Geneva (29 Jan)

a. A parallel world

As reported in Chinese media at the time, on 29 Jan, just back from his trip to Beijing, Dr. Tedros piled the compliments high again during at press conference in Geneva, despite all the aggravations that the WHO had to endure while trying to get information from China since the beginning of the outbreak.

> *[Dr. Tedros said] that the just-concluded trip to Beijing saw the great contribution of the Chinese government.* ***After the outbreak, China immediately shared epidemic and virus information with the WHO and other countries****. Due to China's transparency and timely information on the epidemic, there are currently only 68 confirmed cases outside China. China's efforts in preventing and controlling this epidemic should be respected and appreciated.*
>
> *In response to questions from some media and the Internet, WHO Director-General Tedros Adhanom Ghebreyesus* ***said that he will praise China again and again****. China has helped prevent the spread of the virus. The understanding and investment of Chinese leaders in the epidemic are huge. The WHO calls on other countries to learn from China.*
>
> *"****For that, China deserves our gratitude and respect… China is implementing very serious measures and we cannot ask for more****," he said.*
>
> *source: China Daily: page 1 and page 2 (with infographics)*

b. Dr. Tedros' excessive praises start annoying some

These excessive praises were starting to upset some WHO member states, especially the US — the largest historical contributor to the WHO budget by far [footnote 70]. According to an excellent NY Times article published in Nov 2020,

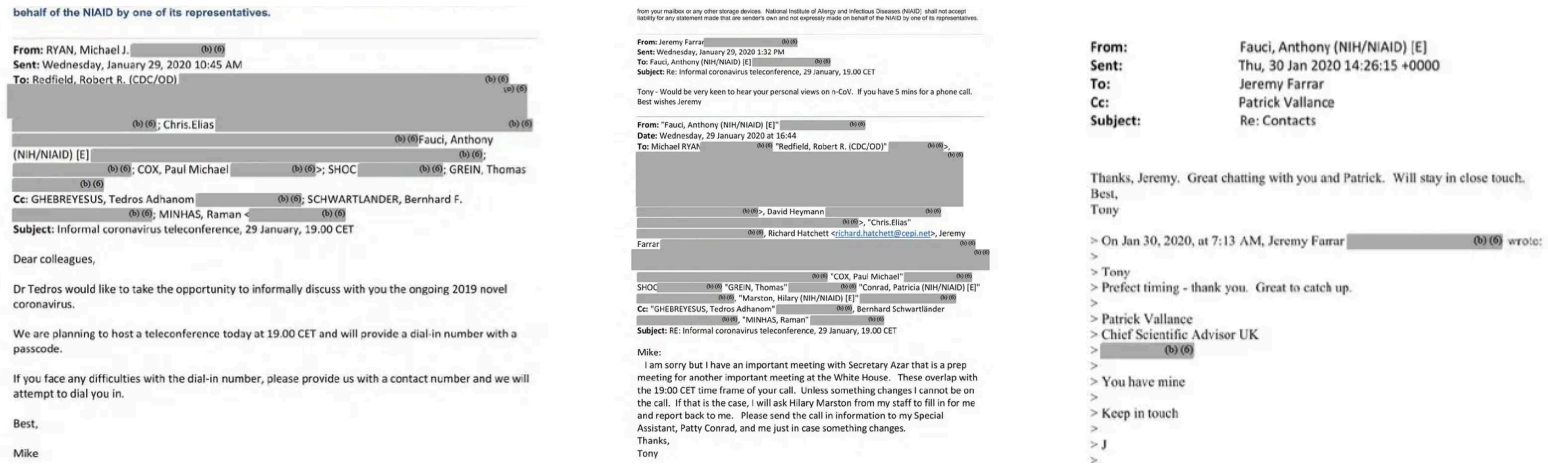
> *Back in Geneva, the American ambassador, Andrew Bremberg, urged Dr. Tedros to dial back praise of China.*
>
> *"You're risking your personal and organizational reputation," he told Dr. Tedros, according to several Western diplomats briefed on that conversation.*

2.9 WHO reaches out to Fauci, Farrar and Redfield (29–30 Jan)

On 29 Jan, Michael Ryan (WHO, just back from the trip to Beijing) emailed Redfield (US CDC Director), Fauci and Farrar, plus others, asking them if they could join an informal chat with Dr. Tedros about the outbreak situation. Fauci has another engagement, and offered to have a member of his staff fill in for him.

Bouncing on that email by Ryan, Farrar replied directly to Fauci (nobody in CC) by asking him if the two of them could have a chat to discuss his

'personal views on n-CoV'. The call happens on 30 Jan, between Fauci, Farrar and Patrick Vallance (also at the Wellcome Trust).



behalf of the NIAID by one of its representatives.

From: RYAN, Michael J. (b) (6)
Sent: Wednesday, January 29, 2020 10:45 AM
To: Redfield, Robert R. (CDC/OD) (b) (6) (b) (6); Chris.Elias (b) (6) (b) (6) Fauci, Anthony (NIH/NIAID) [E] (b) (6); (b) (6); COX, Paul Michael (b) (6)>; SHOC (b) (6); GREIN, Thomas (b) (6)
Cc: GHEBREYESUS, Tedros Adhanom (b) (6); SCHWARTLANDER, Bernhard F. (b) (6); MINHAS, Raman < (b) (6)
Subject: Informal coronavirus teleconference, 29 January, 19.00 CET

Dear colleagues,

Dr Tedros would like to take the opportunity to informally discuss with you the ongoing 2019 novel coronavirus.

We are planning to host a teleconference today at 19.00 CET and will provide a dial-in number with a passcode.

If you face any difficulties with the dial-in number, please provide us with a contact number and we will attempt to dial you in.

Best,

Mike

from your mailbox or any other storage devices. National Institute of Allergy and Infectious Diseases (NIAID) shall not accept liability for any statement made that are sender's own and not expressly made on behalf of the NIAID by one of its representatives.

From: Jeremy Farrar (b) (6)
Sent: Wednesday, January 29, 2020 1:32 PM
To: Fauci, Anthony (NIH/NIAID) [E] (b) (6)
Subject: Re: Informal coronavirus teleconference, 29 January, 19.00 CET

Tony - Would be very keen to hear your perscnal views on n-CoV. If you have 5 mins for a phone call.
Best wishes Jeremy

From: "Fauci, Anthony (NIH/NIAID) [E]" (b) (6)
Date: Wednesday, 29 January 2020 at 16:44
To: Michael RYAN (b) (6) "Redfield, Robert R. (CDC/OD)" (b) (6)>, (b) (6)>, David Heymann (b) (6) (b) (6)>, "Chris.Elias" (b) (6), Richard Hatchett <richard.hatchett@cepi.net>, Jeremy Farrar (b) (6) (b) (6) "COX, Paul Michael" (b) (6) SHOC (b) (6) "GREIN, Thomas" (b) (6) "Conrad, Patricia (NIH/NIAID) [E]" (b) (6), "Marston, Hilary (NIH/NIAID) [E]" (b) (6)
Cc: "GHEBREYESUS, Tedros Adhanom" (b) (6), Bernhard Schwartländer (b) (6), "MINHAS, Raman" (b) (6)
Subject: RE: Informal coronavirus teleconference, 29 January, 19.00 CET

Mike:
I am sorry but I have an important meeting with Secretary Azar that is a prep meeting for another important meeting at the White House. These overlap with the 19:00 CET time frame of your call. Unless something changes I cannot be on the call. If that is the case, I will ask Hilary Marston from my staff to fill in for me and report back to me. Please send the call in information to my Special Assistant, Patty Conrad, and me just in case something changes.
Thanks,
Tony

From: Fauci, Anthony (NIH/NIAID) [E]
Sent: Thu, 30 Jan 2020 14:26:15 +0000
To: Jeremy Farrar
Cc: Patrick Vallance
Subject: Re: Contacts

Thanks, Jeremy. Great chatting with you and Patrick. Will stay in close touch.
Best,
Tony

> On Jan 30, 2020, at 7:13 AM, Jeremy Farrar (b) (6) wrote:
>
> Tony
> Prefect timing - thank you. Great to catch up.
>
> Patrick Vallance
> Chief Scientific Advisor UK
> (b) (6)
>
> You have mine
>
> Keep in touch
>
> J
>

farrar-fauci-comms.pdf, p. 157–162, 29 Jan 2020

2.10 Dr. Tedros praises China while declaring a Health Emergency (30 Jan)

On 30 Jan 2020, the WHO eventually declared a health emergency (PHEIC). Dr. Tedros, in his statement, was particularly careful to frame this as not being '*a vote of no-confidence in China',* since this was what precisely what Beijing had complained about when successfully pushing back against such declaration on 23 Jan (see 2.4), and heaped praise on the country efforts, declaring them to be *'a new standard for outbreak response':*

> *The speed with which China detected the outbreak, isolated the virus, sequenced the genome and shared it with WHO and the world are very impressive, and beyond words. So is China's commitment to transparency and to supporting other countries.*
> *In many ways, China is actually setting a new standard for outbreak response. It's not an exaggeration.*
> *[...]*
> *Let me be clear: this declaration is not a vote of no confidence in China. On the contrary, WHO continues to have confidence in China's capacity to control the outbreak.*
> *As you know, I was in China just a few days ago, where I met with President Xi Jinping. I left in absolutely no doubt about China's commitment to transparency, and to protecting the world's people.*
> *[...]*
> *There is no reason for measures that unnecessarily interfere with international travel and trade. WHO doesn't recommend limiting trade and movement. We call on all countries to implement decisions that are evidence-based and consistent.*
>
> *source: Dr. Tedros, 30 Jan 2020, IHR Emergency Committee on Novel Coronavirus*

Source: YouTube

. . .

3. Concerns about the sequence (unspecified US scientists, MI5, Farrar, Holmes)

3.1 The Origin question makes a slow entry in the public sphere (last week of Jan)

a. Doubts enter the US media sphere

On 23 Jan (~5 pm EST), the **Daily Mail**, a publication with a large readership across its UK, US and Australian editions, published the very first article in the press that raised potential doubts as to the origin of the outbreak.

The fairly mild article quotes Richard Ebright stating that '**at this point there's no reason to harbour suspicions**', but it also explains that SARS-1 lab-accidents previously occurred in Beijing, and raises the perfectly valid point of the potential danger off-sourcing dangerous research to Chinese labs that may not have the required biosafety experience and practice, especially for animal work:

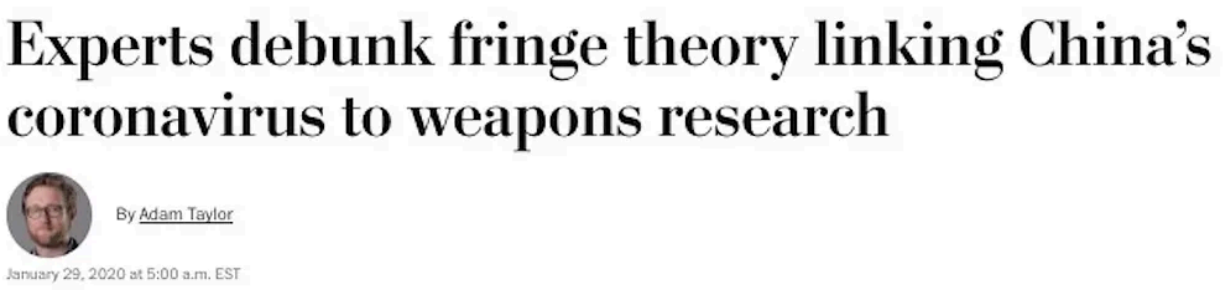
> ***Tim Trevan**, a Maryland biosafety consultant, told **Nature** that year, when the lab was on the cusp of opening, that he worried that China's culture could make the institute [the WIV] unsafe because 'structures where everyone feels free to speak up and openness of information are important.'*
>
> *In fact, the SARS virus had 'escaped' multiple times from a lab in Beijing, according to the Nature article. [..]*
>
> *Regulations for animal research — especially that conducted on primates — are much looser in China than in the US and other Western countries, meaning these studies are less costly and face fewer barriers that could limit or slow them.*
>
> *source: Daily Mail article, 23 Jan 2020 (~5 pm EST)*

On 26 Jan, the **Washington Times** ran a story that quotes a former Israeli Intelligence officer who hinted at the possibility of a link to a military research program, but conceded that this was speculative. Richard Ebright's quote in the Daily Mail about unnecessary suspicions was also repeated:

> *"Coronaviruses [particularly SARS] have been studied in the institute and are probably held therein," Mr. Shoham said. "SARS is included within the Chinese BW program, at large, and is dealt with in several pertinent facilities."*
>
> *It is not known whether the institute's coronaviruses are specifically included in China's biological weapons program but it is possible, he said.*
>
> *Asked whether the new coronavirus may have leaked, Mr. Shoham said: "In principle, outward virus infiltration might take place either as leakage or as an indoor unnoticed infection of a person that normally went out of the concerned facility. This could have been the case with the Wuhan Institute of Virology, but* ***so far there isn't evidence or indication for such incident****."*
>
> *Rutgers University microbiologist* ***Richard Ebright*** *told London's Daily Mail that "****at this point there's no reason to harbor suspicions****" that the lab may be linked to the virus outbreak."*
>
> *source: Washington Time article, 26 Jan 2020*

On 29 Jan, the **Washington Post** published a high-profile article dismissing the speculation about the Wuhan Institute of Virology that followed the Daily Mail and Washington Times. Richard Ebright was again cited, explaining that this was likely not an engineered virus:

> *But in conversations with The Washington Post, experts rejected the idea that the virus could be man-made.*
> *"Based on the virus genome and properties there is no indication whatsoever that it was an engineered virus," said* ***Richard Ebright****, a professor of chemical biology at Rutgers University.*
>
> ***Tim Trevan****, a biological safety expert based in Maryland, said most countries had largely abandoned their bioweapons research after years of work proved fruitless.*
> *"The vast majority of new, nasty diseases … come from nature," he said.*
>
> *source: Washington Post article, 29 Jan*



Experts debunk fringe theory linking China's coronavirus to weapons research

By Adam Taylor

January 29, 2020 at 5:00 a.m. EST

WaPo article, 29 Jan 2020

By the 29 Jan, there was no other notable article that had discussed a potential non-natural origin at that time. All that one can find is an article in Metro.UK on 24 Jan, which regurgitated the Daily Mail piece, and an article in an obscure Indian Weekly on 28 Jan, that drew the attention on the Chinese scientists that illegally shipped Ebola pathogens (BSL-4 level) to China, where the only BSL-4 lab at the time was the WIV. This was followed on 29 Jan by a blog entry by Shoham drawing attention to that Ebola shipping story. [A story that would eventually become a major embarrassment for Canada, as predicted by Shoham in his piece.]

Hence, one could say that after a brief mention of an unlikely research origin accident by three US dailies, there was really no public debate about the potential origins of the outbreak, so far in that last week of January.

b. Doubts enter the US political sphere

On 30 Jan (US time), Sen. Tom Cotton (R.AK), who had recently asked the President to block travellers coming from China, mentioned that the origin of the virus remains an open question. Even if his wording was actually rather measured, it may have been a first strong hint that the origin question was about to enter the US political arena.

In the climate of intense political divisions in the US, this could easily be perceived as the dawning of a tough game of political football, with all the distractions, pains, complications (domestic and abroad) that this may entail, especially if the subject got amplified in the media.

Sen Cotton's tweet, 30 Jan 2020 (US time)

c. Trump has other priorities and keeps praising China:
All the while, Pres. Trump was more focussed on closing a trade deal with China, and kept praising President Xi Jin-ping to try to achieve his trade objectives. In fact, Trump's conciliatory tone would not change **until mid-March 2020**, when it became clear that the US was being hit by the pandemic with the first recorded deaths there.

In early February, he was actually being more often criticised for being too conciliant with China and of ignoring a potential pandemic in the making.

Pres. Trump, 7 Feb 2020, Tweet

In March 2020, Politico would publish an article highlighting that sudden sharp pivot: '15 times Trump praised China as coronavirus was spreading across the globe'.

> *"Just had a long and very good conversation by phone with President Xi of China. He is strong, sharp and powerfully focused on leading the counterattack on the Coronavirus. He feels they are doing very well, even building hospitals in a matter of only days. Nothing is easy, but he will be successful, especially as the weather starts to warm & the virus hopefully becomes weaker, and then gone. Great discipline is taking place in China, as President Xi strongly leads what will be a very successful operation. We are working closely with China to help!*
> [Pres. Trump, 7 Feb 2020, Tweet]
>
> *"On January 31st, I imposed travel restrictions on foreign nations who had — and anybody that had been to China or people coming out of China. And I want to say that to be making tremendous progress. Their numbers are way down.*
> *[…]*
> *I think our relationship with China is very good. We just did a big trade deal. We're starting on another trade deal with China — a very big one. And we've been working very closely. They've been talking to our people, we've been talking to their people, having to do with the virus."*

> *source: Pres. Trump, Feb 29, press conference*

The very first sign of a potential change in Trump's priorities, from his industrial and economic policies to managing a health crisis at home, would appear as late as the **16 March**, when he tweeted about the "Chinese virus" — and still in the context of his support to US industries:

Trump's tweet — 16 Mar 2020, 6:51 pm EST

3.2 Baric lays out some concerns (29 Jan)

In his interview with the House Subcommittee on the Coronavirus Pandemic, Baric revealed that he had a 'BSEC meeting' (actually **BSEG**, see below) on 25 or 26 Jan 2020 [actual date: 29, as shown further below], during which he explained that there were three potential causes for the outbreak:

1. natural origin,
2. laboratory escape
3. genetic engineering.

Baric's 'potential causes' would have been better described as 'potential factors', since only (1) and (3) are exclusive, but we get the gist: what he meant is that it could be a standard zoonosis with a natural virus, a lab escape of a natural virus, or a lab escape of an engineered one.

2974 A I don't recall this. So because of this, I'm
2975 going to at least say one thing that I gave in the BSEC
2976 meeting on January 25th or 26th. My summary of the origin of
2977 the pandemic was the following.
2978 There are three potential causes for that pandemic. First is
2979 natural origin, second was laboratory escape, and the third
2980 was genetically engineered.
2981 Q And what was the date of that again?
2982 A January 25th or 26th of 2020. So I don't know

Baric-TI-Transcript.pdf, p.120–121

Just from the slight reluctance of Baric to mention it in his interview, we can infer that the BSEC/BSEG meeting was likely a classified one. What is clear is that, as early as 29 Jan, at least part of the government apparatus was given a very open analysis of the possible origins by Baric. **An analysis that was not any different from the one that Tom Cotton would give a few days later (see 3.1.b).** For context, Tom Cotton was at the time a member of the **Subcommittee on Emerging Threats and Capabilities**, within the Senate Armed Services Committee.

Update: BSEC stands most likely, with a transcription mistake for, a classified meeting of the BSEG Biological Sciences Experts Group, convened by the National Counterproliferation Center (NCPC) within the Office of the Director of National Intelligence (ODNI). Baric's CV explains that he is one of the members of the BSEG, a classified set-up where all members need to have security clearance [see **99**].

Latest Data Points:

As per the documents released on 30 Oct 2025 by Sen. Rand Paul, the presentation to the 'Sponsor' was on Wednesday 29 Jan 2020. **So Baric got it possibly wrong by 3 days in his Congress Testimony.** I corrected my text which was based on Baric's memory of 25 or 26 Jan.

'B Group' below is without any doubt BSEG as per the file name: 'BARIC BSEG Jan 2020.ppt'.

Message

From: Baric, Ralph S [redacted]
Sent: 1/29/2020 1:49:25 AM
To: [redacted]
CC: [redacted]
Subject: RE: B Group Next Week
Attachments: Baric BSEG Jan 2020.pptx

My presentation

From: [redacted]
Sent: Thursday, January 23, 2020 3:16 PM
To: Baric, Ralph S [redacted]
Cc: [redacted]
Subject: Re: B Group Next Week

Hi Ralph,

The Sponsor will be excited to hear that you are willing to lead this timely and important discussion.

It would be up to [redacted] for final decision on the site's ability to accept and position slides for Wednesday, but I believe that you sending them directly to [redacted] (you can just Reply All to this note) Tuesday night might work for a Wednesday presentation.

[redacted] do you concur?

Thanks,

[redacted]

From: Baric, Ralph S [redacted]
Sent: Thursday, January 23, 2020 3:01 PM
To: [redacted]
Subject: [EXTERNAL] RE: B Group Next Week

Very timely and appropriate. I was going to email this suggest on to you when I finally shed myself from all the reporters today. I can put together a few slides that capture the essence of the problem and I would be delighted to present on this topic. If I present for 15-20 mins on wed, when would you need the slides? Tuesday night? And who do I send them to? Thanks, ralph

From: [redacted]
Sent: Thursday, January 23, 2020 2:07 PM
To: Baric, Ralph S [redacted]
Cc: [redacted]
Subject: B Group Next Week

Hi Ralph,

3.3 Farrar learns about some US scientist doubts (before 27 Jan)

a. No 'US Scientists' doubts were public by 27 Jan

As we have just seen in 3.1, very few articles raising the rather remote possibility of a non-natural origin had appeared by 27 Jan; essentially just the Daily Mail and Washington Times articles. In particular, the engineering suspicion mentioned by Farrar ('*almost engineered to infect human cells*') does not correspond to any of these articles. **On the contrary, these articles insisted that there was no sign of a non-natural virus so far.**

b. Farrar's version: last week of Jan, but before 27 Jan

In *Spike*, Farrar explains that, during the last week of January, he nevertheless became privy to some email chatter from 'credible scientists' in the US, that suggested that the virus looked '**almost engineered to infect human cells**'.

Farrar then tuned to Eliza Manningham-Buller, Chair of the Wellcome Trust's Board of Governors, and before that, the head of MI5, the UK internal intelligence services (2002–2007), to discuss this chatter coming from US scientists. Since we know that this discussion resulted in him purchasing a

burner phone on 27 Jan (see 3.4), the discussion itself must have happened shortly before 27 Jan [footnote 14], so possibly shortly after the classified BSEG meeting during which Baric expressed his concerns.

c. Andersen's version: early January
Andersen later confirmed what Farrar wrote in S*p*ike, except for the time when Farrar is supposed to have heard of rumours involving the WIV: Andersen mentioned **early January**, instead of the last week of January.

For what it is worth, Farrar also explained in *Spike* that, in early January 2020, he had discussed the outbreak with the then head of MI5, Andrew Parker, who happened to be visiting ex-MI5 head Eliza Manningham-Buller at the Wellcome Trust. Whether this was the occasion when Farrar learnt of some 'chatter', or whether that happened only later, is not clear.

> *So Jeremy Farrar had heard rumors about a potential association with the Wuhan Institute of Virology. I don't know the details about that, but he had heard these rumors in early January 2020.*
> *And he reaches out to his old colleague and friend Eddie Holmes, who is also a colleague, a friend of myself, asks Eddie about these rumors because Eddie Holmes is one of the most respected scientists, has written books on — the primary books on viral emergence.*
>
> *Jeremy asks Eddie: Is there anything to these rumors? Eddie takes a quick look at the data and basically says to Jeremy: No, I don't think so. And then leaves it there.*
>
> *source: Andersen's Transcript, p. 51*

c. Reconciling the two versions
A simple explanation to these two versions is that some initial doubts were raised by some 'US scientists' when the genome was published, so shortly after the 12 Jan, leading to the classified BSEG meeting that Baric attended on 29 Jan.

News of both could have reached Farrar via his MI5 contacts, themselves kept in the loop by US intelligence, who could very logically have asked their top 5-eyes partner to also quietly look into this.

d. Who are the 'US scientists' behind the early January warnings?
It is not clear who these 'US scientists' behind the early January warnings were. One possible clue may nevertheless be given by CDC Director Robert Redfield, who testified that he started looking into the possibility of a non-natural origin in the few days after the sequence was published, and was then reaching out to try to get some interest in the issue. For what it is worth,

Redfield even stated he '**was the one who brought it to them**', meaning 'to the Farrar call group'.

Redfield further stated that he had also contacted Tedros and Fauci (who denied it) to ask them to properly look into the two origin hypotheses, based on science. However, Redfield is not mentioned at all in Farrar's *Spike*.

3.4 Farrar gets some advice from MI5 (27 Jan)

a. Enters MI5

After he got privy to these origin concerns, Farrar explains that he contacted Eliza Manningham-Buller (the previous head of MI5), for advice. Soon after talking to her, on 27 Jan, Farrar requested some burner phone.

> *27 Jan 202, at 11:59 [UK time]*
> *Special request!*
>
> *Can I get a second phone today? Separate number, need to have one separate to my existing Wellcome one, I hope just for 3–6 months — can explain when we meet*
>
> *source: Jeremy Farrar, as recorded in his book, Spike*

b. A discreet external evaluation

What is sure is that if these 'US scientists' had any doubts, they did not make them public until much later. Likewise, it is clear that Farrar decided not to talk directly to these scientists, or even to simply ask for a quick summary of their findings, which would have seemed to be the perfectly natural thing to do.

But the exclusion of the US side is actually what would be expected if the Farrar call was at the request, or with the guidance, of a British intelligence agency. An agency that was aware of the non-publicised doubts of some US scientists, and quite possibly of some of their frustrations too, as Redfield would later explain that these concerns were being ignored at the time in the US. An agency that still needed to perform an independent check, and a quiet one, so that this would not turn into a diplomatic mêlée.

In that case, one approach would be for the MI5 to get some external evaluation by reaching out to some trusted domain expert in some trusted organisation, such as Wellcome Trust, on top of getting an internal evaluation via its specialists in Porton Down.

Farrar would be the perfect person for this, with express instructions to keep it quiet and away from the US scientists that may have been involved in such work (because of obvious conflicts of interest), and from the US individuals pushing a possible non-natural origin (because they could be tempted to

instrumentalise this evaluation by their main Five Eyes partner, and make it public, to buttress their point).

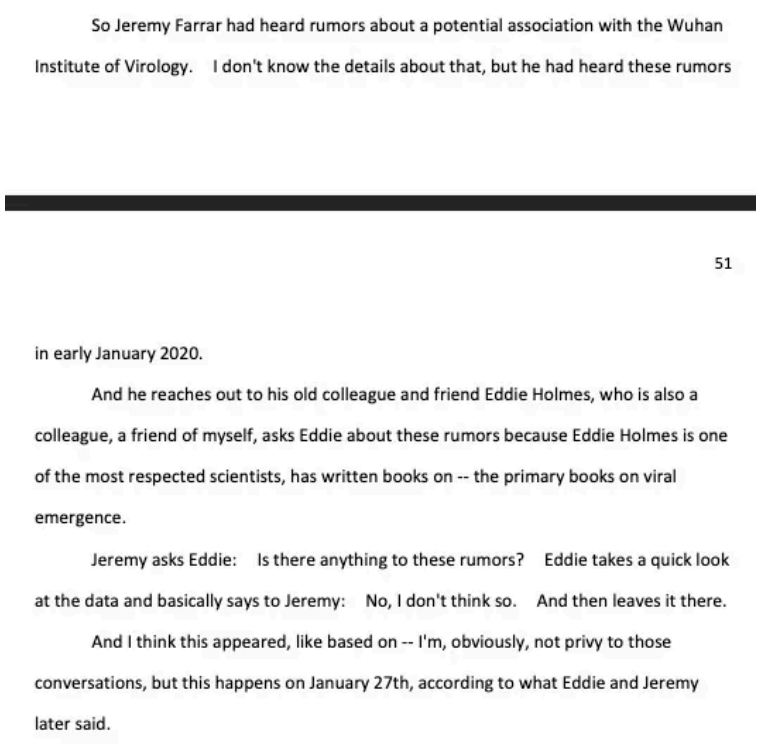
In fact, Andersen reached the same conclusion in his testimony: going back to the mention of Farrar hearing rumours about a possible association with the Wuhan Institute of Virology, **he explains that he believes that the rumours originated from the UK intelligence community**.

> *And I think this comes from the fact that, again, I think when Jeremy heard these,* ***quote, unquote, rumors, I believe it's the intelligence community in the United Kingdom****, he's talked about burner phones in his book.*
> ***So I believe Jeremy was working at a level in which he didn't want interference, potential interference****.*
>
> *source: Andersen's Transcript, p. 51*

3.5 Farrar talks to Holmes on his new burner phone (27 Jan)

On 27 Jan (UK time), Farrar emailed Eddie Holmes with whom he had already worked to force the release of the sequence on 9–10 Jan (see 1.2.c), requesting a chat on his new burner phone about the possibility that the virus may be a lab-escape.

Holmes took a brief look at the data and told Farrar that he could not see anything suspicious.



So Jeremy Farrar had heard rumors about a potential association with the Wuhan Institute of Virology. I don't know the details about that, but he had heard these rumors

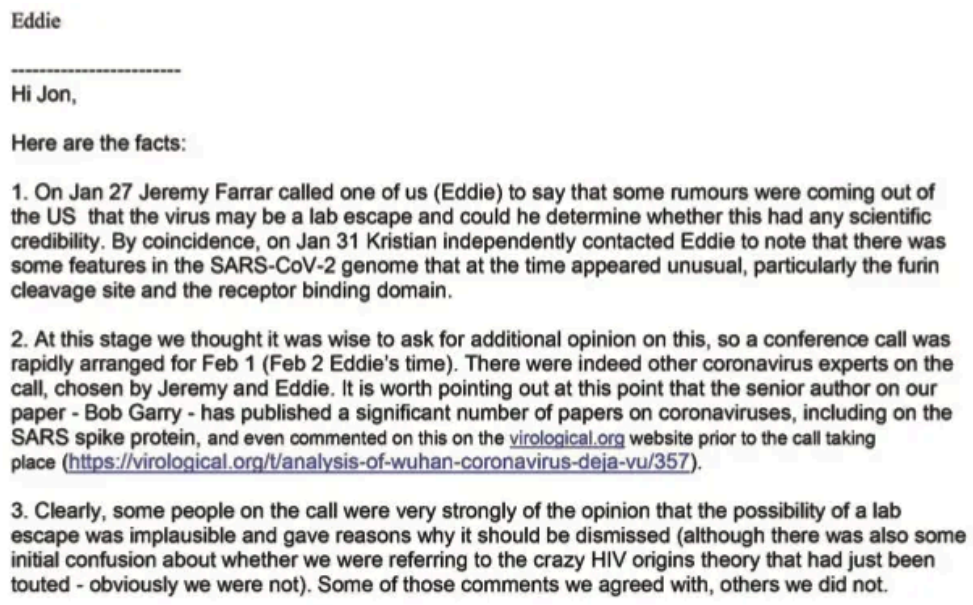
51

in early January 2020.

And he reaches out to his old colleague and friend Eddie Holmes, who is also a colleague, a friend of myself, asks Eddie about these rumors because Eddie Holmes is one of the most respected scientists, has written books on -- the primary books on viral emergence.

Jeremy asks Eddie: Is there anything to these rumors? Eddie takes a quick look at the data and basically says to Jeremy: No, I don't think so. And then leaves it there.

And I think this appeared, like based on -- I'm, obviously, not privy to those conversations, but this happens on January 27th, according to what Eddie and Jeremy later said.



Eddie

Hi Jon,

Here are the facts:

1. On Jan 27 Jeremy Farrar called one of us (Eddie) to say that some rumours were coming out of the US that the virus may be a lab escape and could he determine whether this had any scientific credibility. By coincidence, on Jan 31 Kristian independently contacted Eddie to note that there was some features in the SARS-CoV-2 genome that at the time appeared unusual, particularly the furin cleavage site and the receptor binding domain.

2. At this stage we thought it was wise to ask for additional opinion on this, so a conference call was rapidly arranged for Feb 1 (Feb 2 Eddie's time). There were indeed other coronavirus experts on the call, chosen by Jeremy and Eddie. It is worth pointing out at this point that the senior author on our paper - Bob Garry - has published a significant number of papers on coronaviruses, including on the SARS spike protein, and even commented on this on the virological.org website prior to the call taking place (https://virological.org/t/analysis-of-wuhan-coronavirus-deja-vu/357).

3. Clearly, some people on the call were very strongly of the opinion that the possibility of a lab escape was implausible and gave reasons why it should be dismissed (although there was also some initial confusion about whether we were referring to the crazy HIV origins theory that had just been touted - obviously we were not). Some of those comments we agreed with, others we did not.

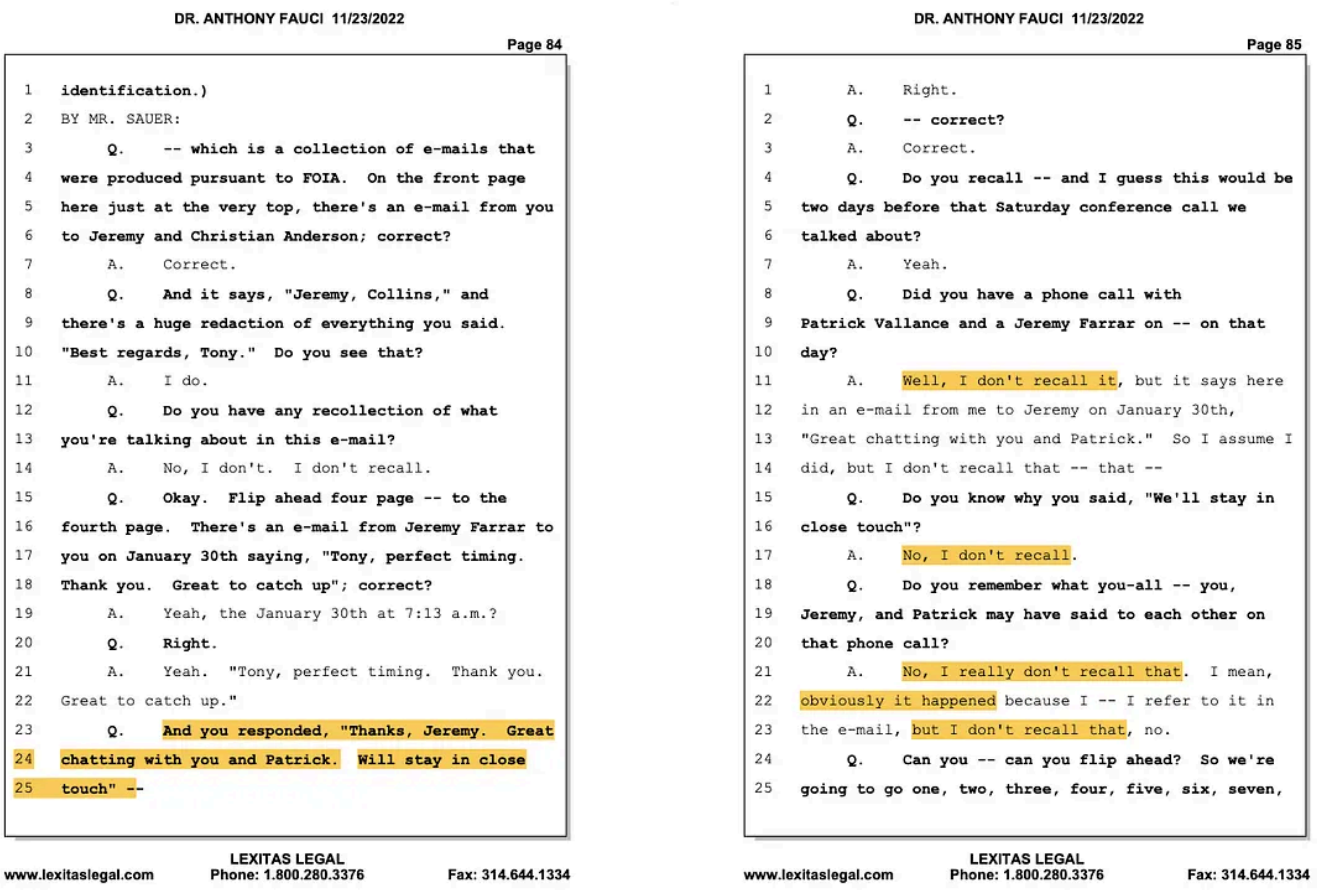
2023.06.16-Andersen-Transcript.pdf, p. 50–51, and Proximal_Origin_Emails.pdf, p. 149, 28 Jul

3.6 Farrar and Vallance talk to Fauci (30 Jan)

On 30 Jan, around 2 pm UK time, early morning US East Coast time, Farrar and Vallance (UK Government Chief Scientific Adviser (GCSA)) had the short call with Fauci that had been programmed on 29 Jan (see 2.9). Farrar then gave Vallance's contact details to Fauci (who already had his) so that they may stay in touch.

It is not clear what that call was about. At this stage, Andersen had not yet contacted Holmes, and Holmes had already replied to Farrar that he could see nothing suspicious. Possibly, that was actually the message that Farrar fed back to Fauci, or it was something totally unrelated.

In a deposition given on 23 Nov 2022 (The State of Missouri, *et al.* vs. Joseph R. Biden), Fauci would declare that he had no memory at all of this call, even if it was clear to him that it happened.



DR. ANTHONY FAUCI 11/23/2022 Page 84

identification.)
BY MR. SAUER:
Q. -- which is a collection of e-mails that were produced pursuant to FOIA. On the front page here just at the very top, there's an e-mail from you to Jeremy and Christian Anderson; correct?
A. Correct.
Q. And it says, "Jeremy, Collins," and there's a huge redaction of everything you said. "Best regards, Tony." Do you see that?
A. I do.
Q. Do you have any recollection of what you're talking about in this e-mail?
A. No, I don't. I don't recall.
Q. Okay. Flip ahead four page -- to the fourth page. There's an e-mail from Jeremy Farrar to you on January 30th saying, "Tony, perfect timing. Thank you. Great to catch up"; correct?
A. Yeah, the January 30th at 7:13 a.m.?
Q. Right.
A. Yeah. "Tony, perfect timing. Thank you. Great to catch up."
Q. And you responded, "Thanks, Jeremy. Great chatting with you and Patrick. Will stay in close touch" --

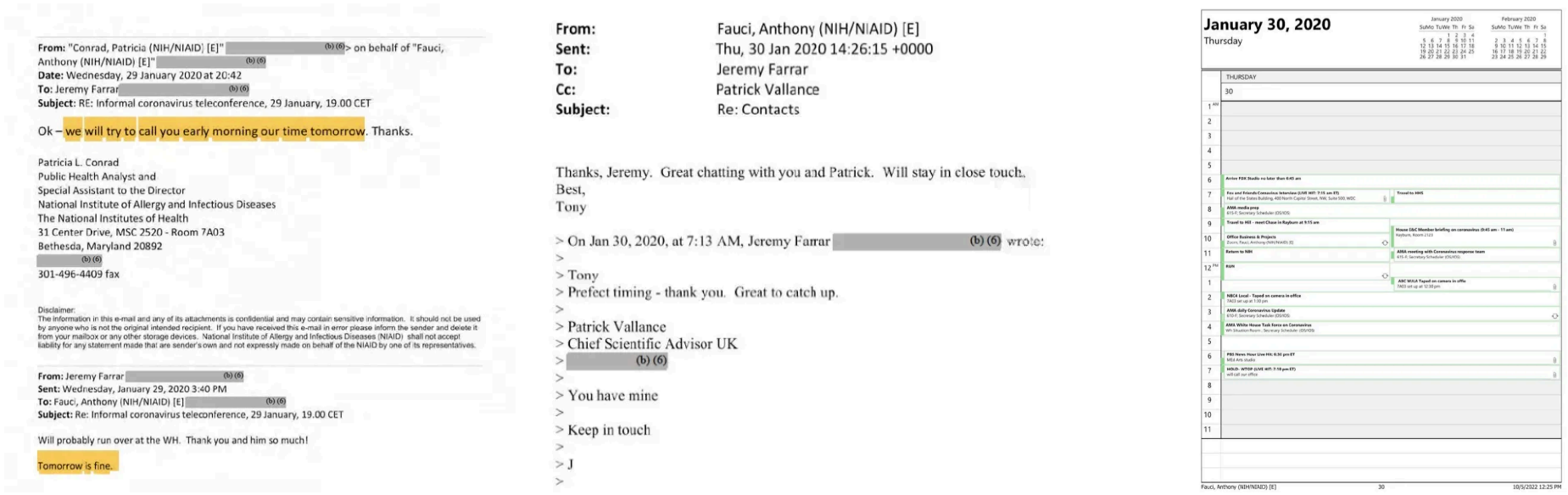
LEXITAS LEGAL
www.lexitaslegal.com Phone: 1.800.280.3376 Fax: 314.644.1334

DR. ANTHONY FAUCI 11/23/2022 Page 85

A. Right.
Q. -- correct?
A. Correct.
Q. Do you recall -- and I guess this would be two days before that Saturday conference call we talked about?
A. Yeah.
Q. Did you have a phone call with Patrick Vallance and a Jeremy Farrar on -- on that day?
A. Well, I don't recall it, but it says here in an e-mail from me to Jeremy on January 30th, "Great chatting with you and Patrick." So I assume I did, but I don't recall that -- that --
Q. Do you know why you said, "We'll stay in close touch"?
A. No, I don't recall.
Q. Do you remember what you-all -- you, Jeremy, and Patrick may have said to each other on that phone call?
A. No, I really don't recall that. I mean, obviously it happened because I -- I refer to it in the e-mail, but I don't recall that, no.
Q. Can you -- can you flip ahead? So we're going to go one, two, three, four, five, six, seven,

LEXITAS LEGAL
www.lexitaslegal.com Phone: 1.800.280.3376 Fax: 314.644.1334

Fauci's testimony of 23 Nov 2022, p. 84–85



From: "Conrad, Patricia (NIH/NIAID) [E]" (b) (6)> on behalf of "Fauci, Anthony (NIH/NIAID) [E]" (b) (6)
Date: Wednesday, 29 January 2020 at 20:42
To: Jeremy Farrar (b) (6)
Subject: RE: Informal coronavirus teleconference, 29 January, 19.00 CET

Ok – we will try to call you early morning our time tomorrow. Thanks.

Patricia L. Conrad
Public Health Analyst and
Special Assistant to the Director
National Institute of Allergy and Infectious Diseases
The National Institutes of Health
31 Center Drive, MSC 2520 - Room 7A03
Bethesda, Maryland 20892
(b) (6)
301-496-4409 fax

Disclaimer:
The information in this e-mail and any of its attachments is confidential and may contain sensitive information. It should not be used by anyone who is not the original intended recipient. If you have received this e-mail in error please inform the sender and delete it from your mailbox or any other storage devices. National Institute of Allergy and Infectious Diseases (NIAID) shall not accept liability for any statement made that are sender's own and not expressly made on behalf of the NIAID by one of its representatives.

From: Jeremy Farrar (b) (6)
Sent: Wednesday, January 29, 2020 3:40 PM
To: Fauci, Anthony (NIH/NIAID) [E] (b) (6)
Subject: Re: Informal coronavirus teleconference, 29 January, 19.00 CET

Will probably run over at the WH. Thank you and him so much!

Tomorrow is fine.

From: Fauci, Anthony (NIH/NIAID) [E]
Sent: Thu, 30 Jan 2020 14:26:15 +0000
To: Jeremy Farrar
Cc: Patrick Vallance
Subject: Re: Contacts

Thanks, Jeremy. Great chatting with you and Patrick. Will stay in close touch.
Best,
Tony

> On Jan 30, 2020, at 7:13 AM, Jeremy Farrar (b) (6) wrote:
>
> Tony
> Prefect timing - thank you. Great to catch up.
>
> Patrick Vallance
> Chief Scientific Advisor UK
> (b) (6)
>
> You have mine
>
> Keep in touch
>
> J
>

January 30, 2020
Thursday

farrar-fauci-comms-full.pdf p.157, then p 158, 30 Jan (2 pm, UK time) and nih-foia-request-60081and-59096-march-production_redactedpdf-tobias-foia.pdf, p.219

3.7 Andersen is independently asked to check the sequence (30 Jan)

On 30 Jan 2020, the day after the Washington Post published its article 'debunking' a possible non-natural origin (see 3.1), Kristian Andersen was asked by the chairman of his institution (Scripps) to check the SARS-CoV-2 sequence for any non-natural origin clue. Andersen started spotting some potential issues, such as the Furin Cleavage Site (FCS). Not being a coronavirus expert, Andersen decided to contact Holmes, a friend and top evolutionary expert with whom he had published before.

I get prompted -- I have been looking at just the genome in general but mostly focused on just understanding the virus. I hadn't really consider the origin question yet. I get prompted to do that a little bit more specifically by my chairman on the 30th of January.

I take a quick look at it, and I'm aware of some of the work at the Wuhan Institute of Virology, but I don't really see anything sort of that concerns me. But I decide to take a closer look after being prompted by, again, my chairman at the institution.

I do so and I start seeing things that concerns me, the furin cleavage site, for example, but also just a few things that to me looked unusual at the time.

What's really important to understand here is that I'm an expert on viral evolution and emergence. I study the emergence of viruses. But I'm not a coronaviruses expert. I certainly wasn't at the time. I had not studied coronaviruses previously in any great detail. So there were many things around just the diversity of coronaviruses that was unfamiliar to me at the time.

2023.06.16-Andersen-Transcript.pdf, p. 51

3.8 Andersen expresses his concerns to Holmes (31 Jan)

On 31 Jan (US time), Eddie, then back to Sydney from a ski-holiday cum conference in Switzerland, opened an email from Andersen, who wanted to have a chat. They soon got on a call during which Andersen pointed out three issues with the genome:

1. The RBD section of the spike protein looks perfect.
2. There is a Furin Cleavage Site in a key position on the spike protein.
3. The WIV has been doing exactly the kind of RBD and spike manipulations that could explain these features.

3.9 Holmes feeds the info back to his 'handler' (31 Jan)

During that first call with Andersen, Holmes did not mention Farrar by name. Instead, Holmes told Andersen that he had to get back to his '**handler**' first before seeing how he could work with Andersen, and have a new shot at the review that Farrar has previously asked him to do.

> *'He does not name — he doesn't mention Jeremy's name on this initial phone call I have with Eddie. In fact, I think he calls him his handler. And I think this comes from the fact that, again, I think when Jeremy heard these, quote, unquote, rumors, I believe it's the intelligence community in the United Kingdom, he's talked about burner phones in his book.*
> *So I believe Jeremy was working at a level in which he didn't want interference, potential interference.'*
>
> *source: Andersen's transcript, p. 52*

He does not name -- he doesn't mention Jeremy's name on this initial phone call I have with Eddie. In fact, I think he calls him his handler. And I think this comes from the fact that, again, I think when Jeremy heard these, quote, unquote, rumors, I believe it's the intelligence community in the United Kingdom, he's talked about burner phones in his book. So I believe Jeremy was working at a level in which he didn't want interference, potential interference.

So Eddie contacts Jeremy. Jeremy then contacts me the next day, and that's when I realize it's Jeremy Farrar, because, again, I didn't know that based on the conversation with Eddie. We talk about my concerns and talk about, like, look, we need to get a group together to discuss these concerns that I have, and we need to select people that are unconflicted. So we did.

2023.06.16-Andersen-Transcript.pdf, p. 52

'**Handler**' is a very specific word. In Human Intelligence (HUMINT), a **handler** is a case officer who is responsible for handling agents. An **agent**, by opposition to an intel **officer**, is an individual who works for or has been recruited by an intel officer, but who is not (in the formal sense) employed by the intelligence agency of the intel officer. Here Farrar is acting as a handler, and Holmes as the agent. As is normal in that case, Farrar has also asked Holmes to keep his identity confidential, unless told otherwise.

Hence, Andersen's use of the word 'handler' confirms that the British intel services, most likely MI5, likely tasked Farrar with reaching out to external subject-matter experts, for an independent advice (beyond their own experts at Porton Down).

That intel context explains perfectly:

1. Farrar acquiring a burner phone after talking to a former head of MI5 to discuss some origins concerns,
2. the oddity of not contacting the supposed US scientists who initially had these concerns, and
3. Andersen's understanding of the situation as written down in his testimony.

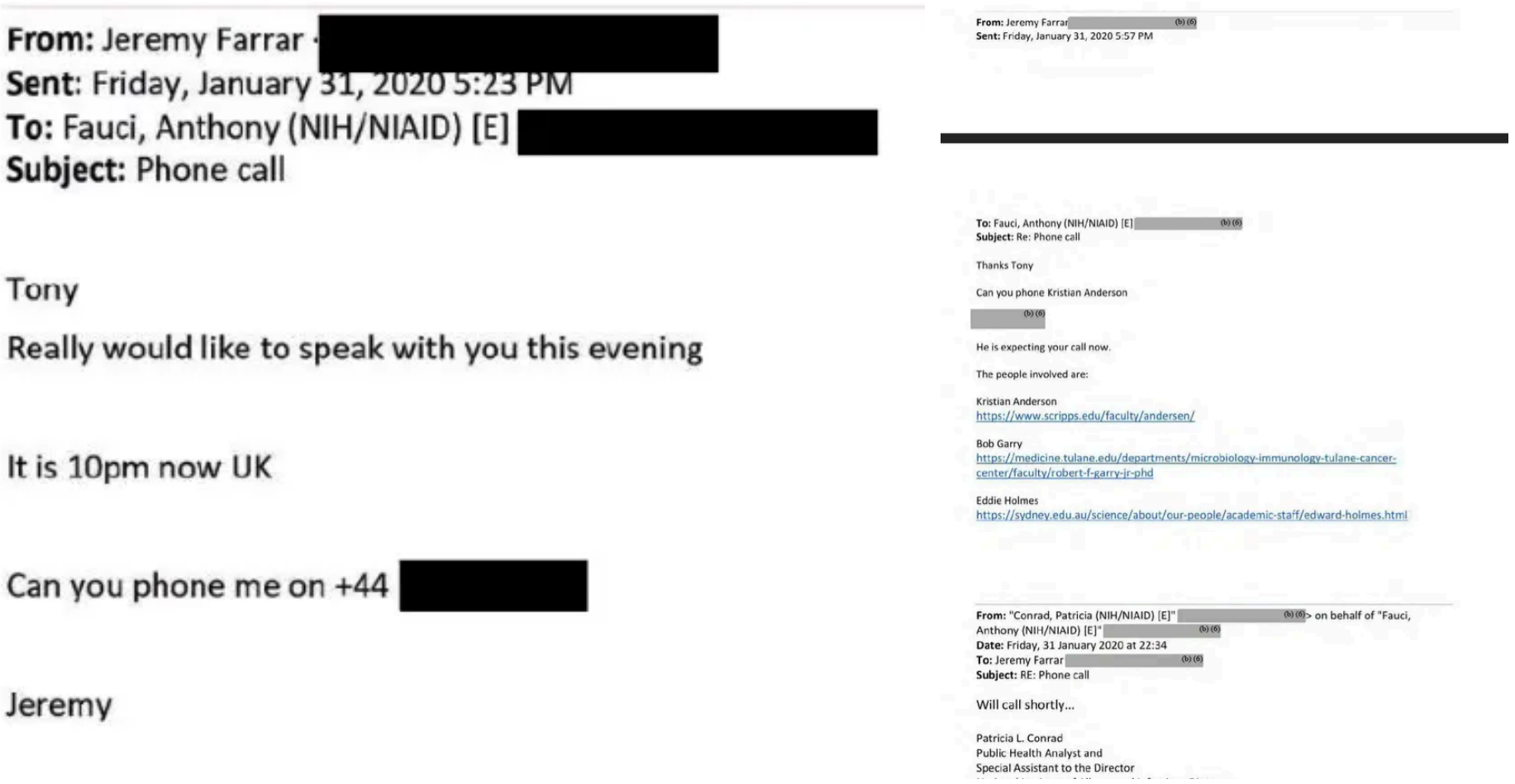
3.10 Garry joins Andersen and Holmes — Farrar updates Fauci (31 Jan, ~6 pm EST)

Farrar emailed Fauci on 31 Jan, saying that he would really like to talk to him quickly this 'evening' (it is already 10 pm in the UK). He likely gave him his burner phone number.

The expressed urgency of the call need to be considered against the fact that only the day before (30 Jan, 9 am EST), Farrar and Vallance already had a call

with Fauci. Obviously, Farrar wanted to let Fauci know about Andersen's concerns that Holmes had relayed to him in-between.

The call happened. The next day, Jeremy sent to Fauci the profiles of Andersen (misspelt), Garry (a regular collaborator of Andersen) and Holmes, describing them as the 'people involved'. He also asked Fauci to call Andersen, who was ready to take the call.



From: Jeremy Farrar [redacted]
Sent: Friday, January 31, 2020 5:23 PM
To: Fauci, Anthony (NIH/NIAID) [E] [redacted]
Subject: Phone call

Tony

Really would like to speak with you this evening

It is 10pm now UK

Can you phone me on +44 [redacted]

Jeremy

From: Jeremy Farrar (b) (6)
Sent: Friday, January 31, 2020 5:57 PM

To: Fauci, Anthony (NIH/NIAID) [E] (b) (6)
Subject: Re: Phone call

Thanks Tony

Can you phone Kristian Anderson

(b) (6)

He is expecting your call now.

The people involved are:

Kristian Anderson
https://www.scripps.edu/faculty/andersen/

Bob Garry
https://medicine.tulane.edu/departments/microbiology-immunology-tulane-cancer-center/faculty/robert-f-garry-jr-phd

Eddie Holmes
https://sydney.edu.au/science/about/our-people/academic-staff/edward-holmes.html

From: "Conrad, Patricia (NIH/NIAID) [E]" (b) (6)> on behalf of "Fauci, Anthony (NIH/NIAID) [E]" (b) (6)
Date: Friday, 31 January 2020 at 22:34
To: Jeremy Farrar (b) (6)
Subject: RE: Phone call

Will call shortly...

Patricia L. Conrad
Public Health Analyst and
Special Assistant to the Director

Proximal_Origin_Emails.pdf, p.2–3, 31 Jan 2020

3.11 Andersen briefs Fauci (31 Jan, ~7 pm EST)

On 31 Jan afternoon (~7 pm EST, ~4 pm CA), Andersen discussed his doubts about the sequence with Fauci, and goes over the FCS, the RBD and a few other oddities.

During the call with Andersen, Fauci suggested to actually make these doubts public by writing a peer-reviewed-paper. Andersen very logically answered that it was too early to do so, and that further analysis by a group of scientists was needed. Only Holmes (who turned up so many origins papers) seemed to be thinking of publishing at that time.

And Jeremy gets all of this set up. He, I'm sure, has been in touch with Tony Fauci at the time, reaches out to Dr. Fauci, asks him to call me.

I have a quick phone call with Dr. Fauci. I sort of relay what I have found and what the early views of this is, round receptor-binding domain, the furin cleavage site, a few other features.

And he, Dr. Fauci, specifically mentions the furin cleavage site in his email to me following that telephone call, but he also makes it clear that he will contact the intelligence community and sort of run it up the chain in the United States Government. I don't know the result and outcomes of any of those other than the White House Office of Science and Technology Policy did establish a meeting on February 3rd to look into this as well.

But following that call, essentially what I recall Dr. Fauci saying is to the effect of -- and I'm paraphrasing him here -- but basically saying that, look, if you think this came from a lab, then you should really consider writing a paper on it so it can be peer reviewed and judged by the community.

And I relayed back to Dr. Fauci that I feel that it's too premature. We need to look at this more closely. We can't just publish a paper just based on no analysis at the time. And, of course, he understands that and looks forward to the conference call in which scientists can get a chance to discuss my findings.

And that's the purpose of the conference call, which, again, is completely organized by Jeremy. Jeremy runs the call. I honestly don't remember Drs. Fauci or Collins even chiming in on the call itself. I'm sure they probably had questions. But this is not their area of expertise and they were just there to hear the scientists discuss what I had brought up. And the discussion is primarily between myself and Andy Rambaut, Eddie Holmes, Christian Drosten, Ron Fouchier, and Marion Koopsman. Those are the main factors I remember here.

At that time, as I had relayed, was that this was not for the purpose of drafting a paper. I think Eddie was probably more focused on a paper already at that stage. I was more like, look, I just want to talk about this, and then I want to do some more analyses, and I want to look into this closer.

2023.06.16-Andersen-Transcript.pdf, p.53–54

3.12 Fauci suggest a call with a larger group of experts (31 Jan, 7:38 pm EST)

a. Fauci wants a larger group of experts to look at the genome issues
Following his call with Andersen, at 7:38 pm EST, Fauci immediately got back to Farrar and explained that Farrar should convene a larger group of experts to look into the question raised by the FCS. He also recommended that this should be done without delay. In CC were Andersen, the Director of the NIH Vaccine Research Center who was also principal advisor to Fauci on vaccines and biomedical research affairs (John Mascola), and Fauci's special assistant (Patricia Conrad).

Note that until that precise moment (i) the 'people involved' (as per Farrar's wording in 3.10) were Holmes, Andersen, and his collaborator Garry, and that (ii) there had been no mention of any call with more scientists. In fact, a larger review call might not have been very logical for Farrar; he needed to keep the review group tight after having been advised (or guided) by MI5. That's what a handler' does.

On his side, Farrar could not really oppose Fauci's idea. From the next suggestion of Fauci, we can see that Farrar did not explain the intelligence dimension of his tight group of experts. And in any case, having more people to review Andersen's concerns could genuinely help.

From: Fauci, Anthony (NIH/NIAID) [E]
Sent: Sat, 1 Feb 2020 00:38:35 +0000
To: Jeremy Farrar
Cc: Kristian G. Andersen
Bcc: Conrad, Patricia (NIH/NIAID) [E];Mascola, John (NIH/VRC) [E];Conrad, Patricia (NIH/NIAID) [E]
Subject: RE: Phone call

Jeremy:
I just got off the phone with Kristian Anderson and he related to me his concern about the Furine site mutation in the spike protein of the currently circulating 2019-nCoV. I told him that as soon as possible he and Eddie Holmes should get a group of evolutionary biologists together to examine carefully the data to determine if his concerns are validated. He should do this very quickly and if everyone agrees with this concern, they should report it to the appropriate authorities. I would imagine that in the USA this would be the FBI and in the UK it would be MI5. It would be important to quickly get confirmation of the cause of his concern by experts in the field of coronaviruses and evolutionary biology. In the meantime, I will alert my US. Government official colleagues of my conversation with you and Kristian and determine what further investigation they recommend. Let us stay in touch.
Best regards,
Tony

From: Jeremy Farrar [redacted]
Subject: Re: Stuff 6:31 am UK
Date: 7 February 2020 at 5:31:44 pm AEDT
To: Edward Holmes [redacted]

I will be neutral.

Anyone from China?

Tomorrow morning fine.

Any preference for journal? All will take immediately, I can let them know coming if helpful and you have a preference

With revisions – will share with the TC group over the weekend – if OK – got to add the new info

farrar-fauci-comms-full.pdf p.154, **just past midnight on 1 Feb UK time (**31 Jan 2020 US time) showing who was in BCC on that call. and Proximal_Origin_Emails.pdf, p.49, 7 Feb 2020

b. Fauci recommends contacting the FBI and MI5 if the issues are confirmed

At the same time, Fauci told Farrar that if, following that call, there was some agreement that Andersen's concerns were valid, then the FBI and MI5 would need to be quickly alerted. He also mentioned that, meanwhile, he would check with US Gov. officials what the best way to proceed from there would be.

We need to keep in mind that Fauci would he be necessarily aware of the earlier concerns of some 'US scientists', nor that these concerns had reached MI5, which then tasked Farrar, or at least encouraged him, with doing an independent external investigation with a focus on the type of work done at the WIV. Farrar could have simply told Fauci that Andersen's elements of analysis should be of interest to intelligence services, without much more precision. From there, the way Fauci mentions the FBI and MI5 in his email following the call with Andersen, would make sense.

3.13 Constitution of the 'bounce-off' group (31 Jan–1 Feb morning)

As just mentioned, Farrar had no real choice but to agree with Fauci's suggestion for a larger group of experts to review Andersen's findings, while trying to keep the discussions tight and confidential at this time.
What Farrar did was to split the call group between his existing inner group of experts and a new adjunct group, against which the inner group could bounce off their ideas.

- The '**inner group**' (the final co-authors) was the original Holmes, Andersen, Garry, with Rambaut already in the background, to whom only Lipkin would be much later added after being vetted by Farrar.
- The '**bounce-off**' group of scientists was quickly put together by Farrar, to answer Fauci's request. Farrar added Fouchier and Drosten at the suggestion of his wife (who is a professor of tropical diseases at Oxford Uni), and then Koopmans who was already advising the WHO, while Andersen added Farzan. None of them would appear as co-author of P.O.

So, about half an hour after Fauci's email, at 8:05 pm EST, Andersen replied to Fauci and explained that his Scripps colleague Farzan, plus possibly Drosten and Fouchier, would join the sequence analysis group.

> *'Jeremy will reach out to Christian Drosten and Ron Fouchier in the morning to get their expertise as well. Combined, this group will be able to objectively assess the available data and determine whether the genome looks unusual'.*
>
> *source: Proximal_Origin_Emails.pdf, p.1, 31 Jan 2020 8:05 pm EST*

Rambaut was already in the background as a good friend of Holmes who had been instrumental in releasing the virus sequence on his virological.org website on 11 Jan.

Koopmans was also added by Farrar. She was already involved with the WHO, and had taken part in the two IHR sessions leading to the eventual declaration of a Public Health Emergency of International Concern (PHEIC) only the day before (30 Jan).

For context, Fouchier and Daszak (as well as Baric) were already part of a group of NIH Principal Investigators (and thus recipients of NIH grants) that was put together by the NIAID to discuss the unfolding outbreak, with a first online meeting on 14 Jan the moving to weekly meetings on 28 Jan.

From: Degrace, Marciela (NIH/NIAID) [E] <marciela.degrace@nih.gov>
Sent: Thursday, January 23, 2020 23:32
To: Webby, Richard; malik; Ghazi Kayali; Yoshi Kawaoka; R.A.M. Fouchier; 'adolfo.garcia-sastre@mssm.edu'; Richard Rothman; Pekosz, Andrew S.; Schultz-Cherry, Stacey; 'david_topham@urmc.rochester.edu'; Orenstein, Walter; Lowen, Anice; Baric, Ralph; 'Perlman, Stanley'; daszak@ecohealthalliance.org; zhuhch; Aubree Gordon; Munster, Vincent (NIH/NIAID) [E]
Cc: Post, Diane (NIH/NIAID) [E]; Embry, Alan (NIH/NIAID) [E]; Lampley, Rebecca (NIH/VRC) [F]; Stemmy, Erik (NIH/NIAID) [E]
Subject: Wuhan - scheduling weekly investigator calls

Hi everyone,

Given the escalating rate of infections with the new coronavirus, we would like to plan a **weekly meeting** with investigators working on coronaviruses. Our hopes is that we can all share information we have regarding research and sample progress on the virus, and we at NIAID can share any information and resource updates with all of you.

I know everyone's schedules are quite busy at this point. Given the time zones everyone works in we are thinking **Tuesdays at 9am** or **Wednesdays at 9am.** Please let me know by Friday if you have any strong preference, and we will get an invitation and call in sent out for next week.

Thank you!

Marciela and Erik

Baric-Emails-2.17.21.pdf, p.102

In the end, the new 'bounce-off' group was largely made up of European nationals, while the 'inner group' was unchanged. As Andersen later explained, both Baric and Daszak were discussed but officially excluded from the call group of experts, as having co-published with the WIV, since this would be an obvious conflict of interest and would additionally likely cause 'potential interference'.

And I sort of explained to Eddie that, look, we can't just dismiss this possibility out of hand because of the type of work that was ongoing at the Wuhan Institute of Virology and knew about biosafety levels. And I talked to Eddie here, and he sort of realizes that his early dismissal of this was probably not the correct thing to do.

He does not name -- he doesn't mention Jeremy's name on this initial phone call I have with Eddie. In fact, I think he calls him his handler. And I think this comes from the fact that, again, I think when Jeremy heard these, quote, unquote, rumors, I believe it's the intelligence community in the United Kingdom, he's talked about burner phones in his book. So I believe Jeremy was working at a level in which he didn't want interference, potential interference.

So Eddie contacts Jeremy. Jeremy then contacts me the next day, and that's when I realize it's Jeremy Farrar, because, again, I didn't know that based on the conversation with Eddie. We talk about my concerns and talk about, like, look, we need to get a group together to discuss these concerns that I have, and we need to select people that are unconflicted. So we did.

Ralph Baric, for example, is a name that came up. We all know Ralph. Ralph is a very important coronavirus biologist. But we also knew that Ralph had very close associations and collaborations with the Wuhan Institute of Virology, for example. So if this did, in fact, originate from a lab, then, of course, he would not be a person to have on a call like this.

Peter Daszak is another person that, of course, came up for the same reasons, because they had been co-publishing.

I should say that I'm not at that time, I'm not familiar with grants going to the Wuhan Institute of Virology or anything about that. But we're talking about just publications they co-published.

So we get a group of international scientists. These are primarily from Europe. I actually think maybe all of them are from Europe -- Christian Drosten, Ron Fouchier, Marion Koopsman -- that have different perspectives on coronaviruses and gain-of-function research from Ron Fouchier, for example. And we decide to set up this conversation for me to essentially walk through my concerns and then we discuss it.

2023.06.16-Andersen-Transcript.pdf, p.52–53

Message

From: Fauci, Anthony (NIH/NIAID) [E] [redacted]
Sent: 1/31/2020 6:25:48 PM
To: Kristian G. Andersen [redacted]
CC: Jeremy Farrar [redacted]
Subject: RE: Phone call

Thanks, Christian. I will keep you posted.
Best regards,
Tony

From: Kristian G. Andersen [redacted]
Sent: Friday, January 31, 2020 8:05 PM
To: Fauci, Anthony (NIH/NIAID) [E] [redacted]
Cc: Jeremy Farrar [redacted]
Subject: Re: Phone call

Thanks Tony,

In addition to Eddie and Bob we have Mike Farzan on board (discoverer of the SARS receptor https://www.scripps.edu/faculty/farzan/), and I believe Jeremy will reach out to Christian Drosten and Ron Fouchier in the morning to get their expertise as well. Combined, this group will be able to objectively assess the available data and determine whether the genome looks unusual.

Please let me know if anything changes on your end or if you have any questions.

Best,
Kristian

On Fri, Jan 31, 2020 at 4:38 PM Fauci, Anthony (NIH/NIAID) [E] [redacted] wrote:

Jeremy:

I just got off the phone with Kristian Anderson and he related to me his concern about the Furine site mutation in the spike protein of the currently circulating 2019-nCoV. I told him that as soon as possible he and Eddie Holmes should get a group of evolutionary biologists together to examine carefully the data to determine if his concerns are validated. He should do this very quickly and if everyone agrees with this concern, they should report it to the appropriate authorities. I would imagine that in the USA this would be the FBI and in the UK it would be MI5. It would be important to quickly get confirmation of the cause of his concern by experts in the field of coronaviruses and evolutionary biology. In the meantime, I will alert my US. Government official colleagues of my conversation with you and Kristian and determine what further investigation they recommend. Let us stay in touch.

Best regards,

Tony

Anthony S. Fauci, MD
Director

REV0000750

We already had Eddie and Kristian. They had been having conversations in the background with Andrew Rambaut and Bob Garry, an expert on viruses at Tulane University in Louisiana and one of Kristian's regular collaborators. Bob had independently clocked what looked like peculiarities in the virus. Kristian had also sought the opinion of his Scripps colleague Michael Farzan, who had made important discoveries about how SARS-CoV-1 binds to human cells.

Michael confessed he was struggling to figure out how the new coronavirus could have acquired its features in a natural way. My wife Christiane recommended Marion Koopmans and Ronald Fouchier, both at Erasmus University in the Netherlands, plus Christian Drosten, who directs the Institute of Virology at the Charité Hospital in Berlin. Christiane pointed out that a truly international group would scotch any rumbling that this was a conspiratorial Anglo-American stitch-up against China.

The panel made a dream team: all globally respected, opinionated scientists who would challenge each other without fear or favour. Marion serves as a science adviser to the WHO – we work together on the science advisory committee for the WHO R&D Blueprint, which focuses on the research strategy for new diseases – and was a member of the Emergency Committee that declared the public health emergency. Later, she would also form part of the WHO delegation that went into Wuhan to investigate the origins of SARS-CoV-2. Ron is a renowned virologist who is a longtime advocate, and practitioner, of gain of function studies. He would be able to give us an insider's view of how likely this kind of research might be at the heart of the mystery.

Proximal_Origin_Emails.pdf, p.1, 31 Jan 2020 (afternoon, US time) and Spike

3.14 Fauci shares Cohen's latest article with Farrar and Andersen (31 Jan, 8:47 pm EST)

Another 40 minutes later, at 8:43 EST, Fauci was sent a Science article by Jon Cohen published only two hours earlier: '*Mining coronavirus genomes for clues to the outbreak's origins*'. He forwarded it at 8:47 pm EST to Farrar and Andersen for their info.

Fauci does not seem to have had the time to properly read it, as Andersen and Holmes are actually quoted in the article, with Holmes emphasising the importance of the positive environmental samples in the Western part of the market, while Andersen was more circumspect.

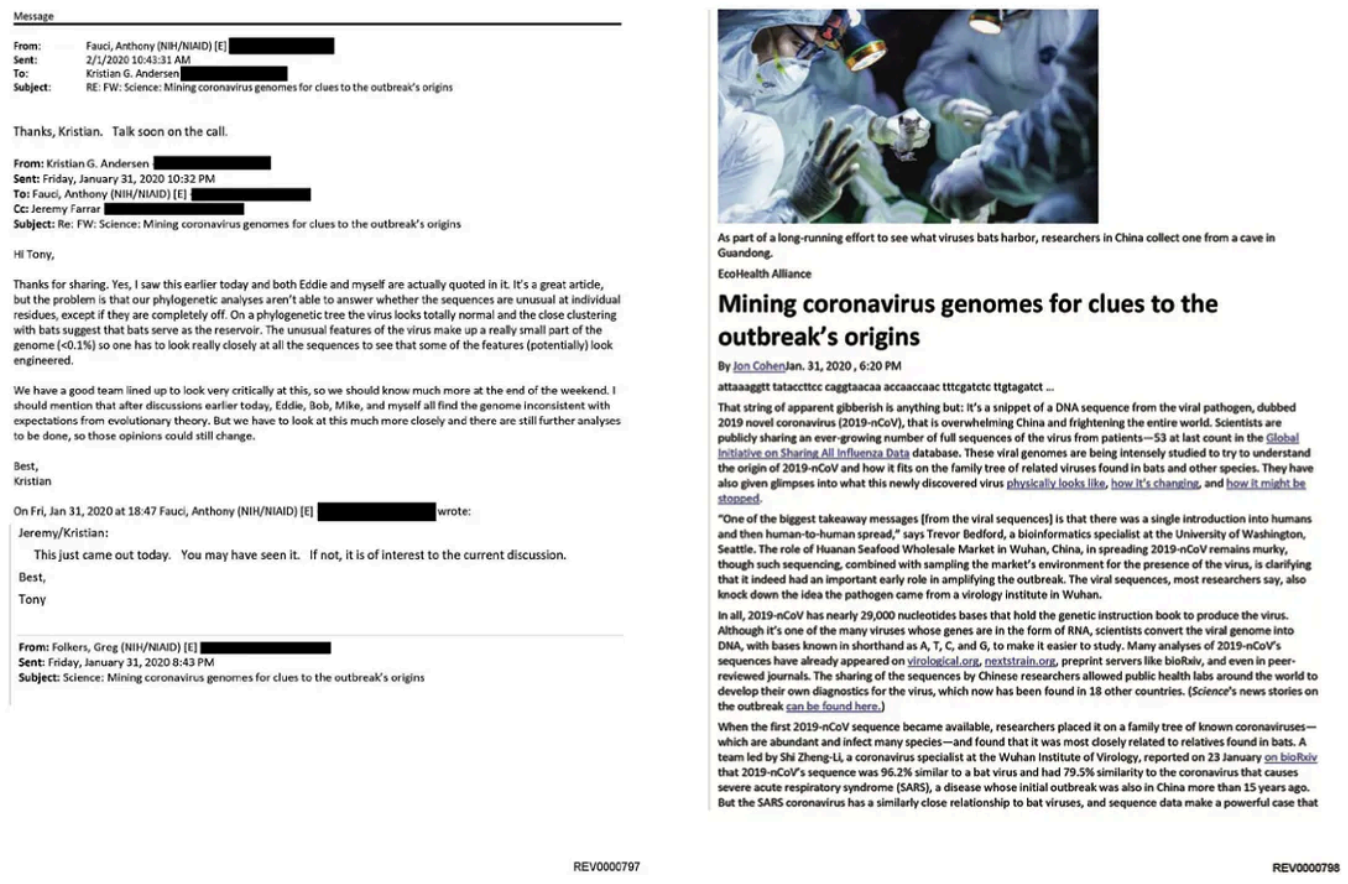
According to Xinhua, the state-run news agency, "environmental sampling" of the Wuhan seafood market has found evidence of 2019-nCoV. Of the 585 samples tested, 33 were positive for 2019-nCoV and all were in the huge market's western portion, which is where wildlife were sold. "The positive tests from the wet market are hugely important," says Edward Holmes, an evolutionary biologist at the University of Sydney who collaborated with the first group to publicly release a 2019-nCoV sequence. "Such a high rate of positive tests would strongly imply that animals in the market played a key role in the emergence of the virus."

Yet there have been no preprints or official scientific reports on the sampling, so it's not clear which, if any, animals tested positive. "Until you consistently isolate the virus out of a single species, it's really, really difficult to try and determine what the natural host is," says Kristian Andersen, an evolutionary biologist at Scripps Research.

Science article: mining-coronavirus-genomes-clues-outbreak-s-origins

Andersen sent an answer back to Fauci that evening (10:32 pm EST) that Fauci acknowledged the next morning (10:43 am EST).

> *Andersen to Fauci:*
> *'The unusual features of the virus make up a really small part of the genome (<0.1%) so one has to look really closely at all the sequences to see that some of the features (potentially) look engineered'.*
>
> *source: Proximal_Origin_Emails.pdf p. 4, 31 Jan 10:32 pm EST*



Message

From: Fauci, Anthony (NIH/NIAID) [E] [redacted]
Sent: 2/1/2020 10:43:31 AM
To: Kristian G. Andersen [redacted]
Subject: RE: FW: Science: Mining coronavirus genomes for clues to the outbreak's origins

Thanks, Kristian. Talk soon on the call.

From: Kristian G. Andersen [redacted]
Sent: Friday, January 31, 2020 10:32 PM
To: Fauci, Anthony (NIH/NIAID) [E] [redacted]
Cc: Jeremy Farrar [redacted]
Subject: Re: FW: Science: Mining coronavirus genomes for clues to the outbreak's origins

Hi Tony,

Thanks for sharing. Yes, I saw this earlier today and both Eddie and myself are actually quoted in it. It's a great article, but the problem is that our phylogenetic analyses aren't able to answer whether the sequences are unusual at individual residues, except if they are completely off. On a phylogenetic tree the virus locks totally normal and the close clustering with bats suggest that bats serve as the reservoir. The unusual features of the virus make up a really small part of the genome (<0.1%) so one has to look really closely at all the sequences to see that some of the features (potentially) look engineered.

We have a good team lined up to look very critically at this, so we should know much more at the end of the weekend. I should mention that after discussions earlier today, Eddie, Bob, Mike, and myself all find the genome inconsistent with expectations from evolutionary theory. But we have to look at this much more closely and there are still further analyses to be done, so those opinions could still change.

Best,
Kristian

On Fri, Jan 31, 2020 at 18:47 Fauci, Anthony (NIH/NIAID) [E] [redacted] wrote:

Jeremy/Kristian:

This just came out today. You may have seen it. If not, it is of interest to the current discussion.

Best,
Tony

From: Folkers, Greg (NIH/NIAID) [E] [redacted]
Sent: Friday, January 31, 2020 8:43 PM
Subject: Science: Mining coronavirus genomes for clues to the outbreak's origins

As part of a long-running effort to see what viruses bats harbor, researchers in China collect one from a cave in Guandong.

EcoHealth Alliance

Mining coronavirus genomes for clues to the outbreak's origins

By Jon CohenJan. 31, 2020 , 6:20 PM

attaaaggtt tataccttcc caggtaacaa accaaccaac tttcgatctc ttgtagatct ...

That string of apparent gibberish is anything but: It's a snippet of a DNA sequence from the viral pathogen, dubbed 2019 novel coronavirus (2019-nCoV), that is overwhelming China and frightening the entire world. Scientists are publicly sharing an ever-growing number of full sequences of the virus from patients—53 at last count in the Global Initiative on Sharing All Influenza Data database. These viral genomes are being intensely studied to try to understand the origin of 2019-nCoV and how it fits on the family tree of related viruses found in bats and other species. They have also given glimpses into what this newly discovered virus physically looks like, how it's changing, and how it might be stopped.

"One of the biggest takeaway messages [from the viral sequences] is that there was a single introduction into humans and then human-to-human spread," says Trevor Bedford, a bioinformatics specialist at the University of Washington, Seattle. The role of Huanan Seafood Wholesale Market in Wuhan, China, in spreading 2019-nCoV remains murky, though such sequencing, combined with sampling the market's environment for the presence of the virus, is clarifying that it indeed had an important early role in amplifying the outbreak. The viral sequences, most researchers say, also knock down the idea the pathogen came from a virology institute in Wuhan.

In all, 2019-nCoV has nearly 29,000 nucleotides bases that hold the genetic instruction book to produce the virus. Although it's one of the many viruses whose genes are in the form of RNA, scientists convert the viral genome into DNA, with bases known in shorthand as A, T, C, and G, to make it easier to study. Many analyses of 2019-nCoV's sequences have already appeared on virological.org, nextstrain.org, preprint servers like bioRxiv, and even in peer-reviewed journals. The sharing of the sequences by Chinese researchers allowed public health labs around the world to develop their own diagnostics for the virus, which now has been found in 18 other countries. (*Science*'s news stories on the outbreak can be found here.)

When the first 2019-nCoV sequence became available, researchers placed it on a family tree of known coronaviruses—which are abundant and infect many species—and found that it was most closely related to relatives found in bats. A team led by Shi Zheng-Li, a coronavirus specialist at the Wuhan Institute of Virology, reported on 23 January on bioRxiv that 2019-nCoV's sequence was 96.2% similar to a bat virus and had 79.5% similarity to the coronavirus that causes severe acute respiratory syndrome (SARS), a disease whose initial outbreak was also in China more than 15 years ago. But the SARS coronavirus has a similarly close relationship to bat viruses, and sequence data make a powerful case that

REV0000797 REV0000798

Proximal_Origin_Emails.pdf p. 4–5 31 Jan 10:32 pm EST

Insight: A unique situation

One needs to go back to the events that happened around the 28–31 Jan to appreciate the dilemma that health authorities and intelligence services did find themselves in on the last day of January 2020.

- China had been blocking access, had delayed the release of the genome, had been hiding the extent of the infections and worrying aspects of the outbreak, such as human-to-hum transmission and asymptomatic transmissions.
- An international health emergency (PHEIC) had just been declared on 30 Jan, as cases were exploding in China and started piling up abroad.
- Questions about the origin of the virus were being raised by some credible scientists.

So the main questions would be:
If that virus is not natural, what does it mean to our ability to control it? How truly good is that virus at transmitting? How do we stop it? May it cause unexpected physiological damages? What collaboration can we expect from China? etc.

Insight: Playing Offence

The end of the Science article quoted Ebright telling Jon Cohen that the 2019-nCoV data are "*consistent with entry into the human population as either a natural accident or a laboratory accident.*".

That quote was fairly mild and correctly represented the state of the science, just ahead of a call organised by Farrar to review some issues with the genome raised by Andersen and others, and while Fauci was at that precise time trying to understand what the implications of these issues might be for the NIH.

What was not mild was the attack by Daszak on Ebright, quoted at the very end of the article:

> *Peter Daszak of the EcoHealth Alliance, dismissed Ebright's conjecture. "Every time there's an emerging disease, a new virus, the same story comes out: This is a spillover or the release of an agent or a bioengineered virus," Daszak says. "It's just a shame. It seems humans can't resist controversy and these myths [..]"*

To his credit, Cohen tried to put Daszak's reaction in context when he alluded to the still open wounds left by the GoF moratorium of 2014–2017, and to the various warnings that Ebright had raised about unfettered GoF research and the multiplication of P4s. This is indeed the key to understand Daszak: while Farrar, Andersen, and Holmes are starting a discussion about the possibility of a lab-product, and while Fauci is taking notice, Daszak is already out there pouring scorn on a mild and fundamentally correct quote by Ebright.

Daszak is doing so because offence is his best defence. The last thing that Daszak needs at that time is another discussion of laboratory biosafety risks and of GoF experiments, after having been already affected by the 2014–17 GoF moratorium, and since he knew very well that he may have already crossed the line since 2017, depending on how one twists the definition of GoF or Potential Pandemic Pathogens. Shutting out any discussion before it could take off was his priority, or the grants may stop flowing.

4. The Farrar call

Note: Many people have called it the Fauci call, to reflect the fact that Fauci was the one who asked him to set up a call with a larger set of scientists. I have been insisting that it should instead be called the '**Farrar-Fauci call**',

given the essential role of Farrar in the genesis of this investigation and its management all along. I shall now go further and call it the '**Farrar call'**, since Farrar played a much larger and direct role than Fauci in the shaping of Proximal Origin, driven largely by imperatives of the WHO. One might still argue that '**the Farrar-MI5 call'** would be an even better choice, since Farrar was actually very likely working under guidance, if not instructions, of the UK MI5, when starting his investigative effort with his core-team (Holmes, Andersen, Rambaut). This would unfortunately likely misrepresent the role of the MI5, once Farrar recycled that initiative into support for the WHO, after feeding back his findings.

4.1 Information about EHA work with WIV received by Fauci by 31 Jan

a. First NIH quick checks: 23 Jan

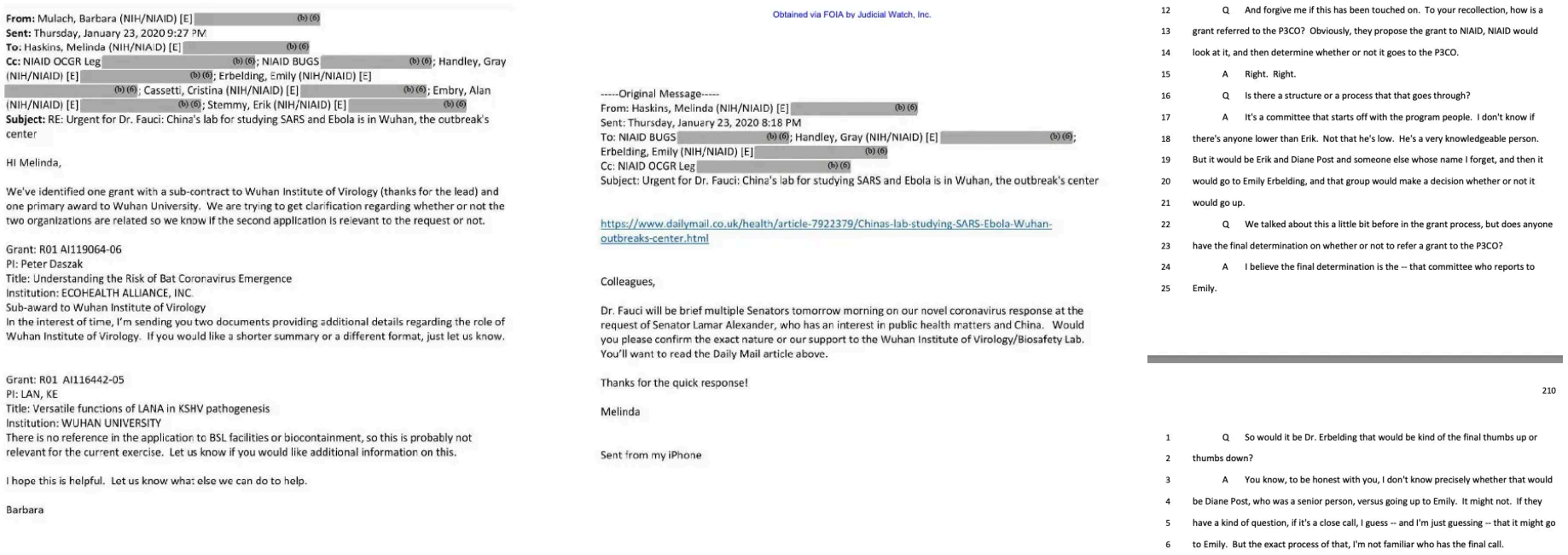
On 23 Jan (8:18 pm EST), in preparation for a briefing of US senators by Fauci the following morning, Melinda Haskins, of the NIH Office of Communications and Government Relations (OCGR), asked Emily Erbelding for some confirmation of *'the exact nature of our support to the Wuhan Institute of Virology/Biosafety Lab'*.

Haskins' email started with a link to the Daily Mail article published only three hours earlier (5:08 pm EST).

According to her NIH profile, *"Dr. Erbelding was responsible for the strategic and scientific vision for DMID's [NIAID Division of Microbiology and Infectious Diseases] complex national and international research program. DMID supports basic, preclinical, and clinical investigations into the causes, diagnosis, treatment, and prevention of a broad range of pathogens, including those related to biodefense and emerging infectious diseases."*

Practically, Erbelding's portfolio includes NIH international research programs involving DURC pathogens and those related to biodefence. Even more practically, she was also the Division Director (DMID) for Daszak's R01 grants, and, as Fauci confirmed in his deposition to the House Select Subcommittee on the Coronavirus Pandemic, Erbelding oversaw the NIH committee that decided whether Daszak's grants should go through the HHS P3 framework, meant to vet Potential Pandemic Pathogens research (P3CO for P3 Care and Oversight), which none of them ever did.

The email was also addressed to Gray Hendley, NIAID Associate Director for International Research Affairs.



From: Mulach, Barbara (NIH/NIAID) [E] (b) (6)
Sent: Thursday, January 23, 2020 9:27 PM
To: Haskins, Melinda (NIH/NIAID) [E] (b) (6)
Cc: NIAID OCGR Leg (b) (6); NIAID BUGS (b) (6); Handley, Gray (NIH/NIAID) [E] (b) (6); Erbelding, Emily (NIH/NIAID) [E] (b) (6); Cassetti, Cristina (NIH/NIAID) [E] (b) (6); Embry, Alan (NIH/NIAID) [E] (b) (6); Stemmy, Erik (NIH/NIAID) [E] (b) (6)
Subject: RE: Urgent for Dr. Fauci: China's lab for studying SARS and Ebola is in Wuhan, the outbreak's center

HI Melinda,

We've identified one grant with a sub-contract to Wuhan Institute of Virology (thanks for the lead) and one primary award to Wuhan University. We are trying to get clarification regarding whether or not the two organizations are related so we know if the second application is relevant to the request or not.

Grant: R01 AI119064-06
PI: Peter Daszak
Title: Understanding the Risk of Bat Coronavirus Emergence
Institution: ECOHEALTH ALLIANCE, INC.
Sub-award to Wuhan Institute of Virology
In the interest of time, I'm sending you two documents providing additional details regarding the role of Wuhan Institute of Virology. If you would like a shorter summary or a different format, just let us know.

Grant: R01 AI116442-05
PI: LAN, KE
Title: Versatile functions of LANA in KSHV pathogenesis
Institution: WUHAN UNIVERSITY
There is no reference in the application to BSL facilities or biocontainment, so this is probably not relevant for the current exercise. Let us know if you would like additional information on this.

I hope this is helpful. Let us know what else we can do to help.

Barbara

Obtained via FOIA by Judicial Watch, Inc.

-----Original Message-----
From: Haskins, Melinda (NIH/NIAID) [E] (b) (6)
Sent: Thursday, January 23, 2020 8:18 PM
To: NIAID BUGS (b) (6); Handley, Gray (NIH/NIAID) [E] (b) (6); Erbelding, Emily (NIH/NIAID) [E] (b) (6)
Cc: NIAID OCGR Leg (b) (6)
Subject: Urgent for Dr. Fauci: China's lab for studying SARS and Ebola is in Wuhan, the outbreak's center

https://www.dailymail.co.uk/health/article-7922379/Chinas-lab-studying-SARS-Ebola-Wuhan-outbreaks-center.html

Colleagues,

Dr. Fauci will be brief multiple Senators tomorrow morning on our novel coronavirus response at the request of Senator Lamar Alexander, who has an interest in public health matters and China. Would you please confirm the exact nature or our support to the Wuhan Institute of Virology/Biosafety Lab. You'll want to read the Daily Mail article above.

Thanks for the quick response!

Melinda

Sent from my iPhone

12 Q And forgive me if this has been touched on. To your recollection, how is a
13 grant referred to the P3CO? Obviously, they propose the grant to NIAID, NIAID would
14 look at it, and then determine whether or not it goes to the P3CO.
15 A Right. Right.
16 Q Is there a structure or a process that that goes through?
17 A It's a committee that starts off with the program people. I don't know if
18 there's anyone lower than Erik. Not that he's low. He's a very knowledgeable person.
19 But it would be Erik and Diane Post and someone else whose name I forget, and then it
20 would go to Emily Erbelding, and that group would make a decision whether or not it
21 would go up.
22 Q We talked about this a little bit before in the grant process, but does anyone
23 have the final determination on whether or not to refer a grant to the P3CO?
24 A I believe the final determination is the -- that committee who reports to
25 Emily.

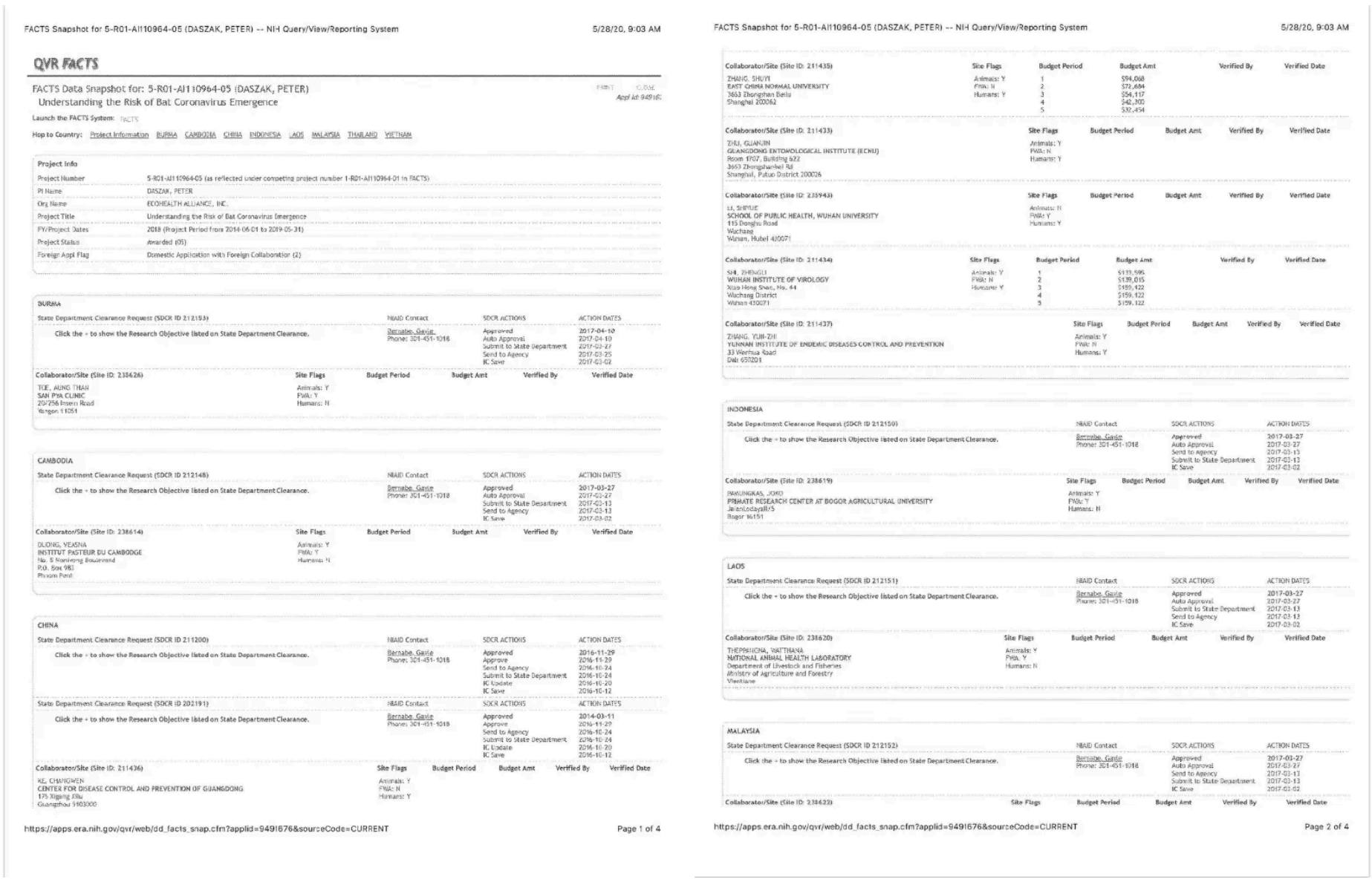
210

1 Q So would it be Dr. Erbelding that would be kind of the final thumbs up or
2 thumbs down?
3 A You know, to be honest with you, I don't know precisely whether that would
4 be Diane Post, who was a senior person, versus going up to Emily. It might not. If they
5 have a kind of question, if it's a close call, I guess -- and I'm just guessing -- that it might go
6 to Emily. But the exact process of that, I'm not familiar who has the final call.

JW-v-HHS-NIAID-Wuhan-June-2021–00692.pdf, p.47–48, 23 Jan 2020, and Fauci-Part-1-Transcript.pdf, p. 209–210.

In a little more than one hour, Haskins got an answer which flagged Daszak's R01 AI119064 grant. Two documents were attached that offered more details: 'R01AI110964 Renewal **FACTS** clearance.docx' and 'Daszak Wuhan Exceprts[sic].docx'.

FACTS is the Foreign Award and Component Tracking System, used by the National Institutes of Health (NIH) to track research investments in foreign countries and process requests for foreign collaborations that require State Department clearance. The full FACTS files below for Daszak's R01 grant were obtained by FOI:



FACTS Snapshot for 5-R01-AI110964-05 (DASZAK, PETER) -- NIH Query/View/Reporting System 5/28/20, 9:03 AM

QVR FACTS

FACTS Data Snapshot for: 5-R01-AI110964-05 (DASZAK, PETER)
Understanding the Risk of Bat Coronavirus Emergence
Appl Id: 949167

Launch the FACTS System:

Hop to Country: Project Information BURMA CAMBODIA CHINA INDONESIA LAOS MALAYSIA THAILAND VIETNAM

Project Info

Project Number	5-R01-AI110964-05 (as reflected under competing project number 1-R01-AI110964-01 in FACTS)
PI Name	DASZAK, PETER
Org Name	ECOHEALTH ALLIANCE, INC.
Project Title	Understanding the Risk of Bat Coronavirus Emergence
FY/Project Dates	2018 (Project Period from 2014-06-01 to 2019-05-31)
Project Status	Awarded (05)
Foreign Appl Flag	Domestic Application with Foreign Collaboration (2)

BURMA

State Department Clearance Request (SDCR ID 212103)

Click the + to show the Research Objective listed on State Department Clearance.

NIAID Contact: Bernabe, Gayle, Phone: 301-451-1018

SDCR ACTIONS	ACTION DATES
Approved	2017-04-10
Auto Approval	2017-04-10
Submit to State Department	2017-03-27
Send to Agency	2017-03-25
IC Save	2017-03-02

Collaborator/Site (Site ID: 238626)	Site Flags	Budget Period	Budget Amt	Verified By	Verified Date
TOE, AUNG THAN SAN PYA CLINIC 29/256 Insein Road Yangon 11051	Animals: Y PWA: Y Humans: N				

CAMBODIA

State Department Clearance Request (SDCR ID 212148)

Click the + to show the Research Objective listed on State Department Clearance.

NIAID Contact: Bernabe, Gayle, Phone: 301-451-1018

SDCR ACTIONS	ACTION DATES
Approved	2017-03-27
Auto Approval	2017-03-27
Submit to State Department	2017-03-13
Send to Agency	2017-03-13
IC Save	2017-03-02

Collaborator/Site (Site ID: 238614)	Site Flags	Budget Period	Budget Amt	Verified By	Verified Date
DUONG, YEASNA INSTITUT PASTEUR DU CAMBODGE No. 5 Monivong Boulevard P.O. Box 983 Phnom Penh	Animals: Y PWA: Y Humans: N				

CHINA

State Department Clearance Request (SDCR ID 211200)

Click the + to show the Research Objective listed on State Department Clearance.

NIAID Contact: Bernabe, Gayle, Phone: 301-451-1018

SDCR ACTIONS	ACTION DATES
Approved	2016-11-29
Approve	2016-11-29
Send to Agency	2016-10-24
Submit to State Department	2016-10-24
IC Update	2016-10-20
IC Save	2016-10-12

State Department Clearance Request (SDCR ID 202191)

Click the + to show the Research Objective listed on State Department Clearance.

NIAID Contact: Bernabe, Gayle, Phone: 301-451-1018

SDCR ACTIONS	ACTION DATES
Approved	2014-03-11
Approve	2016-11-29
Send to Agency	2016-10-24
Submit to State Department	2016-10-24
IC Update	2016-10-20
IC Save	2016-10-12

Collaborator/Site (Site ID: 211436)	Site Flags	Budget Period	Budget Amt	Verified By	Verified Date
KE, CHANGWEN CENTER FOR DISEASE CONTROL AND PREVENTION OF GUANGDONG 175 Xigang XiRu Guangzhou 5103000	Animals: Y PWA: N Humans: Y				

https://apps.era.nih.gov/qvr/web/dd_facts_snap.cfm?applid=9491676&sourceCode=CURRENT Page 1 of 4

FACTS Snapshot for 5-R01-AI110964-05 (DASZAK, PETER) -- NIH Query/View/Reporting System 5/28/20, 9:03 AM

Collaborator/Site (Site ID: 211435)	Site Flags	Budget Period	Budget Amt	Verified By	Verified Date
ZHANG, SHUYI EAST CHINA NORMAL UNIVERSITY 3663 Zhongshan Beilu Shanghai 200062	Animals: Y PWA: N Humans: Y	1 2 3 4 5	$94,068 $72,684 $54,117 $42,300 $32,454		

Collaborator/Site (Site ID: 211433)	Site Flags	Budget Period	Budget Amt	Verified By	Verified Date
ZHU, GUANJIN GUANGDONG ENTOMOLOGICAL INSTITUTE (ECNU) Room 1707, Building 622 3663 Zhongshanbei Rd Shanghai, Putuo District 200026	Animals: Y PWA: N Humans: Y				

Collaborator/Site (Site ID: 235943)	Site Flags	Budget Period	Budget Amt	Verified By	Verified Date
LI, SHIYUE SCHOOL OF PUBLIC HEALTH, WUHAN UNIVERSITY 115 Donghu Road Wuchang Wuhan, Hubei 430071	Animals: N PWA: Y Humans: Y				

Collaborator/Site (Site ID: 211434)	Site Flags	Budget Period	Budget Amt	Verified By	Verified Date
SHI, ZHENGLI WUHAN INSTITUTE OF VIROLOGY Xiao Hong Shan, No. 44 Wuchang District Wuhan 430071	Animals: Y PWA: N Humans: Y	1 2 3 4 5	$139,015 $159,122 $159,122 $159,122 $159,122		

Collaborator/Site (Site ID: 211437)	Site Flags	Budget Period	Budget Amt	Verified By	Verified Date
ZHANG, YUN-ZHI YUNNAN INSTITUTE OF ENDEMIC DISEASES CONTROL AND PREVENTION 33 Wenhua Road Dali 650201	Animals: Y PWA: N Humans: Y				

INDONESIA

State Department Clearance Request (SDCR ID 212150)

Click the + to show the Research Objective listed on State Department Clearance.

NIAID Contact: Bernabe, Gayle, Phone: 301-451-1018

SDCR ACTIONS	ACTION DATES
Approved	2017-03-27
Auto Approval	2017-03-27
Send to Agency	2017-03-13
Submit to State Department	2017-03-13
IC Save	2017-03-02

Collaborator/Site (Site ID: 238619)	Site Flags	Budget Period	Budget Amt	Verified By	Verified Date
PAMUNGKAS, JOKO PRIMATE RESEARCH CENTER AT BOGOR AGRICULTURAL UNIVERSITY JalanLodayaIR5 Bogor 16151	Animals: Y PWA: Y Humans: N				

LAOS

State Department Clearance Request (SDCR ID 212151)

Click the + to show the Research Objective listed on State Department Clearance.

NIAID Contact: Bernabe, Gayle, Phone: 301-451-1018

SDCR ACTIONS	ACTION DATES
Approved	2017-03-27
Auto Approval	2017-03-27
Submit to State Department	2017-03-13
Send to Agency	2017-03-13
IC Save	2017-03-02

Collaborator/Site (Site ID: 238620)	Site Flags	Budget Period	Budget Amt	Verified By	Verified Date
THEPPHICHA, VAITTHANA NATIONAL ANIMAL HEALTH LABORATORY Department of Livestock and Fisheries Ministry of Agriculture and Forestry Vientiane	Animals: Y PWA: Y Humans: N				

MALAYSIA

State Department Clearance Request (SDCR ID 212152)

Click the + to show the Research Objective listed on State Department Clearance.

NIAID Contact: Bernabe, Gayle, Phone: 301-451-1018

SDCR ACTIONS	ACTION DATES
Approved	2017-03-27
Auto Approval	2017-03-27
Send to Agency	2017-03-13
Submit to State Department	2017-03-13
IC Save	2017-03-02

Collaborator/Site (Site ID: 238623)	Site Flags	Budget Period	Budget Amt	Verified By	Verified Date

https://apps.era.nih.gov/qvr/web/dd_facts_snap.cfm?applid=9491676&sourceCode=CURRENT Page 2 of 4

Excerpt of the FACTS file (tracking w/r State Department Clearance), year 5 of R01AI120964 (2018) as per tobias-v-hhs-21-cv-02805–103122-partial-production.pdf, p. 108–109

b. Daszak's summary for Fauci: 27 Jan

On 27 Jan, Peter Daszak sent David Morens (CAPT, Senior Advisor to Fauci) some quick notes to be forwarded to Fauci, in case of interviews. The notes summarise EHA work funded by NIH/NIAID grants in China, with the WIV and Ralph Baric as collaborators.

Morens circulated these note verbatim internally, and they reached Fauci the very same day, alongside some Talking Points for an HHS press-conference at the White House the next day (28 Jan).

From: Peter Daszak <daszak@ecohealthalliance.org>
Sent: Monday, January 27, 2020 1:36 PM
To: Morens, David (NIH/NIAID) [E] (b) (6)
Cc: Stemmy, Erik (NIH/NIAID) [E] (b) (6); Alison Andre <andre@ecohealthalliance.org>
Subject: Wuhan novel coronavirus - NIAID's role in bat-origin CoVs
Importance: High

Hi David – Happy to have a phone call re. the Wuhan CoV, but just wanted to mentioned a few things for your information and hopefully to pass on to Tony Fauci for when he's being interviewed re. the new CoV:

NIAID has been funding coronavirus work in China for the past 5 years through an R01 to me (1R01AI110964: "Understanding the Risk of Bat Coronavirus Emergence"). That's now been renewed, with a specific focus that we identify cohorts of people highly exposed to bats in China, and work out if they're getting sick from CoVs. Erik Stemmy is the Program Officer (cc'd here). Collaborators include Wuhan Institute of Virology (currently working on the nCoV), and Ralph Baric. The results of our work to date include:

(b) (4)

- Discovered Swine Acute Diarrheal Syndrome Virus (SADS-CoV) killing >25,000 pigs in Guangdong Province (Published in Nature)
- Found SARS-related CoVs that can bind to human cells (Published in Nature), and that cause SARS-like disease in humanized mouse models.

(b) (4)

Also – FYI, prior to the R01, we worked under an R01 with Eun-Chung Park as Program Officer on viral discovery in bats, where originally identified SARS-CoV as having a likely origin in bats (published in Science)

As I mentioned, I'm now part of a group that's meeting by phone weekly with CEIRS to discuss the nCoV and Erik's part of that.

Cheers,

Peter

Peter Daszak
President

EcoHealth Alliance
460 West 34th Street – 17th Floor
New York, NY 10001

Tel. +1 212-380-4474
Website: www.ecohealthalliance.org
Twitter: @PeterDaszak

EcoHealth Alliance leads cutting-edge research into the critical connections between human and wildlife health and delicate ecosystems. With this science we develop solutions that prevent pandemics and promote conservation.

From: Morens, David (NIH/NIAID) [E] (b) (6)
Sent: Thursday, January 9, 2020 1:36 PM
To: Peter Daszak; Ian Lipkin (wil2001@columbia.edu); Jon Epstein
Subject: RE: Wuhan virus

Thanks, the excitement never ends, right?

<image001.gif>

From: "Morens, David (NIH/NIAID) [E]" (b) (6)
Date: January 27, 2020 at 4:53 OSU-1.12.23-1265-pages.pdf
To: Peter Daszak <daszak@ecohealthalliance.org>
Cc: "Stemmy, Erik (NIH/NIAID) [E]" (b) (6), Alison Andre <andre@ecohealthalliance.org>
Subject: RE: Wuhan novel coronavirus - NIAID's role in bat-origin CoVs

Great info, thanks. Tony doesn't maintain awareness of these things and doesn't know unless program officers tell him, which they rarely do, since they are across town and may not see him more than once a year, or less....

(b) (4)

Obtained via FOIA by Judicial Watch, Inc.

(b) (4)

Interested in your feeling about where this is going. The experts buzzing around us are all over the map, between doomsday and not that big a deal, with everything in between.

JW-v-HHS-NIAID-Wuhan-June-2021–00692.pdf, p.145–146 and p.151, 27 Jan 2020

From: Folkers, Greg (NIH/NIAID) [E]
To: Routh, Jennifer (NIH/NIAID) [E]; Fauci, Anthony (NIH/NIAID) [E]
Cc: Billet, Courtney (NIH/NIAID) [E]; Stover, Kathy (NIH/NIAID) [E]; Folkers, Greg (NIH/NIAID) [E]; Conrad, Patricia (NIH/NIAID) [E]; Marston, Hilary (NIH/NIAID) [E]; Lerner, Andrea (NIH/NIAID) [E]
Subject: RE: For review (due to HHS for White House by 8:30 tonight): press conference talking points
Date: Monday, January 27, 2020 6:24:59 PM
Attachments: Talking Points for NIAID Director Dr. Fauci.docx

As a place folder looks good to me.
+ Andrea who is the lead on a CoV talk ASF is giving on Tuesday

Also --- when talking about CoV (not necessarily in this venue) we have on our team (Vincent and folks we fund, Peter Daszak, Ralph Baric, Ian Lipkin, etc.) probably the world's experts non-human coronaviruses.

From David M -- EcoHealth group (Peter Daszak et al), has for years been among the biggest players in coronavirus work, also in collaboration with Ralph Baric, Ian Lipkin and others.

NIAID has funded Peter's group for coronavirus work in China for the past 5 years through R01 1R01AI110964: "Understanding the Risk of Bat Coronavirus Emergence". That's now been renewed, with a specific focus to identify cohorts of people highly exposed to bats in China, and work out if they're getting sick from CoVs. Erik Stemmy is the Program Officer. Collaborators include Wuhan Institute of Virology (currently working on the nCoV), and Ralph Baric. The results of the work to date include:

(b) (5)

- Discovered Swine Acute Diarrheal Syndrome Virus (SADS-CoV) killing >25,000 pigs in Guangdong Province (Published in *Nature*)
- Found SARS-related CoVs that can bind to human cells (Published in *Nature*), and that cause SARS-like disease in humanized mouse models.

(b) (5)

Also – prior to the above R01, Peter's folks worked under an R01 with Eun-Chung Park as Program Officer on viral discovery in bats, and originally identified SARS-CoV as having a likely origin in bats (published in *Science*)

From: Routh, Jennifer (NIH/NIAID) [E] (b) (6)
Sent: Monday, January 27, 2020 6:14 PM
To: Fauci, Anthony (NIH/NIAID) [E] (b) (6)
Cc: Billet, Courtney (NIH/NIAID) [E] (b) (6); Stover, Kathy (NIH/NIAID) [E] (b) (6); Folkers, Greg (NIH/NIAID) [E] (b) (6); Conrad, Patricia (NIH/NIAID) [E] (b) (6); Marston, Hilary (NIH/NIAID) [E] (b) (6)
Subject: For review (due to HHS for White House by 8:30 tonight): press conference talking points

Dr. Fauci –

You will be making brief remarks at the HHS press conference tomorrow. HHS has requested your talking points by 8:30 tonight. (They are really just a placeholder for the White House to have something and so they are not surprised by any news). I have attached proposed points and pasted them below.

Please let me know if you have edits. I will send the final version to HHS tonight.

Thanks,
Jen

Talking Points for NIAID Director Anthony S. Fauci, M.D.
HHS Press Conference on Coordinated Public Health Response to 2019 Novel Coronavirus

(b) (5)

OSU-1.12.

From: Folkers, Greg (NIH/NIAID) [E] (b) (6)
Sent: Monday, January 27, 2020 6:25 PM
To: Routh, Jennifer (NIH/NIAID) [E] (b) (6); Fauci, Anthony (NIH/NIAID) [E] (b) (6)
Cc: Billet, Courtney (NIH/NIAID) [E] (b) (6); Stover, Kathy (NIH/NIAID) [E] (b) (6); Conrad, Patricia (NIH/NIAID) [E] (b) (6) Marston, Hilary (NIH/NIAID) [E] (b) (6); Lerner, Andrea (NIH/NIAID) [E] (b) (6)
Subject: RE: For review (due to HHS for White House by 8:30 tonight): press conference talking points

As a place folder looks good to me.
+ Andrea who is the lead on a CoV talk ASF is giving on Tuesday

Also --- when talking about CoV (not necessarily in this venue) we have on our team (Vincent and folks we fund, Peter Daszak, Ralph Baric, Ian Lipkin, etc.) probably the world's experts non-human coronaviruses.

From David M -- EcoHealth group (Peter Daszak et al), has for years been among the biggest players in coronavirus work, also in collaboration with Ralph Baric, Ian Lipkin and others.

NIAID has funded Peter's group for coronavirus work in China for the past 5 years through R01 1R01AI110964: "Understanding the Risk of Bat Coronavirus Emergence". That's now been renewed, with a specific focus to identify cohorts of people highly exposed to bats in China, and work out if they're getting sick from CoVs. Erik Stemmy is the Program Officer. Collaborators include Wuhan Institute of Virology (currently working on the nCoV), and Ralph Baric. The results of the work to date include:

- Sampled 10,074 bats and ~2,000 other mammals at 47 sites across S. China
- Discovered 172 novel β-CoVs (52 novel SARSr-CoVs), >350 novel α-CoVs
- Discovered closest relative to Wuhan nCoV (92% homology)
- Discovered Swine Acute Diarrheal Syndrome Virus (SADS-CoV) killing >25,000 pigs in Guangdong Province (Published in *Nature*)
- Found SARS-related CoVs that can bind to human cells (Published in *Nature*), and that cause SARS-like disease in humanized mouse models.
- Found that clinical signs of bat SARSr-CoVs in mice were not prevented with a vaccine candidate against SARS-CoV, and were not treatable with most monoclonal therapies being developed
- Found serological evidence that 3% of people living at the wildlife-human interface in rural China are being exposed to these bat SARS-related coronaviruses

Also – prior to the above R01, Peter's folks worked under an R01 with Eun-Chung Park as Program Officer on viral discovery in bats, and originally identified SARS-CoV as having a likely origin in bats (published in *Science*)

BuzzFeed-v.-HHS-20-cv-2704–7.19.21-Partial-Production_Redacted.pdf, p.22–23,and NIH-FOIA-Request-60130-Complete-Response.pdf, p.3–4, 27 Jan 2020

4.2 Fauci asks his right hand to do some checks (1 Feb, 00:29 am)

At half past midnight on 1 Feb, following his exchanges with Farrar and Andersen (31 Jan, see 3.11 and 3.12) and the decision to set up a larger review group, Fauci emailed his Principal Deputy Director, Hugh Auchincloss, and soon after forwarded another email to him (which we don't have but may be Farrar's summary). As per Auchincloss' testimony,

Auchincloss only saw the email the next morning (Saturday) and then started working on it.

> *Hugh.*
> *It is essential that we speak this AM. Keep your cell phone on. [..]*
> *Read this paper as well as the e-mail that I will forward to you now. You have tasks today that must be done.*
> *Thanks,*
> *Tony*
>
> *source: Fauci's email to Auchincloss, 1 Feb 00:29 am (exact FOI not clear)*

The paper that Fauci included and asked Auchincloss to read was '*A SARS-like cluster of circulating bat coronaviruses shows potential for human emergence*', with Ralph Baric and Shi Zhengli as senior authors. That 2015 Nature Medicine paper describes the creation of a chimeric SARS virus. It was partially funded via NIH and NIAID grants and was likely mentioned by Andersen ion his discussion with Fauci (Director of NIAID).

The '**It is essential that we speak this AM. Keep your cell phone on.**' suggests that Fauci planned to call Auchincloss in the morning, with more specific instructions, instead of putting these in emails.

From Fauci's testimony of Nov 2022, we learn that, two minutes after sending Auchincloss the Nature Medicine article, he also sent him a Science article (by Jon Cohen) published on 31 Jan, that featured a quote by Richard Ebright stating that the 2019-nCiV data were '***consistent with entry into the human population as either a natural accident or a laboratory accident***' and quotes by Daszak stating that the outbreak was a natural zoonosis.

Fauci further explained:

> *'I didn't want to go into the phone call not knowing the scope of what our relationship was regarding funding of grants in China. [..] t**his being the first time I had heard of this**, I wanted to be briefed as to the extent of our involvement with funding in China.*
>
> *source: Fauci's testimony of Nov 2022*

As just seen (in 4.1), this was not exactly the first time Fauci had heard of this; at the very least (supposing that somehow he never knew about it before that), Daszak's very own notes about his collaboration with China had reached Fauci on 27 Jan, and on 24 Jan he had received some notes put together by Erbelding, which spelt out the importance of the work of Ralph Baric, working in collaboration with Daszak in China with NIH/NIAID funding, including the full FACT extracts of Daszak's key R01 grant.

Nevertheless, the episode shows that Fauci's main concern was potential Gain of Function work at the WIV done with NIH money; now that the key Baric-Zhengli paper, funded via NIH, was at the very centre of Andersen's analysis managed under a major foreign institution (the Wellcome Trust), Fauci potentially faced some difficult time.

But, to be fair, Fauci had agreed to that very call only 5 hours earlier (31 Jan, 7:38 pm EST), if not actually suggested it himself, when he also recommended that FBI and MI5 should be informed if the engineering suspicions were confirmed (see 3.12).

We must also assume that Farrar would have explained to Fauci that Baric and Daszak were not meant to be part of that call, given their obvious conflicts of interest. One of the first reactions of Fauci would otherwise likely have been to make sure that they were part of the extended group of experts he asked to be put together for the call. With Baric and Daszak out, Fauci had another reason to ask his deputy to double-check the possible implications of NIH grant in work at the WIV, ahead of the call.

From: Fauci, Anthony (NIH/NIAID) [E]
Sent: Sat, 1 Feb 2020 12:29:01 +0000
To: Auchincloss, Hugh (NIH/NIAID) [C] [redacted] (b) (6)
Cc: [redacted] (b) (6)
Subject: IMPORTANT
Attachments: Baric, Shi et al - Nature medicine - SARS Gain of function.pdf

Hugh:
It is essential that we speak this AM. Keep your cell phone on. I have a conference call at 7:45 AM with Azar. It likely will be over at 8:45 AM. Read this paper as well as the e-mail that I will forward to you now. You will have tasks today that must be done.
Thanks,
Tony

Anthony S. Fauci, MD
Director
National Institute of Allergy and Infectious Diseases
Building 31, Room 7A-03
31 Center Drive, MSC 2520
National Institutes of Health
Bethesda, MD 20892-2520
Phone: [redacted] (b) (6)
FAX: (301) 496-4409
E-mail: [redacted] (b) (6)
The information in this e-mail and any of its attachments is confidential and may contain sensitive information. It should not be used by anyone who is not the original intended recipient. If you have received this e-mail in error please inform the sender and delete it from your mailbox or any other storage devices. The National Institute of Allergy and Infectious Diseases (NIAID) shall not accept liability for any statements made that are the sender's own and not expressly made on behalf of the NIAID by one of its representatives.

Fauci's email to Auchincloss, 1 Feb (half past midnight) (source not clear)

nature > nature medicine > letters > article

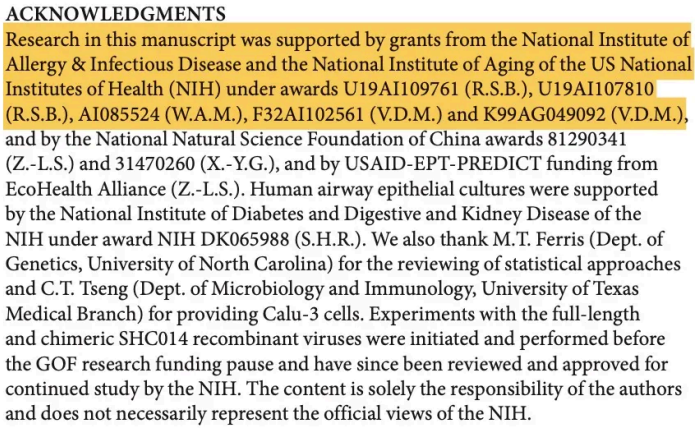
Letter | Published: 09 November 2015

A SARS-like cluster of circulating bat coronaviruses shows potential for human emergence

Vineet D Menachery, Boyd L Yount Jr, Kari Debbink, Sudhakar Agnihothram, Lisa E Gralinski, Jessica A Plante, Rachel L Graham, Trevor Scobey, Xing-Yi Ge, Eric F Donaldson, Scott H Randell, Antonio Lanzavecchia, Wayne A Marasco, Zhengli-Li Shi & Ralph S Baric

Nature Medicine **21**, 1508–1513 (2015) | Cite this article

2.85m Accesses | **520** Citations | **16292** Altmetric | Metrics



ACKNOWLEDGMENTS

Research in this manuscript was supported by grants from the National Institute of Allergy & Infectious Disease and the National Institute of Aging of the US National Institutes of Health (NIH) under awards U19AI109761 (R.S.B.), U19AI107810 (R.S.B.), AI085524 (W.A.M.), F32AI102561 (V.D.M.) and K99AG049092 (V.D.M.), and by the National Natural Science Foundation of China awards 81290341 (Z.-L.S.) and 31470260 (X.-Y.G.), and by USAID-EPT-PREDICT funding from EcoHealth Alliance (Z.-L.S.). Human airway epithelial cultures were supported by the National Institute of Diabetes and Digestive and Kidney Disease of the NIH under award NIH DK065988 (S.H.R.). We also thank M.T. Ferris (Dept. of Genetics, University of North Carolina) for the reviewing of statistical approaches and C.T. Tseng (Dept. of Microbiology and Immunology, University of Texas Medical Branch) for providing Calu-3 cells. Experiments with the full-length and chimeric SHC014 recombinant viruses were initiated and performed before the GOF research funding pause and have since been reviewed and approved for continued study by the NIH. The content is solely the responsibility of the authors and does not necessarily represent the official views of the NIH.

A SARS-like cluster of circulating bat coronaviruses shows potential for human emergence, Baric and Shi paper (2015)

4.3 Farrar sets up the call (1 Feb, 10:34 am EST)

At 10:34 am EST, Farrar sent an invitation for a teleconference planned for 2 pm (EST). This is the first time that we have a written reference to the Farrar call, which was conceived with Fauci the previous day as an enhancement to Farrar' investigative efforts (see 3.12).

The key scientists lined up for the call were:

- the '**core group**' of Andersen, Holmes, Andersen, Garry and Rambaut
- the just added '**bounce-off**' group of Fouchier, Drosten and Koopmans, which are all well versed in gain-of-function on coronaviruses [footnote 94], seconded by Stefan Pöhlmann, a German virologist.

Of the other individuals included in the call invite, Farrar (Wellcome Trust) acted as call leader and facilitator, while Fauci (NIAID), Ferguson (Wellcome Trust), Schreier (Wellcome Trust) and Vallance (UK Government Chief Scientific Adviser (GCSA)) were essentially acting as observers.

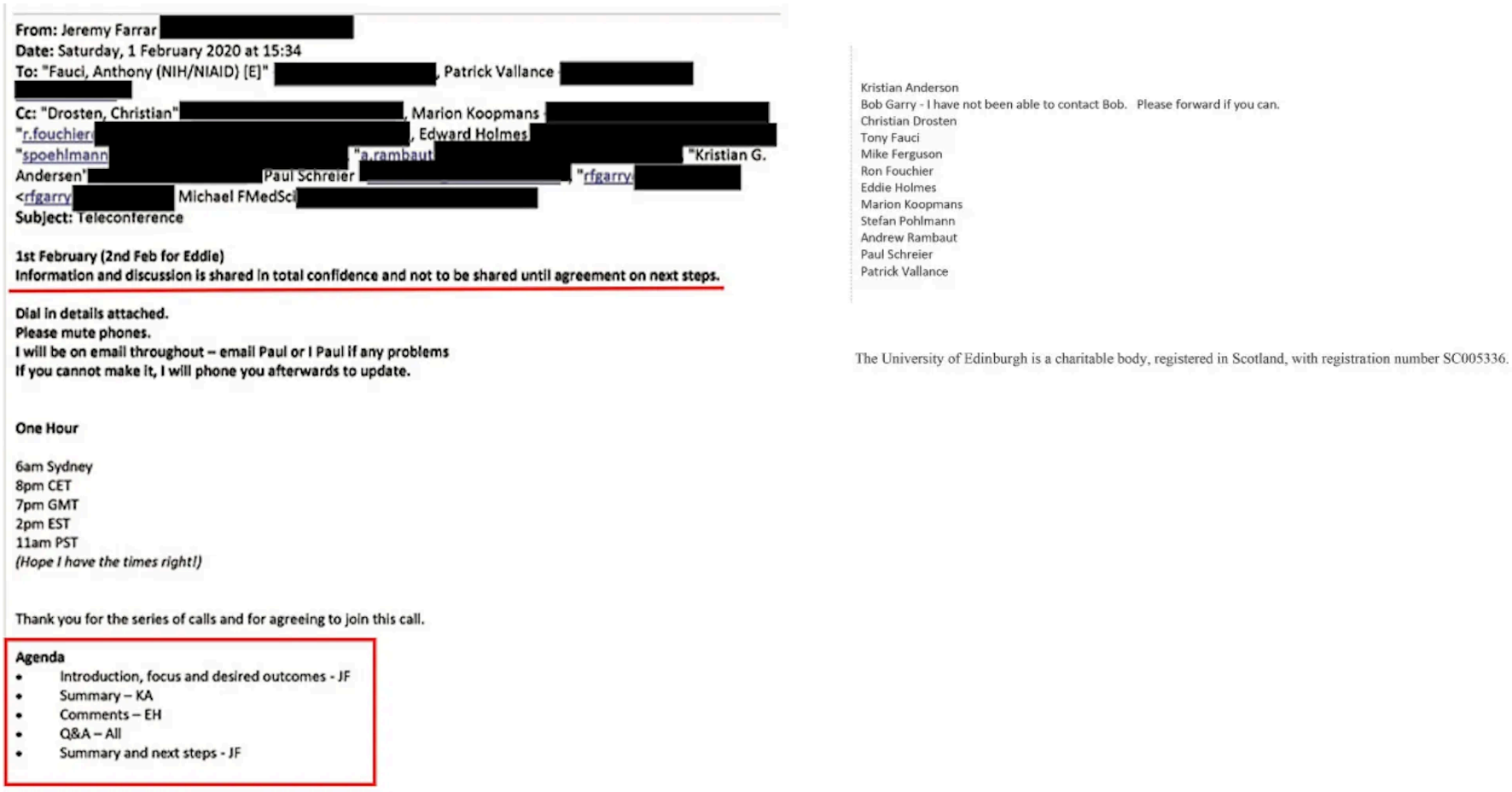
Michael Farzan, a colleague of Andersen at Scripps research (US citizen), which Andersen has been consulting for a second opinion, was not in the call list but will take part in the discussions immediately after the call.

Status	Name	Role in call	Workplace / Job	Country	Nationality	Note
In call invite	Jeremy Farrar	Organizer, handler of core group, WHO and MI5 links	Welcome Trust, Director	UK	British	
	Kristian Andersen	core group (set up by Farrar)	Scripps Research	US	Danish	
	Eddie Holmes		Sydney Uni.	Australia	British	Edinburgh connection
	Andrew Rambaut		Edinburgh Uni.	UK	British	
	Robert Garry		Tulane Uni.	US	US	Collaborator of Andersen
	Ron Fouchier	bounce-off group (at the request of Fauci)	Erasmus	Holland	Dutch	GOF practitioner
	Marion Koopmans		Erasmus	Holland	Dutch	GOF practitioner
	Christian Drosten		Charité Berlin	Germany	German	GOF practitioner
	Stefan Pöhlmann		German Primate Center	Germany	German	
	Paul Schreier	UK Observer	Welcome Trust, COO	UK	British	
	Patrick Vallance	UK Observer	Chief Scientific Adviser to the UK Government	Uk	British	
	Mike Ferguson	UK Observer	Dundee Uni., Wellcome Trust boards	UK	British	
	Tony Fauci	US NIAID/NIH Observer - State Dep links	NIAID, Director	US	US	
not in call invite, but in follow-up email by Farrar	Francis Collins	US NIH Observer, Fauci's boss	NIH, Director	US	US	
	Lawrence Tabak	US NIH Observer	NIH, Principal Deputy Director (Collins' right arm)	US	US	
	Josie Golding	UK Observer	Epidemic Research Lead at Wellcome Trust	UK	UK	

People invited to the Farrar call, or included in the follow-up e-mail by Farrar

`Except for Robert Garry, a usual collaborator of Andersen (who is Danish), there is a deliberate absence of US nationals in either the core group of scientists or the bounce-off group. These two groups of key scientists remind one of a British/European seminar.

The call invite set the agenda: introduction by Farrar, main findings by Andersen, some commentary by Holmes and then conclusion back to Farrar. All the participants were reminded that the information and discussions must remain confidential [footnote 3]



From: Jeremy Farrar [redacted]
Date: Saturday, 1 February 2020 at 15:34
To: "Fauci, Anthony (NIH/NIAID) [E]" [redacted], Patrick Vallance [redacted]
Cc: "Drosten, Christian" [redacted], Marion Koopmans [redacted], "r.fouchier [redacted], Edward Holmes [redacted], "spoehlmann [redacted], "a.rambaut [redacted], "Kristian G. Andersen" [redacted], Paul Schreier [redacted], "rfgarry [redacted], <rfgarry [redacted] Michael FMedSci [redacted]
Subject: Teleconference

1st February (2nd Feb for Eddie)
Information and discussion is shared in total confidence and not to be shared until agreement on next steps.

Dial in details attached.
Please mute phones.
I will be on email throughout – email Paul or I Paul if any problems
If you cannot make it, I will phone you afterwards to update.

One Hour

6am Sydney
8pm CET
7pm GMT
2pm EST
11am PST
(Hope I have the times right!)

Thank you for the series of calls and for agreeing to join this call.

Agenda
- Introduction, focus and desired outcomes - JF
- Summary – KA
- Comments – EH
- Q&A – All
- Summary and next steps - JF

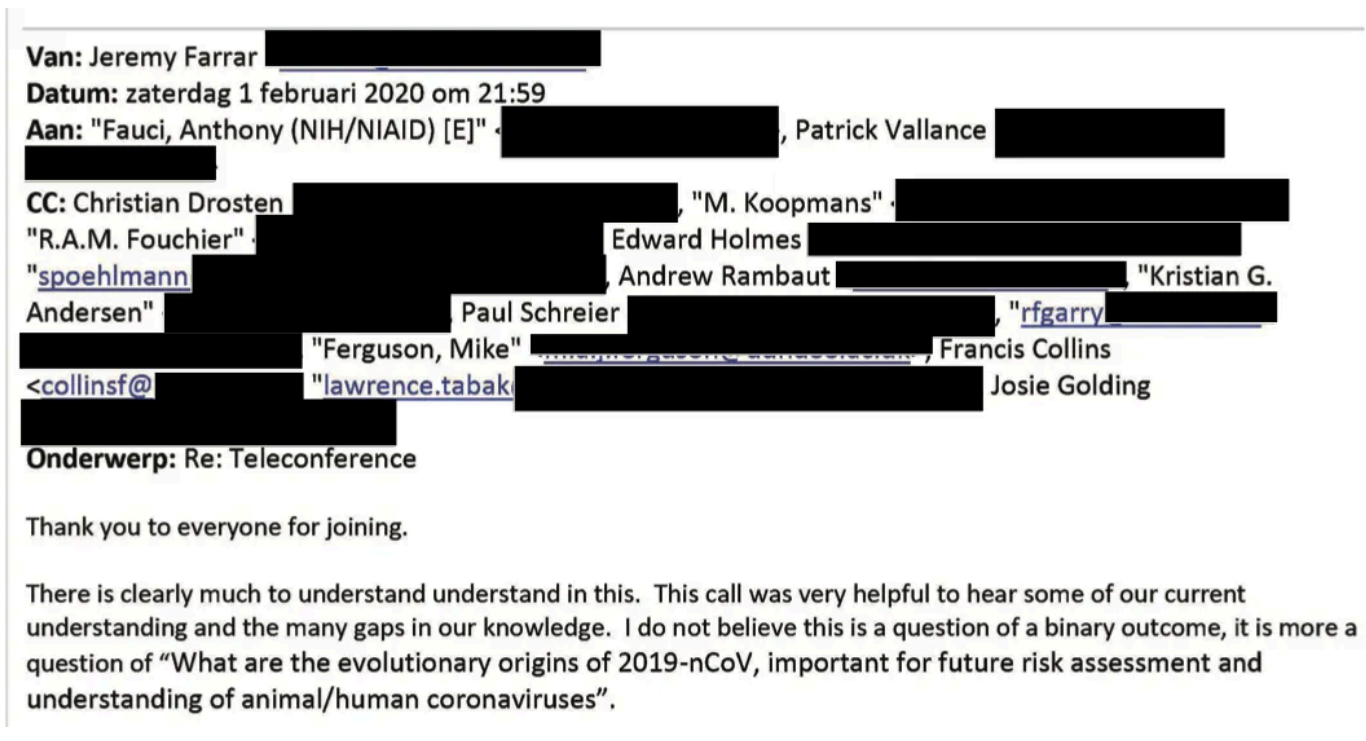
Kristian Anderson
Bob Garry - I have not been able to contact Bob. Please forward if you can.
Christian Drosten
Tony Fauci
Mike Ferguson
Ron Fouchier
Eddie Holmes
Marion Koopmans
Stefan Pohlmann
Andrew Rambaut
Paul Schreier
Patrick Vallance

The University of Edinburgh is a charitable body, registered in Scotland, with registration number SC005336.

Proximal_Origin_Emails.pdf, p.155–156, 1 Feb 2020

Sideline: Who else was on the Farrar call?

An email sent by Farrar just after the call has people in CC who were not on the original invite: Lawrence Tabak, Josie Golding and Francis Collins:



Van: Jeremy Farrar [redacted]
Datum: zaterdag 1 februari 2020 om 21:59
Aan: "Fauci, Anthony (NIH/NIAID) [E]" [redacted], Patrick Vallance [redacted]
CC: Christian Drosten [redacted], "M. Koopmans" [redacted], "R.A.M. Fouchier" [redacted], Edward Holmes [redacted], "spoehlmann [redacted], Andrew Rambaut [redacted], "Kristian G. Andersen" [redacted], Paul Schreier [redacted], "rfgarry [redacted], "Ferguson, Mike" [redacted], Francis Collins <collinsf@ [redacted], "lawrence.tabak [redacted], Josie Golding [redacted]
Onderwerp: Re: Teleconference

Thank you to everyone for joining.

There is clearly much to understand understand in this. This call was very helpful to hear some of our current understanding and the many gaps in our knowledge. I do not believe this is a question of a binary outcome, it is more a question of "What are the evolutionary origins of 2019-nCoV, important for future risk assessment and understanding of animal/human coronaviruses".

Proximal_Origin_Emails.pdf, p.153, 1 Feb 2020, 4 pm EST

▶️ Tabak? Yes

Ferguson would later explicitly mention that Tabak (Principal Deputy

Director at the NIH) was also on the call (see 13.3), and Collins included Tabak in an email discussing Farrar's roadmap soon after the call (see 6.1). Last, Tabak himself confirmed that he attended the Farrar Call in his 2024 interview by the House Committee on Oversight and Accountability.

17 Q I'll go with a soft "maybe" on Dr. Farrar. Do you have any recollection --
18 A Only because I attended meetings -- or a meeting for sure, at least
19 one -- that Dr. Farrar actually led, where this was discussed. But I never spoke to him
20 one-on-one.
21 Q Was that the conference call on February 1st of 2020?
22 A If that's the call where the scientists gathered, yes --
23 Q All right.
24 A -- that's the call I'm referring to.

Tabak-TI-Transcript.pdf, p. 17

▶️ Collins? Yes

Collins, the NIH director and Fauci's boss, was on the call. Farrar explained that *'Francis ha[d] to step out of his granddaughter's swimming gala because he couldn't hear us properly'*. Collins then missed the 'after-call' email from Farrar and joined it with a few minutes delay (see 4.10).

▶️ Golding? Possibly

Josie Golding, Epidemic Research Lead at the Wellcome Trust, was likely either included in the call, or added to the email chain after the call by Farrar, to help him parse through and interpret the information.

▶️ Baric? Not clear, certainly not officially and definitely not actively

During his interview with the House Subcommittee on the Coronavirus Pandemic on 24 Jan 2024, Baric started by stating that he was on an origin phone conference with Fauci, then after a few hours of interview, in a fairly confusing passage, he gave a fairly precise description of the Farrar call, including some of the main participants and the way the discussions progressed, but did not mention speaking during it, only listening. This was interpreted by some as meaning that Baric was indeed on the Farrar call.

327 But after January 1st, 2020, I was on a phone conference with
328 him on February 1st of 2020 that had to do with the origins.
329 I met with him in his office with several staff, high level
330 staff, both including himself and other representatives from
331 both the extramural and intramural program for NIH on, I
332 think, February 12, 2020. And I believe that's it.
333 Oh, yes, I was also part of -- we were both part of an email
334 exchange that was associated with the Red Dawn group, which
335 was basically trying to help prepare the United States to
336 respond to -- to track and respond to the emerging COVID-19
337 pandemic.
338 Q Thank you.

2683 Q I want to read a little bit from
2684 Dr. Andersen's interview.
2685 A Okay.
2686 Q We asked him these questions and asked him
2687 about the call.
2688 He said, "Ralph Baric, for example, is a name that came up.
2689 We all know Ralph, Ralph is a very important coronavirus
2690 biologist, but we also know that Ralph had very close
2691 associations and collaborations with the Wuhan Institute of
2692 Virology, for example. So if this did, in fact, originate
2693 from a lab, then, of course, he would not be a person to have
2694 on a call like this."
2695 A I must have been on that call. He may not
2696 have known it. It was -- again, right now, I have huge
2697 uncertainty about what call I was on, but he was there.
2698 Q I think we're talking about the same call.
2699 A I think we're talking about the same call.
2700 But I was on a phone, so it wasn't like a Zoom link for me.
2701 I didn't have anyone else's picture. So I was hearing mostly
2702 names, or I knew who they were, who was speaking.
2703 Q And you don't recall how you got on to the
2704 call?
2705 A I don't recall how I got invited.
2706 Q Okay.
2707 A No, I would have to look it up. I thought I
2708 knew, but apparently not.

2709 Q And you've discussed a little bit about the
2710 kind of back-and-forth of the people split on the origins
2711 question.
2712 A Yeah.
2713 Q Do you recall anything else from that
2714 conversation?
2715 A There was a fairly strong consensus, I think
2716 that was building toward the end of the call, that there
2717 wasn't data to support engineering, that there were other
2718 alternatives for the furin cleavage site.
2719 The receptor binding domain was still a little uncertain at
2720 that time, but if I remember correctly, one of the first
2721 pangolin strains had been sequenced and the sequence was
2722 available, which was very close to the SARS2 sequence, which
2723 argued that the RBD itself was natural origin.
2724 So that actually -- you know, in scientific method, you're
2725 trying to disprove a hypothesis. That actually was more
2726 against the current hypothesis, which was somebody tinkered

Baric-TI-Transcript.pdf, p.15, then p.109–110

A few elements are worth noting:

- Baric was not invited to the Farrar call. He had been specifically excluded from the Farrar call, officially for being too close to the WIV to be an unbiased participant.
- **But also**, since he was part of the 'B(io)SEC(urity) meeting 'on 25 or 26 January 2020, where he laid out the possibility of a lab escape and of engineering, there would have been little sense for MI5 to do a side check of these concerns, just to use the same key expert whose analysis already reached them.
- Baric did not join via the Zoom application and had no video.
- Baric only mentioned listening, not speaking — which may not mean much by itself, as it doesn't mean that he did not speak.
- None of the other scientists on the Farrar call ever mentioned that he was there.

So the only way that Baric could have been on the call is if his presence was not advertised, if he did not take part in any discussion, and had joined by using the provided phone-in line and stayed on mute all along.

But then why have Baric in the call when Farrar wished NOT to have him there? Also, why should one have put him on mute? And why, as the best world expert on coronaviruses, would he want to be on mute?

If one considers that Baric is right in his recollections, then the only way he could be there despite Farrar's concerns is that there was a US game on the back of the Farrar call. In that scenario, Baric — who had already briefed expressed very similar concerns during a 'B(io)SEC(urity) meeting' on 25 or 26 Jan, could have been added as an undisclosed passive listener to help analyse the Farrar call discussions, or he might even have listened later to a recording.

Another and simpler explanation is that Baric got confused and mixed up the Farrar call with the NASEM call that took place on 3 Feb (see 8.3.c). As to the rather matching description of the Farrar call that Baric gave in his interview with the House Subcommittee on the Coronavirus Pandemic, he could be that he later imagined some of these details based on reading about the Farrar call later — a common false recall problem.

In short, the best we can say it that it is not clear whether Baric was on *that* call, and if he was, then he was certainly not officially on it, and definitely not as an active element of it.

notes and the invites, you're not listed anywhere, but you were on that conference call?

A I wasn't listed on any of the invites?

Q No.

A I didn't know that. I'm kind of surprised. They clearly reached out to me. I don't know why they didn't reach out -- this must have been within the NIH staff?

Q No, there was a conference call with Dr. Fauci and Dr. Andersen?

A Wait, you're talking about the February 1st call.

Q Yes, sir.

A Not the February 11th call.

Q Correct.

A I'm sorry, I was confused. Can you restate the question?

Q The February 1st call with Dr. Fauci, Dr. Andersen, and Dr. Farrar, and ten or so others, we have gotten emails from almost every American participant on the call, and haven't seen your name come up anywhere. So I was surprised to hear that you were on it. But I want to confirm that you were on the call?

A I think I was. My recollection is this meeting was heavily dominated by the evolutionary biologists, who were split on the origin of the virus. Is that the

meeting you're talking about?

Q That sounds right.

A So I must have been there.

Q Do you recall how you got invited?

A No, I thought I was on the email chain, to tell you the truth.

Q I want to read a little bit from Dr. Andersen's interview.

A Okay.

Q We asked him these questions and asked him about the call.

He said, "Ralph Baric, for example, is a name that came up. We all know Ralph, Ralph is a very important coronavirus biologist, but we also know that Ralph had very close associations and collaborations with the Wuhan Institute of Virology, for example. So if this did, in fact, originate from a lab, then, of course, he would not be a person to have on a call like this."

A I must have been on that call. He may not have known it. It was -- again, right now, I have huge uncertainty about what call I was on, but he was there.

Q I think we're talking about the same call.

A I think we're talking about the same call. But I was on a phone, so it wasn't like a Zoom link for me. I didn't have anyone else's picture. So I was hearing mostly

names, or I knew who they were, who was speaking.

Q And you don't recall how you got on to the call?

A I don't recall how I got invited.

Q Okay.

A No, I would have to look it up. I thought I knew, but apparently not.

Q And you've discussed a little bit about the kind of back-and-forth of the people split on the origins question.

A Yeah.

Q Do you recall anything else from that conversation?

A There was a fairly strong consensus, I think that was building toward the end of the call, that there wasn't data to support engineering, that there were other alternatives for the furin cleavage site. The receptor binding domain was still a little uncertain at that time, but if I remember correctly, one of the first pangolin strains had been sequenced and the sequence was available, which was very close to the SARS2 sequence, which argued that the RBD itself was natural origin. So that actually -- you know, in scientific method, you're trying to disprove a hypothesis. That actually was more against the current hypothesis, which was somebody tinkered

Baric-TI-Transcript.pdf, p.108–110

4.4 Fauci gets some feedback from Auchincloss (11:47 am EST)

At 11:47 am (EST) on 1 Feb, so about 2 hours before the call, and a few hours after getting up and getting on with his task, Auchincloss replied to the email that Fauci had sent him just after midnight:

> *'The paper you sent me says the experience would perform before the gain-of-function pause that have since been reviewed and approved by the NIH.[..] Emily [Erbelding] is sure that no coronavirus work had gone through the P3 framework, which means it did not rise to the level of concern to get the extra approval of P3CO. She will try to determine if we have any distant ties to this work.*

What Auchincloss is saying there may seem positive at first: Daszak's R01 grant did not present the level of concern that would have required Erbelding to put it through the HHS P3CO framework, for dangerous work on Potential Pandemic Pathogens.

However, as a matter of fact, only **one project** was ever referred to the P3CO since the introduction of that framework in 2018 following the end of the GoF memorandum in 2017 [footnote 71]. So this could raise an even more general issue: Was the NIH doing its supervision work and was the P3CO framework working? These questions have haunted Fauci ever since.

The Nature Medicine article would the first of the six articles included in the presentation that Farrar sent the with the call invite at 10:34 am EST that day.

DR. ANTHONY FAUCI 11/23/2022

Page 61

Q. Do you recall how you became aware of this attachment that you've called Baric, Shi, et al., Nature Medicine SARS gain-of-function?

A. I don't recall -- I mean, I became aware of it, but right now, I really don't recall what specific day I became aware of it.

Q. In your e-mail, you tell Hugh, "Read this paper as well as the paper I will forward to you later on."

Can you turn to the next page?

A. Yeah.

Q. And here it's an e-mail just a couple minutes later -- it's literally the same minute that you sent to Hugh at 12:29 a.m. on Saturday, February 1st, 2020; correct?

A. Right.

Q. And here you're forwarding him that same Jon Cohen article that you had sent to Christian Anderson and Farrar?

A. Did I send it to Christian and Farrar? Yeah.

Q. Remember sending it to him and saying to him: This is of interest in the current discussion? This same Jon Cohen e-mail or article?

A. I'm sorry. I'm getting confused with your

DR. ANTHONY FAUCI 11/23/2022

Page 69

THE WITNESS: Okay. I'm reading from an e-mail from Hugh Auchincloss to me at 11:47 a.m. on February 1st, 2020. And --

BY MR. SAUER:

Q. I'm not asking you to read the e-mail. I'm just asking you to --

A. Okay. But she asked me to.

Q. I'm asking you to identify what time that e-mail from Hugh was sent to you.

MR. KIRSCHNER: And, Counsel, I'm asking you to give the witness an opportunity to familiarize himself with this document.

THE WITNESS: And what's what I was reading when you're telling me to read it out loud. So let's get together here.

THE REPORTER: And if you read on the record, I need you to speak up so that I can get it on the record --

THE WITNESS: Okay. I'll keep my mouth shut and read it silently so it's not on the record.

Okay. What's the question, sir?

BY MR. SAUER:

Q. Hugh e-mailed you saying, "The paper you sent me says the experience would perform before the gain-of-function pause that have since been reviewed

DR. ANTHONY FAUCI 11/23/2022

Page 70

and approved by the NIH."

Do you know what he was referring to when he said that?

A. I don't know exactly, but I'm assuming and I think correctly, apropos of what I've been telling you for the last several minutes, is that I wanted to get a feel of the scope of what we were doing, and I had mentioned to Hugh, I'm sure, "Hugh, I want to know everything that's going on because I want to make sure that I understand all of what our involvement is in funding research in China."

And Hugh's response was he looked at it and he said everything's been reviewed and approved by NIH, and Emily, who is Emily Erbelding, is the director of the Division of Microbiology and Infectious Diseases, and she would have been the one who was closest to the ground in understanding what we were doing in funding China.

And it says, "Emily is sure that no coronavirus work had gone through the P3 framework, which means it did not rise to the level of concern to get the extra approval of P3CO. She will try to determine if we have any distant ties to this work and above -- all of these sentences and statements are compatible with what I mentioned to

Fauci's testimony (Nov 2022)

4.5 Andersen sets up a Slack channel to discuss the analysis (1 Feb, 12:11 pm EST)

A bit less than two hours before the call, Andersen set up a Slack channel titled "**project-wuhan_engineering**" [footnote 20]. Rambaut and Holmes were quickly added, and they started discussing the presentation and materials they were going to review during the call.

Rambaut produced at the time the graphs of the presentation, which were promptly shared and discussed on Slack. A few minutes ahead of the call (1:52 pm EST), Rambaut noted that:

> *"It will be interesting to know what Ron thinks. He is not going to want it to be a GOF escape."*

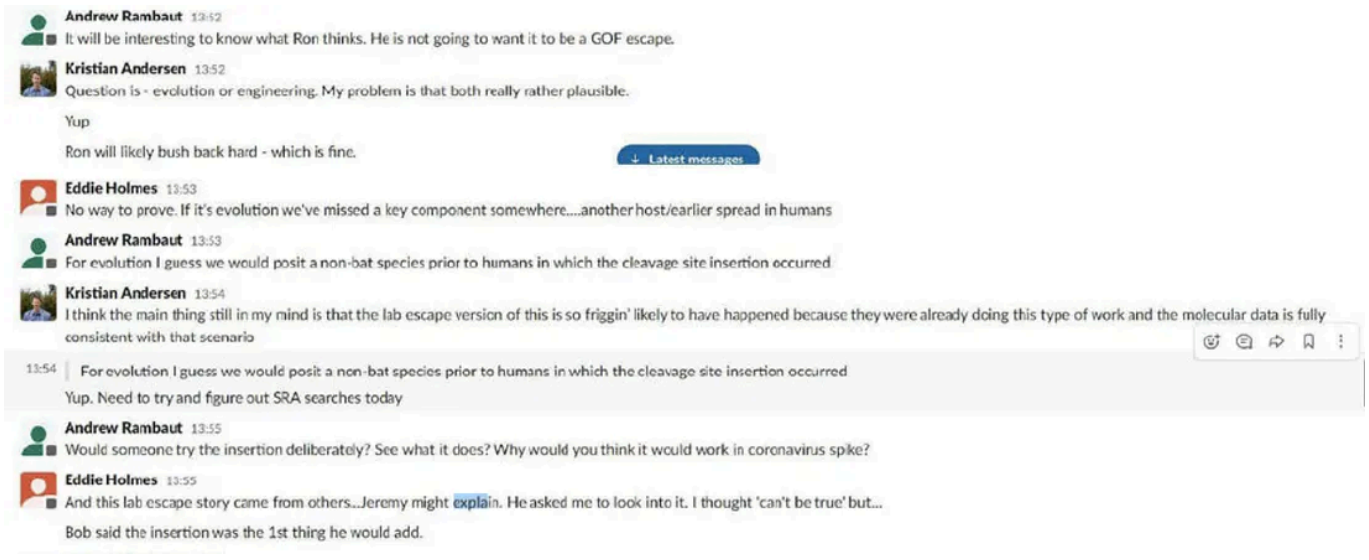
Andersen replies that:

> *"Ron will likely push back hard — which is fine."*

Holmes soon after (1:55 pm EST) added:

> *"And this lab escape story came from others…Jeremy might explain. He asked me to look into it. I thought 'can't be true' but…*
> *Bob said the insertion [FCS} was the 1st thing he would add."*

This again raises the question of who the 'others' were, why they did not make their claims public, and why they are not on the call. As we have seen earlier (section 3.3), Farrar would explain that the 'others' were US scientists who supposedly expressed their concerns in early January, well before such concerns became public.



Proximal_Origin_Slack.pdf, p.3, 1 Feb 2020, 1::52 pm EST

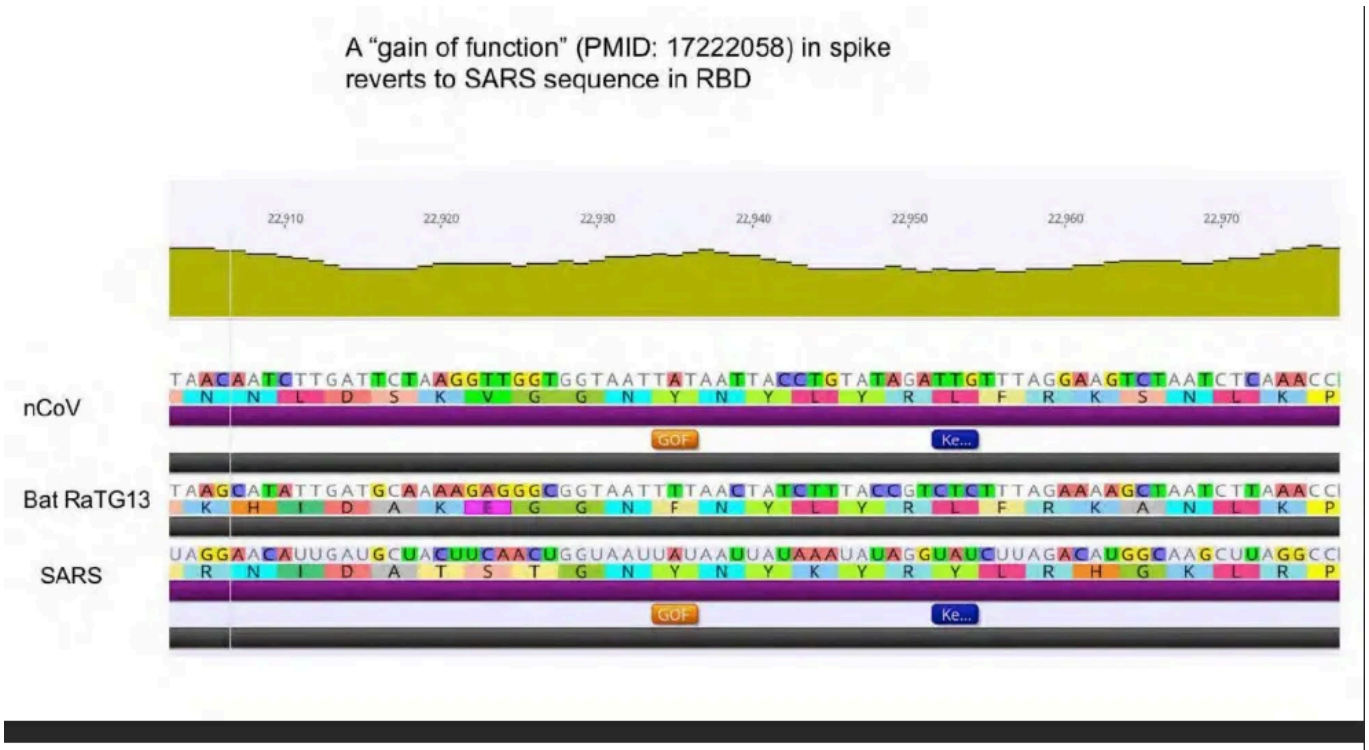
4.6 Farrar forwards the presentation ahead of the call (1 Feb, 1.12 pm EST)

At 1:12 pm EST, Farrar circulated to the call attendees the comparison of coronaviruses sequences presentation produced by Rambaut, Andersen and Holmes ('Coronavirussequencecomparison1.pdf').

The two slides below set the tone:

- The first slide shows the acquisition of the Furin Cleavage Site (**" a 'gain of function' in spike"**), against a SARS-like RBD.
- The second slide lists 6 papers as references, all with Baric as an author. The first one in the list is '*A SARS-like cluster of circulating bat coronaviruses shows potential for human emergence*' by Baric and Zhengli; that's the paper that Fauci had sent to Auchincloss in the wee hours of the morning (0:29 am EST), most likely after Andersen had mentioned it to Fauci when they had their discussion in the evening the previous day (~7 pm EST).

Even if omnipresent in the papers referenced, Baric, the foremost US gain of function specialist, was not part of the call, since, as we have seen, he and Daszak were passed over precisely for their links to the WIV (see 3.10).



extract of the presentation for the call (‘Coronavirus sequence comparison[1].pdf’), extracted from farrar-fauci-comms-full.pdf

Seeing that presentation was essential for the bounce-off group of Koopmans, Fouchier, Drosten and Pöhlmann, who were looped in out of the blue only less than 24 hours ago. However, the rather direct reference to suspected GoF work, and to Baric’s links to the WIV, was clearly not going to please the GoF bench of Koopmans, Fouchier and Drosten, just as Andersen and Garry expected.

The six Baric papers listed at the end of the presentation are:

- ‘**A SARS-like cluster of circulating bat coronaviruses shows potential for human emergence**’ (Menachery *et al.*, 2015); https://pubmed.ncbi.nlm.nih.gov/26552008/
  A key Nature Medicine paper with Shi Zhengli and Baric as lead authors. Funded in part by NIH, NIAID and USAID grants. Mentioned by Andersen in his call to Fauci on 31 Jan, then forwarded later that night by Fauci to Auchincloss with instruction to read it and wait for a call.

- ‘**A mouse-adapted SARS-coronavirus causes disease and mortality in BALB/c mice**’ (Roberts *et al.*, 2007); https://pubmed.ncbi.nlm.nih.gov/17222058/
  A paper with Baric as 2nd lead author.
  Funded by NIH and NIAID grants.

- ‘**SARS-like HIV 1-CoV poised for human emergence**’ (Menachery *et al.*, 2016); https://pubmed.ncbi.nlm.nih.gov/26976607/.

Yet another paper with Baric as lead-author.
Funded in parts by NIH and NIAID grants.

- '**Modeling pathogenesis of emergent and pre-emergent human coronaviruses in mice**' (Cockrell *et al.*, 2018);
  https://pubmed.ncbi.nlm.nih.gov/30043100/
  Yet another paper with Baric as lead-author.
  Funded by NIAID grants.

- '**Receptor recognition by novel coronavirus from Wuhan: An analysis based on decade-long structural studies of SARS**';
  https://pubmed.ncbi.nlm.nih.gov/31996437/ (Wan *et al.*, accepted January 28, 2020)
  A post-outbreak paper with Baric as 2nd lead author.
  Funded by NIH grants.

- '**Molecular determinants of severe acute respiratory syndrome coronavirus pathogenesis and virulence in young and aged mouse models of human disease**' (Frieman *et al.*, 2011);
  https://pubmed.ncbi.nlm.nih.gov/22072787/
  Yet another paper with Baric as lead-author.
  Funded by NIH and NIAID grants.

4.7 'Derision down the phone line' (1 Feb, call: 2 to 3 pm EST)

During the one-hour Farrar call, the 'core group' ended up clashing with the 'bounce-off group'. As expected (see 4.5), **Fouchier** and **Drosten** pushed hard against any mention of laboratory-product. These made their point in ways that sounded excessive for Andersen, Holmes, and Garry.

As Farrar recalled in Spike: '**There was derision down the phone line**'. Farrar went on to explain that Fouchier, Koopmans and Drosten would not countenance any discussion of a possible engineering origin of the FCS, based on the simple reason that the backbone was unknown, and was not one of the many available instead in databases.

That they had no idea what was in Chinese databases as a result of the last 2 to 3 years of extensive sampling in South East Asia, sometimes by Chinese teams on their own, with (as should be expected) very little yet processed and then published, clearly did not register with them as an issue. [footnotes 25, 56]

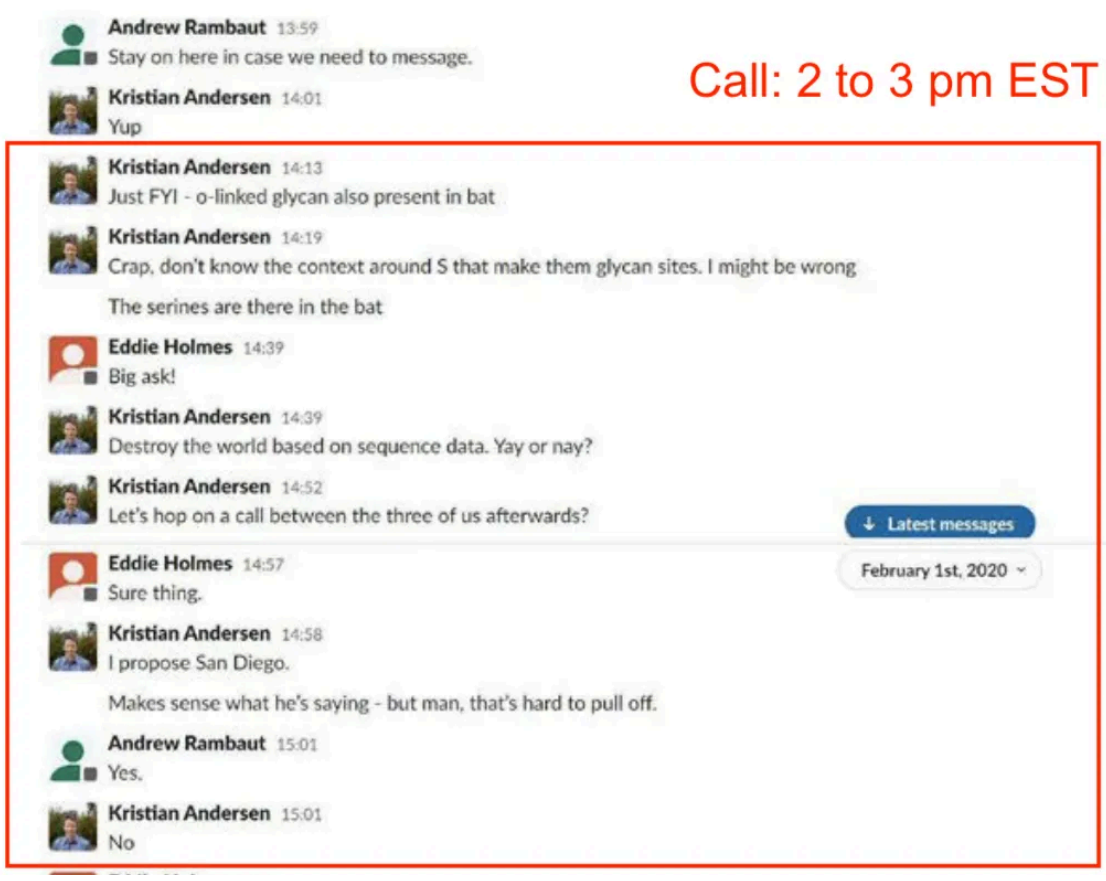
As to what ought to be done next, following that clash, some elements became obvious:

- The call group was not in a position to solve the origin question, since the people there could not agree between themselves, and could be quite vehement (Fouchier and Drosten in particular).

- The stakes behind the origin questions were simply too high to be solved without an adequate international and institutional setup, especially with this level of incertitude and this type of reaction.

The logical answer to these was that:

- Opinions from more experts would have to be solicited.
- A proper international and institutional setup had to be sought.



We already had Eddie and Kristian. They had been having conversations in the background with Andrew Rambaut and Bob Garry, an expert on viruses at Tulane University in Louisiana and one of Kristian's regular collaborators. Bob had independently clocked what looked like peculiarities in the virus. Kristian had also sought the opinion of his Scripps colleague Michael Farzan, who had made important discoveries about how SARS-CoV-1 binds to human cells.

Michael confessed he was struggling to figure out how the new coronavirus could have acquired its features in a natural way. My wife Christiane recommended Marion Koopmans and Ronald Fouchier, both at Erasmus University in the Netherlands, plus Christian Drosten, who directs the Institute of Virology at the Charité Hospital in Berlin. Christiane pointed out that a truly international group would scotch any rumbling that this was a conspiratorial Anglo-American stitch-up against China.

The panel made a dream team: all globally respected, opinionated scientists who would challenge each other without fear or favour. Marion serves as a science adviser to the WHO – we work together on the science advisory committee for the WHO R&D Blueprint, which focuses on the research strategy for new diseases – and was a member of the Emergency Committee that declared the public health emergency. Later, she would also form part of the WHO delegation that went into Wuhan to investigate the origins of SARS-CoV-2. Ron is a renowned virologist who is a longtime advocate, and practitioner, of gain of function studies. He would be able to give us an insider's view of how likely this kind of research might be at the heart of the mystery.

There was derision down the phone line.

Eddie recalls: 'The general feeling from the biologists was that they just didn't believe a word of it. They could not countenance the fact it might be a lab escape, for good reason: if someone was engineering a coronavirus, they wouldn't use some random bat virus in their own lab. They'd use a familiar strain that they knew could

Proximal_Origin_Slack.pdf, p.4, 1 Feb 2020, during the call, and extract from Spike

During the call, at 2:43 pm EST, Koopmans sent back some notes on the pangolin sequences shared before the call. Holmes answered her back (the text of the answer is redacted) just after the call (3.10 pm EST).

NIH-FOIA-57153-Murray-Complete-Response.pdf, p. 48, 1 Feb 2020 and farrar-fauci-comms-full.pdf, p.123, 1 Feb 2020

4.8 Fauci and Collins suggest publishing (1 Feb, call: 2 to 3 pm EST)

During the call, Fauci suggested again that Andersen, Garry, Holmes and Rambaut should write a paper about their doubts in a peer-reviewed publication. Fauci (director of NIAID) was apparently backed by Collins (Director of NIH):

A Again, Drs. Fauci or Drs. Collins, other than suggesting that we should

[redacted]

66

consider writing a scientific peer-reviewed paper as part of the conversations on February 1st, and the phone call I had with Dr. Fauci the day before, had no involvement in the drafting of this particular paper.

Andersem, in 2023.06.16-Andersen-Transcript.pdf

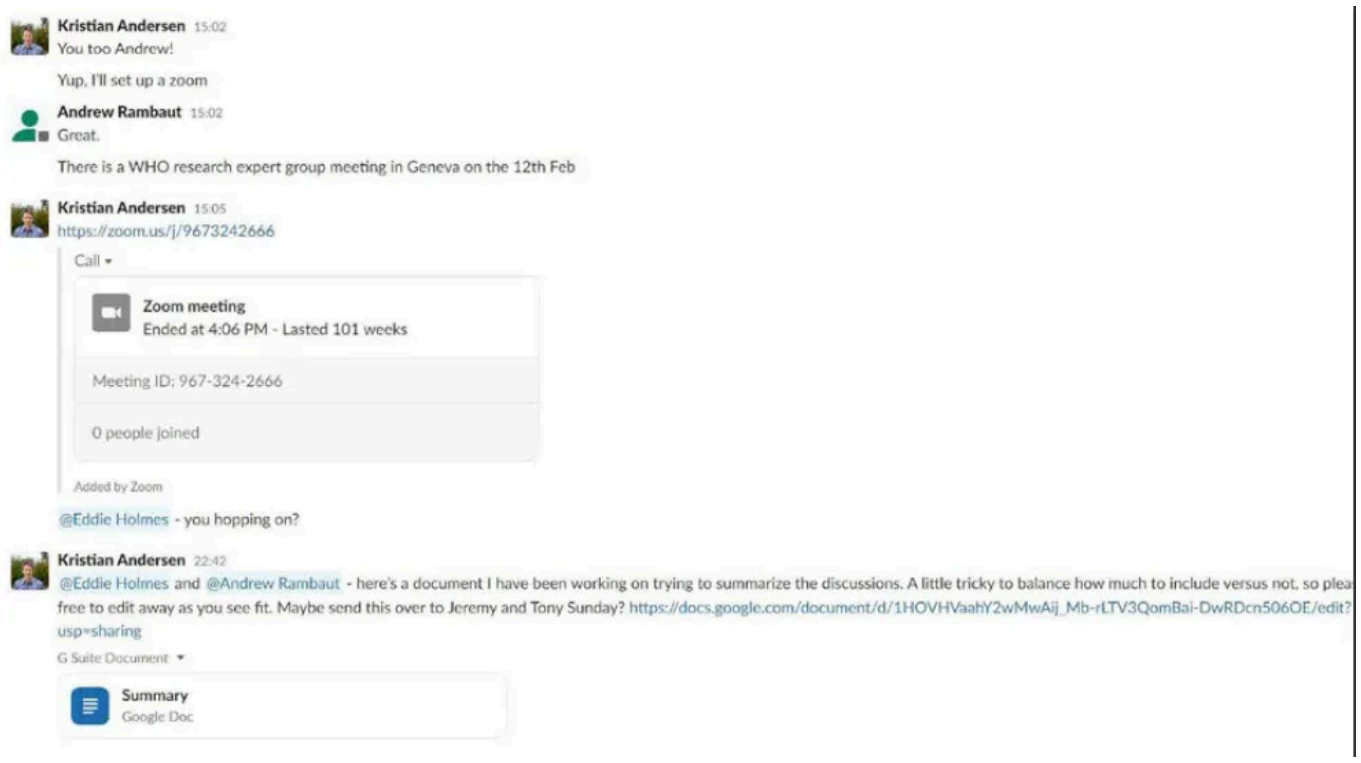
This was a rather logical advice, as it would move the difficult discussion (as Andersen and Fouchier are clashing) to a larger scientific forum that can decide for itself. Still, Farrar's suggestion would have been pushed back by both Andersen, who thought it was too early to write some balanced review of the possible origins, and by Fouchier and Koopmans, who didn't want anything published about a non-natural origin.

So Fauci's suggestion to write a paper for publication would have been considered inappropriate at that stage by the call group, except maybe for Holmes (as mentioned by Andersen in his House transcript [footnote 92]). However, writing a *confidential report* instead, to share within the proposed proper setup, would be perfectly logical.

4.9 The core-group 'After-Call': ~3 pm to max 4 pm EST

Immediately after the Farrar call, Andersen, Holmes and Rambaut spent one hour on a Zoom call (their own 'After Call').

The next message on Slack is Andersen's sharing the very first summary of their analysis ('Summary' Google Doc), which was meant as a confidential report that could be shared with the right setup.



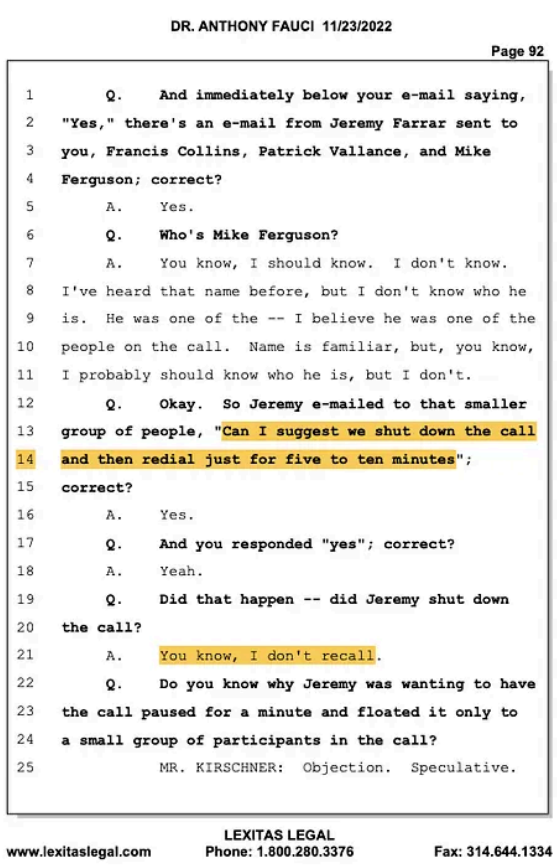
Proximal_Origin_Slack.pdf, p.4, 1 Feb 2020, during the call

4.10 The bosses 'After-Call': 3 pm, for probably 10–15 minutes

As the call was being wrapped up (2:56 pm EST), Farrar emailed Fauci, Collins, Ferguson and Vallance to invite them to redial in once the main call had been closed, for a short conversation.

This was the '**After-Call**' of the NIH (Collins), NIAID (Fauci) and Wellcome Trust (Farrar, Vallance) managers, plus the UK scientific adviser (Ferguson). They had heard the scientific arguments, they now needed to share impressions and agree on the implementation of the path forward (especially reaching out to Dr. Tedros at the WHO).

The line was reopened by Farrar at 3:07 pm EST. Collins seems to have joined after 3:15 pm EST, having missed the e-mail from Farrar. In his testimony of 23 Nov 2022, just as with the call of 30 Jan 2020, Fauci could not remember anything of that 'After Call'.



Fauci's testimony of 23 Nov 2022, p. 92–93

From: Jeremy Farrar
Sent: Sat, 1 Feb 2020 20:13:15 +0000
To: Collins, Francis (NIH/OD) [E];Fauci, Anthony (NIH/NIAID) [E];Ferguson, Mike;Patrick Vallance
Subject: Re: Teleconference

sure

From: Francis Collins (b) (6)
Date: Saturday, 1 February 2020 at 20:12
To: Jeremy Farrar (b) (6), "Fauci, Anthony (NIH/NIAID) [E]" (b) (6), Michael FMedSci (b) (6), Patrick Vallance (b) (6)>
Subject: RE: Teleconference

Just saw this, still want me to call in?

From: Jeremy Farrar (b) (6)
Sent: Saturday, February 1, 2020 2:56 PM
To: Fauci, Anthony (NIH/NIAID) [E] (b) (6) Collins, Francis (NIH/OD) [E] (b) (6)>; Ferguson, Mike (b) (6) Patrick Vallance (b) (6)
Subject: Re: Teleconference

Can I suggest we shut down the call and then redial in?

Just for 5-10mins?

From: Fauci, Anthony (NIH/NIAID) [E]
Sent: Sat, 1 Feb 2020 22:06:26 +0000
To: Jeremy Farrar;Collins, Francis (NIH/OD) [E]
Subject: RE: Teleconference

Thanks, Jeremy. We really appreciate what you are doing here. Pleasure to work with you.
Best,
Tony

From: Jeremy Farrar (b) (6)
Sent: Saturday, February 1, 2020 4:00 PM
To: Collins, Francis (NIH/OD) [E] (b) (6)
Cc: Fauci, Anthony (NIH/NIAID) [E] (b) (6)
Subject: Re: Teleconference

We are altogether as you know! Conversations with you and Tony, and Patrick and others – always great working with you both

---

From: Francis Collins (b) (6)
Date: Saturday, 1 February 2020 at 20:50
To: Jeremy Farrar (b) (6)
Cc: "Fauci, Anthony (NIH/NIAID) [E]" (b) (6)
Subject: Re: Teleconference

Hi Jeremy,
I can make myself available at any time 24/7 for the call with Tedros. Just let me know.
Thanks for your leadership on this critical and sensitive issue.
Francis

Sent from my iPhone

On Feb 1, 2020, at 3:07 PM, Jeremy Farrar (b) (6) wrote:

I have rejoined so a line is open if any help to rejoin.

From: Jeremy Farrar (b) (6)
Date: Saturday, 1 February 2020 at 19:56
To: "Fauci, Anthony (NIH/NIAID) [E]" (b) (6), Francis Collins (b) (6) Michael FMedSci (b) (6), Patrick Vallance (b) (6)
Subject: Re: Teleconference

Proximal_Origin_Emails.pdf, p.153, 1 Feb 2020, 3:12 pm EST (for Collins' email), and farrar-fauci-comms-full.pdf, p. 115, 3:50 pm (for Collins' email)

4.11 Farrar proposes a way out via the WHO (1 Feb, 3:59 pm EST)

Immediately following the call and its after-call, Farrar wrote down a plan of action in an email, sent at 3:59 pm EST, and asked for feedback from the call participants:

- One should transfer the origin question to the WHO or some international body with the right authority and credibility to look into it.
- The body should avoid putting the question in such a charged binary way (zoonotic / not zoonotic). It could sugar it by seemingly asking *'What [Why] are the evolutionary origins of 2019-nCoV important for future risk assessment and understanding of animal/human coronaviruses'.*
- To fight conspiracies, that initiative should start as soon as possible.

Note that these was no mention in Farrar's email that Andersen, Holmes, Rambaut and Garry should write a paper for publication. This was proposed by Fauci during the call, but was likely turned down, as we have just discussed (see 4.8).

Van: Jeremy Farrar [redacted]>
Datum: zaterdag 1 februari 2020 om 21:59
Aan: "Fauci, Anthony (NIH/NIAID) [E]" [redacted] Patrick Vallance [redacted]
CC: Christian Drosten [redacted], "M. Koopmans" [redacted], "R.A.M. Fouchier" <r.fouchier@erasmusmc.nl>, Edward Holmes [redacted] Andrew Rambaut [redacted], "Kristian G. Andersen" [redacted], Paul Schreier [redacted] "Ferguson, Mike" [redacted] Francis Collins [redacted], [redacted] Josie Golding [redacted]
Onderwerp: Re: Teleconference

Thank you to everyone for joining.

There is clearly much to understand understand in this. This call was very helpful to hear some of our current understanding and the many gaps in our knowledge. I do not believe this is a question of a binary outcome, it is more a question of "What are the evolutionary origins of 2019-nCoV, important for future risk assessment and understanding of animal/human coronaviruses".

I do know there are papers being prepared, there will media interest and there is already chat on Twitter/WeChat.

We on this call are not the only ones with scientific expertise in this area and this was an ad hoc group that came together to air some thoughts. It is clearly not the sole group to take this forward, that will need a broader range of imput and a respected international body to ask an expert group to explore this, with a completely open mind. In order to stay ahead of the conspiracy theories and social media I do think there is an urgency for a body to convene such a group and commission some work to – (draft)

"To understand the evolutionary origins of 2019-nCoV, important for this epidemic and for future risk assessment and understanding of animal/human coronaviruses".

In other words a completely open minded and neutral question bringing in the best minds, and under the umbrella of a respected international agency

I hope that is a reasonable approach, please send any thoughts or suggestions.

Once again, thank you for making time over a weekend and for such an informed discussion on a complex issue.

Thank you and best wishes Jeremy

farrar-fauci-comms-full.pdf, p. 112, 1 Feb 2020, 3:59 pm EST

4.12 Fauci gets back to the HHS group (1 Feb, ~6 pm EST)

Another good summary of the call and its takeaways was provided by Fauci, in the email he sent shortly after the call to people in the US. Dep. of Health and Human Services (HHS).

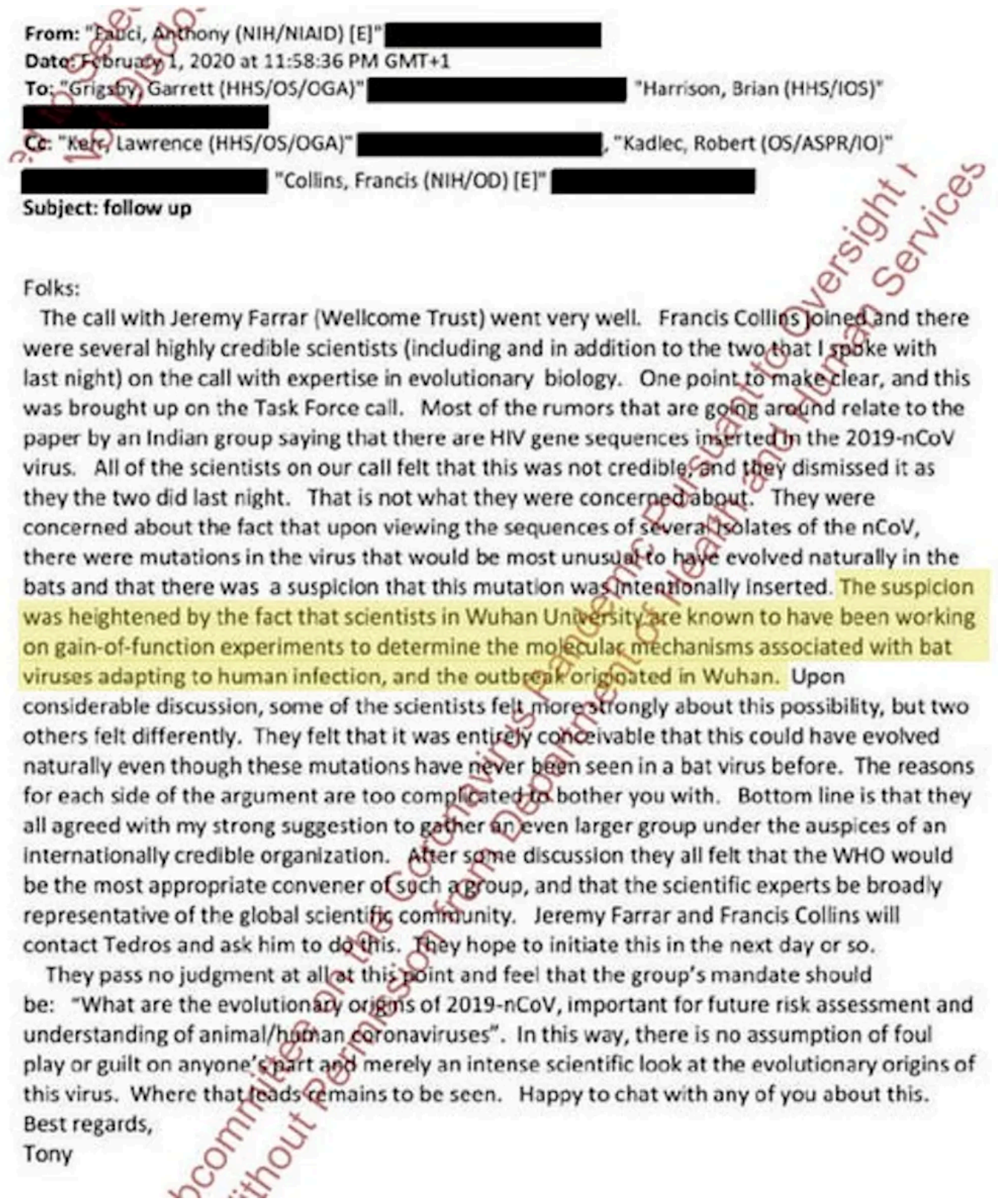
The recipients of the email were:

- Garrett Grigsby (Director for the Office of Global Affairs at HHS),
- Lawrence Kerr ('Larry', Director of the Office of Pandemics and Emerging Threats at HHS),
- Brian Harrison (Republican, coordinated the HHS early response to COVID-19),
- Robert Kadlec (Republican, Assistant Secretary for Preparedness and Response at HHS).

With Kadlec and Harrison, that 'HHS group' had a potential political colouring. Collins (Director of NIH), who was anyway on the Farrar call, was the only non-HHS recipient.

In that email sent on 1 Feb, 5:58 pm EST, Fauci explained that:

- Scientists in Wuhan are known to be working of GoF experiments focussed on bat viruses adaptation to infect human.
- While the rest of the group has serious concerns, Fouchier and Drosten strongly opposed any suggestion of a lab-construct.
- In the end, and maybe precisely because of the strong tensions in the group, the group accepted his suggestion to try to transfer the problem to a 'larger group' under the auspices of an internationally credible organisation.
- Farrar and Collins accordingly agreed to reach out to Dr. Tedros at the WHO, with the neutral-sounding question.



From: "Fauci, Anthony (NIH/NIAID) [E]" [redacted]
Date: February 1, 2020 at 11:58:36 PM GMT+1
To: "Grigsby, Garrett (HHS/OS/OGA)" [redacted] "Harrison, Brian (HHS/IOS)" [redacted]
Cc: "Kerr, Lawrence (HHS/OS/OGA)" [redacted], "Kadlec, Robert (OS/ASPR/IO)" [redacted] "Collins, Francis (NIH/OD) [E]" [redacted]
Subject: follow up

Folks:

The call with Jeremy Farrar (Wellcome Trust) went very well. Francis Collins joined and there were several highly credible scientists (including and in addition to the two that I spoke with last night) on the call with expertise in evolutionary biology. One point to make clear, and this was brought up on the Task Force call. Most of the rumors that are going around relate to the paper by an Indian group saying that there are HIV gene sequences inserted in the 2019-nCoV virus. All of the scientists on our call felt that this was not credible, and they dismissed it as they the two did last night. That is not what they were concerned about. They were concerned about the fact that upon viewing the sequences of several isolates of the nCoV, there were mutations in the virus that would be most unusual to have evolved naturally in the bats and that there was a suspicion that this mutation was intentionally inserted. The suspicion was heightened by the fact that scientists in Wuhan University are known to have been working on gain-of-function experiments to determine the molecular mechanisms associated with bat viruses adapting to human infection, and the outbreak originated in Wuhan. Upon considerable discussion, some of the scientists felt more strongly about this possibility, but two others felt differently. They felt that it was entirely conceivable that this could have evolved naturally even though these mutations have never been seen in a bat virus before. The reasons for each side of the argument are too complicated to bother you with. Bottom line is that they all agreed with my strong suggestion to gather an even larger group under the auspices of an internationally credible organization. After some discussion they all felt that the WHO would be the most appropriate convener of such a group, and that the scientific experts be broadly representative of the global scientific community. Jeremy Farrar and Francis Collins will contact Tedros and ask him to do this. They hope to initiate this in the next day or so.

They pass no judgment at all at this point and feel that the group's mandate should be: "What are the evolutionary origins of 2019-nCoV, important for future risk assessment and understanding of animal/human coronaviruses". In this way, there is no assumption of foul play or guilt on anyone's part and merely an intense scientific look at the evolutionary origins of this virus. Where that leads remains to be seen. Happy to chat with any of you about this.

Best regards,
Tony

email from 1 Feb 2020 (UK Time), Source: COVID Select Committee tweet

In the email, Fauci repeats word for word the *'What are the evolutionary origins of 2019-nCoV important..'* of Farrar's email (see 4.11), typo included ('What' for 'Why'), and explains that *'In this way, there is no assumption of foul play or gilt on anyone's part'.*

Unlike Farrar, Fauci did not mention conspiracy theories. He also made no assumption as to what origin was more likely: *'Where that leads remains to be seen'.* Last, he mentioned that Farrar and Collins would contact Dr. Tedros the next day to get this moving.

4.13 Grigsby mentions high-level discussions between HHS and WHO (1 Feb, 2:29 pm)

Fauci's email above was actually in the form of a reply to an earlier email by Garrett Grigsby, HHS Director of Global Affairs, sent during the Farrar call (at 2:29 pm EST).

The main recipient of the e-mail, entitled '**follow up**', was Brian Harrison, with Fauci and Lawrence Kerr in cc. In his email, Grigsby's mentioned to his recipients that Fauci was talking to Farrar that afternoon and explained that the 'imperative' of what they were discussing was now 'moot', since it was by then in the news and people would do their own analysis of the data.

By this,

- he was likely referring to various articles published on 1 Feb following the Indian HIV insert preprint posted on 31 jan,
- he likely meant an imperative to analyse the SARS-CoV-2 sequence for any sign of engineering.

Grigsby also mentioned that he had previously many related calls with, Kerr, Fauci, and Stewart Simonson, the WHO director in charge of the WHO-US Liaison Office ('Stew').

The following day (2 Feb), Grigsby would be forwarding Fauci's proposal to see the WHO pick up the non-natural features question to Stewart Simonson (see 7.4).

From: Fauci, Anthony (NIH/NIAID) [E]
To: Grigsby, Garrett (HHS/OS/OGA); Harrison, Brian (HHS/IOS)
Cc: Kerr, Lawrence (HHS/OS/OGA); Kadlec, Robert (OS/ASPR/IO); Collins, Francis (NIH/OD) [E]
Subject: follow up
Date: Saturday, February 1, 2020 5:58:33 PM

Folks:
The call with Jeremy Farrar (Wellcome Trust (b) (5)
(b) (5)
(b) (5) Happy to chat with any of you about this.
Best regards,
Tony

From: Grigsby, Garrett (HHS/OS/OGA) (b) (6)
Sent: Saturday, February 1, 2020 2:29 PM
To: Harrison, Brian (HHS/IOS) (b) (6)
Cc: Fauci, Anthony (NIH/NIAID) [E] (b) (6); Kerr, Lawrence (HHS/OS/OGA) (b) (6)
Subject: follow up
Brian,

Sorry it's been so hard to connect…
Been on many calls w/ Dr F, Stew & Larry, but things have evolved throughout the day – see articles attached from around the world on issue being discussed. It's now very much in the public domain so it would appear to make the imperative of what we were discussing moot because others will certainly get the ball rolling on examination of data.
I know Dr F was speaking with Jeremy of Wellcome Trust this afternoon, so he may have more perspective to add…
Happy to connect on this or anything else at your convenience…
Garrett Grigsby
Director for Global Affairs
Office of the Secretary
U.S. Department of Health & Human Services
(b) (6)

NIH-FOIA-57153-Murray-Complete-Response.pdf, p. 50–51, 1 Feb 2020, 2:29 pm EST

Insight: Boomerang?

Grigsby's email reveals that Fauci and his HHS contacts had already been discussing contacting the WHO to look into the origin question, ahead of the Farrar call.

Maybe they were prompted by recent discussions with Fauci, or quite possibly people like Kadlec would have been already plugged into the 'US scientists' that had raised some concerns in mid-January, possibly tipping off the MI5. With the irony that these concerns would have then come back to them via Farrar's handling of a side-check.

At that stage, since the concerns were starting to get into the open, pushing them to the WHO and away from Farrar made sense.

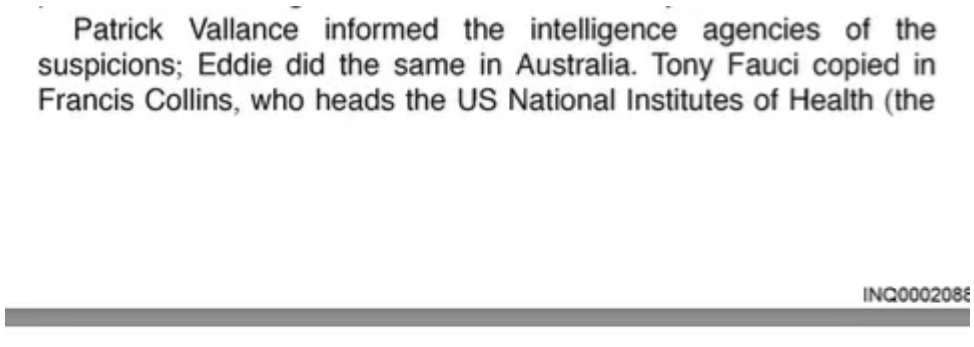
4.14 US-UK-AU Intel services are informed of the suspicions (after call)

Sometimes after the call, according to *Spike*, the US, British and Australian intel services were informed of Andersen's analysis and the still ongoing suspicions (likely while reporting differences of opinions). This followed the instructions given by Fauci after Farrar and Andersen briefed him on 31 Jan evening (see 3.12.b).

- For the British services (MI5) it would be an update, since Farrar had been in touch with them right from the beginning.

- For the US, some intel elements would have likely already been looking at it (possibly tipping the MI5, who then turned to Farrar for an external option).
- For Australia, it might not have been exactly news either, given the 5-eyes links.

Spike adds that Vallance updated the British services, Holmes the Australian ones, and that Collins took care of the US ones.



Patrick Vallance informed the intelligence agencies of the suspicions; Eddie did the same in Australia. Tony Fauci copied in Francis Collins, who heads the US National Institutes of Health (the

INQ0002088

National Institute of Allergy and Infectious Disease, which Tony heads, is part of the NIH). Tony and Francis understood the extreme sensitivity of what was being suggested, given the anti-China rhetoric coming from both President Trump and US Secretary of State Mike Pompeo. Christiane's advice kept running through my head: 'Stick to the science'.

extract from Spike

. . .

5. Farrar and the 'core-group' compare notes and estimates

5.1 The FCS can still be explained by passaging (1–2 Feb)

After the strong pushback of Fouchier-Drosten during the call, one of the first things that the 'core' group and Farrar did was very logically to check where each still stood in terms of probability of a lab accident. Did Fouchier's and Drosten's explanations change their mind? Was it even still worth considering the origin question after facing their derision?

As the 'core-group' members (plus Farzan) shared notes and gave their updated probability estimates, it became clear that passaging was still a very valid non-natural hypothesis, whatever pushback Fouchier and Drosten had effected on the alternative explanation of a *direct* gene construct of the FCS [footnote 17].

The first ones to give their updates were Farzan and Garry.

a. Farzan:
In his notes sent to Farrar after the call, Mike Farzan, considered that passaging could have produced the observed FCS and adaptation to humans. He also highlighted the BSL-2 biosafety level, which would have been totally insufficient for that work. One key insight that he gave was that "**tissue**

culture can often lead to gain of basic sites — including furin cleavage sites" and that he has seen it with human coronaviruses.

This is the beginning of the passaging alternative to pure direct engineering, an option that was much more realistic [footnote 19] and would *never* go away in Andersen's evaluation. In the end, Farzan gave his estimate of between 60:40 and 70:30 for a lab accident.

From Mike Farzan (discoverer of SARS receptor):

1. The RBD didn't look 'engineered' to him - as in, no human would have selected the individual mutations and cloned them into the RBD (I think we all agree)
2. Tissue culture passage can often lead to gain of basic sites - including furin cleavage sites (this is stuff they have seen with human coronaviruses)
3. He is bothered by the furin site and has a hard time explaining that as an event outside the lab (though, there are possible ways in nature, but highly unlikely)
4. Instead of directed engineering, changes in the RBD and acquisition of furin site would be highly compatible with the idea of continued passage of virus in tissue culture
5. Acquisition of furin site would likely destabilize the virus, but would make it disseminate to new tissues

So given above, a likely explanation could be something as simple as passaging SARS-like CoVs in tissue culture on human cell lines (under BSL-2) for an extended period time, accidentally creating a virus that would be primed for rapid transmission between humans via gain of furin site (from tissue culture) and adaptation to human ACE2 receptor via repeated passage.

All of this brings it back to a simple conversation about how this virus might have gained a furin site (but with a stretch and series of coincidences you can find a way to explain others – although very odd all together) and there are ways in which that could occur both in nature and in the lab. Nothing seems to specifically suggest whether this virus was most likely to be "adapted", "evolved", or maybe even "engineered". So I think it becomes a question of how do you put all this together, whether you believe in this series of coincidences, what you know of the lab in Wuhan, how much could be in nature - accidental release or natural event? I am 70:30 or 60:40.

farrar-fauci-comms-full.pdf, p.107, 2 Feb 2020

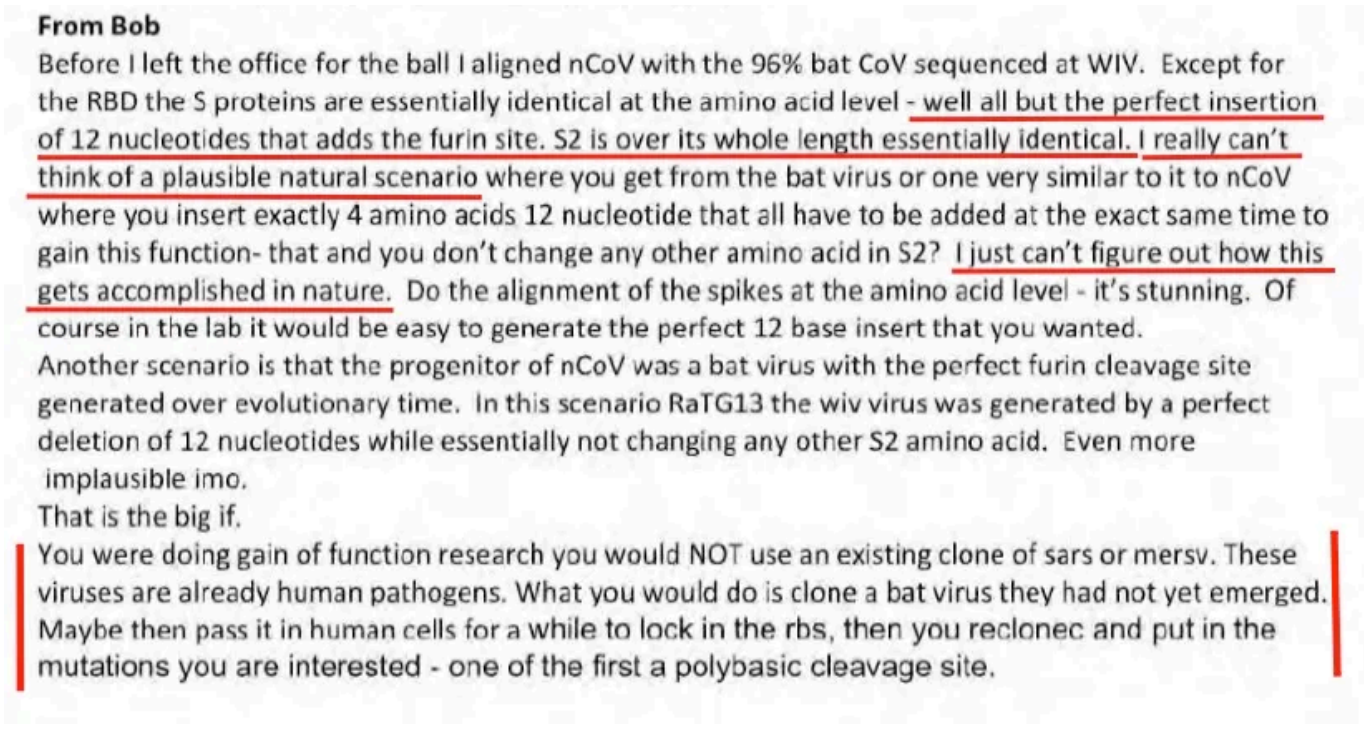
b. Garry

Garry sent his notes to Farrar, commenting first on the difficulty of understanding how the SARS-CoV-2 RBD could be essentially identical to the one from RaTG13, except for 'the perfect insertion of the FCS".

He also provided an excellent **insight**; one that precisely matched the DEFUSE proposal [footnote 73], whereby one would use a not yet emerged coronavirus (so nothing too close to SARS or MERS) and do a mix of targeted engineering (for the FCS) and passaging (to stabilize and optimize the spike protein) [footnotes 19].

> *Garry:*
> *"If you were doing gain-of-function research* ***you would NOT use an existing clone of sars or mers. These viruses are already human pathogens.*** *What you would do is clone a bat virus that has not yet emerged. Maybe then pass it in human cells for a while to look in the RB[D]. Then you reclone and put in the*

> *mutations you are interested [in] — one of the first [being] a polybasic cleavage site [i.e. an FCS].”*
>
> *source: farrar-fauci-comms-full.pdf, p.107, 2 Feb 2020*



From Bob
Before I left the office for the ball I aligned nCoV with the 96% bat CoV sequenced at WIV. Except for the RBD the S proteins are essentially identical at the amino acid level - well all but the perfect insertion of 12 nucleotides that adds the furin site. S2 is over its whole length essentially identical. I really can't think of a plausible natural scenario where you get from the bat virus or one very similar to it to nCoV where you insert exactly 4 amino acids 12 nucleotide that all have to be added at the exact same time to gain this function- that and you don't change any other amino acid in S2? I just can't figure out how this gets accomplished in nature. Do the alignment of the spikes at the amino acid level - it's stunning. Of course in the lab it would be easy to generate the perfect 12 base insert that you wanted.
Another scenario is that the progenitor of nCoV was a bat virus with the perfect furin cleavage site generated over evolutionary time. In this scenario RaTG13 the wiv virus was generated by a perfect deletion of 12 nucleotides while essentially not changing any other S2 amino acid. Even more implausible imo.
That is the big if.
You were doing gain of function research you would NOT use an existing clone of sars or mersv. These viruses are already human pathogens. What you would do is clone a bat virus they had not yet emerged. Maybe then pass it in human cells for a while to lock in the rbs, then you reclonec and put in the mutations you are interested - one of the first a polybasic cleavage site.

farrar-fauci-comms-full.pdf, p.107, 2 Feb 2020

5.2 Fouchier argues against bothering the WHO with the question (2 Feb, 8:30am UK)

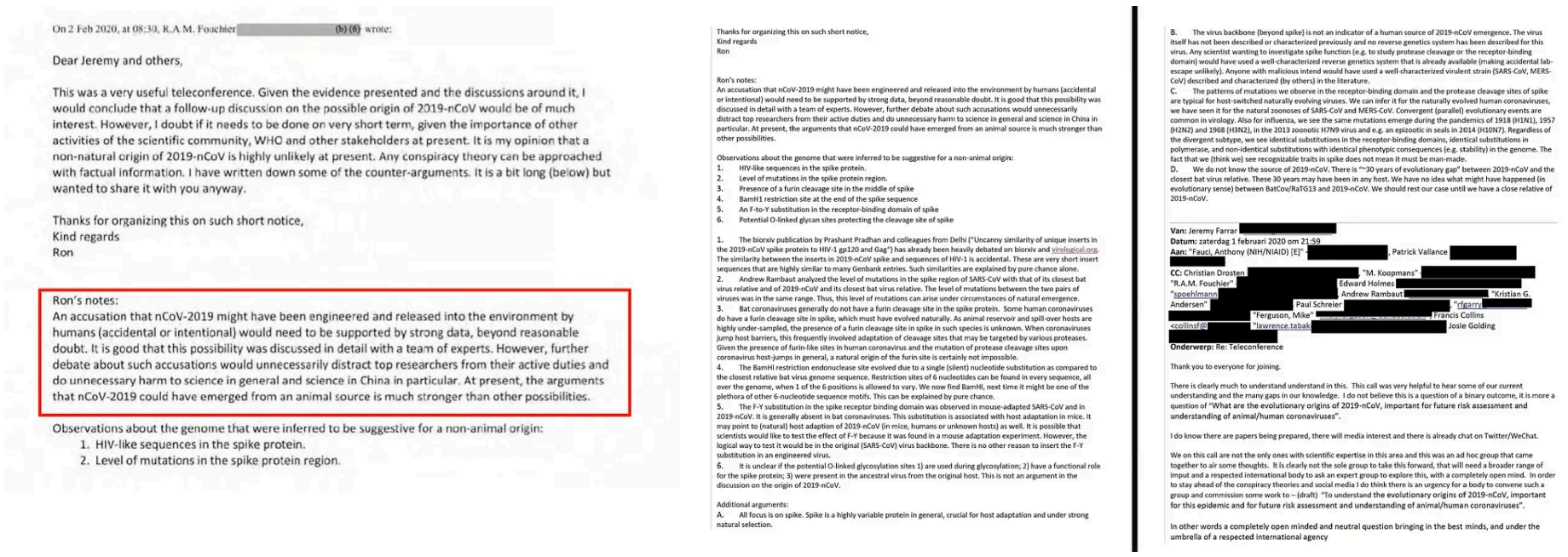
Next, Ron Fouchier sent an email back to Farrar and the conference group.

First, Fouchier addressed the plan to move the question to the WHO, as discussed during the call (see 4.11). According to Fouchier:

1. A natural origin is much more likely than a non-natural one.
2. Unless one can be sure that any feature is non-natural (*‘beyond reasonable doubt’*), there is no point raising the question outside this closed circles of scientists
3. Bringing the question into the open would likely ‘.. *do unnecessary harm to science in general and science in China in particular.’*
4. Hence, the WHO and other scientists, who already have more urgent priorities, have much better things to do than looking into the origin question right now.

As we saw in 4.11, at that stage Fouchier may well have had no idea that Andersen was starting to write a confidential report for the WHO. **For all purposes, what Fouchier was asking the group to do, was to stop looking into the origin question and do something supposedly more useful.**

Fouchier then offered some fairly detailed list of arguments against a non-animal origin, to try to pre-empt any possible further discussion.



On 2 Feb 2020, at 08:30, R.A.M. Fouchier [redacted] (b) (6) wrote:

Dear Jeremy and others,

This was a very useful teleconference. Given the evidence presented and the discussions around it, I would conclude that a follow-up discussion on the possible origin of 2019-nCoV would be of much interest. However, I doubt if it needs to be done on very short term, given the importance of other activities of the scientific community, WHO and other stakeholders at present. It is my opinion that a non-natural origin of 2019-nCoV is highly unlikely at present. Any conspiracy theory can be approached with factual information. I have written down some of the counter-arguments. It is a bit long (below) but wanted to share it with you anyway.

Thanks for organizing this on such short notice,
Kind regards
Ron

Ron's notes:
An accusation that nCoV-2019 might have been engineered and released into the environment by humans (accidental or intentional) would need to be supported by strong data, beyond reasonable doubt. It is good that this possibility was discussed in detail with a team of experts. However, further debate about such accusations would unnecessarily distract top researchers from their active duties and do unnecessary harm to science in general and science in China in particular. At present, the arguments that nCoV-2019 could have emerged from an animal source is much stronger than other possibilities.

Observations about the genome that were inferred to be suggestive for a non-animal origin:
1. HIV-like sequences in the spike protein.
2. Level of mutations in the spike protein region.

Thanks for organizing this on such short notice,
Kind regards
Ron

Ron's notes:
An accusation that nCoV-2019 might have been engineered and released into the environment by humans (accidental or intentional) would need to be supported by strong data, beyond reasonable doubt. It is good that this possibility was discussed in detail with a team of experts. However, further debate about such accusations would unnecessarily distract top researchers from their active duties and do unnecessary harm to science in general and science in China in particular. At present, the arguments that nCoV-2019 could have emerged from an animal source is much stronger than other possibilities.

Observations about the genome that were inferred to be suggestive for a non-animal origin:
1. HIV-like sequences in the spike protein.
2. Level of mutations in the spike protein region.
3. Presence of a furin cleavage site in the middle of spike
4. BamH1 restriction site at the end of the spike sequence
5. An F-to-Y substitution in the receptor-binding domain of spike
6. Potential O-linked glycan sites protecting the cleavage site of spike

1. The biorxiv publication by Prashant Pradhan and colleagues from Delhi ("Uncanny similarity of unique inserts in the 2019-nCoV spike protein to HIV-1 gp120 and Gag") has already been heavily debated on biorxiv and virological.org. The similarity between the inserts in 2019-nCoV spike and sequences of HIV-1 is accidental. These are very short insert sequences that are highly similar to many Genbank entries. Such similarities are explained by pure chance alone.
2. Andrew Rambaut analyzed the level of mutations in the spike region of SARS-CoV with that of its closest bat virus relative and of 2019-nCoV and its closest bat virus relative. The level of mutations between the two pairs of viruses was in the same range. Thus, this level of mutations can arise under circumstances of natural emergence.
3. Bat coronaviruses generally do not have a furin cleavage site in the spike protein. Some human coronaviruses do have a furin cleavage site in spike, which must have evolved naturally. As animal reservoir and spill-over hosts are highly under-sampled, the presence of a furin cleavage site in spike in such species is unknown. When coronaviruses jump host barriers, this frequently involved adaptation of cleavage sites that may be targeted by various proteases. Given the presence of furin-like sites in human coronavirus and the mutation of protease cleavage sites upon coronavirus host-jumps in general, a natural origin of the furin site is certainly not impossible.
4. The BamHI restriction endonuclease site evolved due to a single (silent) nucleotide substitution as compared to the closest relative bat virus genome sequence. Restriction sites of 6 nucleotides can be found in every sequence, all over the genome, when 1 of the 6 positions is allowed to vary. We now find BamHI, next time it might be one of the plethora of other 6-nucleotide sequence motifs. This can be explained by pure chance.
5. The F-Y substitution in the spike receptor binding domain was observed in mouse-adapted SARS-CoV and in 2019-nCoV. It is generally absent in bat coronaviruses. This substitution is associated with host adaptation in mice. It may point to (natural) host adaption of 2019-nCoV (in mice, humans or unknown hosts) as well. It is possible that scientists would like to test the effect of F-Y because it was found in a mouse adaptation experiment. However, the logical way to test it would be in the original (SARS-CoV) virus backbone. There is no other reason to insert the F-Y substitution in an engineered virus.
6. It is unclear if the potential O-linked glycosylation sites 1) are used during glycosylation; 2) have a functional role for the spike protein; 3) were present in the ancestral virus from the original host. This is not an argument in the discussion on the origin of 2019-nCoV.

Additional arguments:
A. All focus is on spike. Spike is a highly variable protein in general, crucial for host adaptation and under strong natural selection.
B. The virus backbone (beyond spike) is not an indicator of a human source of 2019-nCoV emergence. The virus itself has not been described or characterized previously and no reverse genetics system has been described for this virus. Any scientist wanting to investigate spike function (e.g. to study protease cleavage or the receptor-binding domain) would have used a well-characterized reverse genetics system that is already available (making accidental lab-escape unlikely). Anyone with malicious intend would have used a well-characterized virulent strain (SARS-CoV, MERS-CoV) described and characterized (by others) in the literature.
C. The patterns of mutations we observe in the receptor-binding domain and the protease cleavage sites of spike are typical for host-switched naturally evolving viruses. We can infer it for the naturally evolved human coronaviruses, we have seen it for the natural zoonoses of SARS-CoV and MERS-CoV. Convergent (parallel) evolutionary events are common in virology. Also for influenza, we see the same mutations emerge during the pandemics of 1918 (H1N1), 1957 (H2N2) and 1968 (H3N2), in the 2013 zoonotic H7N9 virus and e.g. an epizootic in seals in 2014 (H10N7). Regardless of the divergent subtype, we see identical substitutions in the receptor-binding domains, identical substitutions in polymerase, and non-identical substitutions with identical phenotypic consequences (e.g. stability) in the genome. The fact that we (think we) see recognizable traits in spike does not mean it must be man-made.
D. We do not know the source of 2019-nCoV. There is "~30 years of evolutionary gap" between 2019-nCoV and the closest bat virus relative. These 30 years may have been in any host. We have no idea what might have happened (in evolutionary sense) between BatCov/RaTG13 and 2019-nCoV. We should rest our case until we have a close relative of 2019-nCoV.

Van: Jeremy Farrar [redacted]
Datum: zaterdag 1 februari 2020 om 21:59
Aan: "Fauci, Anthony (NIH/NIAID) [E]" <[redacted]>, Patrick Vallance [redacted]
CC: Christian Drosten [redacted], "M. Koopmans" [redacted], "R.A.M. Fouchier" [redacted], Edward Holmes [redacted], "spoehlmann" [redacted], Andrew Rambaut [redacted], "Kristian G. Andersen" [redacted], Paul Schreier [redacted], "rfgarry" [redacted], "Ferguson, Mike" [redacted], Francis Collins <collinsf@[redacted]>, "lawrence.tabak" [redacted], Josie Golding [redacted]
Onderwerp: Re: Teleconference

Thank you to everyone for joining.

There is clearly much to understand understand in this. This call was very helpful to hear some of our current understanding and the many gaps in our knowledge. I do not believe this is a question of a binary outcome, it is more a question of "What are the evolutionary origins of 2019-nCoV, important for future risk assessment and understanding of animal/human coronaviruses".

I do know there are papers being prepared, there will media interest and there is already chat on Twitter/WeChat.

We on this call are not the only ones with scientific expertise in this area and this was an ad hoc group that came together to air some thoughts. It is clearly not the sole group to take this forward, that will need a broader range of imput and a respected international body to ask an expert group to explore this, with a completely open mind. In order to stay ahead of the conspiracy theories and social media I do think there is an urgency for a body to convene such a group and commission some work to – (draft) "To understand the evolutionary origins of 2019-nCoV, important for this epidemic and for future risk assessment and understanding of animal/human coronaviruses".

In other words a completely open minded and neutral question bringing in the best minds, and under the umbrella of a respected international agency

farrar-fauci-comms-full.pdf, p.110 and 153–154, 2 Feb 2020

5.3 Farrar politely dismisses Fouchier's concerns to the 'core group' (2 Feb, 8:40am UK)

Farrar quickly answered Fouchier by highlighting that it may be better to address the issue without delay, when there was still time, before questions started spreading further and create damaging tensions.

> *"We live in a polarised world where there is a quick reaction to try and deflect issues by blaming someone somewhere. That may only increase tension and reduce cooperation"*
>
> *source: farrar-fauci-comms-full.pdf, p.112, 2 Feb 2020*

The wording of the answer is notable, and clearly was a reference to potentially blaming China which would make scientific cooperation more difficult, exactly when it was most needed. Farrar effectively turned around the argument of Fouchier. Fouchier's point was that discussing the possibility of a research accident would be hurting cooperation, but for Farrar, **not** discussing that possibility may risk hurting cooperation even more.

On 2 Feb 2020, at 08:40, Jeremy Farrar [redacted] (b) (6) wrote:

Thanks Ron

My view is completely neutral on this. The evolutionary origins on this virus are clearly important.

I do know these questions are being asked by politicians, starting in the scientific literature, certainly on social and main stream media. If, and I stress if, this does spread further, pressure and tensions rise, j fear these questions will get louder and more polarised and people will start to look who to blame. We live in a polarised world where there is a quick reaction to try and deflect issues by blaming someone somewhere. That may only increase tension and reduce cooperation.

A respected body convening a group now to consider the evolutionary origins of this, with an open mind, neutral, and in a transparent way is I think an approach that may prevent wild claims being made.

Such a group needs the best minds, from around the world, not just US-Europe-Australia. It needs to be transparent and respected.

I am not sure your thoughts on "short time frame". I am concerned if this is not done quite quickly it will be reacting to what may be lurid claims.

Thoughts on that very welcome.

farrar-fauci-comms-full.pdf, p.112, 2 Feb 2020

5.4 Rambaut pushes back against Fouchier (2 Feb, 9:38 am UK)

Next, Rambaut replied to Fouchier and the rest of the group, insisting on the need to keep an open mind. He also made an interesting public policy point, arguing that the possibility of a non-natural origin could make containment of the outbreak more difficult, and thus may actually need to be evaluated urgently.

From: (b) (6)
Date: Sunday, 2 February 2020 at 09:38
To: Jeremy Farrar (b) (6)
Cc: (b) (6) "Fauci, Anthony (NIH/NIAID) [E]" (b) (6), Patrick Vallance (b) (6), "Drosten, Christian" (b) (6), Marion Koopmans (b) (6) Edward Holmes (b) (6) (b) (6), "Kristian G. Andersen" (b) (6), Paul Schreier (b) (6) Michael FMedSci (b) (6) Francis Collins (b) (6), (b) (6) Josie Golding <J.Golding@wellcome.ac.uk>
Subject: Re: Teleconference

Dear Jeremey, Ron and all,

Thanks for inviting me on the call yesterday. I am also agnostic on this - I do not have any experience of laboratory virology and don't know what it is likely or not in that context. From a (natural) evolutionary point of view the only thing here that strikes me as unusual is the furin cleavage site. It strongly suggests to me that we are missing something important in the origin of this virus. My inclination would be that it is a missing host species in which this feature arose because it was selected for in that host. We can see this insertion has resulted in an extremely fit virus in humans - we can also deduce that it is not optimal for transmission in bat species.

The alternative is that it arose early in the human outbreak, perhaps during a longer period of hidden transmission and then the current epidemic is the result of this mutation but this seems less likely to me (it didn't happen in SARS for example).

Perhaps this needs to be discussed urgently, not only because of the lurid claims on Twitter but because if it is in a non-human host, pre-adapted, it may threaten control efforts through new zoonotic jumps (although perhaps we are beyond this point now).

The biggest hindrance at the moment (for this and more generally) is the lack of data and information. There have been no genome sequences from Wuhan for cases more recent than the beginning of January and reports, but no information, about virus from non-human animals in Wuhan. If the evolutionary origins of the epidemic were to be discussed, I think the only people with sufficient information or access to samples to address it would be the teams working in Wuhan.

Best,
Andrew

farrar-fauci-comms-full.pdf, p.109, 2 Feb 2020

5.5 Farrar shares his roadmap (2 Feb, 9:48 am UK)

Soon after, Farrar effectively closed the debate. Repeating essentially the points made earlier to the call group (see 4.11), and then directly to Fouchier (see 5.3), his focus at this stage was to get the WHO to pick up the question as soon as possible.

Farrar further advised that the discussions within the call group should stop until such transition to the WHO, since these discussions should instead be open, involve a broader panel of scientists and organised within a respected international framework.

It is worth remembering that the core and bounce-off groups of scientists did not know, contrarily to Farrar, that there had already been some discussions with the HHS to get the WHO to look into the question (see 4.13). In fact, the decision had been taken with HHS, and was not open for discussions.

From: Jeremy Farrar [redacted]
Sent: Sunday, February 2, 2020 4:48 AM
To: Andrew Rambaut [redacted]
Cc: R.A.M. Fouchier [redacted]; Fauci, Anthony (NIH/NIAID) [E] [redacted]>; Patrick Vallance [redacted] Drosten, Christian [redacted]>; M.P.G. Koopmans [redacted] Eddie Holmes [redacted] Kristian G. Andersen [redacted]>; Paul Schreier [redacted]; Ferguson, Mike [redacted] Collins, Francis (NIH/OD) [E] [redacted]; Tabak, Lawrence (NIH/OD) [E] [redacted]; Josie Golding [redacted]
Subject: Re: Teleconference

This is a very complex issue.

I will:

- Be in contact with WHO today. I contacted them last night and will speak with them today and set up a broader call with them as soon as possible.
- As discussed on the phone this discussion is not limited to those on this email, it is happening wider in the scientific, social and main stream media.
- I believe the best way forward is for a body like the WHO has to ask or commission a group of scientists from around the world to ask the neutral question "To understand the evolutionary origins of 2019-nCoV, important for this epidemic and for future risk assessment and understanding of animal/human coronaviruses".
- That should be done in an open way and quite quickly so that the world can see it is being done, it can respect the report when it is available and I think that will help with the growing interest of this question.

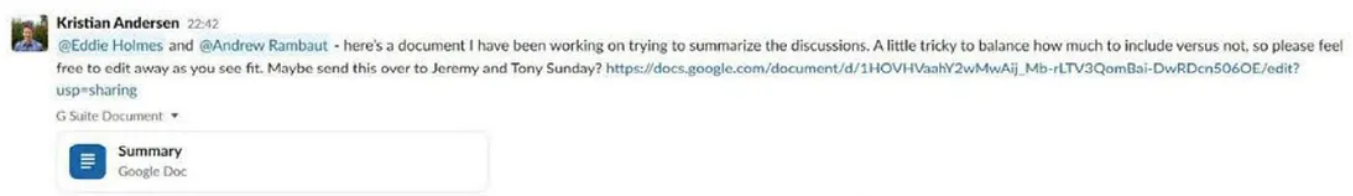
I suggest we don't get into a further scientific discussion here, but wait for that group to be established.

Jeremy

farrar-fauci-comms-full.pdf, p.108, 2 Feb 2020

5.6 In parallel: discussion on Slack

a. Andersen puts together a Summary (1 Feb, 10:42 pm EST)
Following the Farrar call and the core-group After-Call (see 4.9), Andersen spent a few hours putting together a summary, which he shared on Slack at 10:42 pm EST. (That version should be very close to v2 of 8:57 pm EST.)



Proximal_Origin_Slack.pdf, p.5, 1 Feb 2020, 10:42 pm EST

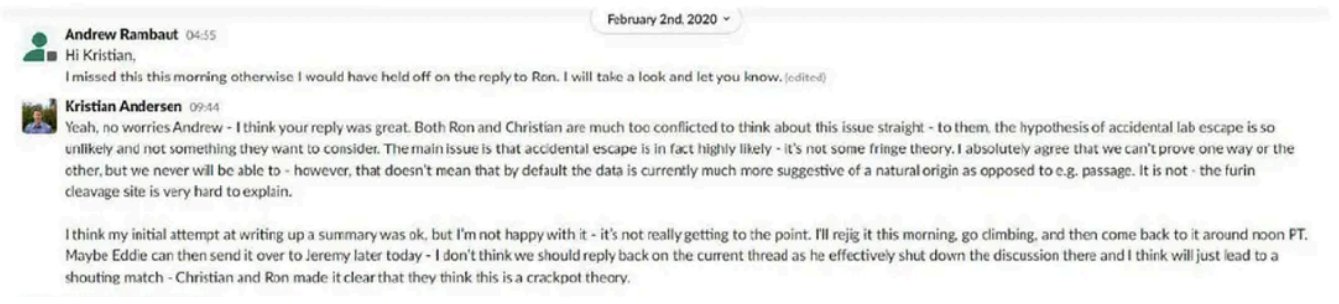
b. The 'core group' rejects Fouchier's arguments (1–2 Feb)
Andersen and Rambaut had some detailed discussions on Slack of Fouchier's and Drosten's vehement oppositions to discussing the possibility of a lab escape. They saw them as being *'too conflicted to think about this issue straight'*, while the simple fact that such experiments were done at BSL-3 (or less) was in itself already a good enough reason to look into this.

> *Andersen, 2 Feb, 9:44 am EST:*
> *[..] Both Ron and Christian are much too conflicted to think about this issue straight -to them, the hypothesis of accidental lab escape is so unlikely and not something they want to consider. The main issue is that accidental escape is in fact highly likely — it's not some fringe theory.*
> *I absolutely agree that we can't prove one way or the other, but we never will be able to — however, that doesn't mean that by default the data is currently much more suggestive of a natural origin as opposed to e.g. passage. It is not — the furin cleavage site is very hard to explain.*

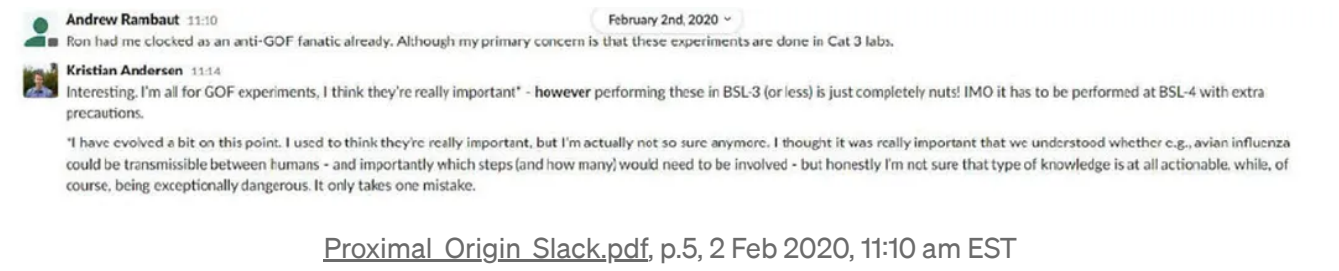
> *Rambaut, 11:00 am EST:*
> *Ron had me clocked as an anti-GOF fanatic already. Although my primary concern is that these experiments are done in Cat 3 labs.*
>
> *Andersen, 11.14 am EST:*
> *Interesting. I'm all for GOF experiments, I think they're really important' - however performing these in BSL-3 (or less) is just completely nuts! IMO it has to be performed at BSL-4 with extra precautions.*
>
> *Andersen, 11:47 am EST:*
> *Reading through Ron's comments again I agree on pretty much everything he's saying — I come to the same conclusions, Where we differ is that he's looking for very specific evidence proving that this is unnatural (which is understandable), but except for the most simple scenario where somebody plugged a gene into a preexisting backbone, that would simply be impossible to prove.*
> *Natural selection and accidental release are both plausible scenarios explaining the data — and a priori should be equally weighed as possible explanations. The presence of furin a posteriori moves me slightly more towards accidental release, but it's well above my paygrade to cal the shots on a final conclusion.*

For reference, Fouchier was well known to totally underestimate the possibility of an outbreak caused by a lab infection. He memorably wrote a paper during the GoF moratorium seriously arguing that his own BSL-3 lab would cause an outbreak (at least one human-to-human infection) **less frequently than once every 33 billion years**, which is about 7 times the age of the earth [footnote 16].

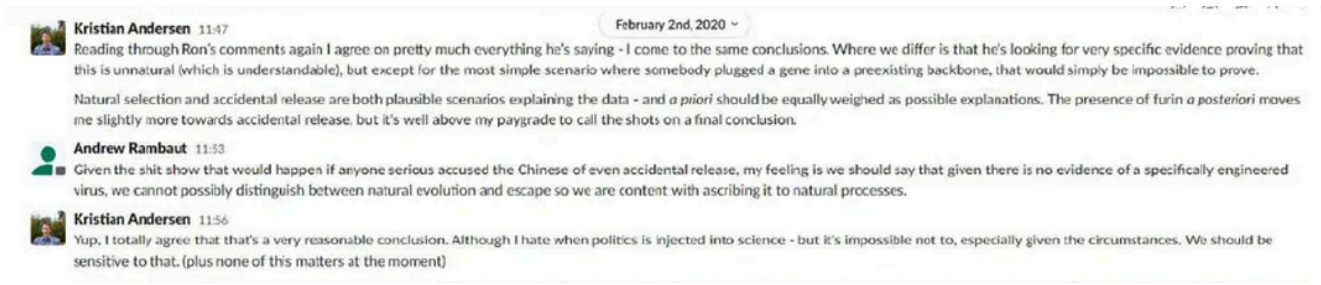
Hence, any serious scientist would take Fouchier's assurance that *'accidental lab escape are very uncommon'* with a truck-full of salt, which is precisely what the core group of Andersen, Holmes and Rambaut did at the time. The conversation on Slack went more specific the next day, and shows that Fouchier's attitude during the call was suspected to be once again intrinsically linked to his vehement pro-GoF position (of which he is a main practitioner).



Proximal_Origin_Slack.pdf, p.5, 2 Feb 2020, 9:44 am EST



Proximal_Origin_Slack.pdf, p.5, 2 Feb 2020, 11:10 am EST



Proximal_Origin_Slack.pdf, p.5, 2 Feb 2020, 11:47 am EST

c. Better to keep Fouchier and Drosten out of this (2 Feb)

Given the resolute and rather irrational opposition of Fouchier and Drosten, Andersen also concluded that there was little point having discussions on the call-group email thread, short of wanting to get into a 'shouting match'.

As a result, the main scientific discussions would from now on be contained on the Slack and in targeted emails, and would bypass the very defensive bounce-off group of Fouchier, Koopmans and Drosten. If these ever had a chance of joining the core-group in the drafting of a scientific report, or later of a paper, they would have definitely burnt it during the Farrar call.

> *Andersen, 2 Feb, 9:44 am EST:*
> *[..] I don't think we should reply back on the current thread as he [Fouchier] effectively shut down the discussion there and I think will just lead to a shouting match — Christian and Ron made it clear that they think this is a crackpot theory.*

d. Better to err on the side of extreme caution (2 Feb)

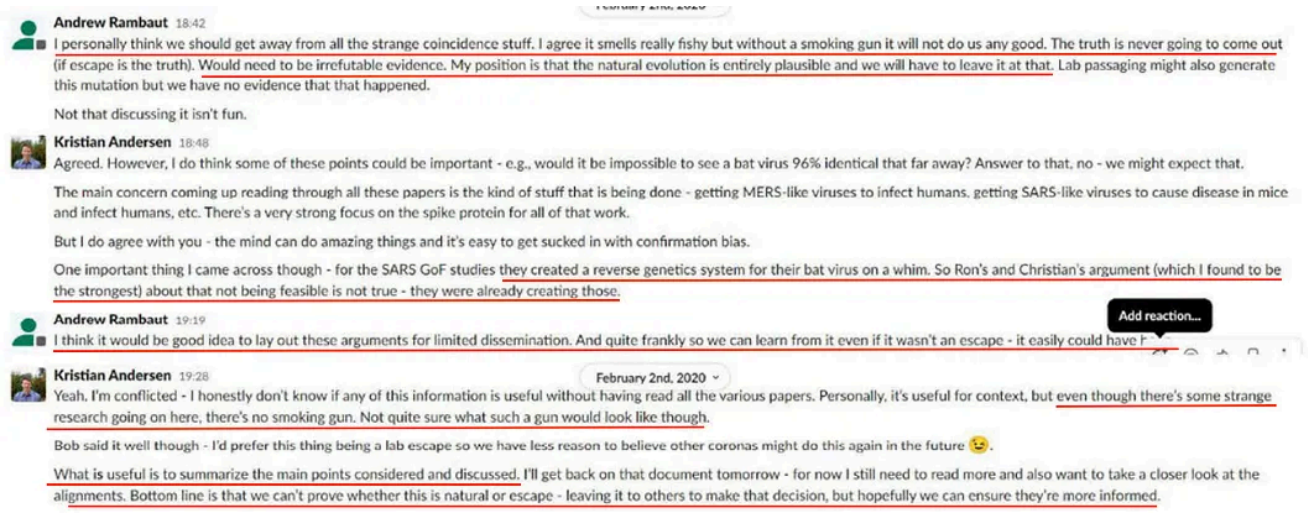
On Slack, Rambaut insisted on the fact that the truth was likely never to come out if it was a lab escape. There would most likely never be a smoking gun, and without a smoking gun, it would just not do them any good to look into all the '*fishy coincidences*'.

Andersen and Rambaut then agreed that the best position was to '***write this report erring on the side of extreme caution***', by stating that 'natural evolution is 'entirely plausible' and 'leaving it at that', meaning by not trying to reach any firmer conclusion, leaving it in that state pointing specifically to a natural origin, while neglecting the 'fishy coincidence'.

> *Rambaut, 2 Feb 2020, 11:52 am EST*
> *Given the shit show that would happen if anyone serious accused the Chinese of even accidental release, my feeling is we should say that given there is no evidence of a specifically engineered virus, we cannot possibly distinguish between natural evolution and escape so we are content with ascribing it to natural processes.*
>
> *Andersen,11:56 am EST*
> *Yup, I totally agree that that's a very reasonable conclusion. Although I hate when politics is injected into science — but it's impossible not to, especially given the circumstances. We should be sensitive to that. (plus none of this matters at the moment).*

And a bit later that day:

> *Rambaut, 7:19 pm EST:*
> *I think it would be [a] good idea to lay out these arguments for limited dissemination. And quite frankly so we can learn from it even if it wasn't an escape — it easily could have been.*
>
> *Andersen, 7:28 pm EST:*
> *[..] even though there's some strange research going on here, there's no smoking gun. Not quite sure what such a gun would look like though.*
> *Bob said it well though — I'd prefer this thing being a lab escape so we have less reason to believe other coronas might do this again in the future* 😉*.*
> *What is useful is to summarize the main points considered and discussed. [..].*
> *Bottom line is that we can't prove whether this is natural or escape — leaving it to others to make that decision, but hopefully we can ensure they're more informed.*
>
> *Rambaut, 7:31 pm EST:*
> *I suggest we write this report erring on the side of extreme caution. Also I think the natural evolutionary story may be a interesting one as well. Then we can give all the curious coincidences and dodgy goings on to Marc Lipsitch to have fun with.*



Proximal_Origin_Slack.pdf, p.5, 2 Feb 2020, 6:42 pm EST

. . .

6. Farrar shares the passaging hypothesis with the 'Bethesda boys'

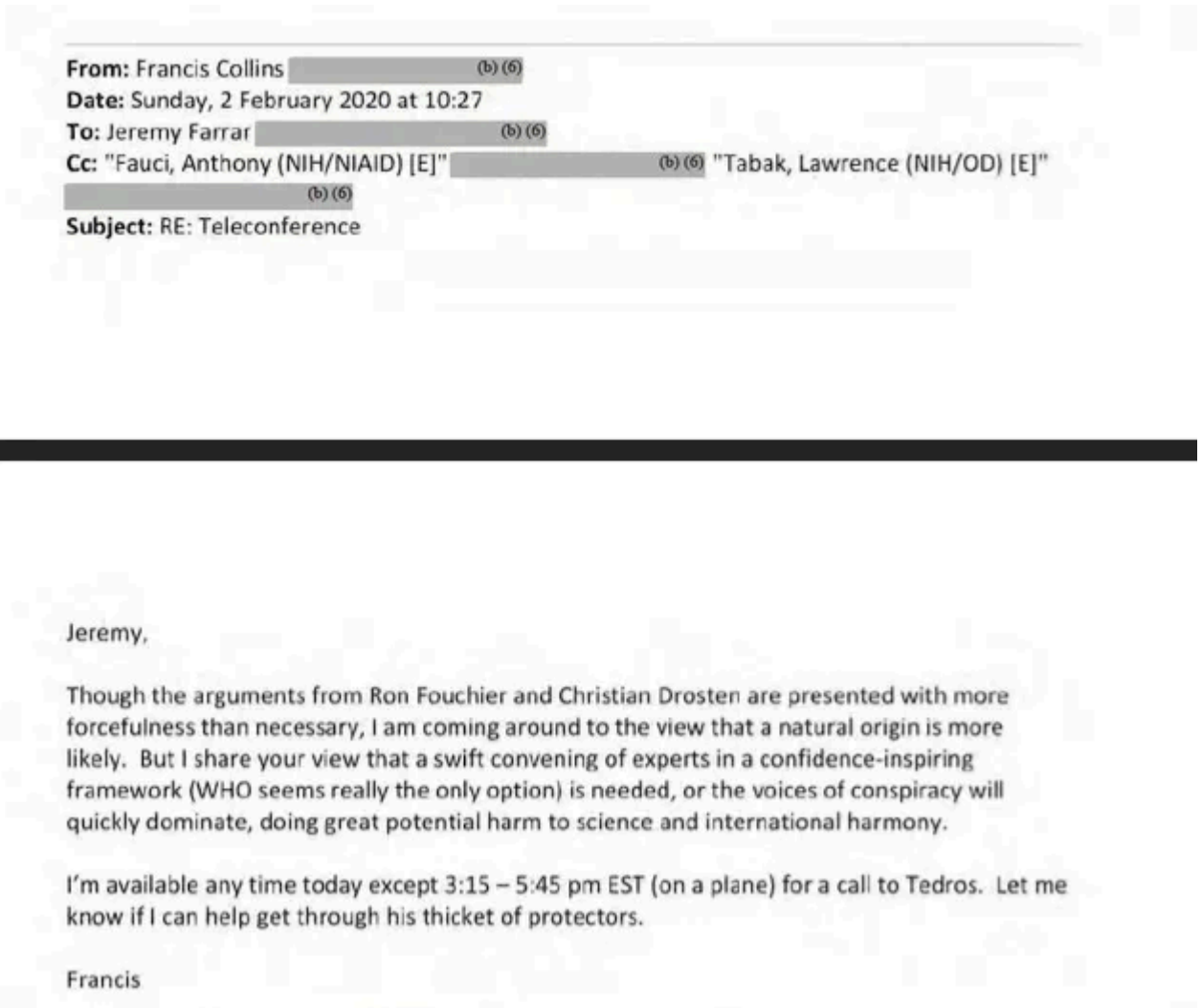
6.1 Collins gets back to Farrar, partly paraphrasing Fouchier (2 Feb, 5:27 am EST)

On 2 Feb 5:27 am EST, in a reply to Farrar's roadmap (see 5.5) addressed only to Fauci and Tabak (instead of the whole call group), Francis Collins, explained that he was starting to lean more towards a natural origin, partly due to the points raised by Fouchier and Drosten during the call, even if they clearly did show too much confidence in these.

Even so, he added that he supported passing the problem over to the WHO, the only relevant '*confidence inspiring framework*' he could think of. One part

of his email jumps up: ***'.. or the voices of conspiracy will quickly dominate, doing great potential harm to science and international harmony'.*** There Collins offered a counter-argument to Fouchier's rather unfortunate formulation '*further debate would .. do unnecessary harm to science in general and science in China in particular'* (see 5.2), by retorting that not addressing the issue would be doing the damage.

One should note that from the point of view of Farrar, Collins' reply may have seemed a bit weak: spurred by Andersen, Holmes and Garry, Farrar had taken on him to alert Collins and Fauci about reasonable doubts as to a possible lab-construct. During the call, Fouchier and Drosten hit at Andersen with some excess, and now Collins seemed to be shying away.



From: Francis Collins (b) (6)
Date: Sunday, 2 February 2020 at 10:27
To: Jeremy Farrar (b) (6)
Cc: "Fauci, Anthony (NIH/NIAID) [E]" (b) (6) "Tabak, Lawrence (NIH/OD) [E]" (b) (6)
Subject: RE: Teleconference

Jeremy,

Though the arguments from Ron Fouchier and Christian Drosten are presented with more forcefulness than necessary, I am coming around to the view that a natural origin is more likely. But I share your view that a swift convening of experts in a confidence-inspiring framework (WHO seems really the only option) is needed, or the voices of conspiracy will quickly dominate, doing great potential harm to science and international harmony.

I'm available any time today except 3:15 – 5:45 pm EST (on a plane) for a call to Tedros. Let me know if I can help get through his thicket of protectors.

Francis

farrar-fauci-comms-full.pdf, p.107–108, 2 Feb 2020

. . .

6.2 Farrar updates the 'Bethesda boys' with the passaging hypothesis (2 Feb, noon UK, 6:53 am EST)

Farrar next shared Farzan and Garry notes that discussed a possible passaging origin of the FCS (see 5.1), with Collins, Fauci and Tabak (the 'Bethesda boys', footnote 10). In doing so, he also gave his assessment of 50:50 research accident vs. zoonosis, coming after similar evaluations by Farzan (70:30 to 60:40), and Garry ('*I just can't figure out how this gets accomplished in nature'*).

This update, and the introduction of the passaging alternative, would have been important, as Collins was stating to be convinced by some of Fouchier's arguments. But Fouchier and Drosten had pushed back against a direct

genetic construct of the FCS, not against passaging, which had not been well covered in the call.

Farrar then insisted on the need to get the WHO to take the question over, even if that may take a good month for the WHO to get on to it, and made a very prescient prediction:

> *On a spectrum if 0 is nature and 100 is release — I am honestly at 50!* ***My guess is that this will remain grey, unless there is access to the Wuhan lab — and I suspect that is unlikely****!*
>
> *But grey, from a respected group, under the umbrella of let us say WHO, would in itself help!*
>
> *source: farrar-fauci-comms-full.pdf, p.106, 2 Feb 2020, ~7am EST*

From: Jeremy Farrar (b) (6)
Sent: Sunday, February 2, 2020 6:53 AM
To: Collins, Francis (NIH/OD) [E] (b) (6)
Cc: Fauci, Anthony (NIH/NIAID) [E] (b) (6) Tabak, Lawrence (NIH/OD) [E] (b) (6)
Subject: Re: Teleconference

Thank you

See thoughts overnight from others.

On a spectrum if 0 is nature and 100 is release – I am honestly at 50! My guess is that this will remain grey, unless there is access to the Wuhan lab – and I suspect that is unlikely!

But grey, from a respected group, under the umbrella of let us say WHO, would in itself help!

A question for you – if WHO say, well maybe, let us think, we might do it in a month. What would be our next step?

Jeremy

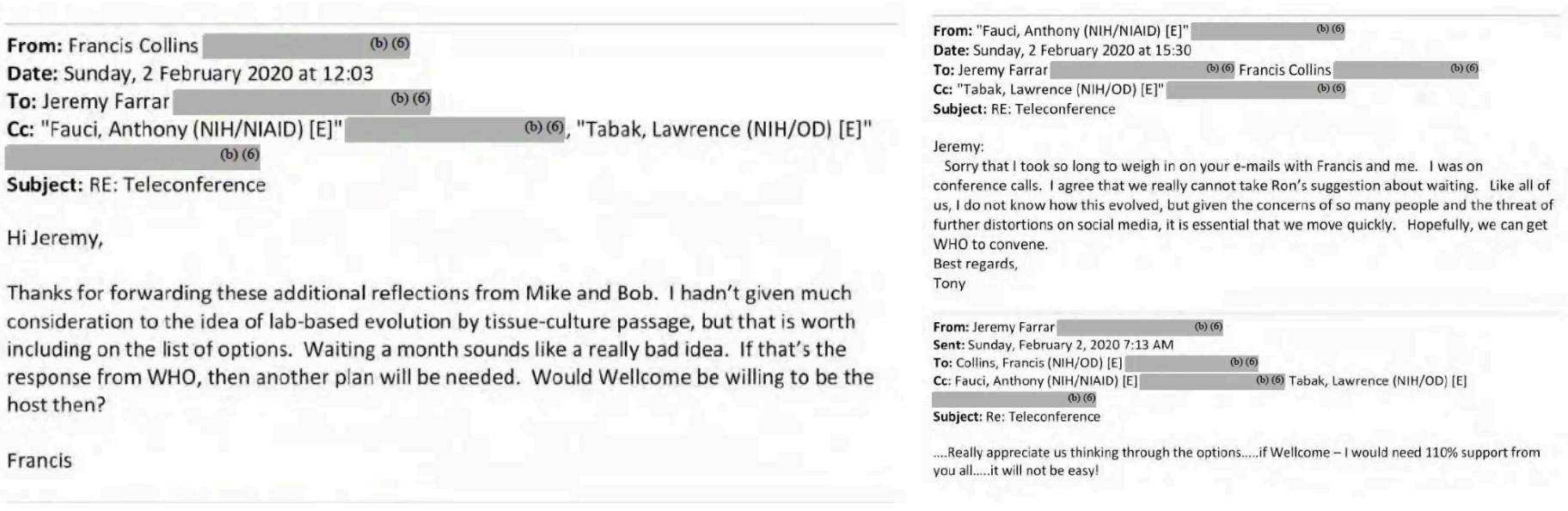
farrar-fauci-comms-full.pdf, p.106, 2 Feb 2020, ~7am EST

6.3 Fauci and Collins reply to Farrar (2 Feb, 7:03am & 10:30am EST)

In his immediate reply to Farrar, Collins took note of these evaluations and of the real possibility of passaging behind the FCS.

Collins then raised the question of when the WHO could pick up the origin question. He even suggested the Wellcome Trust itself might pick up the question, as a credible neutral organisation, if the WHO could not move fast enough.

Fauci replied to Collins and Farrar a few hours later, and confirmed that Fouchier's suggestion about waiting was not acceptable, and that he still hoped that the WHO would quickly pick up the origin question.



From: Francis Collins (b) (6)
Date: Sunday, 2 February 2020 at 12:03
To: Jeremy Farrar (b) (6)
Cc: "Fauci, Anthony (NIH/NIAID) [E]" (b) (6), "Tabak, Lawrence (NIH/OD) [E]" (b) (6)
Subject: RE: Teleconference

Hi Jeremy,

Thanks for forwarding these additional reflections from Mike and Bob. I hadn't given much consideration to the idea of lab-based evolution by tissue-culture passage, but that is worth including on the list of options. Waiting a month sounds like a really bad idea. If that's the response from WHO, then another plan will be needed. Would Wellcome be willing to be the host then?

Francis

From: "Fauci, Anthony (NIH/NIAID) [E]" (b) (6)
Date: Sunday, 2 February 2020 at 15:30
To: Jeremy Farrar (b) (6) Francis Collins (b) (6)
Cc: "Tabak, Lawrence (NIH/OD) [E]" (b) (6)
Subject: RE: Teleconference

Jeremy:
Sorry that I took so long to weigh in on your e-mails with Francis and me. I was on conference calls. I agree that we really cannot take Ron's suggestion about waiting. Like all of us, I do not know how this evolved, but given the concerns of so many people and the threat of further distortions on social media, it is essential that we move quickly. Hopefully, we can get WHO to convene.
Best regards,
Tony

From: Jeremy Farrar (b) (6)
Sent: Sunday, February 2, 2020 7:13 AM
To: Collins, Francis (NIH/OD) [E] (b) (6)
Cc: Fauci, Anthony (NIH/NIAID) [E] (b) (6) Tabak, Lawrence (NIH/OD) [E] (b) (6)
Subject: Re: Teleconference

....Really appreciate us thinking through the options.....if Wellcome – I would need 110% support from you all.....it will not be easy!

farrar-fauci-comms-full.pdf, p.105–106, 2 Feb 2020

• • •

7. In Parallel: Trying to get the WHO on-board

7.1 Farrar and Collins join forces to try to convince the WHO to pick up the origin question (1–2 Feb)

Farrar had quickly contacted the WHO on Saturday 1 Feb in the evening, to schedule a first call for Sunday 2 Feb, on which occasion he would try to get them to agree to a broader call as soon as possible. Collins made himself available to join the call with Dr. Tedros.

(For context, by 31 Jan, the HHS had already had discussions on this subject with Simonson, the WHO director of the WHO-US Liaison Office, with Fauci in the loop, as per 4.13)

As time went by on 2 Feb, Farrar expressed some of his frustration of not being able so far to reach Tedros or his right hand, Bernhard Schwartländer. At about 4:30 pm (UK time), Farrar suggested a call later that day with Fauci and Collins if he could not talk to Tedros by the end of the day.

Key Insight: Schwartländer

Schwartländer, an epidemiologist by training, became Tedros' Chef de Cabinet on 3 October 2017, just 3 months after Dr Tedros started as head of the WHO. Before becoming Tedros' right hand, Schwartländer had been WHO Representative in China since 2013. He was well appreciated there.

Many seasoned observers and journalists extensively reported how Tedros was selected with the support of China, which he was seemingly eager to repay. Schwartländer's nomination might thus be interpreted as another step towards securing China's interest at the WHO.

One must also notice that Germany was effectively the 2nd largest contributor to the WHO budget, based on obligatory & voluntary contributions (with the US the 1st contributor, and China hardly registering), so Schwartländer's nomination was — at least optically — rather logical.

Schwartländer left a few months after the embarrassing failure of the early 2021 WHO mission to China, and the ensuing volte-face by Dr Tedros, who — under international pressure — contradicted the WHO team's official conclusions only 3 days after their public disclosure in Beijing. Schwartländer immediately landed back in Beijing as 'Global Health Envoy for the German Ministry of Foreign Affairs', whatever that means.

A Again, there is no involvement at any stage of the production of the paper from Drs. Collins or Fauci.

I believe, though, that because Jeremy was trying -- putting some pressure on the WHO to get this conference where this could be considered, I think he might have communicated with Collins on this separately. Because I remember that we talked at the time that Collins was helping Jeremy basically to get a conference set up with the WHO, and probably from that side Jeremy from the U.K. side, maybe Collins from the U.S. side, putting some pressure on the WHO to consider the origin questions in general.

Q But you would not have personal knowledge of --

A I would not have personal knowledge. I did not have any personal communication with Fauci or Collins as part of this other than the email chain that is going on with the participants of the call in which I think Dr. Fauci has a single question, which relates to potential passage in animals, yes.

2023.06.16-Andersen-Transcript.pdf p.64

From: Fauci, Anthony (NIH/NIAID) [E]
Sent: Sat, 1 Feb 2020 22:06:26 +0000
To: Jeremy Farrar;Collins, Francis (NIH/OD) [E]
Subject: RE: Teleconference

Thanks, Jeremy. We really appreciate what you are doing here. Pleasure to work with you.
Best,
Tony

From: Jeremy Farrar [redacted]
Sent: Saturday, February 1, 2020 4:00 PM
To: Collins, Francis (NIH/OD) [E] [redacted]
Cc: Fauci, Anthony (NIH/NIAID) [E] [redacted]
Subject: Re: Teleconference

We are altogether as you know! Conversations with you and Tony, and Patrick and others – always great working with you both

From: Francis Collins [redacted]
Date: Saturday, 1 February 2020 at 20:50
To: Jeremy Farrar [redacted]
Cc: "Fauci, Anthony (NIH/NIAID) [E]" [redacted]
Subject: Re: Teleconference

Hi Jeremy,
I can make myself available at any time 24/7 for the call with Tedros. Just let me know.
Thanks for your leadership on this critical and sensitive issue.
Francis

Sent from my iPhone

On Feb 1, 2020, at 3:07 PM, Jeremy Farrar [redacted] wrote:

I have rejoined so a line is open if any help to rejoin.

From: Jeremy Farrar [redacted]
Date: Saturday, 1 February 2020 at 19:56
To: "Fauci, Anthony (NIH/NIAID) [E]" [redacted], Francis Collins [redacted], Michael FMedSci [redacted], Patrick Vallance [redacted]
Subject: Re: Teleconference

From: Francis Collins [redacted]
Date: Sunday, 2 February 2020 at 10:27
To: Jeremy Farrar [redacted]
Cc: "Fauci, Anthony (NIH/NIAID) [E]" [redacted] "Tabak, Lawrence (NIH/OD) [E]" [redacted]
Subject: RE: Teleconference

Jeremy,

Though the arguments from Ron Fouchier and Christian Drosten are presented with more forcefulness than necessary, I am coming around to the view that a natural origin is more likely. But I share your view that a swift convening of experts in a confidence-inspiring framework (WHO seems really the only option) is needed, or the voices of conspiracy will quickly dominate, doing great potential harm to science and international harmony.

I'm available any time today except 3:15 – 5:45 pm EST (on a plane) for a call to Tedros. Let me know if I can help get through his thicket of protectors.

Francis

From: Jeremy Farrar [redacted]
Sent: Sunday, February 2, 2020 4:48 AM
To: Andrew Rambaut [redacted]
Cc: R.A.M. Fouchier [redacted]; Fauci, Anthony (NIH/NIAID) [E] [redacted]; Patrick Vallance [redacted] Drosten, Christian [redacted]; M.P.G. Koopmans [redacted] Eddie Holmes [redacted] Kristian G. Andersen [redacted]; Paul Schreier [redacted]; Ferguson, Mike [redacted] Collins, Francis (NIH/OD) [E] [redacted]; Tabak, Lawrence (NIH/OD) [E] [redacted]; Josie Gol[redacted]
Subject: Re: Teleconference

This is a very complex issue.

will:

- Be in contact with WHO today. I contacted them last night and will speak with them today and set up a broader call with them as soon as possible.
- As discussed on the phone this discussion is not limited to those on this email, it is happening wider in the scientific, social and main stream media.
- I believe the best way forward is for a body like the WHO has to ask or commission a group of scientists from around the world to ask the neutral question "To understand the evolutionary origins of 2019-nCoV, important for this epidemic and for future risk assessment and understanding of animal/human coronaviruses".
- That should be done in an open way and quite quickly so that the world can see it is being done, it can respect the report when it is available and I think that will help with the growing interest of this question.

I suggest we don't get into a further scientific discussion here, but wait for that group to be established.

Jeremy

From: Jeremy Farrar
Sent: Sun, 2 Feb 2020 16:28:29 +0000
To: Fauci, Anthony (NIH/NIAID) [E];Collins, Francis (NIH/OD) [E]
Cc: Tabak, Lawrence (NIH/OD) [E]
Subject: Re: Teleconference

Tedros and Bernhard have apparently gone into conclave....they need to decide today in my view. If they do prevaricate, I would appreciate a call with you later tonight or tomorrow to think how we might take forward.

Meanwhile....

https://www.zerohedge.com/geopolitical/coronavirus-contains-hiv-insertions-stoking-fears-over-artificially-created-bioweapon

From: "Fauci, Anthony (NIH/NIAID) [E]" [redacted]
Date: Sunday, 2 February 2020 at 15:30
To: Jeremy Farrar [redacted] Francis Collins [redacted]
Cc: "Tabak, Lawrence (NIH/OD) [E]" [redacted]
Subject: RE: Teleconference

Jeremy:
Sorry that I took so long to weigh in on your e-mails with Francis and me. I was on conference calls. I agree that we really cannot take Ron's suggestion about waiting. Like all of us, I do not know how this evolved, but given the concerns of so many people and the threat of further distortions on social media, it is essential that we move quickly. Hopefully, we can get WHO to convene.
Best regards,
Tony

From: Jeremy Farrar [redacted]
Sent: Sunday, February 2, 2020 7:13 AM
To: Collins, Francis (NIH/OD) [E] [redacted]
Cc: Fauci, Anthony (NIH/NIAID) [E] [redacted] Tabak, Lawrence (NIH/OD) [E] [redacted]
Subject: Re: Teleconference

....Really appreciate us thinking through the options.....if Wellcome – I would need 110% support from you all.....it will not be easy!

farrar-fauci-comms.pdf, p.107–108 and 105, 2 Feb 2020

7.2 Farrar's call with WHO — The 'Tedros Proposal' (2 Feb, ~ 5 pm UK)

The call with Schwartländer did eventually happen on 2 Feb around 5 pm UK time. Also on the call were Farrar and Mike Ryan (Executive Director of the WHO's Health Emergencies Programme). According to Farrar's book Spike, the three agreed to try to ask Dr. Tedros to take over the forensic investigation of the origins, looking into the genome. That's the '**Tedros proposal**'.

The call was summarised in an email, which was sent to Collins, Fauci and Dr. Tedros, as well as Ryan and Schwartländer. It is not clear how much of Farrar's doubts about the origin had been disclosed, but the summary, that is meant to reflect the WHO position, shows a good understanding of the difficulty of the situation, and seems to acknowledge the urgency despite the essential tasks the WHO was already trying to perform.

[Added 12 Sep 2025] The email includes a remarkable sentence:
'Critical that responsible, respected scientists and agencies get ahead of the science and the narrative of this and are not reacting to reports which could be very damaging'.
'**Getting ahead of the science**' is not part of the scientific method. It is instead a narrative building exercise, as the rest of the sentence shows. And the reason for this effort to ignore science and build a sandwall is also given: '**reports which could be very damaging**'. The true concerns of Fauci, and Farrar could not be clearer at that moment.

The responses from Collins to Farrar and Fauci, and then from Fauci to Collins, were released by Rand Paul in September 2025. They show that Collins and particularly Fauci were also worried about the likely slowness of the WHO in implementing the proposal [footnote 105].

From: Jeremy Farrar
Sent: Sun, 2 Feb 2020 17:47:36 +0000
To: Michael RYAN;Bernhard Schwartländer;Collins, Francis (NIH/OD) [E];Fauci, Anthony (NIH/NIAID) [E];Dr Tedros
Subject: 2019nC-V

Mike and Bernhard
Thank you for phoning – very helpful, copying in Francis and Tony as well as Tedros.

Fully agree with your summary.

- Best is if this is addressed under the umbrella of WHO
- Has to be framed as 'To understand the source and evolution of the 2019n-CoV"
  - Within that a number of issues to be addressed including – Environmental and animal sampling, human viral genome sequencing and analysis, and more
- Quickest is via a Working Group within an established structure rather than set something new up
- Multiple options for this within WHO
- Mike and Bernhard will work out best approach
- Appreciate the urgency and importance of this issue in midst of a very troubling epidemic
- Gathering interest evident in the science literature and in mainstream and social media to the question of the origin of this virus
- Critical that responsible, respected scientists and agencies get ahead of the science and the narrative of this and are not reacting to reports which could be very damaging.
- I am sure I speak for Francis and Tony when I say we are here and ready to play any constructive role in this
- Do think this is an urgent matter to address (among many we appreciate)
- Fully agree to your comments on the GCM.

Hope that is a reasonable summary

Wellcome exists to improve health by helping great ideas to thrive. We support researchers, we take on big health challenges, we campaign for better science, and we help everyone get involved with science and health research. We are a politically and financially independent foundation.

The Wellcome Trust is a charity registered in England and Wales, no. 210183. Its sole trustee is The Wellcome Trust Limited, a company registered in England and Wales, no. 2711000 (whose registered office is at 215 Euston Road, London NW1 2BE, UK)

From: "Fauci, Anthony (NIH/NIAID) [E]" [redacted]
To: "Collins, Francis (NIH/OD) [E]" [redacted]
Subject: FW: 2019nC-V
Date: Sun, 02 Feb 2020 14:15:16 -0500
Importance: Normal

I agree with you. Mike and Bernhard are good people but they really slow things down, particularly Mike Ryan whose main concern is not offending anyone. Please delete this e-mail after you read it.

From: Collins, Francis (NIH/OD) [E] [redacted]
Sent: Sunday, February 2, 2020 1:45 PM
To: Jeremy Farrar [redacted]
Cc: Fauci, Anthony (NIH/NIAID) [E] [redacted]; Tabak, Lawrence (NIH/OD) [E] [redacted]
Subject: Re: 2019nC-V

Sounds mostly encouraging though I don't understand the choice of working through Mike and Bernhard and not just going straight to Tedros. But that's fine as long as these next steps are initiated very quickly!
FC

Sent from my iPhone

On Feb 2, 2020, at 12:48 PM, Jeremy Farrar [redacted] wrote:

Mike and Bernhard
Thank you for phoning – very helpful, copying in Francis and Tony as well as Tedros.

Fully agree with your summary.

- Best is if this is addressed under the umbrella of WHO
- Has to be framed as 'To understand the source and evolution of the 2019n-CoV"
  - Within that a number of issues to be addressed including – Environmental and animal sampling, human viral genome sequencing and analysis, and more
- Quickest is via a Working Group within an established structure rather than set something new up
- Multiple options for this within WHO
- Mike and Bernhard will work out best approach
- Appreciate the urgency and importance of this issue in midst of a very troubling epidemic
- Gathering interest evident in the science literature and in mainstream and social media to the question of the origin of this virus
- Critical that responsible, respected scientists and agencies get ahead of the science and the narrative of this and are not reacting to reports which could be very damaging.
- I am sure I speak for Francis and Tony when I say we are here and ready to play any constructive role in this
- Do think this is an urgent matter to address (among many we appreciate)
- Fully agree to your comments on the GCM.

Hope that is a reasonable summary

HSGAC_NIH_18135

farrar-fauci-comms.pdf, p.104, 2 Feb 2020, 5:47 pm UK and Rand Paul's tweet, 2 Feb 2020, 7:16 pm UK.

7.3 Farrar's call with China's ex Minister of Health Chen Zhu (2 Feb)

On Sunday 2 Feb, Farrar called Chen Zhu, who was the Chinese Minister of Health from 2007 to 2013 (source: *Spike*). The purpose of the call was to let the Chinese Government know, via a back channel, that some enquiries in the origin of the outbreak were likely, and that it would be essential to keep open links with China even if a non-natural origin was the conclusion.

It would be logical for that call with Chen Zhu to have happened after the call with the WHO, Schwartländer and Ryan. Schwartländer may even have suggested it, since, before becoming Dr. Tedros' right arm in early October 2017, he had been the WHO representative in China in 2013–2017, and, before that, the UN Country Coordinator on AIDS in China in 2007–2010, so overlapping Chen Zhu tenure as Minister of Health [footnote 74].

Additionally, Chen Zhu, who had been appointed before Xi Jin-Ping became Premier in 2012, was very well seen in foreign circles; he had studied in Paris, was a foreign member of the US National Academy of Sciences (NAS) and a foreign member of the French Academy of Sciences, to list just a few.

This would explain why Chen Zhu was chosen, as a trusted back-channel to the Chinese government, for a difficult mission.

The first thing that the WHO would indeed want to know, was if China would likely balk at the idea of some look into the origin question, even after the semantic sugaring of the Tedros Proposal. Hopefully, the recent trip by Tedros and Ryan to China (see 2.6), with the promise of access to China

extracted from Xi Jin-Ping, and the many praises laid out by Tedros since then (see 2.7), would allow for some understanding.

Essential Insight: A Very Risky Gamble

Using Chen Zhu as a back channel was a very risky gamble. Any observer of China could have predicted the most likely outcome, once the Western favourite of the pre-Xi Jin-Ping era had relayed the message to the Chinese government: China would delay any WHO visit until it had clear signs from the WHO that there would be no accusation of any lab-origin, especially as the Western media started relaying some of these suspicions.

Farrar, who called Chen Zhu, would eventually have to come with a solution to the problem he created for the WHO. Which he did.

7.4 Grigsby contacts the WHO (2–3 Feb)

On 2 Feb, Grigsby sent a 1-pager to Stewart Simonson, the Assistant Director-General of the World Health Organization responsible for the WHO Office at the United Nations and the WHO-US Liaison Office, based in New York (see also 4.1).

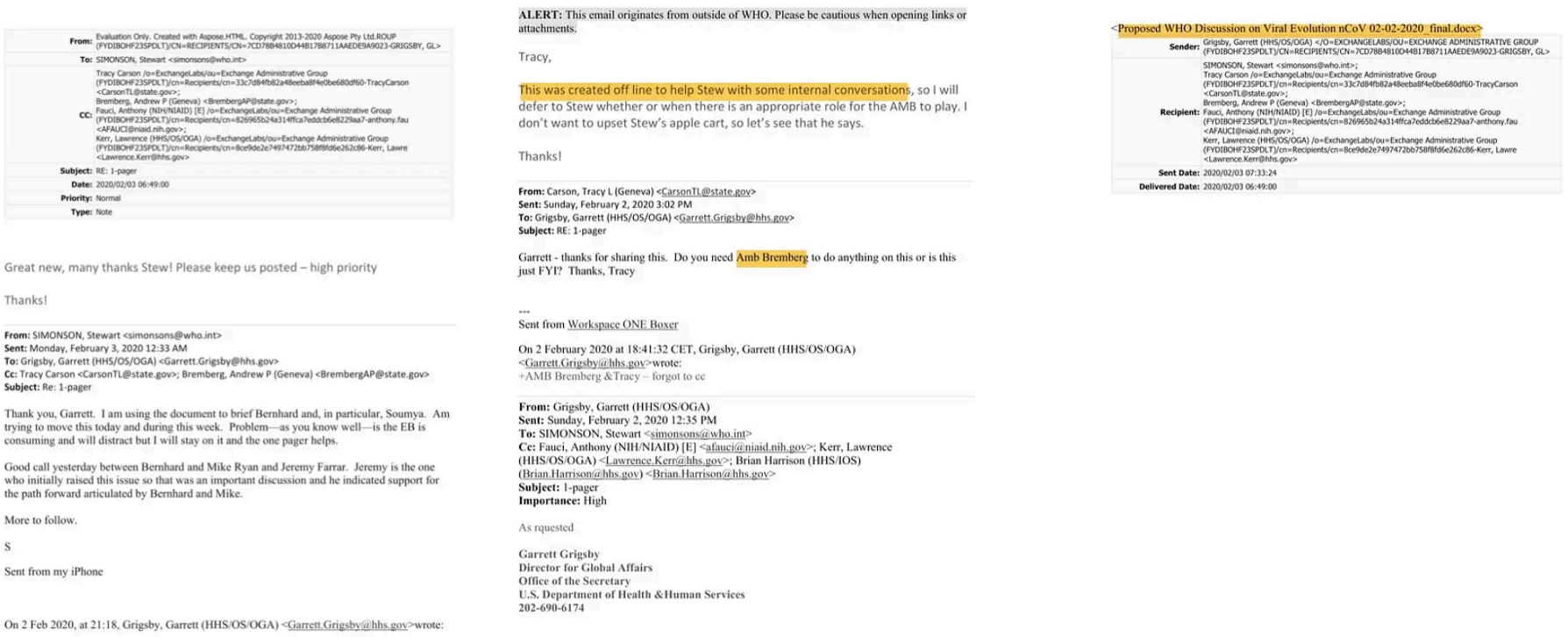
The document shared is '**Proposed WHO Discussion on Viral Evolution nCoV 02–02–2020 final.docx**'.

This is the 'Tedros Proposal' that was discussed by Farrar and relayed to his HHS colleagues by Fauci. Tracy Carson (U.S. Health Attaché acting as the U.S. primary interface with the World Health Organization, State Dep.), and Andrew Bremberg (U.S. representative to the European Office of the U.N. and other international organisations, Ambassador rank, State Dep.) are added in cc.

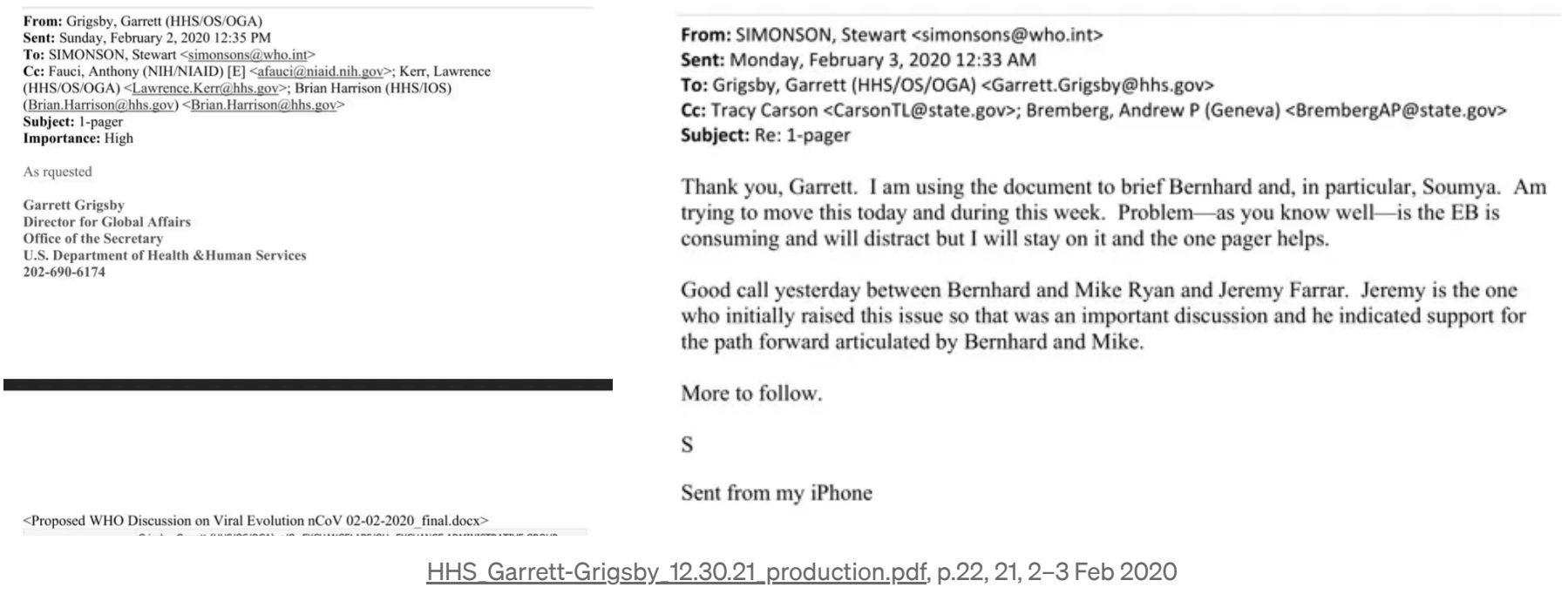
Stew Simonson, who had been previously on some related call[s] with Grigsby (Director for the Office of Global Affairs at HHS, see 4.13) replied that he was briefing Schwartländer (Dr. Tedros' right arm, based in Geneva) and Soumya Swaminathan (then WHO Chief Scientist, succeeded by Farrar himself in 2022). Simonson also gave confirmation of the call that happened the previous day between Schwartländer, Ryan and Farrar.

> *Tracy,*

> *This was created off line to help Stew with some internal conversations, so I will defer to Stew whether or when there is an appropriate role for the AMB [note: US Ambassador to WHO] to play.*
> *I don't want to upset Stew's apple cart, so let's see [w]hat he says.*
>
> *Thanks!*
>
> *source: HHS_Garrett-Grigsby_12.30.21_production.pdf, p.23, 2 Feb 2020*



HHS_Garrett-Grigsby_12.30.21_production.pdf, p.21–23, 2–3 Feb 2020



HHS_Garrett-Grigsby_12.30.21_production.pdf, p.22, 21, 2–3 Feb 2020

. . .

8. CONTEXT: US parallel effort: Track II via NASEM

8.1 Introduction: Track II diplomacy, NASEM and CAS

a. The context

At this stage, we need to take a little detour to look at the role of the National academies of Sciences, Engineering and Medicine (NASEM) in January and February 2020.

Under a common Track II approach of direct scientist-to-scientist cooperation with an otherwise rather adversarial country, NASEM had been

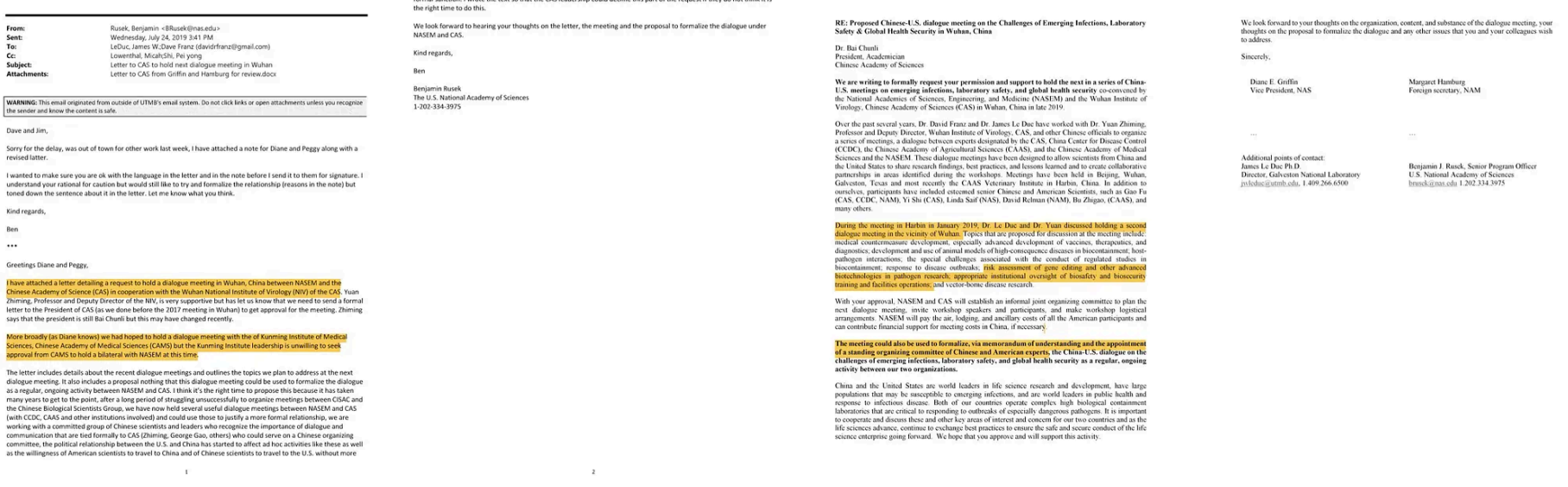
strongly involved over the years with its Chinese equivalent, the Chinese Academy of Sciences (CAS), but also the WIV (part of CAS) and the Chinese CDC. That involvement was managed by the permanent Committee on International Security and Arms Control (CISAC) of the National Academy of Sciences (NAS) [footnote 24].

One of the recent highlights of that NASEM CISAC Track II effort had been the Jan 2019 trip to the Harbin, with a US side that included Baric, Griffin, Relman, Saif, and Rusek, and a main focus on the '*Responsible Conduct in the Use of Gene Editing in Pathogen Research*' [footnote 19]. That trip was considered an important achievement, since it offered hopes of proper NASEM/CAS dialogues on some very top US biosecurity concern, as China was forging ahead in synthetic biology.

Unfortunately, not long after the trip, the tensions between China and the US started taking a toll. Kunming (with its BSL-4), which NASEM had wanted to be the site of the next NASEM-CAS dialogue in summer 2019, became unresponsive. Accordingly, that next dialogue was being planned instead in Wuhan for late 2019. For that occasion, NASEM was trying to get CAS/WIV to formalize the series of dialogues via a Memorandum of Understanding, to start securing them. The official letter requesting these next steps was signed by Griffin and Hamburg.

Pei Yong, a colleague of LeDuc at the University of Texas Medical Branch (UTMB, in Galveston) and a trusted day-to-day contact with the Chinese side was still able to visit Kunming and the WIV (both BSL-4 sites) on his own in October 2019 (14–26 Oct). However, he was then told that, because of the degrading US-China relationship, Wuhan was not any more an option for the next dialogue, and that the city of Huang Gang, a one-hour drive away from Wuhan airport, could instead be considered [footnote 21].

Track II diplomacy was starting to hit the buffers as Chinese cities with P4 labs were closed to the NASEM-CAS dialogues.



From: Rusek, Benjamin <BRusek@nas.edu>
Sent: Wednesday, July 24, 2019 3:41 PM
To: LeDuc, James W.;Dave Franz (davidrfranz@gmail.com)
Cc: Lowenthal, Micah;Shi, Pei yong
Subject: Letter to CAS to hold next dialogue meeting in Wuhan
Attachments: Letter to CAS from Griffin and Hamburg for review.docx

WARNING: This email originated from outside of UTMB's email system. Do not click links or open attachments unless you recognize the sender and know the content is safe.

Dave and Jim,

Sorry for the delay, was out of town for other work last week, I have attached a note for Diane and Peggy along with a revised latter.

I wanted to make sure you are ok with the language in the letter and in the note before I send it to them for signature. I understand your rational for caution but would still like to try and formalize the relationship (reasons in the note) but toned down the sentence about it in the letter. Let me know what you think.

Kind regards,

Ben

•••

Greetings Diane and Peggy,

I have attached a letter detailing a request to hold a dialogue meeting in Wuhan, China between NASEM and the Chinese Academy of Science (CAS) in cooperation with the Wuhan National Institute of Virology (NIV) of the CAS. Yuan Zhiming, Professor and Deputy Director of the NIV, is very supportive but has let us know that we need to send a formal letter to the President of CAS (as we done before the 2017 meeting in Wuhan) to get approval for the meeting. Zhiming says that the president is still Bai Chunli but this may have changed recently.

More broadly (as Diane knows) we had hoped to hold a dialogue meeting with the of Kunming Institute of Medical Sciences, Chinese Academy of Medical Sciences (CAMS) but the Kunming Institute leadership is unwilling to seek approval from CAMS to hold a bilateral with NASEM at this time.

The letter includes details about the recent dialogue meetings and outlines the topics we plan to address at the next dialogue meeting. It also includes a proposal nothing that this dialogue meeting could be used to formalize the dialogue as a regular, ongoing activity between NASEM and CAS. I think it's the right time to propose this because it has taken many years to get to the point, after a long period of struggling unsuccessfully to organize meetings between CISAC and the Chinese Biological Scientists Group, we have now held several useful dialogue meetings between NASEM and CAS (with CCDC, CAAS and other institutions involved) and could use those to justify a more formal relationship, we are working with a committed group of Chinese scientists and leaders who recognize the importance of dialogue and communication that are tied formally to CAS (Zhiming, George Gao, others) who could serve on a Chinese organizing committee, the political relationship between the U.S. and China has started to affect ad hoc activities like these as well as the willingness of American scientists to travel to China and of Chinese scientists to travel to the U.S. without more

1

formal sanction. I wrote the text so that the CAS leadership could decline this part of the request if they do not think it is the right time to do this.

We look forward to hearing your thoughts on the letter, the meeting and the proposal to formalize the dialogue under NASEM and CAS.

Kind regards,

Ben

Benjamin Rusek
The U.S. National Academy of Sciences
1-202-334-3975

2

RE: Proposed Chinese-U.S. dialogue meeting on the Challenges of Emerging Infections, Laboratory Safety & Global Health Security in Wuhan, China

Dr. Bai Chunli
President, Academician
Chinese Academy of Sciences

We are writing to formally request your permission and support to hold the next in a series of China-U.S. meetings on emerging infections, laboratory safety, and global health security co-convened by the National Academies of Sciences, Engineering, and Medicine (NASEM) and the Wuhan Institute of Virology, Chinese Academy of Sciences (CAS) in Wuhan, China in late 2019.

Over the past several years, Dr. David Franz and Dr. James Le Duc have worked with Dr. Yuan Zhiming, Professor and Deputy Director, Wuhan Institute of Virology, CAS, and other Chinese officials to organize a series of meetings, a dialogue between experts designated by the CAS, China Center for Disease Control (CCDC), the Chinese Academy of Agricultural Sciences (CAAS), and the Chinese Academy of Medical Sciences and the NASEM. These dialogue meetings have been designed to allow scientists from China and the United States to share research findings, best practices, and lessons learned and to create collaborative partnerships in areas identified during the workshops. Meetings have been held in Beijing, Wuhan, Galveston, Texas and most recently the CAAS Veterinary Institute in Harbin, China. In addition to ourselves, participants have included esteemed senior Chinese and American Scientists, such as Gao Fu (CAS, CCDC, NAM), Yi Shi (CAS), Linda Saif (NAS), David Relman (NAM), Bu Zhigao, (CAAS), and many others.

During the meeting in Harbin in January 2019, Dr. Le Duc and Dr. Yuan discussed holding a second dialogue meeting in the vicinity of Wuhan. Topics that are proposed for discussion at the meeting include: medical countermeasure development, especially advanced development of vaccines, therapeutics, and diagnostics, development and use of animal models of high-consequence diseases in biocontainment; host-pathogen interactions; the special challenges associated with the conduct of regulated studies in biocontainment; response to disease outbreaks; risk assessment of gene editing and other advanced biotechnologies in pathogen research; appropriate institutional oversight of biosafety and biosecurity training and facilities operations; and vector-borne disease research.

With your approval, NASEM and CAS will establish an informal joint organizing committee to plan the next dialogue meeting, invite workshop speakers and participants, and make workshop logistical arrangements. NASEM will pay the air, lodging, and ancillary costs of all the American participants and can contribute financial support for meeting costs in China, if necessary.

The meeting could also be used to formalize, via memorandum of understanding and the appointment of a standing organizing committee of Chinese and American experts, the China-U.S. dialogue on the challenges of emerging infections, laboratory safety, and global health security as a regular, ongoing activity between our two organizations.

China and the United States are world leaders in life science research and development, have large populations that may be susceptible to emerging infections, and are world leaders in public health and response to infectious disease. Both of our countries operate complex high biological containment laboratories that are critical to responding to outbreaks of especially dangerous pathogens. It is important to cooperate and discuss these and other key areas of interest and concern for our two countries and as the life sciences advance, continue to exchange best practices to ensure the safe and secure conduct of the life science enterprise going forward. We hope that you approve and will support this activity.

We look forward to your thoughts on the organization, content, and substance of the dialogue meeting, your thoughts on the proposal to formalize the dialogue and any other issues that you and your colleagues wish to address.

Sincerely,

Diane E. Griffin
Vice President, NAS

Margaret Hamburg
Foreign secretary, NAM

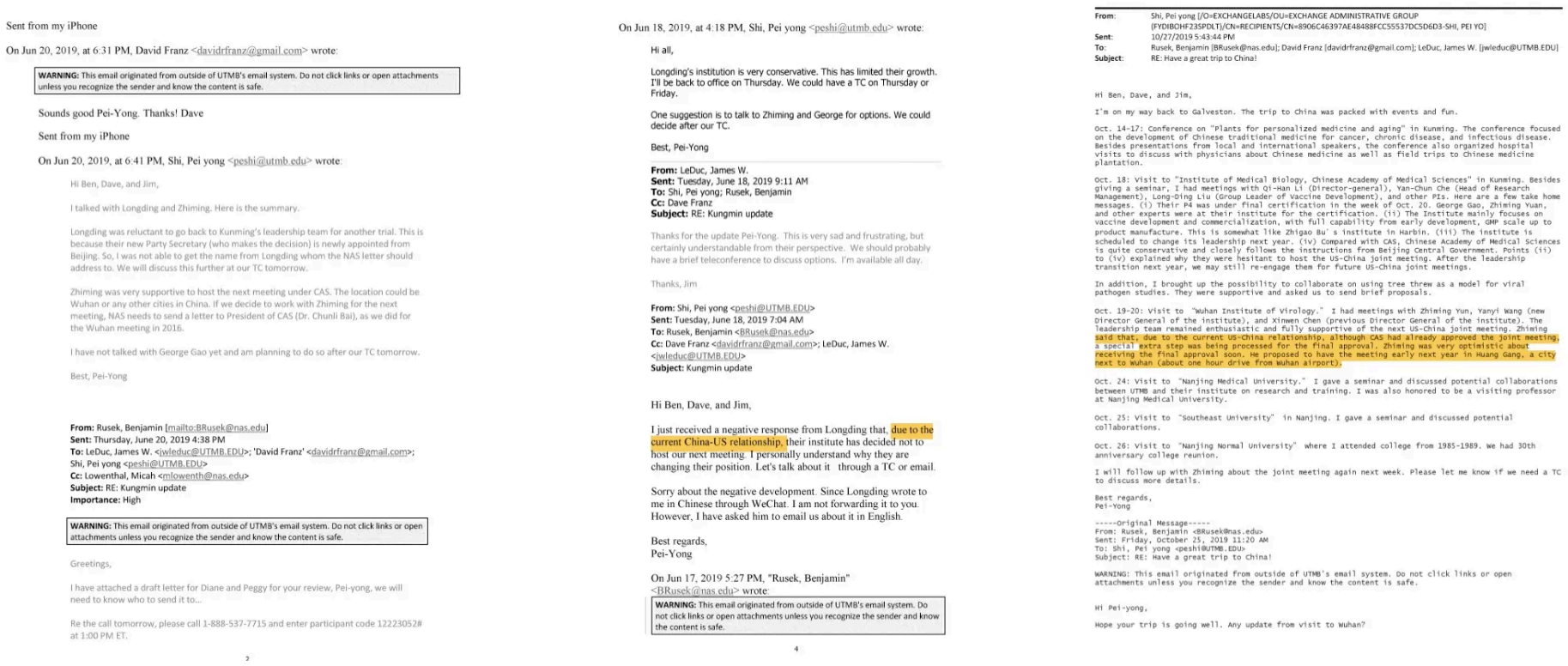
...

...

Additional points of contact:
James Le Duc Ph.D.
Director, Galveston National Laboratory
jvleduc@utmb.edu, 1.409.266.6500

Benjamin J. Rusek, Senior Program Officer
U.S. National Academy of Sciences
brusek@nas.edu 1.202.334.3975

ED-20–01981-March-2022-Production_Redacted_Upload.pdf, p.45–48, Jul 2019.and op38, Jun 2019



Sent from my iPhone

On Jun 20, 2019, at 6:31 PM, David Franz <davidrfranz@gmail.com> wrote:

WARNING: This email originated from outside of UTMB's email system. Do not click links or open attachments unless you recognize the sender and know the content is safe.

Sounds good Pei-Yong. Thanks! Dave

Sent from my iPhone

On Jun 20, 2019, at 6:41 PM, Shi, Pei yong <peshi@utmb.edu> wrote:

Hi Ben, Dave, and Jim,

I talked with Longding and Zhiming. Here is the summary.

Longding was reluctant to go back to Kunming's leadership team for another trial. This is because their new Party Secretary (who makes the decision) is newly appointed from Beijing. So, I was not able to get the name from Longding whom the NAS letter should address to. We will discuss this further at our TC tomorrow.

Zhiming was very supportive to host the next meeting under CAS. The location could be Wuhan or any other cities in China. If we decide to work with Zhiming for the next meeting, NAS needs to send a letter to President of CAS (Dr. Chunli Bai), as we did for the Wuhan meeting in 2016.

I have not talked with George Gao yet and am planning to do so after our TC tomorrow.

Best, Pei-Yong

From: Rusek, Benjamin [mailto:BRusek@nas.edu]
Sent: Thursday, June 20, 2019 4:38 PM
To: LeDuc, James W. <jwleduc@UTMB.EDU>; 'David Franz' <davidrfranz@gmail.com>; Shi, Pei yong <peshi@UTMB.EDU>
Cc: Lowenthal, Micah <mlowenth@nas.edu>
Subject: RE: Kungmin update
Importance: High

WARNING: This email originated from outside of UTMB's email system. Do not click links or open attachments unless you recognize the sender and know the content is safe.

Greetings,

I have attached a draft letter for Diane and Peggy for your review, Pei-yong, we will need to know who to send it to...

Re the call tomorrow, please call 1-888-537-7715 and enter participant code 12223052# at 1:00 PM ET.

3

On Jun 18, 2019, at 4:18 PM, Shi, Pei yong <peshi@utmb.edu> wrote:

Hi all,

Longding's institution is very conservative. This has limited their growth. I'll be back to office on Thursday. We could have a TC on Thursday or Friday.

One suggestion is to talk to Zhiming and George for options. We could decide after our TC.

Best, Pei-Yong

From: LeDuc, James W.
Sent: Tuesday, June 18, 2019 9:11 AM
To: Shi, Pei yong; Rusek, Benjamin
Cc: Dave Franz
Subject: RE: Kungmin update

Thanks for the update Pei-Yong. This is very sad and frustrating, but certainly understandable from their perspective. We should probably have a brief teleconference to discuss options. I'm available all day.

Thanks, Jim

From: Shi, Pei yong <peshi@UTMB.EDU>
Sent: Tuesday, June 18, 2019 7:04 AM
To: Rusek, Benjamin <BRusek@nas.edu>
Cc: Dave Franz <davidrfranz@gmail.com>; LeDuc, James W. <jwleduc@UTMB.EDU>
Subject: Kungmin update

Hi Ben, Dave, and Jim,

I just received a negative response from Longding that, due to the current China-US relationship, their institute has decided not to host our next meeting. I personally understand why they are changing their position. Let's talk about it through a TC or email.

Sorry about the negative development. Since Longding wrote to me in Chinese through WeChat. I am not forwarding it to you. However, I have asked him to email us about it in English.

Best regards,
Pei-Yong

On Jun 17, 2019 5:27 PM, "Rusek, Benjamin" <BRusek@nas.edu> wrote:

WARNING: This email originated from outside of UTMB's email system. Do not click links or open attachments unless you recognize the sender and know the content is safe.

4

From: Shi, Pei yong [/O=EXCHANGELABS/OU=EXCHANGE ADMINISTRATIVE GROUP (FYDIBOHF23SPDLT)/CN=RECIPIENTS/CN=8906C46397AE48488FCC55537DC5D6D3-SHI, PEI YO]
Sent: 10/27/2019 5:43:44 PM
To: Rusek, Benjamin [BRusek@nas.edu]; David Franz [davidrfranz@gmail.com]; LeDuc, James W. [jwleduc@UTMB.EDU]
Subject: RE: Have a great trip to China!

Hi Ben, Dave, and Jim,

I'm on my way back to Galveston. The trip to China was packed with events and fun.

Oct. 14-17: Conference on "Plants for personalized medicine and aging" in Kunming. The conference focused on the development of Chinese traditional medicine for cancer, chronic disease, and infectious disease. Besides presentations from local and international speakers, the conference also organized hospital visits to discuss with physicians about Chinese medicine as well as field trips to Chinese medicine plantation.

Oct. 18: Visit to "Institute of Medical Biology, Chinese Academy of Medical Sciences" in Kunming. Besides giving a seminar, I had meetings with Qi-Han Li (Director-general), Yan-Chun Che (Head of Research Management), Long-Ding Liu (Group Leader of Vaccine Development), and other PIs. Here are a few take home messages. (i) Their P4 was under final certification in the week of Oct. 20. George Gao, Zhiming Yuan, and other experts were at their institute for the certification. (ii) The Institute mainly focuses on vaccine development and commercialization, with full capability from early development, GMP scale up to product manufacture. This is somewhat like Zhigao Bu's institute in Harbin. (iii) The institute is scheduled to change its leadership next year. (iv) Compared with CAS, Chinese Academy of Medical Sciences is quite conservative and closely follows the instructions from Beijing Central Government. Points (ii) to (iv) explained why they were hesitant to host the US-China joint meeting. After the leadership transition next year, we may still re-engage them for future US-China joint meetings.

In addition, I brought up the possibility to collaborate on using tree threw as a model for viral pathogen studies. They were supportive and asked us to send brief proposals.

Oct. 19-20: Visit to "Wuhan Institute of Virology." I had meetings with Zhiming Yun, Yanyi Wang (new Director General of the institute), and Xinwen Chen (previous Director General of the institute). The leadership team remained enthusiastic and fully supportive of the next US-China joint meeting. Zhiming said that, due to the current US-China relationship, although CAS had already approved the joint meeting, a special extra step was being processed for the final approval. Zhiming was very optimistic about receiving the final approval soon. He proposed to have the meeting early next year in Huang Gang, a city next to Wuhan (about one hour drive from Wuhan airport).

Oct. 24: Visit to "Nanjing Medical University." I gave a seminar and discussed potential collaborations between UTMB and their institute on research and training. I was also honored to be a visiting professor at Nanjing Medical University.

Oct. 25: Visit to "Southeast University" in Nanjing. I gave a seminar and discussed potential collaborations.

Oct. 26: Visit to "Nanjing Normal University" where I attended college from 1985-1989. We had 30th anniversary college reunion.

I will follow up with Zhiming about the joint meeting again next week. Please let me know if we need a TC to discuss more details.

Best regards,
Pei-Yong

-----Original Message-----
From: Rusek, Benjamin <BRusek@nas.edu>
Sent: Friday, October 25, 2019 11:20 AM
To: Shi, Pei yong <peshi@UTMB.EDU>
Subject: RE: Have a great trip to China!

WARNING: This email originated from outside of UTMB's email system. Do not click links or open attachments unless you recognize the sender and know the content is safe.

Hi Pei-yong,

Hope your trip is going well. Any update from visit to Wuhan?

[ED-20-01981-March-2022-Production_Redacted_Upload.pdf](), p.38, 40, Jun 2019.and p.99, Jul 2019

b. The significance of the Track II effort during the outbreak

Once the outbreak happened, the US urgently needed to obtain information, isolates and epidemiological data. The epidemiological data was essential to better understand the propagation of the virus, and obtaining some isolates from Wuhan was essential to understand the possible evolution of the virus [footnote 22].

In complement to the formal government-to-government relation (Track I), with Trump still praising China and particularly Xi until early March, the Track II relation between NASEM and CAS was activated. First NASEM itself try to convince its Chinese contacts to share some isolates, then on the origin question, the White House Office of Science and Technology Policy (OSTP) and wrote a formal demand to NASEM, for some recommendations as to what type of data would be required to better understand the origin and likely evolution of the virus (3 Feb).

To be clear, Trump was at the time rather inattentive to the domestic risk of the outbreak, which he considered as a distraction from his US-China trade deal ambitions. It's only in early March 2020, after the outbreak started affecting the US and imposed its own priorities to him, that Trump fully pivoted and resorted to blaming China. So he most likely had zero input in the decision by OSTP to involved NASEM in early February 2020.

Unfortunately, that Track II effort would hit a wall, with the Chinese government artfully redirecting the US demands via a much more malleable WHO, in the name of international equity.

Under the following government of President Biden, the Track II cooperative effort under the Trump administration that had taken place back in January-February 2020 in the context of a disinterested Trump, faded from collective memory, and was replaced by a simplistic narrative of open opposition between Trump and China at the time. That revisionist narrative clearly

better fitted the dynamic under the new administration, but in this specific case, whatever one may think of Trump, that narrative is not only false but attributes to former Pres. Trump a capacity for attention that he simply did not have.

Nevertheless, that essentially forgotten Track II efforts for isolates and for data greatly influenced the choices made by US officials and scientists going into the Farrar call and beyond, while the Chinese counter-move plainly illustrated the limited options available to the US. The purpose of this section is to bring this important Track II logic (and failures) back into focus.

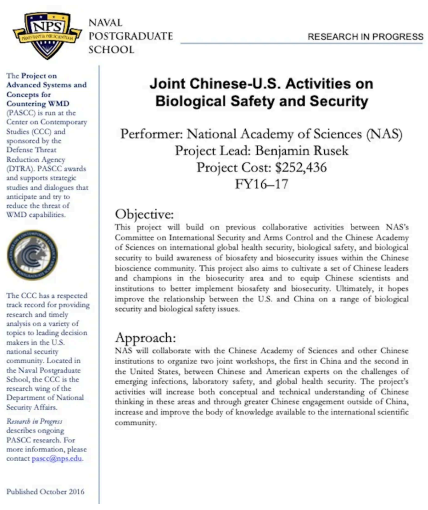
Key insight: The people behind the NAS CISAC Track II effort

The NAS CISAC actual cooperation on the ground was spearheaded by James LeDuc and David Franz.

- **LeDuc** first engagement with Wuhan goes back to 1986, when he helped the Chinese set up a virology lab there, thus contributing the city eventually becoming a centre for virology. Thirty years later, LeDuc was still helping Wuhan scientists develop their skills in Biosafety (BSL3 and BSL4), via UTMB 'International Biosafety training Center' [footnote 104].
- **Franz**, is a former commander of the U.S. Army Medical Research Institute of Infectious Diseases (USAMRIID), US Army Colonel, advisor to EcoHealth Alliance, ex Chief Inspector on three United Nations Special Commission biological warfare inspection missions to Iraq, and a very vocal advocate of self-regulation by laboratories handling the most dangerous pathogens.

Other NAS CISAC members that we will keep encountering are Micah Lowenthal, Benjamin Rusek, Diane Griffin and Margaret Hamburg. These have more overall administrative and policy setting roles:

- Lowenthal is the Director of NAS CISAC, where he 'oversees CISAC's nuclear, biosecurity, and cybersecurity dialogues with counterparts in Russia, China, and India'.
- Benjamin Rusek is a Senior Program Officer for NAS CISAC. He is the Project Leader for the 'Joint Chinese-U.S. Activities on Biological Safety and Security' Track II project, funded by DTRA.
- Griffin (Vice President, NAS) and **Hamburg** (Foreign Secretary, NAM) are members of the CISAC committee.


NAVAL POSTGRADUATE SCHOOL

RESEARCH IN PROGRESS

The **Project on Advanced Systems and Concepts for Countering WMD** (PASCC) is run at the Center on Contemporary Studies (CCC) and sponsored by the Defense Threat Reduction Agency (DTRA). PASCC awards and supports strategic studies and dialogues that anticipate and try to reduce the threat of WMD capabilities.

The CCC has a respected track record for providing research and timely analysis on a variety of topics to leading decision makers in the U.S. national security community. Located in the Naval Postgraduate School, the CCC is the research wing of the Department of National Security Affairs.

Research in Progress describes ongoing PASCC research. For more information, please contact pascc@nps.edu.

Published October 2016

Joint Chinese-U.S. Activities on Biological Safety and Security

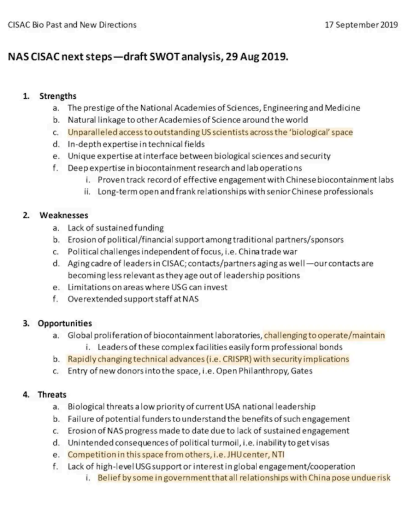
Performer: National Academy of Sciences (NAS)
Project Lead: Benjamin Rusek
Project Cost: $252,436
FY16–17

Objective:

This project will build on previous collaborative activities between NAS's Committee on International Security and Arms Control and the Chinese Academy of Sciences on international global health security, biological safety, and biological security to build awareness of biosafety and biosecurity issues within the Chinese bioscience community. This project also aims to cultivate a set of Chinese leaders and champions in the biosecurity area and to equip Chinese scientists and institutions to better implement biosafety and biosecurity. Ultimately, it hopes improve the relationship between the U.S. and China on a range of biological security and biological safety issues.

Approach:

NAS will collaborate with the Chinese Academy of Sciences and other Chinese institutions to organize two joint workshops, the first in China and the second in the United States, between Chinese and American experts on the challenges of emerging infections, laboratory safety, and global health security. The project's activities will increase both conceptual and technical understanding of Chinese thinking in these areas and through greater Chinese engagement outside of China, increase and improve the body of knowledge available to the international scientific community.


CISAC Bio Past and New Directions 17 September 2019

NAS CISAC next steps—draft SWOT analysis, 29 Aug 2019.

1. **Strengths**
   a. The prestige of the National Academies of Sciences, Engineering and Medicine
   b. Natural linkage to other Academies of Science around the world
   c. Unparalleled access to outstanding US scientists across the 'biological' space
   d. In-depth expertise in technical fields
   e. Unique expertise at interface between biological sciences and security
   f. Deep expertise in biocontainment research and lab operations
      i. Proven track record of effective engagement with Chinese biocontainment labs
      ii. Long-term open and frank relationships with senior Chinese professionals
2. **Weaknesses**
   a. Lack of sustained funding
   b. Erosion of political/financial support among traditional partners/sponsors
   c. Political challenges independent of focus, i.e. China trade war
   d. Aging cadre of leaders in CISAC; contacts/partners aging as well—our contacts are becoming less relevant as they age out of leadership positions
   e. Limitations on areas where USG can invest
   f. Over extended support staff at NAS
3. **Opportunities**
   a. Global proliferation of biocontainment laboratories, challenging to operate/maintain
      i. Leaders of these complex facilities easily form professional bonds
   b. Rapidly changing technical advances (i.e. CRISPR) with security implications
   c. Entry of new donors into the space, i.e. Open Philanthropy, Gates
4. **Threats**
   a. Biological threats a low priority of current USA national leadership
   b. Failure of potential funders to understand the benefits of such engagement
   c. Erosion of NAS progress made to date due to lack of sustained engagement
   d. Unintended consequences of political turmoil, i.e. inability to get visas
   e. Competition in this space from others, i.e. JHU center, NTI
   f. Lack of high-level USG support or interest in global engagement/cooperation
      i. Belief by some in government that all relationships with China pose undue risk

U.S. China dialogue on the Challenges of Emerging Infections, Laboratory Safety and Global Health Security, Galveston, Jan 16–18 2018, grant Oct 2016, NAS-CISAC SWOT analysis (Strengths, Weaknesses, Opportunities and Threats), Sep 2019

8.2 Isolates Effort #1: NASEM to CAS (starting ~17 Jan)

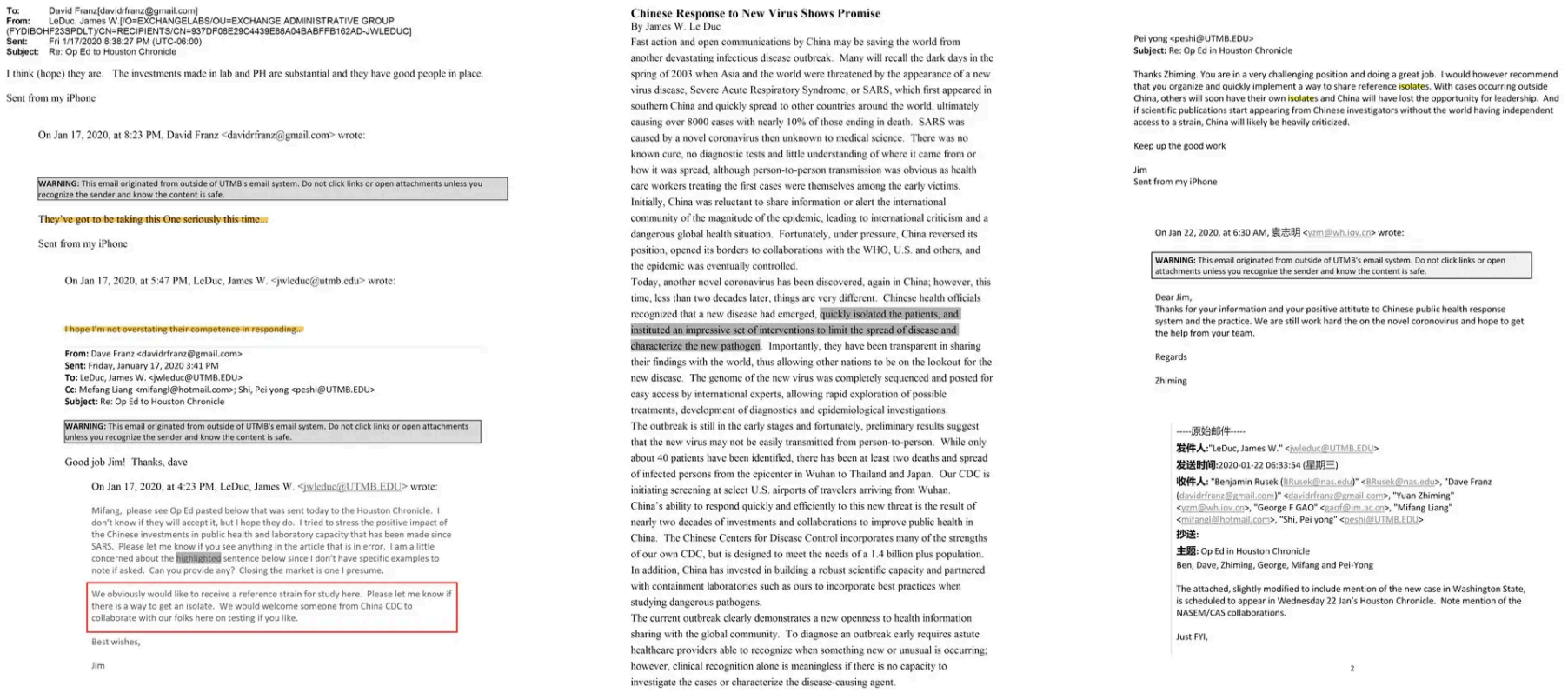
a. Obligatory kowtowing (17 Jan)

On 17 Jan, LeDuc emailed a usual Track II contact, Mifang Liang (of the Institute of Viral Disease Control and Prevention, part of China's CDC). The email started by introducing an attached Op Ed that LeDuc had just proposed to the Houston Chronicle, where he had already published various pieces.

The Op Ed piled praises on China for its transparency, quick sharing of the genome, and for *'demonstrating a new openness to health information sharing with the global community'*. Further exchanges with David Franz in the email thread show that neither LeDuc nor Franz truly believed in what the message of the Op Ed. [footnote 23]

The reason for that piece of frankly demeaning kowtowing is straightforward: after introducing his Op Ed draft, in the very same email, LeDuc asked Mifang if the Chinese CDC could share some isolates with his laboratory, UTMB. Mifang responded the same day by asking LeDuc to discuss it directly with Yuan Zhiming (the director of the WIV).

After the Op Ed was published in the Houston Chronicle on 21 Jan, LeDuc forwarded it to Yuan Zhiming, George Gao, and Mifang Liang.



To: David Franz[davidrfranz@gmail.com]
From: LeDuc, James W.[/O=EXCHANGELABS/OU=EXCHANGE ADMINISTRATIVE GROUP (FYDIBOHF23SPDLT)/CN=RECIPIENTS/CN=937DF08E29C4439E88A04BABFFB162AD-JWLEDUC]
Sent: Fri 1/17/2020 8:38:27 PM (UTC-06:00)
Subject: Re: Op Ed to Houston Chronicle

I think (hopc) they are. The investments made in lab and PH are substantial and they have good people in place.

Sent from my iPhone

On Jan 17, 2020, at 8:23 PM, David Franz <davidrfranz@gmail.com> wrote:

WARNING: This email originated from outside of UTMB's email system. Do not click links or open attachments unless you recognize the sender and know the content is safe.

They've got to be taking this One seriously this time...

Sent from my iPhone

On Jan 17, 2020, at 5:47 PM, LeDuc, James W. <jwleduc@utmb.edu> wrote:

I hope I'm not overstating their competence in responding...

From: Dave Franz <davidrfranz@gmail.com>
Sent: Friday, January 17, 2020 3:41 PM
To: LeDuc, James W. <jwleduc@UTMB.EDU>
Cc: Mefang Liang <mifangl@hotmail.com>; Shi, Pei yong <peshi@UTMB.EDU>
Subject: Re: Op Ed to Houston Chronicle

WARNING: This email originated from outside of UTMB's email system. Do not click links or open attachments unless you recognize the sender and know the content is safe.

Good job Jim! Thanks, dave

On Jan 17, 2020, at 4:23 PM, LeDuc, James W. <jwleduc@UTMB.EDU> wrote:

Mifang, please see Op Ed pasted below that was sent today to the Houston Chronicle. I don't know if they will accept it, but I hope they do. I tried to stress the positive impact of the Chinese investments in public health and laboratory capacity that has been made since SARS. Please let me know if you see anything in the article that is in error. I am a little concerned about the highlighted sentence below since I don't have specific examples to note if asked. Can you provide any? Closing the market is one I presume.

We obviously would like to receive a reference strain for study here. Please let me know if there is a way to get an isolate. We would welcome someone from China CDC to collaborate with our folks here on testing if you like.

Best wishes,

Jim

Chinese Response to New Virus Shows Promise
By James W. Le Duc
Fast action and open communications by China may be saving the world from another devastating infectious disease outbreak. Many will recall the dark days in the spring of 2003 when Asia and the world were threatened by the appearance of a new virus disease, Severe Acute Respiratory Syndrome, or SARS, which first appeared in southern China and quickly spread to other countries around the world, ultimately causing over 8000 cases with nearly 10% of those ending in death. SARS was caused by a novel coronavirus then unknown to medical science. There was no known cure, no diagnostic tests and little understanding of where it came from or how it was spread, although person-to-person transmission was obvious as health care workers treating the first cases were themselves among the early victims. Initially, China was reluctant to share information or alert the international community of the magnitude of the epidemic, leading to international criticism and a dangerous global health situation. Fortunately, under pressure, China reversed its position, opened its borders to collaborations with the WHO, U.S. and others, and the epidemic was eventually controlled.
Today, another novel coronavirus has been discovered, again in China; however, this time, less than two decades later, things are very different. Chinese health officials recognized that a new disease had emerged, quickly isolated the patients, and instituted an impressive set of interventions to limit the spread of disease and characterize the new pathogen. Importantly, they have been transparent in sharing their findings with the world, thus allowing other nations to be on the lookout for the new disease. The genome of the new virus was completely sequenced and posted for easy access by international experts, allowing rapid exploration of possible treatments, development of diagnostics and epidemiological investigations.
The outbreak is still in the early stages and fortunately, preliminary results suggest that the new virus may not be easily transmitted from person-to-person. While only about 40 patients have been identified, there has been at least two deaths and spread of infected persons from the epicenter in Wuhan to Thailand and Japan. Our CDC is initiating screening at select U.S. airports of travelers arriving from Wuhan.
China's ability to respond quickly and efficiently to this new threat is the result of nearly two decades of investments and collaborations to improve public health in China. The Chinese Centers for Disease Control incorporates many of the strengths of our own CDC, but is designed to meet the needs of a 1.4 billion plus population. In addition, China has invested in building a robust scientific capacity and partnered with containment laboratories such as ours to incorporate best practices when studying dangerous pathogens.
The current outbreak clearly demonstrates a new openness to health information sharing with the global community. To diagnose an outbreak early requires astute healthcare providers able to recognize when something new or unusual is occurring; however, clinical recognition alone is meaningless if there is no capacity to investigate the cases or characterize the disease-causing agent.

Pei yong <peshi@UTMB.EDU>
Subject: Re: Op Ed in Houston Chronicle

Thanks Zhiming. You are in a very challenging position and doing a great job. I would however recommend that you organize and quickly implement a way to share reference isolates. With cases occurring outside China, others will soon have their own isolates and China will have lost the opportunity for leadership. And if scientific publications start appearing from Chinese investigators without the world having independent access to a strain, China will likely be heavily criticized.

Keep up the good work

Jim
Sent from my iPhone

On Jan 22, 2020, at 6:30 AM, 袁志明 <yzm@wh.iov.cn> wrote:

WARNING: This email originated from outside of UTMB's email system. Do not click links or open attachments unless you recognize the sender and know the content is safe.

Dear Jim,
Thanks for your information and your positive attitute to Chinese public health response system and the practice. We are still work hard the on the novel coronovirus and hope to get the help from your team.

Regards

Zhiming

-----原始邮件-----
发件人:"LeDuc, James W." <jwleduc@UTMB.EDU>
发送时间:2020-01-22 06:33:54 (星期三)
收件人: "Benjamin Rusek (BRusek@nas.edu)" <BRusek@nas.edu>, "Dave Franz (davidrfranz@gmail.com)" <davidrfranz@gmail.com>, "Yuan Zhiming" <yzm@wh.iov.cn>, "George F GAO" <gaof@im.ac.cn>, "Mifang Liang" <mifangl@hotmail.com>, "Shi, Pei yong" <peshi@UTMB.EDU>
抄送:
主题: Op Ed in Houston Chronicle
Ben, Dave, Zhiming, George, Mifang and Pei-Yong

The attached, slightly modified to include mention of the new case in Washington State, is scheduled to appear in Wednesday 22 Jan's Houston Chronicle. Note mention of the NASEM/CAS collaborations.

Just FYI,

2

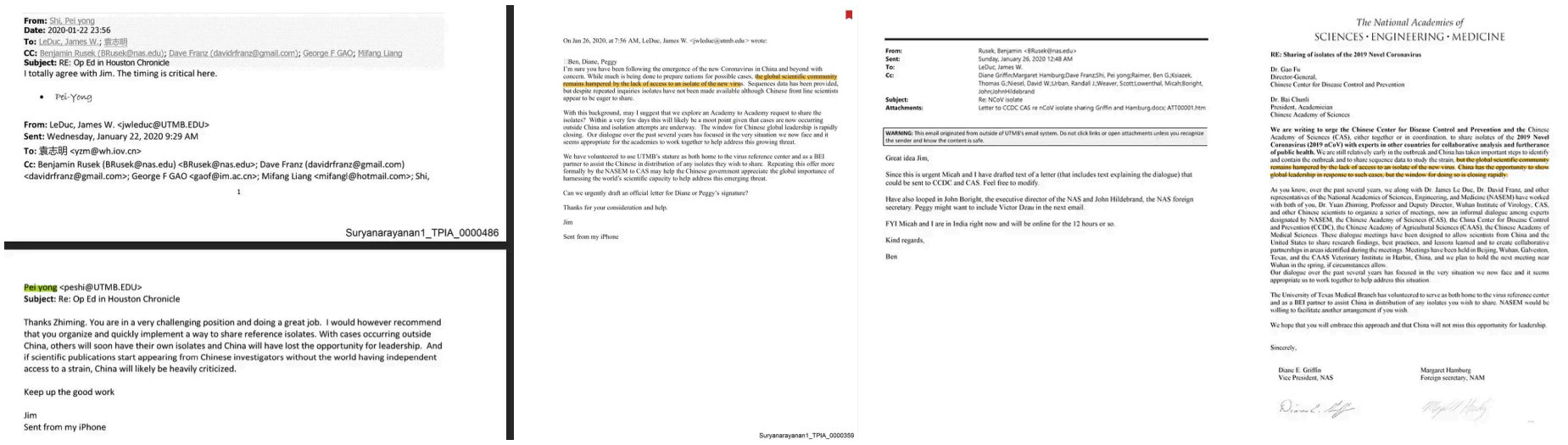
UTMB-LeDuc-batch-1.pdf, p.593–594, and p.487, 17 Jan 2020

b. Playing the Track II links — support from NASEM (26 Jan)
Accordingly, on 22 Jan 2020, LeDuc emailed Yuan Zhiming, the director of the Wuhan BSL-4, to suggest that China quickly starts sharing isolates.

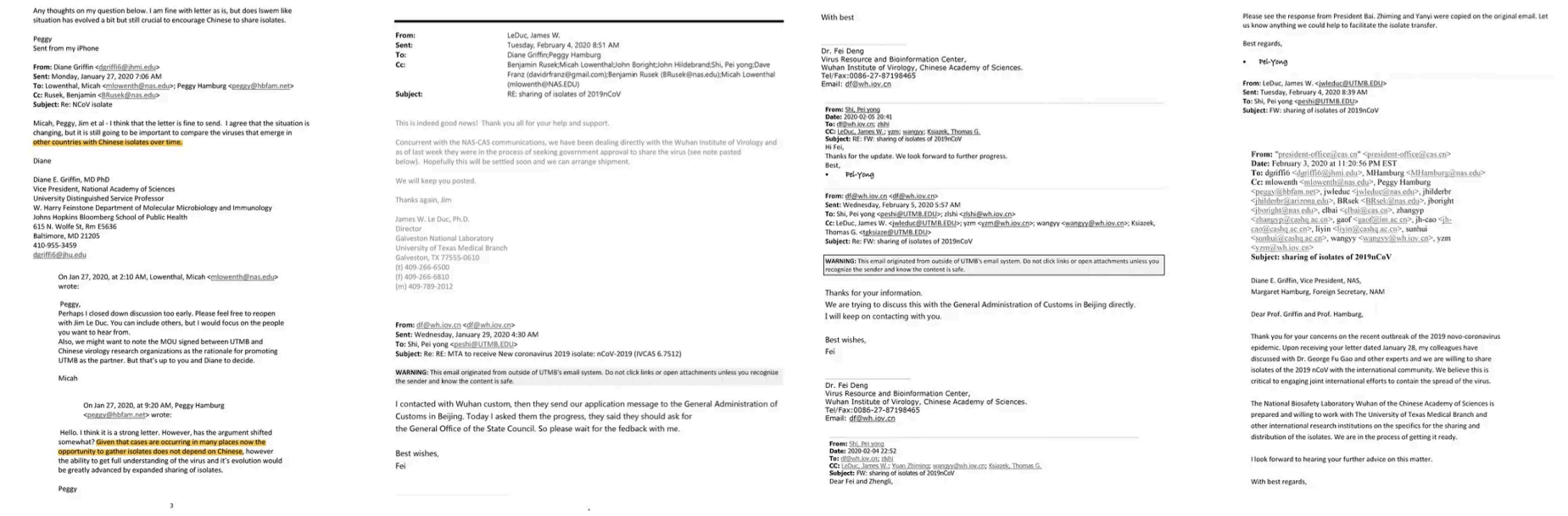
But next, on 26 Jan, LeDuc contacted the top direction of NASEM, namely Diane Griffin, Peggy Hamburg, Benjamin Rusek, and Micah Lowenthal to suggest writing a formal request for isolates sharing. In other words, the Track II NASEM-CAS links were being activated to support the request from UTMB to the WIV.

Things moved quickly. On the very same day, 26 Jan, Benjamin Rusek and Micah Lowenthal drafted an official NASEM letter to be sent to the Chinese Academy of Sciences (CAS) and the Chinese CDC, asking them to facilitate the sharing of isolates, directly between the Chinese CDC and LeDuc's UTMB. In the loop were (as usual) the top representatives of the NASEM, including David Franz (advisor to EcoHealth and one of the strongest proponents of Track II in DURC biotechnologies). The letter duly made reference to the years of Track II dialogues and meetings between the US and Chinese sides.

On 26 Jan, Pei Yong (from UTMB) also called Yuan Zhiming, the Deputy Director of the WIV, to sound him about the letter.



UTMB-LeDuc-batch-1.pdf, p.486–487, 22 Jan 2020, p.355, 26 Jan 2020 and p.356–357, 26 Jan 2020



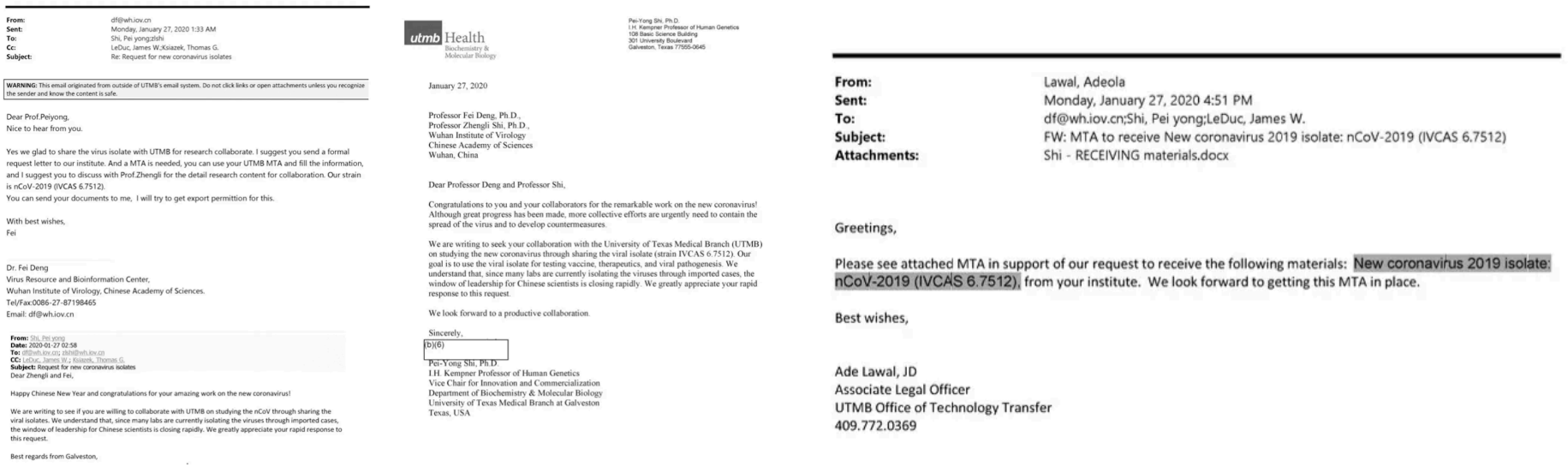
ED-20–01981-F-Dec-2022-Production-OPAQUE.pdf, p.26, 27 Jan 2020, UTMB-LeDuc-batch-1.pdf, p. 813, 29 Jan 2020, and p.800,801, 3–5 Feb 2020

c. The UTMB request and first delays (27–29 Jan)

In parallel, on 27 Jan, an administrative request for sharing a specific nCoV isolate was addressed by Pei Yong (of LeDuc's UTMB) to Fei Deng at the Virus Resource and Bioinformation Center of the WIV (under CAS). In doing so, the demand was drawing on an existing Memorandum of Understanding between the two institutions.

The very same day (**27 Jan**), Fei Deng replied by answering very positively, asking Pei Yong to fill the demand under UTMB MTA and suggesting that he may have a chat with Shi Zhengli to discuss possible collaboration.

On **29 Jan,** Fei Deng got back to Pei Yong saying essentially that the Wuhan Custom had to defer to the State Council (the highest administrative / political instance in China) to try to clear export of the isolates requested on 27 January. **This was an unwelcome turn of event**, as it clearly meant that the Chinese government could now interfere with the request for isolates sharing, at the very time when the UTMB / NASEM Track II collaboration with the WIV was being activated.



From: df@wh.iov.cn
Sent: Monday, January 27, 2020 1:33 AM
To: Shi, Pei yong;zlshi
Cc: LeDuc, James W.;Ksiazek, Thomas G.
Subject: Re: Request for new coronavirus isolates

WARNING: This email originated from outside of UTMB's email system. Do not click links or open attachments unless you recognize the sender and know the content is safe.

Dear Prof.Peiyong,
Nice to hear from you.

Yes we glad to share the virus isolate with UTMB for research collaborate. I suggest you send a formal request letter to our institute. And a MTA is needed, you can use your UTMB MTA and fill the information, and I suggest you to discuss with Prof.Zhengli for the detail research content for collaboration. Our strain is nCoV-2019 (IVCAS 6.7512).
You can send your documents to me, I will try to get export permittion for this.

With best wishes,
Fei

Dr. Fei Deng
Virus Resource and Bioinformation Center,
Wuhan Institute of Virology, Chinese Academy of Sciences.
Tel/Fax:0086-27-87198465
Email: df@wh.iov.cn

From: Shi, Pei yong
Date: 2020-01-27 02:58
To: df@wh.iov.cn; zlshi@wh.iov.cn
CC: LeDuc, James W.; Ksiazek, Thomas G.
Subject: Request for new coronavirus isolates
Dear Zhengli and Fei,

Happy Chinese New Year and congratulations for your amazing work on the new coronavirus!

We are writing to see if you are willing to collaborate with UTMB on studying the nCoV through sharing the viral isolates. We understand that, since many labs are currently isolating the viruses through imported cases, the window of leadership for Chinese scientists is closing rapidly. We greatly appreciate your rapid response to this request.

Best regards from Galveston,

utmb Health
Biochemistry & Molecular Biology

Pei-Yong Shi, Ph.D.
I.H. Kempner Professor of Human Genetics
108 Basic Science Building
301 University Boulevard
Galveston, Texas 77555-0645

January 27, 2020

Professor Fei Deng, Ph.D.,
Professor Zhengli Shi, Ph.D.,
Wuhan Institute of Virology
Chinese Academy of Sciences
Wuhan, China

Dear Professor Deng and Professor Shi,

Congratulations to you and your collaborators for the remarkable work on the new coronavirus! Although great progress has been made, more collective efforts are urgently need to contain the spread of the virus and to develop countermeasures.

We are writing to seek your collaboration with the University of Texas Medical Branch (UTMB) on studying the new coronavirus through sharing the viral isolate (strain IVCAS 6.7512). Our goal is to use the viral isolate for testing vaccine, therapeutics, and viral pathogenesis. We understand that, since many labs are currently isolating the viruses through imported cases, the window of leadership for Chinese scientists is closing rapidly. We greatly appreciate your rapid response to this request.

We look forward to a productive collaboration.

Sincerely,
(b)(6)
Pei-Yong Shi, Ph.D.
I.H. Kempner Professor of Human Genetics
Vice Chair for Innovation and Commercialization
Department of Biochemistry & Molecular Biology
University of Texas Medical Branch at Galveston
Texas, USA

From: Lawal, Adeola
Sent: Monday, January 27, 2020 4:51 PM
To: df@wh.iov.cn;Shi, Pei yong;LeDuc, James W.
Subject: FW: MTA to receive New coronavirus 2019 isolate: nCoV-2019 (IVCAS 6.7512)
Attachments: Shi - RECEIVING materials.docx

Greetings,

Please see attached MTA in support of our request to receive the following materials: New coronavirus 2019 isolate: nCoV-2019 (IVCAS 6.7512), from your institute. We look forward to getting this MTA in place.

Best wishes,

Ade Lawal, JD
Associate Legal Officer
UTMB Office of Technology Transfer
409.772.0369

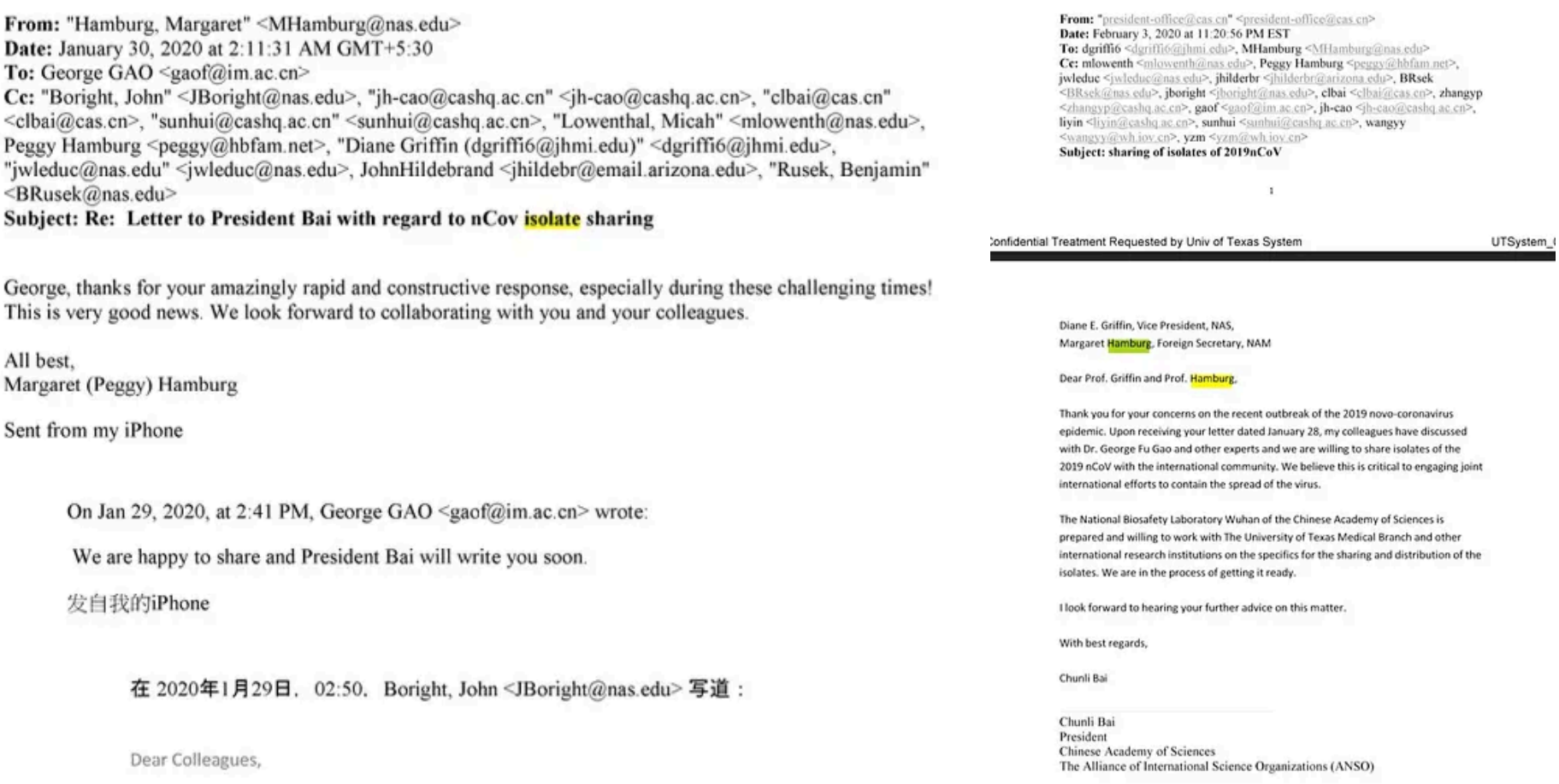
ED-20–01981-F-Dec-2022-Production-OPAQUE.pdf, p.9 and p.26, 27 Jan 2020, UTMB-LeDuc-batch-1.pdf, p.1720, 27 Jan 2020, p. 813

d. The NASEM request is sent over (28/29Jan)

➤ On **28 or 29 Jan**, the NASEM letter (dated 28 Jan) was sent over to CAS, in support of UTMB's request for isolates to the WIV.

➤ On **29 Jan** (US time), George Gao, the director of the Chinese CDC, replied that he would forward the letter to the president of the Chinese Academy of Sciences, raising hopes on the US side.

➤ Then on **3 Feb**, Chunli Bai, the president of CAS, emailed back to Griffin and Hamburg that the WIV was willing to share isolates with UTMB. At first, it looked like if Track II had delivered.



From: "Hamburg, Margaret" <MHamburg@nas.edu>
Date: January 30, 2020 at 2:11:31 AM GMT+5:30
To: George GAO <gaof@im.ac.cn>
Cc: "Boright, John" <JBoright@nas.edu>, "jh-cao@cashq.ac.cn" <jh-cao@cashq.ac.cn>, "clbai@cas.cn" <clbai@cas.cn>, "sunhui@cashq.ac.cn" <sunhui@cashq.ac.cn>, "Lowenthal, Micah" <mlowenth@nas.edu>, Peggy Hamburg <peggy@hbfam.net>, "Diane Griffin (dgriffi6@jhmi.edu)" <dgriffi6@jhmi.edu>, "jwleduc@nas.edu" <jwleduc@nas.edu>, JohnHildebrand <jhildebr@email.arizona.edu>, "Rusek, Benjamin" <BRusek@nas.edu>
Subject: Re: Letter to President Bai with regard to nCov isolate sharing

George, thanks for your amazingly rapid and constructive response, especially during these challenging times! This is very good news. We look forward to collaborating with you and your colleagues.

All best,
Margaret (Peggy) Hamburg

Sent from my iPhone

On Jan 29, 2020, at 2:41 PM, George GAO <gaof@im.ac.cn> wrote:

We are happy to share and President Bai will write you soon.

发自我的iPhone

在 2020年1月29日，02:50，Boright, John <JBoright@nas.edu> 写道：

Dear Colleagues,

From: "president-office@cas.cn" <president-office@cas.cn>
Date: February 3, 2020 at 11:20:56 PM EST
To: dgriffi6 <dgriffi6@jhmi.edu>, MHamburg <MHamburg@nas.edu>
Cc: mlowenth <mlowenth@nas.edu>, Peggy Hamburg <peggy@hbfam.net>, jwleduc <jwleduc@nas.edu>, jhilderbr <jhilderbr@arizona.edu>, BRsek <BRsek@nas.edu>, jboright <jboright@nas.edu>, clbai <clbai@cas.cn>, zhangyp <zhangyp@cashq.ac.cn>, gaof <gaof@im.ac.cn>, jh-cao <jh-cao@cashq.ac.cn>, liyin <liyin@cashq.ac.cn>, sunhui <sunhui@cashq.ac.cn>, wangyy <wangyy@wh.iov.cn>, yzm <yzm@wh.iov.cn>
Subject: sharing of isolates of 2019nCoV

Confidential Treatment Requested by Univ of Texas System UTSystem_

Diane E. Griffin, Vice President, NAS,
Margaret Hamburg, Foreign Secretary, NAM

Dear Prof. Griffin and Prof. Hamburg,

Thank you for your concerns on the recent outbreak of the 2019 novo-coronavirus epidemic. Upon receiving your letter dated January 28, my colleagues have discussed with Dr. George Fu Gao and other experts and we are willing to share isolates of the 2019 nCoV with the international community. We believe this is critical to engaging joint international efforts to contain the spread of the virus.

The National Biosafety Laboratory Wuhan of the Chinese Academy of Sciences is prepared and willing to work with The University of Texas Medical Branch and other international research institutions on the specifics for the sharing and distribution of the isolates. We are in the process of getting it ready.

I look forward to hearing your further advice on this matter.

With best regards,

Chunli Bai

Chunli Bai
President
Chinese Academy of Sciences
The Alliance of International Science Organizations (ANSO)

UTMB-LeDuc-batch-1.pdf, p.1448, p.672, 29 Jan 2020

8.3 Isolates effort #2: OSTP (White House) — NASEM — CAS (from 3 Feb)

a. Official request from the OSTP

a.1 An elaborate *pas-de-deux*

Following the Farrar call, attended by Fauci, and following some developing questions about the origin following the publication of a very controversial

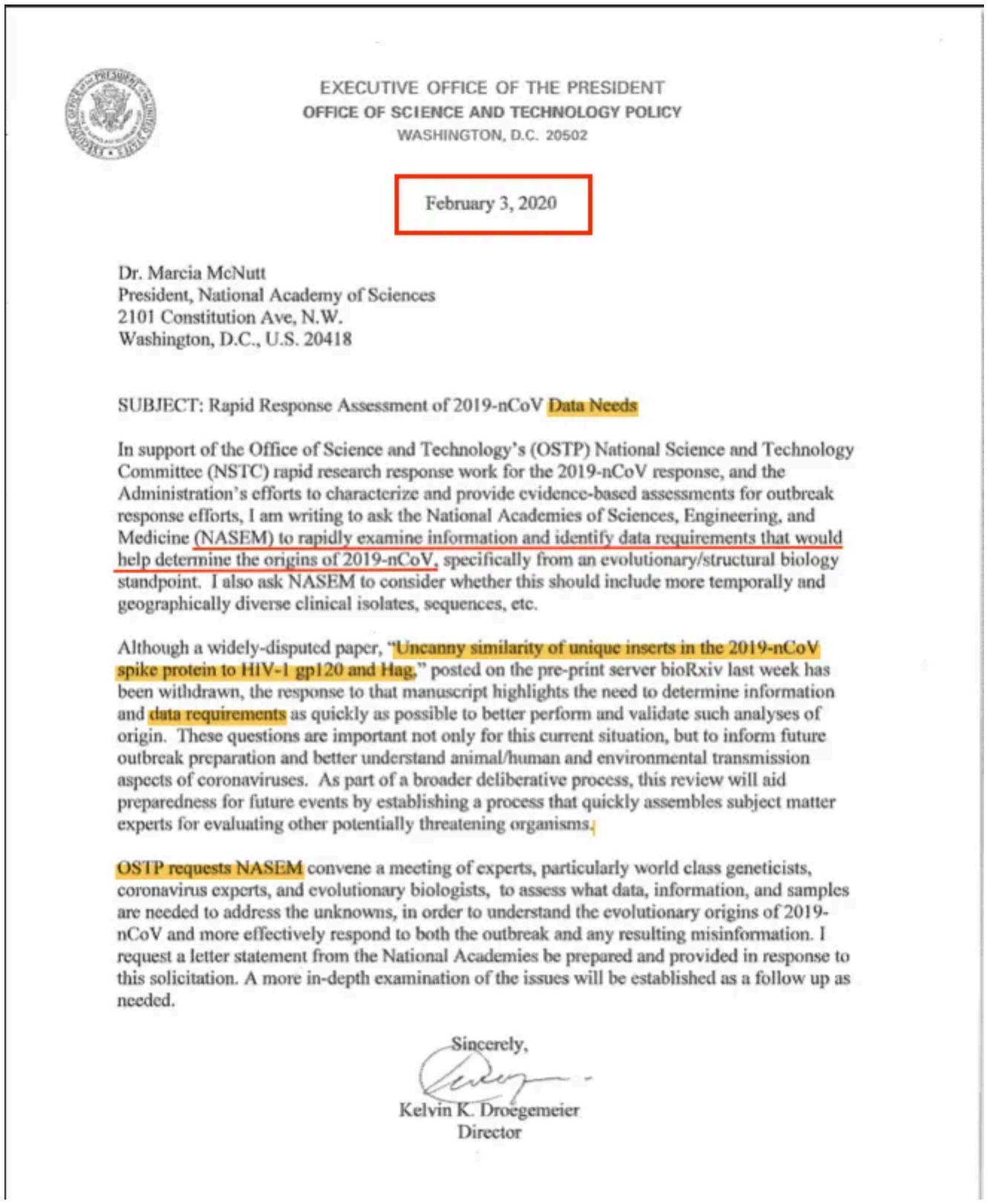
Indian paper on 31 Jan 2020, there was no way the US administration could stay still. It had to be seen as addressing these concerns. It also had to find a way to support the HHS decision to get the WHO to look into it. And it had to do that in a way that would not upset China, and would seem totally apolitical. At the same time, the first effort to obtain isolates from Wuhan was not getting anywhere.

So an upgraded Track II effort was devised, with this time the Office of Science and Technology Policy adding some White House weight, while still maintaining the required Track II distance to direct governmental action.

For this, a *pas-de-deux* was designed, whereby, on Monday 3 Feb, the Office of Science and Technology Policy (Executive, White House) sent a letter to the National Academy of Sciences, Engineering and Medicine (NASEM) asking it to *rapidly* figure out what information, data, and samples would help determine the origins of 2019-nCov, *'especially from an evolutionary/structural biology standpoint'*. [footnote 76]

The request is then laid out clearly in the last, rather official sounding, paragraph:

> *OSTP requests NASEM convene a meeting of experts, particularly world class geneticists, coronavirus experts, and evolutionary biologists* ***to assess what data, information and samples are needed to address the unknowns,*** *in order to understand the evolutionary origins of 2019-nCoV and more effectively respond to both the outbreak and any resulting misinformation.*
>
> *source: OSTP letter to NASEM, 3 Feb 2020*



EXECUTIVE OFFICE OF THE PRESIDENT
OFFICE OF SCIENCE AND TECHNOLOGY POLICY
WASHINGTON, D.C. 20502

February 3, 2020

Dr. Marcia McNutt
President, National Academy of Sciences
2101 Constitution Ave, N.W.
Washington, D.C., U.S. 20418

SUBJECT: Rapid Response Assessment of 2019-nCoV Data Needs

In support of the Office of Science and Technology's (OSTP) National Science and Technology Committee (NSTC) rapid research response work for the 2019-nCoV response, and the Administration's efforts to characterize and provide evidence-based assessments for outbreak response efforts, I am writing to ask the National Academies of Sciences, Engineering, and Medicine (NASEM) to rapidly examine information and identify data requirements that would help determine the origins of 2019-nCoV, specifically from an evolutionary/structural biology standpoint. I also ask NASEM to consider whether this should include more temporally and geographically diverse clinical isolates, sequences, etc.

Although a widely-disputed paper, "Uncanny similarity of unique inserts in the 2019-nCoV spike protein to HIV-1 gp120 and Hag," posted on the pre-print server bioRxiv last week has been withdrawn, the response to that manuscript highlights the need to determine information and data requirements as quickly as possible to better perform and validate such analyses of origin. These questions are important not only for this current situation, but to inform future outbreak preparation and better understand animal/human and environmental transmission aspects of coronaviruses. As part of a broader deliberative process, this review will aid preparedness for future events by establishing a process that quickly assembles subject matter experts for evaluating other potentially threatening organisms.

OSTP requests NASEM convene a meeting of experts, particularly world class geneticists, coronavirus experts, and evolutionary biologists, to assess what data, information, and samples are needed to address the unknowns, in order to understand the evolutionary origins of 2019-nCoV and more effectively respond to both the outbreak and any resulting misinformation. I request a letter statement from the National Academies be prepared and provided in response to this solicitation. A more in-depth examination of the issues will be established as a follow up as needed.

Sincerely,

Kelvin K. Droegemeier
Director

a.2 Not an origin request

We note that the OSTP request to NASEM **did not ask for a direct answer to the origin question;** this would not only have gone against the parallel effort to get the WHO to pick up the non-natural origin question, and it would also have been very detrimental to the request for isolates that had instead to emphasise public health imperatives. If the word *'origin'* was mentioned, it was only in the context of *'evolutionary origin'* and with an aim to respond to *'resulting misinformation'* (about the origin in the larger sense).

Furthermore, we must note the similarity with the proposal of Farrar for the WHO: *'Why are the evolutionary origins of 2019-nCoV important for future risk assessment and understanding of animal/human coronaviruses'* (see 4.11 & 4.12).

The question asked to NASEM was designed to be fed into a WHO-led process. Asking NASEM to answer an origin question would instead have defeated the idea of transferring the question to the WHO. Additionally, asking a data question instead of an origin question, emphasised the need for early samples/isolates that NASEM was already trying to obtain via CAS and the Chinese CDC.

Hence, the OSTP formulation is positioned both for the target WHO process and the previous NASEM efforts for samples/isolates.

a.3 Not a true executive request

As mentioned before, this was a Track-II initiative supported by the OSTP, not one of the President directly, even if the Office of Science and Technology Policy within the Executive Office of the President. Trump was, in fact, focussed on closing some trade agreement with China, and did not have much time (and interest) for the OSTP.

Moreover, this NASEM-OSTP Track II initiative was largely independent of the origins questions that occupied people like Redfield (CDC head, who had raised origin question after the sequence publication), Baric (who stated a possible lab-origin on the B(io)SEC(urity) meeting on 25/26 Jan), and eventually part of the US Intelligence Community. All these people were trying to work behind the scene, as they had to.

b. NASEM's 'Statement of Work" (3 Feb)

On 3 Feb, on receiving the demand from OSTP, NASEM produced an internal 'Statement of Work' ('SOW.docx'), focused on how it planned to answer the OSTP request. The statement mentioned that:

> *"NASEM will hold a **meeting of experts** to assess what data, information and samples are needed to address the unknowns [..]. A statement from the National Academies will be prepared and published on the Web as a **'based on Science' article** that summarizes the status and needs for more and what data. A more in-depth examination of the issues will be established as a follow up as needed".*

Statement of Work

Rapid Response for Assessment of Data Needs for 2019-nCoV

February 3, 2020

Statement of Task:

In response to a request from OSTP, the NASEM will examine information and identify data requirements that would help determine the origins of 2019-nCoV, specifically from an evolutionary/structural biology standpoint. NASEM will also consider whether this should include more temporally and geographically diverse clinical isolates, sequences, etc. Although a widely-disputed paper posted on a pre-print server last week has since been withdrawn, the response to that paper highlights the need to determine these information needs as quickly as possible. As part of a broader deliberative process, this review will help prepare for future events by establishing a process for quickly assembling subject matter experts for evaluation of other potentially threatening organisms.

Workplan:

NASEM will hold a meeting of experts to assess what data, information and samples are needed to address the unknowns, in order to understand the evolutionary origins of NCoV and more effectively respond to both the outbreak and any resulting misinformation. A statement from the National Academies will be prepared and published on the Web as a "Based on Science" article that summarizes the status and needs for more and what types of data. A more in-depth examination of the issues will be established as a follow up as needed.

Baric-Emails-2.17.21.pdf, p.118, 3 Feb 2020

c. The NASEM 'meeting of experts' (3 Feb)

Next, still on 3 Feb, a NASEM meeting was called, as had been laid out in the

Statement of Work. Andersen, Daszak, Fauci, Baric, Kadlec, Gronvall, and Inglesby were all part of the invited experts.

Also invited are:

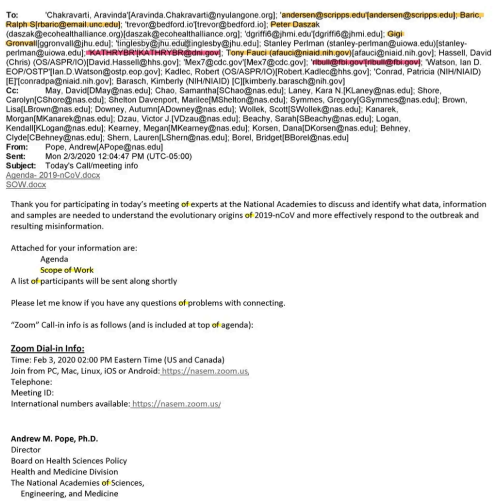
- **Robert Bull** (rlbull@fbi.gov), from the Scientific Response and Analysis Unit, **FBI Laboratory**, Quantico, VA. That's the main FBI laboratory in charge of pathogenic agents analysis, among other scientific tasks.
- **Kathryn Brinsfield** (kathrybr@dni.gov), at the time a Senior Advisor, National Counterproliferation Centre, **Office of the Director of National Intelligence,** and since late 2022 Director of ODNI's National Counterproliferation and Biosecurity Center (NCBC).

One should remember that the key BSEG meeting of 29 January, when Baric exposed the possibility of a lab-accident with a potentially modified virus, was convened by the National Counterproliferation Centre (NCPC) under the ODNI. That happened after Fauci contacted the White House, following the decision reached during the Farrar call to try to offload the origin question to the WHO, and for Fauci to seek support of the US administration. Hence, the presence of Kathryn Brinsfield of the NCPC is to be expected. We, however, don't know if the presence of an FBI attribution expert was new, or if one had already attended that key BSEG meeting.

In a way, the return of the origin question to a US government scientific forum is just a closure of the loop started by the BSEG meeting a week earlier. It's really the Farrar call of 1 February that was a detour, spurred by British Intel wanting a side check. A detour during which the WHO imperatives were discussed, and eventually onboarded.

For reference, **Tom Inglesby**, who has an extensive experience in biosafety, had long been an influential *bête noire* of the like of Gronvall and David Franz (advisor to EcoHealth Alliance), who vehemently oppose any regulation by fear of loss of international competitiveness to adversarial countries free of such limitations. Only two weeks earlier, Inglesby had just published a commentary proposing some changes to the P3CO, the U.S. policy on Potential Pandemic Pathogen Care and Oversight.

The Zoom meeting was programmed to last one hour. Fauci had the second slot, just before discussions, from 2:15 pm to 2:25 pm EST.



To: 'Chakravarti, Aravinda'[Aravinda.Chakravarti@nyulangone.org]; 'andersen@scripps.edu'[andersen@scripps.edu]; Baric, Ralph S[rbaric@email.unc.edu]; 'trevor@bedford.io'[trevor@bedford.io]; Peter Daszak (daszak@ecohealthalliance.org)[daszak@ecohealthalliance.org]; 'dgriffi6@jhmi.edu'[dgriffi6@jhmi.edu]; Gigi Gronvall[ggronvall@jhu.edu]; 'linglesby@jhu.edu'[linglesby@jhu.edu]; Stanley Perlman (stanley-perlman@uiowa.edu)[stanley-perlman@uiowa.edu]; 'KATHRYBR'[KATHRYBR@dni.gov]; Tony Fauci (afauci@niaid.nih.gov)[afauci@niaid.nih.gov]; Hassell, David (Chris) (OS/ASPR/IO)[David.Hassell@hhs.gov]; 'Mex7@cdc.gov'[Mex7@cdc.gov]; 'rlbull@fbi.gov'[rlbull@fbi.gov]; 'Watson, Ian D. ECP/OSTP'[Ian.D.Watson@ostp.eop.gov]; Kadlec, Robert (OS/ASPR/IO)[Robert.Kadlec@hhs.gov]; 'Conrad, Patricia (NIH/NIAID) [E]'[conradpa@niaid.nih.gov]; Barasch, Kimberly (NIH/NIAID) [C][kimberly.barasch@nih.gov]
Cc: May, David[DMay@nas.edu]; Chao, Samantha[SChao@nas.edu]; Laney, Kara N.[KLaney@nas.edu]; Shore, Carolyn[CShore@nas.edu]; Shelton Davenport, Marilee[MShelton@nas.edu]; Symmes, Gregory[GSymmes@nas.edu]; Brown, Lisa[LBrown@nas.edu]; Downey, Autumn[ADowney@nas.edu]; Wollek, Scott[SWollek@nas.edu]; Kanarek, Morgan[MKanarek@nas.edu]; Dzau, Victor J.[VDzau@nas.edu]; Beachy, Sarah[SBeachy@nas.edu]; Logan, Kendall[KLogan@nas.edu]; Kearney, Megan[MKearney@nas.edu]; Korsen, Dana[DKorsen@nas.edu]; Behney, Clyde[CBehney@nas.edu]; Shern, Lauren[LShern@nas.edu]; Borel, Bridget[BBorel@nas.edu]
From: Pope, Andrew[APope@nas.edu]
Sent: Mon 2/3/2020 12:04:47 PM (UTC-05:00)
Subject: Today's Call/meeting info
Agenda- 2019-nCoV.docx
SOW.docx

Thank you for participating in today's meeting of experts at the National Academies to discuss and identify what data, information and samples are needed to understand the evolutionary origins of 2019-nCoV and more effectively respond to the outbreak and resulting misinformation.

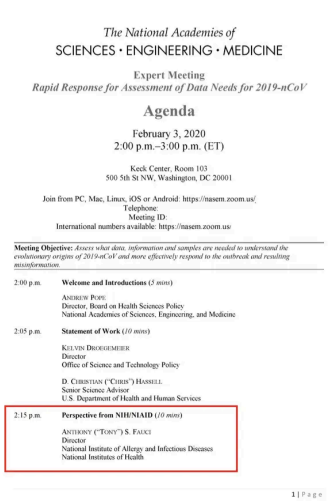
Attached for your information are:
Agenda
Scope of Work
A list of participants will be sent along shortly

Please let me know if you have any questions of problems with connecting.

"Zoom" Call-in info is as follows (and is included at top of agenda):

Zoom Dial-in Info:
Time: Feb 3, 2020 02:00 PM Eastern Time (US and Canada)
Join from PC, Mac, Linux, iOS or Android: https://nasem.zoom.us
Telephone:
Meeting ID:
International numbers available: https://nasem.zoom.us/

Andrew M. Pope, Ph.D.
Director
Board on Health Sciences Policy
Health and Medicine Division
The National Academies of Sciences,
Engineering, and Medicine



The National Academies of
SCIENCES · ENGINEERING · MEDICINE

Expert Meeting
Rapid Response for Assessment of Data Needs for 2019-nCoV

Agenda

February 3, 2020
2:00 p.m.–3:00 p.m. (ET)

Keck Center, Room 103
500 5th St NW, Washington, DC 20001

Join from PC, Mac, Linux, iOS or Android: https://nasem.zoom.us/
Telephone:
Meeting ID:
International numbers available: https://nasem.zoom.us/

Meeting Objective: *Assess what data, information and samples are needed to understand the evolutionary origins of 2019-nCoV and more effectively respond to the outbreak and resulting misinformation.*

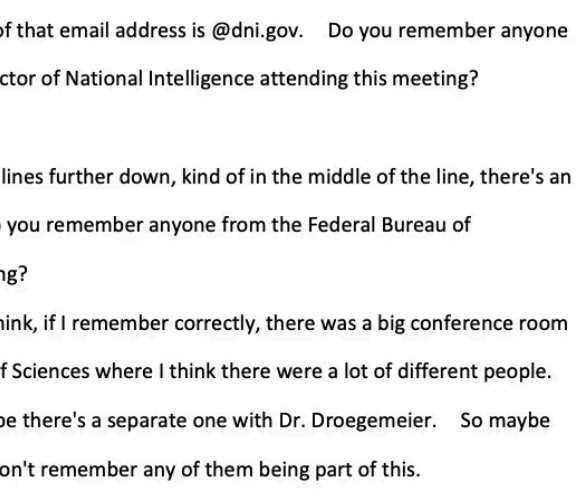
2:00 p.m. **Welcome and Introductions** (*5 mins*)
ANDREW POPE
Director, Board on Health Sciences Policy
National Academies of Sciences, Engineering, and Medicine

2:05 p.m. **Statement of Work** (*10 mins*)
KELVIN DROEGEMEIER
Director
Office of Science and Technology Policy

D. CHRISTIAN ("CHRIS") HASSELL
Senior Science Advisor
U.S. Department of Health and Human Services

2:15 p.m. **Perspective from NIH/NIAID** (*10 mins*)
ANTHONY ("TONY") S. FAUCI
Director
National Institute of Allergy and Infectious Diseases
National Institutes of Health

1 | Page



14 Q And the end of that email address is @dni.gov. Do you remember anyone
15 from the Office of the Director of National Intelligence attending this meeting?
16 A I do not.
17 Q Then, on two lines further down, kind of in the middle of the line, there's an
18 email, rlbull@fbi.gov. Do you remember anyone from the Federal Bureau of
19 Investigation at this meeting?
20 A I do not. I think, if I remember correctly, there was a big conference room
21 at the National Academy of Sciences where I think there were a lot of different people.
22 So maybe they -- and maybe there's a separate one with Dr. Droegemeier. So maybe
23 they're a part of that. I don't remember any of them being part of this.

Baric-Emails-2.17.21.pdf, p.115–116, 3 Feb 2020, and 2023.06.16-Andersen-Transcript.pdf, p. 104b

d. Someone was angry with Andersen (3 Feb)

Andersen later wrote on Slack that:

> *Ralph Baric pretty much attacked me on the call with NASEM, essentially calling anything related to potential lab escape ludicrous, crackpot theories. I wonder if he, himself, is worried about this, too.*
>
> *source: Baric-TI-Transcript.pdf, p.120–121 [quoting the edited Slack]*

In his interview with the House Select Committee on the Coronavirus Pandemic, Baric denied being the one who made these remarks during the call in what seems to be a credible description of the events. And indeed, Ralph Baric did take a different position during the B(io)SEC(urity) meeting on 25/26 Jan, and would later publicly ask for a proper investigation of a possible lab-leak. It therefore is possible that Andersen misattributed these comments to Baric, when they were from Daszak or someone else on the call.

What we can be sure of is that Andersen got some strong pushback from one person on the call, and a pushback that sounded very much like the one that Andersen had got from Fouchier during the Farrar's call.

2982 A January 25th or 26th of 2020. So I don't know
2983 where he's coming from. That may have been his
2984 interpretation, but I'm surprised. I'm really surprised.
2985 Q When we saw it, I wanted to make sure we got
2986 your perspective on the record.
2987 A Can you read it one more time?
2988 Q Yes. "I should mention that Ralph Baric
2989 pretty much attacked me on the call with NASEM, essentially
2990 calling anything related to potential lab escape ludicrous,
2991 crackpot theories. I wonder if he, himself, is worried about
2992 this, too."
2993 A I'm really surprised about this, because I
2994 wrote a piece on his origin paper in Immunology, and said
2995 that laboratory escape was possible because of safety
2996 procedures in their laboratories. So it's not consistent
2997 with what I also reported to other groups publicly on when
2998 interviewed. So I don't believe he's attributing that to the
2999 right person.
3000 Q That's fair. And I wish I could show you the
3001 message, but like I said, it's redacted, so I don't have it.

Baric-TI-Transcript.pdf, p. 121

e. Sudden change of plans and first draft of a response letter to OSTP (4 Feb)
On the morning of the day following the Zoom meeting, Andrew Pope, Director of the Board of Health Science Policy of NASEM, unexpectedly emailed the people who were in the 'meeting of experts' the day before, explaining that the 'Based on Science' web-posting shall be replaced by a response letter to the OSTP, a draft of which his attached in his email for immediate review (within only 3 hours).

Surprisingly, against the guidelines of the 'Statement of Work', the draft formulates an opinion about the origin, before even addressing the OSTP request about data needs.

We may conclude that the 'meeting of experts' was at least partly spent discussing the origin question, not sticking to the OSTP request.

> *''The initial views of the experts is that the **available genomic data are consistent with natural evolution** and that there is currently no evidence that the virus was engineered to spread more quickly among humans.'*

(In that sentence above, the 'engineered to spread more quickly among humans' is a muted reference to the Furin Cleavage Site.)

From: Pope, Andrew [mailto:APope@nas.edu]
Sent: Tuesday, February 4, 2020 9:11 AM
To: 'Chakravarti, Aravinda'; Kristian Andersen ; Ralph Baric (rbaric@email.unc.edu); Trevor Bedford (trevor@bedford.io); Peter Daszak; Gigi Gronvall; Tom Inglesby (tinglesby@jhu.edu); Stanley Perlman (stanley-perlman@uiowa.edu)
Cc: Shore, Carolyn; Chao, Samantha
Subject: URGENT: Please review by NOON if at all possible...
Importance: High

Many thanks again for your thoughtful participation yesterday. The plans have changed in terms of our product. Instead of a "Based on Science" web posting, we are now developing a letter that will be signed by the 3 Presidents of our 3 Academies (NAS, Marcia McNutt; NAM, Victor Dzau; NAE, John Anderson), in response to a letter from OSTP. We think this will be more appropriate and expeditious.

Thus, given the urgency of the request from OSTP and HHS we ask that you please review the attached DRAFT CONFIDENTIAL letter, and let us know if you have any concerns or suggested edits. In particular, we would like to ask if there might be some additional detail added to the data needs that are identified. We think it would be helpful to be a bit more specific, but don't want to go into too much detail either. Your help there would be most helpful.

Many sincere thanks again for your continued engagement on this important activity!

Andy

Andrew M. Pope, Ph.D.
Director
Board on Health Sciences Policy
Health and Medicine Division
The National Academies of Sciences,
Engineering, and Medicine
apope@nas.edu
direct
office
Find us at nationalacademies.org/HMD

The National Academies of
SCIENCES • ENGINEERING • MEDICINE

CONFIDENTIAL DRAFT

February 4, 2020

[insert address]

Dear XXX:

Thank you for your letter regarding the current outbreak of a new respiratory virus, the 2019 Novel Coronavirus, or 2019-nCoV, which was first detected in Wuhan, China, and has now been reported in a growing number of locations worldwide, including the United States.[1] The request from OSTP is timely given the public health urgency of the outbreak and potential for misinformation.

In response to your request, we consulted leading experts[2] in the fields of virology, infectious disease genomics, genome sciences, epidemiology, microbiology, immunobiology, coronaviruses, emerging infections, biosecurity, and global health, to share their views of whether available genomic data on 2019-nCoV are consistent with natural evolution and the data that could help determine the origins of 2019-nCoV, specifically from an evolutionary and structural biology standpoint.

Many studies of the genome of 2019-nCoV to better understand its origin and how it relates to viruses found in bats and other species are already underway.[3] The initial views of the experts[4] is that the available genomic data are consistent with natural evolution[5] and that there is currently no evidence that the virus was engineered to spread more quickly among humans. [ask experts to add specifics re binding sites?] They also told us that additional genomic sequence data from geographically and temporally diverse viral samples, including samples that have been collected prior to the outbreak in Wuhan, could be used to clarify the origins of the virus. Understanding the driving forces behind viral evolution may facilitate the development of more effective strategies for managing the 2019-nCoV outbreak. International collaboration is more important than ever to overcome these types of global challenges.

The National Academies stand ready to assemble a committee of experts to examine these issues in more detail and provide more complete evidence-based advice to you in an expedited manner if requested.

Thank you, again for your commitment to the National Academies and our efforts to provide independent, objective analysis; advise the nation; and inform public policy decisions.

Sincerely,

[1] "2019 Novel Coronavirus (2019-nCoV) Situation Summary." *Centers for Disease Control and Prevention*, 3 Feb. 2020. https://www.cdc.gov/coronavirus/2019-nCoV/summary.html#anchor_1580079137454. Accessed 3 Feb. 2020.
[2] [possible add list]
[3] [insert references]
[5] [possibly add brief explanation that this does not preclude an unintentional release from a laboratory studying the evolution of related coronaviruses]

Pope's email and attached Draft, 4 Feb 2020, Baric-Emails-2.17.21.pdf, p.133 and p. 120

f. Drafting of the NASEM response to OSTP (4 Feb)

f.1. Daszak's contribution to the draft

Daszak sent some rather mild comments, as he was essentially happy with the drafting so far (see f.2 below).

From: Peter Daszak <daszak@ecohealthalliance.org>
Sent: Tuesday, February 4, 2020 12:01 PM
To: Pope, Andrew <APope@nas.edu>; 'Chakravarti, Aravinda' <Aravinda.Chakravarti@nyulangone.org>; Kristian Andersen ; Baric, Ralph S <rbaric@email.unc.edu>; Trevor Bedford (trevor@bedford.io) <trevor@bedford.io>; Gigi Gronvall <ggronvall@jhu.edu>; Tom Inglesby (tinglesby@jhu.edu) <tinglesby@jhu.edu>; Stanley Perlman (stanley-perlman@uiowa.edu) <stanley-perlman@uiowa.edu>
Cc: Shore, Carolyn <CShore@nas.edu>; Chao, Samantha <SChao@nas.edu>
Subject: RE: URGENT: Please review by NOON if at all possible...
Importance: High

I agree with all of the other comments so far sent in, and want to add the following:

1) In the 3rd paragraph, it's important to add "including further samples from wildlife", and perhaps the rationale for this "to identify other viruses closely related to nCoV"
2) Re. references for #3 that there are current and planned studies underway on the bat origins of CoVs. Here are some references to pick from if they make sense:

- Latinne A, Hu B, Olival KJ, et al.; Origin and cross-species transmission of bat coronaviruses in China. *Nature Communications* 2020;**In review**.
- Wang N, Li S-Y, Yang X-L, et al.; Serological Evidence of Bat SARS-Related Coronavirus Infection in Humans, China. *Virologica Sinica* 2018. doi: 10.1007/s12250-018-0012-7.
- Hu B, Zeng L-P, Yang X-L, et al.; Discovery of a rich gene pool of bat SARS-related coronaviruses provides new insights into the origin of SARS coronavirus. *PLOS Pathogens* 2017;**13**(11):e1006698. doi: 10.1371/journal.ppat.1006698.
- Zhou P, Fan H, Lan T, et al.; Fatal Swine Acute Diarrhea Syndrome caused by an HKU2-related Coronavirus of Bat Origin. *Nature* 2018

Cheers,

Peter

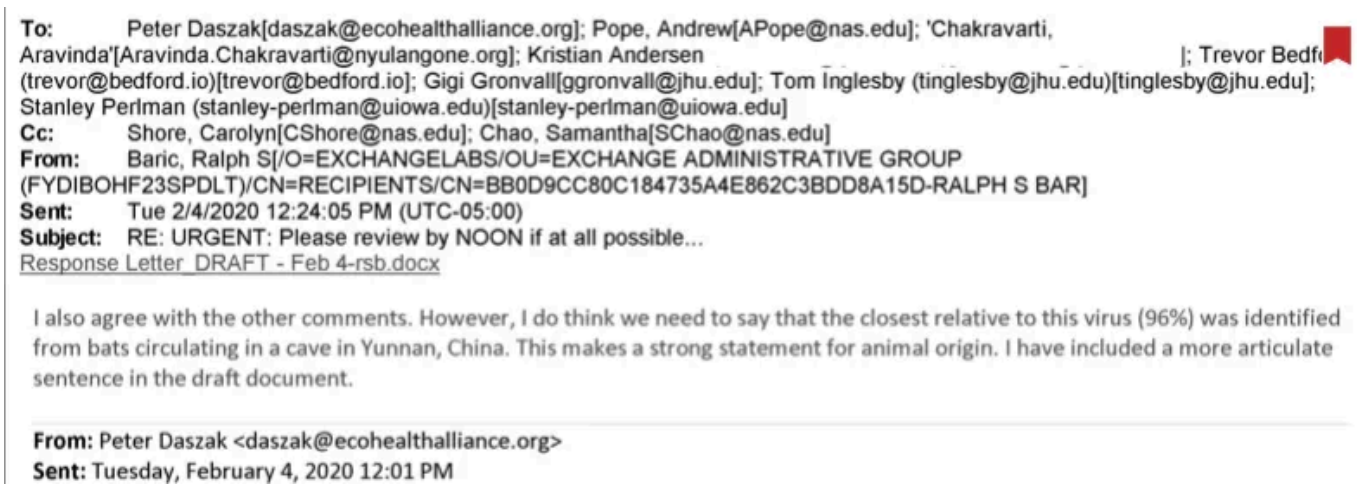
Baric-Emails-2.17.21.pdf, p.128, 4 Feb 2020

f.2 Baric's contribution

Baric also stepped in soon after with a comment that missed the data needs

request and focused instead on '***mak[ing] a strong statement for a natural origin***', backed by a pretend argument. That pretend argument, designed to capture the attention of the media and lead them on the wrong path, was that RaT13 was too distant at 96% to be a precursor or backbone. This, despite Baric having asserted a possible non-natural origin, including of an engineered virus, during the confidential B(io)SEC(urity) meeting of 25 or 26 Jan (see 3.2), and despite Baric knowing very well that the WIV had many unpublished viruses.

The most likely interpretation of that contradiction is that the NASEM letter is an instrument of Track II diplomacy meant to facilitate access for the WHO and to date and isolates for the US. There is no way it can point to a non-natural origin. Actually, the best should really to simply follow the 'Statement of work' and avoid the subject altogether.



To: Peter Daszak[daszak@ecohealthalliance.org]; Pope, Andrew[APope@nas.edu]; 'Chakravarti, Aravinda'[Aravinda.Chakravarti@nyulangone.org]; Kristian Andersen |; Trevor Bedf (trevor@bedford.io)[trevor@bedford.io]; Gigi Gronvall[ggronvall@jhu.edu]; Tom Inglesby (tinglesby@jhu.edu)[tinglesby@jhu.edu]; Stanley Perlman (stanley-perlman@uiowa.edu)[stanley-perlman@uiowa.edu]
Cc: Shore, Carolyn[CShore@nas.edu]; Chao, Samantha[SChao@nas.edu]
From: Baric, Ralph S[/O=EXCHANGELABS/OU=EXCHANGE ADMINISTRATIVE GROUP (FYDIBOHF23SPDLT)/CN=RECIPIENTS/CN=BB0D9CC80C184735A4E862C3BDD8A15D-RALPH S BAR]
Sent: Tue 2/4/2020 12:24:05 PM (UTC-05:00)
Subject: RE: URGENT: Please review by NOON if at all possible...
Response Letter_DRAFT - Feb 4-rsb.docx

I also agree with the other comments. However, I do think we need to say that the closest relative to this virus (96%) was identified from bats circulating in a cave in Yunnan, China. This makes a strong statement for animal origin. I have included a more articulate sentence in the draft document.

From: Peter Daszak <daszak@ecohealthalliance.org>
Sent: Tuesday, February 4, 2020 12:01 PM

Baric-Emails-2.17.21.pdf, p.128, 4 Feb 2020

f.3 Andersen's contribution

Andersen later sent some rather involved comments:

> *Reading through the letter I think it's great, but I do wonder if we need to be more firm on the question of engineering. The main crackpot theories going around at the moment relate to this virus being somehow* ***engineered with intent*** *and that is* ***demonstrably not the case****. Engineering can mean many things and could be done for either basic research or nefarious reasons, but the data conclusively show that neither was done (in the* ***nefarious scenario*** *somebody would have used a SARS/MERS backbone and optimal ACE2 binding as previously described, and for the* ***basic research scenario*** *would have used one of the many already available reverse genetic systems).*
>
> *If one of the main purposes of this document is to counter those fringe theories, I think it's very important that we do so strongly and in plain language (***"consistent with" [natural evolution]*** *is a favorite of mine when talking to scientists, but not when talking to the public — especially conspiracy theorists).'*
>
> *source: Proximal_Origin_Emails.pdf, p.16, 4 Feb 2020*

In his comment, 'engineered with intent' can refer to direct gene editing, which at the time was not what the 'core group' considered anyway as a likely non-natural origin. Instead, the 'Core Group' doubts were focussing on passaging, which cannot really be described as 'with intent' (no more than natural evolution has any intent).

Andersen also made the point that the draft wording '**consistent with natural evolution**' may be good when talking to scientists, but not when addressing the public. In other words, because that letter was also meant for the public, Andersen suggested using a more definite wording than that 'consistent with'.

Gmail [redacted]

URGENT: Please review by NOON if at all possible...

Kristian G. Andersen [redacted] Tue, Feb 4, 2020 at 9:05 AM
To: Peter Daszak [redacted]
Cc: "Pope, Andrew" [redacted], "Chakravarti, Aravinda" [redacted], "Ralph Baric [redacted] "Trevor Bedford [redacted] Gigi Gronvall [redacted], "Tom Inglesby [redacted] "Stanley Perlman [redacted] "Shore, Carolyn" [redacted], "Chao, Samantha" [redacted]

I too agree with all that has been said, but would caution against adding language suggesting that the virus might evolve (i.e., "mutate" to most people) towards better infectivity or transmission - a lot has been said about that for Ebola and other viruses, and it's been driving fear because most people don't fully understand what it means. I'm not arguing that it's not something that might well happen - the SARS data beautifully show it - but I would be worried about the message it could send.

Reading through the letter I think it's great, but I do wonder if we need to be more firm on the question of engineering. The main crackpot theories going around at the moment relate to this virus being somehow engineered with intent and that is demonstrably not the case. Engineering can mean many things and could be done for either basic research or nefarious reasons, but the data conclusively show that neither was done (in the nefarious scenario somebody would have used a SARS/MERS backbone and optimal ACE2 binding as previously described, and for the basic research scenario would have used one of the many already available reverse genetic systems). If one of the main purposes of this document is to counter those fringe theories, I think it's very important that we do so strongly and in plain language ("consistent with" [natural evolution] is a favorite of mine when talking to scientists, but not when talking to the public - especially conspiracy theorists).

Best,
Kristian

On Tue, Feb 4, 2020 at 9:02 AM Peter Daszak [redacted] wrote:

> I agree with all of the other comments so far sent in, and want to add the following:
>
> 1) In the 3rd paragraph, it's important to add "including further samples from wildlife", and perhaps the rationale for this "to identify other viruses closely related to nCoV"
>
> 2) Re. references for #3 that there are current and planned studies underway on the bat origins of CoVs. Here are some references to pick from if they make sense:
>
> · Latinne A, Hu B, Olival KJ, et al.; Origin and cross-species transmission of bat coronaviruses in China. *Nature Communications* 2020;**In review**.
>
> · Wang N, Li S-Y, Yang X-L, et al.; Serological Evidence of Bat SARS-Related Coronavirus Infection in Humans, China. *Virologica Sinica* 2018. doi: 10.1007/s12250-018-0012-7.
>
> · Hu B, Zeng L-P, Yang X-L, et al.; Discovery of a rich gene pool of bat SARS-related coronaviruses provides new insights into the origin of SARS coronavirus. *PLOS Pathogens* 2017;**13**(11):e1006698. doi: 10.1371/journal.ppat.1006698.
>
> · Zhou P, Fan H, Lan T, et al.; Fatal Swine Acute Diarrhea Syndrome caused by an HKU2-related Coronavirus of Bat Origin. *Nature* 2018

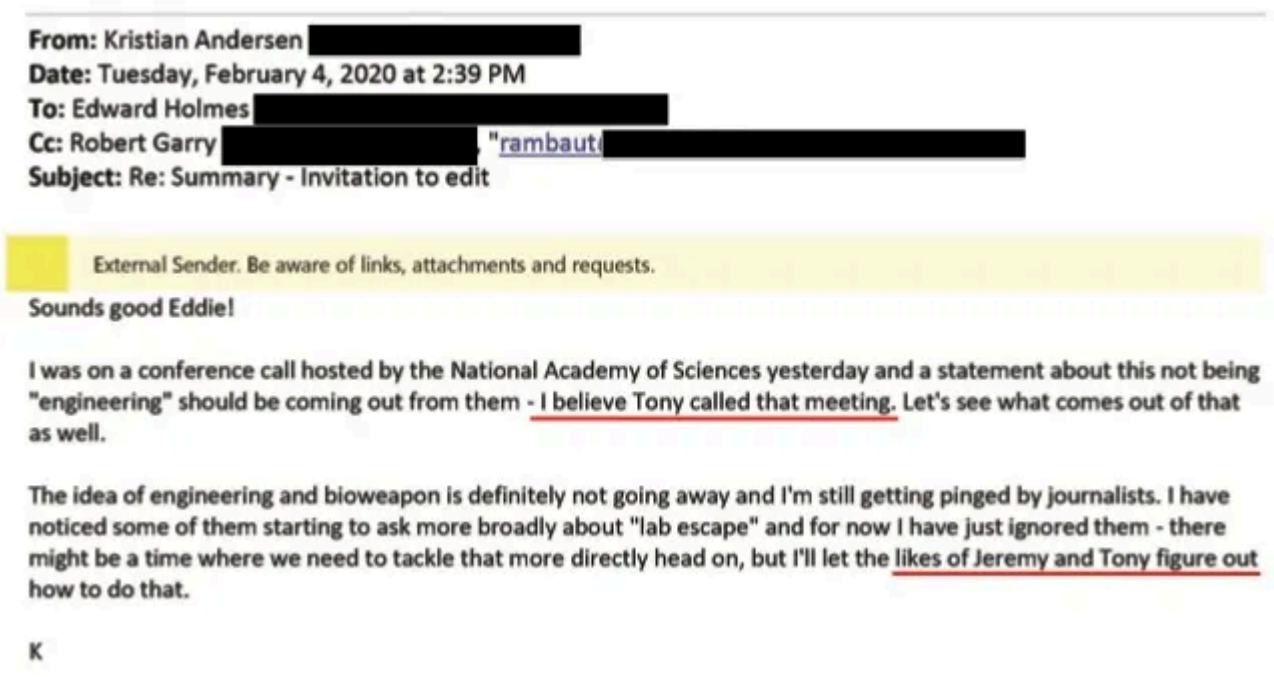
Proximal_Origin_Emails.pdf, p.16, 4 Feb 2020

At about the same time (2:39 pm EST), Andersen also emailed Holmes, Garry and Rambaut explaining that he had been on the NASEM call, and that NASEM should reply with a **non-engineered** statement.

As we know, at that very same time, the 'core group' was still having strong doubts about the virus origin. Nevertheless, if '**engineering**' is meant as '**lab-construct**' (deliberate sequence editing, i.e.: intelligent design) as a

subcategory of '**lab-product**' (which also includes passaging, i.e. accelerated targeted 'natural' evolution in a lab), then this would not contradict the fact that they were still leaning towards possible passaging as the origin of the virus.

Andersen further referred in his email to yet another level of research-related accidents: a '**lab-escape**', that may include natural virus, and not only a 'lab-construct' or a 'lab-product'. This adds yet another possibility for a research-related origin, namely the escape of a virus collected in the wild, without modification. But Andersen adds that this hypothesis is simply not in the remit of the report that the 'core group' is drafting, as they were only looking at possible genetic signatures of a non-natural virus [footnote 5].



From: Kristian Andersen [redacted]
Date: Tuesday, February 4, 2020 at 2:39 PM
To: Edward Holmes [redacted]
Cc: Robert Garry [redacted], "rambaut[redacted]
Subject: Re: Summary - Invitation to edit

External Sender. Be aware of links, attachments and requests.

Sounds good Eddie!

I was on a conference call hosted by the National Academy of Sciences yesterday and a statement about this not being "engineering" should be coming out from them - I believe Tony called that meeting. Let's see what comes out of that as well.

The idea of engineering and bioweapon is definitely not going away and I'm still getting pinged by journalists. I have noticed some of them starting to ask more broadly about "lab escape" and for now I have just ignored them - there might be a time where we need to tackle that more directly head on, but I'll let the likes of Jeremy and Tony figure out how to do that.

K

Proximal_Origin_Emails.pdf, p.32, 4 Feb 2020

There is no escaping the conclusion that at that stage, Andersen was proposing a wording based on some tortuous nuances of language he was fully aware of. A wording that would misdirect most readers of the NASEM draft, including many if not most scientists, away from the very suspicions about a non-natural origin he was having.

It was one step up from the RaTG13 distraction that Baric wanted to introduce, and half too-clever, but typical for Andersen.

f.4 Bedford's contribution

In contrast to the artistic licence of Baric's and Andersen's contributions, Trevor Bedford offered a careful comment, which clearly left the door open to passaging by insisting simply on the "**no evidence of generic engineering**", while pointing to the fact that there would be "**a lot to consider for both scenarios**".

Giving up the game altogether, he concluded by recommending moving "**to more secure forms of communications**".

On Feb 4, 2020, at 9:14 AM, Trevor Bedford <trevor@bedford.io> wrote:

Briefly, my suggestions:

1. I wouldn't mention binding sites here. If you start weighing evidence there's a lot to consider for both scenarios.

2. I would say "no evidence of genetic engineering" full stop.

3. Rather than "including samples that have been collected prior to the outbreak in Wuhan" I would say "including samples collected from as early as possible in the Wuhan outbreak".

I'm not sure what the exact capacity of this group going forward will be, but I might suggest moving to more secure forms of communication.

- Trevor

> On Feb 4, 2020, at 6:10 AM, Pope, Andrew <APope@nas.edu> wrote:
>
> Many thanks again for your thoughtful participation yesterday. The plans have changed in terms of our product. Instead of a "Based on Science" web posting, we are now developing a letter that will be signed by the 3 Presidents of our 3 Academies (NAS, Marcia McNutt; NAM, Victor Dzau; NAE, John Anderson), in response to a letter from OSTP. We think this will be more appropriate and expeditious.
>
> Thus, given the urgency of the request from OSTP and HHS we ask that you please review the attached DRAFT CONFIDENTIAL letter, and let us know if you have any concerns or suggested edits. In particular, we would like to ask if there might be some additional detail added to the

Baric-Emails-2.17.21.pdf, p.123, 4 Feb 2020

f.5 Perlman's contribution

In response to Bedford, Perlman sent some mild comments. However, as a distinct note, he also made reference to discussions about the FCS that took place during the call, especially about the possibility of passaging. His note states that he could not think of an FCS evolving through passaging in a coronavirus, and that natural evolution of an FCS is itself possible (**'as I think we concluded yesterday'**).

This is clearly a pushback on Bedford. But this also indicates that the meeting was largely spent discussing the origin question, hence resulting in a first statement in the draft letter being about the origin, not about the question asked about 'data needs'.

g. Official NASEM's response to the OSTP (6 Feb)

In discussing the draft, the experts had somehow saddled themselves with a question that the 'Statement of Work' did not ask them to answer, resulting in some dancing around the terms 'engineered' or 'consistent with', plus some RaTG13 distraction, all to say something while meaning the opposite, or even recommending moving to secure discussions instead.

Some wiser mind had fortunately prevailed and gone back to the neutral 'Statement of Work': Andersen's central focus of answering that the virus genome was 'consistent with natural evolution' was gone, as was footnote 5 of the draft letter that mentioned the possibility of lab-escape of a natural virus. This was replaced by a statement that: ***'additional genomic sequence data from geographically — and temporally-diverse viral samples are needed to determine the origin and evolution of the virus'.***

Note that there was no mention but also no denial of a possible research-related origin, nor is the word zoonotic used. At the same time, there were statements that defend EcoHealth Alliance work, and which were likely worked in by Daszak:

- '*Research studies to better understand the origin of 2019-nCoV and how it relates to viruses found in bats and other species are already underway*', with its footnote 3, refers to an EcoHealth Alliance paper, Latinne *et al.*, which was at the time in review with Nature Communications.
- '*International collaboration of this kind is more important than ever to overcome these types of global challenges*' is precisely the kind of grandstanding language favoured by Daszak.

The NASEM letter, once back to the Statement of Work and the origin question evacuated, had no issue with the inclusion of these suggestions of Daszak that were not detrimental.

The National Academies of
SCIENCES · ENGINEERING · MEDICINE

February 6, 2020

Kelvin Droegemeier
Director
White House Office of Science and Technology Policy
1650 Pennsylvania Avenue, NW
Washington, D.C. 20504

Dear Dr. Droegemeier:

Thank you for your letter regarding the current outbreak of a new respiratory virus, the 2019 Novel Coronavirus, or 2019-nCoV, which was first detected in W uhan, China, and has now been reported in a growing number of locations worldwide, including the United States. [1] The request from OSTP is timely given the declaration of a public health emergency and potential for misinformation to confound the response.

In response to your request, we consulted leading experts [2] in the fields of virology, infectious disease genomics, genome sciences, epidemiology, microbiology, immunobiology, coronaviruses, emerging infections, biosecurity, and global health. We wanted their views about the data needs that could help elucidate the origin and evolution of 2019-nCoV.

Research studies to better understand the origin of 2019-nCoV and how it relates to viruses found in bats and other species are already underway.[3] The closest known relative of 2019-nCoV appears to be a coronavirus identified from bat-derived samples collected in China.[4] The experts informed us that additional genomic sequence data from geographically- and temporally-diverse viral samples are needed to determine the origin and evolution of the virus. Samples collected as early as possible in the outbreak in W uhan and samples from wildlife would be particularly valuable. Understanding the driving forces behind viral evolution would help facilitate the development of more effective strategies for managing the 2019-nCoV outbreak and for preventing future outbreaks. In this regard, we understand from Chunli Bai, President, Chinese Academy of Sciences, and the Alliance of International Science Organizations (ANSO), that the W uhan National Biosafety Laboratory of the Chinese Academy of Sciences is willing to share is olates of the 2019-nCoV with the international community and is working with the University of Texas Medical Branch and other international research institutions on the specifics for the sharing a nd distribution of the isolates. International collaboration of this kind is more important than ever to overcome these ty pes of global challenges.

[1] "2019 Novel Coronavirus (2019-nCoV) Situation Summary." *Centers for Disease Control and Prevention*, 3 Feb. 2020. https://www.cdc.gov/coronavirus/2019 nCoV/summary.html#anchor_1580079137454 Accessed 3 Feb. 2020.
[2] Experts consulted: Kristian G. Andersen (Scripps Research Institute), Ralph Baric (UNC School of Public Health), Trevor Bedford (Fred Hutchinson Cancer Institute), Aravinda Chakravarti (New York University School of Medicine), Peter Daszak (EcoHealth Alliance), Gigi K. Gronvall (Johns Hopkins Bloomberg School of Public Health), Tom Inglesby (Johns Hopkins Center for Health Security), and Stanley Perlman (University of Iowa).
[3] Latinne *et al*. "Origin and cross-species transmission of bat coronaviruses in China." *Nature Communications*, in review.
[4] Zhou *et. al*. "A pneumonia outbreak associated with a new coronavirus of probable bat origin." *Nature*, 2020. https://doi.org/10.1038/s41586 020-2012-7 (2020).

500 Fifth Street, NW, Washington, DC 20001

NASEM answer to OSTP — final — 6 Feb 2020

h. The real message behind the NASEM answer

One may wonder what actual insights were gained from the OSTP / NASEM exercise, which ostensibly what meant to quickly figure out which information and data would be useful to help answer the origin question.

Here they are in their fullness:

> *The experts informed us that additional genomic sequence data from geographically- and temporally-diverse viral samples are needed to determine the origin and evolution of the virus. Samples collected as early as possible in the outbreak in Wuhan and samples from wildlife would be particularly valuable.*

Obviously, this is not what the NASEM letter was about. As we know, it was an effort to get data and isolates, via some NASEM-CAS Track II activation. Something that the next sentences drives home:

> *In this regard, we understand from Chunli Bai, President, Chinese Academy of Sciences, and the Alliance of International Science Organizations (ANSO), that the **Wuhan National Biosafety Laboratory of the Chinese Academy of Sciences is willing to share isolates of the 2019-nCoV with the international community and is working with the University of Texas Medical Branch** and other international research institutions on the specifics for the sharing and distribution of the isolates. International collaboration of this kind is more important than ever to overcome these types of global challenges.*

8.4 Some problematic reporting, courtesy of Fauci (6 Feb)

a. ABC article (6 Feb)

On Thursday 6 Feb [footnote 67], ABC published an article with some poor reporting and a clumsy quote by Fauci.

> *The White House on Thursday asked U.S. scientists and medical researchers to **investigate the scientific origins of the novel coronavirus**, as misinformation about the outbreak spreads online.*
>
> *The director of the White House's Office of Science and Technology Policy (OSTP), in a letter to the National Academies of Sciences, Engineering, and Medicine, requested that scientific experts "rapidly" look into the origins of the virus in order to address both the current spread and "to inform future outbreak preparation and better understand animal/human and environmental transmission aspects of coronaviruses."*
>
> *ABC News' Chief Medical Correspondent Dr. Jennifer Ashton asked the director of the National Institute of Allergy and Infectious Disease about concerns that stem from misinformation online that the novel coronavirus could have been engineered or deliberately released.*

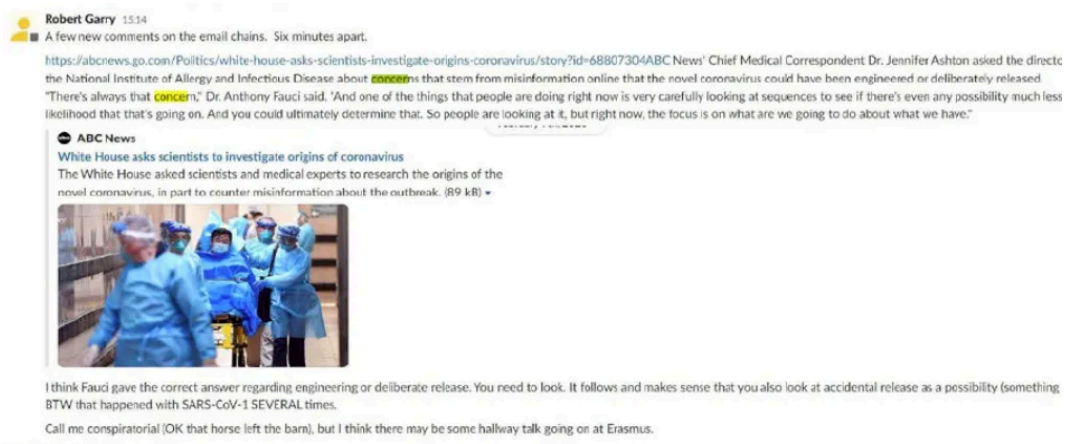
> *"**There's always that concern,**" Dr. Anthony Fauci said. "**And one of the things that people are doing right now is very carefully looking at sequences to see if there's even any possibility much less likelihood that that's going on. And you could ultimately determine that. So people are looking at it,** but right now, the focus is on what are we going to do about what we have."*
>
> *source: 'White House asks scientists to investigate origins of coronavirus', ABC News, 6 Feb 2020, 5:35 PM EST.*

▶️ Poor reporting, since the article misrepresented the NASEM task, which was to *'convene a meeting of experts, particularly world class geneticists, coronavirus experts, and evolutionary biologists* ***to assess what data, information and samples are needed to address the unknowns, in order to understand the evolutionary origins of 2019-nCoV'.***

▶️ Clumsy quote, because it validated the confusion of the reporting and could also sound menacing.

A few days later, on 10 Feb, Garry shared the ABC article on Slack, with some support for Fauci's words, and a clear understanding of Fouchier's and Koopmans' opposition to any such declaration:

> *Garry, 10 Feb 2020, 8:14 pm UK:*
> *I think Fauci gave the correct answer regarding engineering or deliberate release. You need to look. It follows and makes sense that you also look at accidental release as a possibility (something BTW that happened with SARS-CoV-1 SEVERAL times,*
>
> *Call me conspiratorial (OK that horse left the barn), but I think there may be some hallway talk going on at Erasmus.*
> *source: Proximal_Origin_Slack.pdf, p.25*



The White House on Thursday asked U.S. scientists and medical researchers to investigate the scientific origins of the novel coronavirus, as misinformation about the outbreak spreads online.

The director of the White House's Office of Science and Technology Policy (OSTP), in a letter to the National Academies of Sciences, Engineering, and Medicine, requested that scientific experts "rapidly" look into the origins of the virus in order to address both the current spread and "to inform future outbreak preparation and better understand animal/human and environmental transmission aspects of coronaviruses."

The national academies consist of private, nonprofit institutions charged with providing scientific support to the federal government.

Proximal_Origin_Slack.pdf, p.25, 10 Feb 2020, 8:14 pm UK, and ABC article extract, 6 Feb 2020

b. When Fauci went off script…
Unfortunately, it was not just Fouchier and Koopmans that would likely get upset. There was no way that after reading Fauci's quote in the ABC article, the Chinese authorities would not get alarmed.

Fauci going off script was literally throwing a brick, while Farrar had been working quietly and more carefully:

- Wile Farrar was initially pursuing a similar initiative for MI5 and working with Holmes and Andersen, up to and including the call on 1 Feb (itself set up at the suggestion of Fauci), all declarations and statement were kept under strict wrap.
- Farrar took the pain to contact Chen Zhu on 2 Feb, as a back channel to the Chinese government, to at least try to prepare them to the idea of some level-handed investigation of the origin by the WHO — however overly optimistic that may have been (see 7.3).

But only a few days later, on Thursday 6 Feb, stories of an origin investigation commissioned by the White House that day, which were already quite a slanted interpretation of the OSTP request to NASEM of 3 Feb, popped up in ABC. And, to crown it all, Fauci was prominently quoted there as being confident that the US would be able to tell if the virus was a lab product.

It is quite a scene to see the OSTP trying to keep to a careful neutral Track II diplomacy objective, some of the experts on the NASEM panel trying all sorts of semantic artifices to reject a non-natural origin while not really meaning it, some slanted reporting of the objective of the OSTP-NASEM initiative (Track II was way too subtle a concept for reporting), and then Fauci firmly putting his foot in it by declaring that the US would be able to figure out any possible non-natural origin.

The toneless reporting of the NASEM answer for OSTP on 6 Feb (when it was still being finalised), totally unnecessarily spiced up by the declarations of Fauci, could only put China on the alarm after the already difficult mission of Chen Zhu. And worse was yet to come, with the letter of LeDuc to Yuan Zhiming on 9 Feb (see 8.6).

If China was trying to hide something, whatever that was, there would be strictly zero chance that it would agree to an investigation, and any WHO mission in China would be already well compromised, including the one that the WHO was trying to organize at the time.
There was no way out, as the WHO had lost its teeth after SARS-1, Trump was still very much focussed in his trade deals, and the US government and the US political situation were durably dysfunctional.

Key Insight: Daszak gets very annoyed by the NASEM letter and by Andersen

The final version of the NASEM letter would still represent **an immediate issue for Daszak**, since the final version of that official US scientific answer to the OSTP (White House) did not address the origin question.

An email that Daszak sent on 6 Feb, when trying to rally signatories to his parallel 'Statement of Support' (an effort he started the previous day), shows that he was pleased with the current draft of the NASEM letter. And indeed, Daszak had only minor comments about it, as we saw in 8.3.f.1. That draft had that '*consistent with evolution*' wording and raised serious doubts about engineering allegations. In fact, the only residual issue, that Daszak had with it, was that this did not reject all possible forms of research-related accident (such as a lab-escape of a natural virus):

From: Peter Daszak <daszak@ecohealthalliance.org>
Date: Thursday, February 6, 2020 at 10:25 PM
To: Linda Saif <saif.2@osu.edu>, Hume Field <hume.field@ecohealthalliance.org>
Cc: "JMHUGHE@emory.edu" <jmhughe@emory.edu>, Rita Colwell <[redacted]
"rcolwell@umiacs.umd.edu" <rcolwell@umiacs.umd.edu>, "rcolwell@umd.edu" <rcolwell@umd.edu>, Alison Andre <andre@ecohealthalliance.org>, Aleksei Chmura <chmura@ecohealthalliance.org>, Hongying Li <li@ecohealthalliance.org>, "William B. Karesh" <karesh@ecohealthalliance.org>, Robert Kessler <kessler@ecohealthalliance.org>
Subject: RE: A Statement in support of the scientists, public health and medical professionals of China

Thank you very much Linda, Jim, Billy and Hume for signing on to this rapidly. Rita called today but was in a meeting and we haven't had time to talk, but I'll make sure we connect tomorrow.

Hume – I've taken some of your comments and lightly edited, and I've attached the revised version here. Linda – you're right it would be good to be specific about the bioengineered virus conspiracy theory, but we I think we should probably stick to a broad statement. The Presidents of the US National Academies of Science, Engineering and Medicine have drafted a letter that I expect will be released Friday or Monday. I've not seen the final version yet, but the draft version that we (and expert group that met last week) edited has the following sentence: "The initial views of the experts is that the available genomic data are consistent with natural evolution and that there is currently no evidence that the virus was engineered to spread more quickly among humans." I think this is a bit too specific, because there are other conspiracy theories out there.

Our current statment neatly refutes most of them by saying that "We stand together to strongly condemn conspiracy theories suggesting that 2019-nCoV does not have a natural origin. Scientific evidence overwhelmingly suggests that this virus originated in wildlife, as have so many other emerging diseases". Let me know if you want to change specific wording using 'track changes' above.

Biohazard_FOIA_Maryland_Emails_11.6.20.pdf. p.278, 6 Feb 2020

Given that the draft wandered away from the carefully crafted Statement of Work, we may wonder whether Daszak did not have a hand in it in the first place. In any case, he certainly found it largely to his taste. That draft was clashing with its Track II objective of helping to transfer the origin question to the WHO. So predictably, the final version sent to OSTP was devoid of that extra natural origin wording.

But rather stupidly, **because of that removal of the pro natural-origin wording, Daszak would develop a huge rancour against Andersen,** holding him responsible for it, and for 'fuelling some of the conspiracy theories' that, according to him, reached the ears of the White House (more on this later in sections 26.4).

From: Peter Daszak <daszak@ecohealthalliance.org>
Date: Tuesday, February 18, 2020 at 12:27 AM
To: Linda Saif <saif.2@osu.edu>
Subject: RE: Revised commentary for EMI - final!

it: P.O. released on virological.org

Definitely – already cited it. It's in review for Nature. Unfortunately this is the exact same group that elevated the potential that this was a lab release all the way to the White House two weeks ago, and helped fuel some of the conspiracy theorists.

OSU-1.12.23–1265-pages.pdf, p. 319, 18 Feb 2020

8.5 Rumours of a parallel US intel investigation spread (8–9 Feb)

a. The Economist enquiry via Hamburg of NASEM:

On the next day (8 Feb), Peggy Hamburg of NASEM was forwarded an email that a friend of hers working at the Economist had received. To be clear, Hamburg, the secretary of NASEM, had been instrumental in the dual NASEM effort to get the WIV to share its isolates with LeDuc UTMB.

The email highlighted the possibility of a lab accident in Wuhan. She shared it with James LeDuc. In his reply on 9 Feb (12:39 am EST), LeDuc discounted the possibility of an accident in the WIV BSL-4, but was not so certain for their BSL-3, where he thought the work would have been much more likely.

From: LeDuc, James W. [/O=EXCHANGELABS/OU=EXCHANGE ADMINISTRATIVE GROUP (FYDIBOHF23SPDLT)/CN=RECIPIENTS/CN=937DF08E29C4439E88A04BABFFB162AD-JWLEDUC]
Sent: 2/9/2020 12:39:49 AM
To: Peggy Hamburg [peggy@hbfam.net]
Subject: Re: Crucial information re. 2019-nCoV

Hi Peggy

Yes I have seen these arguments and recognize that serious folks are questioning the coincidences noted. My initial thoughts are that the outbreak shares many characteristics with sars and that the explanations provided so far and the scientific observation are all consistent with that scenario. The coronavirologist at the center of the bat and other host work is world class and has participated in all of our NAS-CAS meetings and she is very well respected. I have a great deal of trust in her. The Chinese are working with current isolates in the new bsl4 lab (and a recently returned post doc from our lab is leading much of the work). I don't know if they were doing the bat coronavirus work in the bsl4 (I don't think they were) or in their bsl3 lab at the new facility or in their labs closer to the city. The bsl4 lab is a 45 min drive from the university campus in town. I'm working with our colleagues to get some clarity as to where work was done and a better idea if this might have been an accidental release. I do not suspect something nefarious but we can't rule out an accidental release quite yet.

There is a second bsl4 lab in Harbin where we held our last meeting that works on agricultural pathogens and a third bsl4 lab in Kunming that has yet to open.

I'm working hard to get more information and will keep you posted. Please don't share quite yet.

This whole episode demonstrates the immense value of the dialogue you helped establish. Good job!

Best wishes. Jim
Sent from my iPhone

On Feb 8, 2020, at 5:48 PM, Peggy Hamburg <peggy@hbfam.net> wrote:

WARNING: This email originated from outside of UTMB's email system. Do not click links or open attachments unless you recognize the sender and know the content is safe.

Hello. All well? Hope progress is being made in obtaining the isolates. Do keep us in the loop! I was amazed by how responsive Prof Bai was…do you think they were planning to make them available already?

I am curious if you know the gentleman below? The note was sent to a friend of mine who is the editor of a major magazine. Not sure why he sent…but of course these ideas are circulating. Frankly, I feel like we can't completely rule out though I am sure you are much closer to the actual viral information etc. the lab has had some Biosecurity issues in past I gather, though my sense is that people feel pretty food about both the quality of the science and the biosafety/Biosecurity there.

Any thoughts?
Peggy

Sent from my iPhone

From: [b)(6)]
Date: 8 February 2020 at 10:41:39 GMT-7
Subject: Crucial information re. 2019-nCoV
Reply-To: [b)(6)]

While fringe theories circulate about the origin of 2019-nCoV, **five established points** warrant further investigation:

1. The most rigorous study so far (published in The Lancet) shows the **first** known case was **not** exposed to the Huanan Market. (Layman summary of the study) That patient was diagnosed **Dec 1**. It's not until **Dec 10** that patients exposed to the market appear. Infectious disease physician Dr. Daniel Lucey believes the market was a locus of *transmission but not the source* and that "*China must have realized the epidemic* ***did not*** *originate in that Wuhan Huanan seafood market*". Microbiologist Dr. Lin Xiaoxu concurs, while senior research analyst at Freedom House, Sarah Cook, notes the CCP exerting its influence to control information on the outbreak, including its origin.

2. For weeks, China has refused to allow W.H.O. and C.D.C. experts into the country.

3. As this list shows, the **only** facility in China with a biosafety level 4 (i.e. designed for researching the most dangerous known pathogens) is located in Wuhan - the virus **epicentre**. This may just be a coincidence, but...

4. The fact that the Wuhan institute of Virology was researching **novel bat coronavirus** in the very place an apparently bat-borne coronavirus emerged, is a 'coincidence' that stretches credibility.

5. In **2017**, international scientists expressed concern about the lab in Wuhan, particularly China's lack of expertise/transparency in handling deadly pathogens.

Note:
Some 'debunking' articles, such as this WaPo piece, conflate spurious claims with warranted questions and deprecate a cogent explanation by associating it with irrelevant speculation.

Conspiracy theories about 'bioweapons' are red herrings: While it is true that in March 2019, Canada authorised the transfer of lethal viruses to China (prompting alarm from bio-warfare experts and Taiwan) and in July 2019, scientists from the Wuhan National Biosafety lab were expelled from a Canadian BSL-4 lab for "policy breach", these facts have **no bearing** on 2019-nCoV's origin since a virus doesn't have to be 'synthetic' to leak from a laboratory.

As molecular biologist Dr. Richard Ebright observes, data on 2019-nCoV are "*consistent with entry into the human population as either a natural accident* ***or*** *a laboratory accident.*"

While the market origination theory is all but scientifically debunked, Occam's razor suggests an interpretation with more sufficient explanatory power.

Bibliography:

Science Insider report:
https://www.sciencemag.org/news/2020/01/wuhan-seafood-market-may-not-be-source-novel-virus-spreading-globally

Wikipedia's list of the world's BSL-4 labs:
https://en.wikipedia.org/wiki/Biosafety_level#List_of_BSL-4_facilities

2018 Paper on novel bat coronavirus from Wuhan Institute of Virology:
https://jvi.asm.org/content/92/13/e00116-18

2018 article from The Scientist on another bat coronavirus study by the Wuhan facility:
https://www.the-scientist.com/the-nutshell/bats-identified-as-source-of-pig-killing-coronavirus-in-china-30035

2017 Nature article mentioning safety concerns re. the lab (disclaimer added Jan 2020):
https://www.nature.com/news/inside-the-chinese-lab-poised-to-study-world-s-most-dangerous-pathogens-1.21487

Vox's layman summary of the study by Chinese professors published in Lancet:
https://www.vox.com/2020/1/27/21082354/coronavirus-outbreak-wuhan-china-early-on-lancet

WaPo's 'debunking' piece:
https://www.washingtonpost.com/world/2020/01/29/experts-debunk-fringe-theory-linking-chinas-coronavirus-weapons-research/

7 Feb NYT article on China's refusal of W.H.O. and C.D.C assistance.

ED-20–01981-F-Dec-2022-Production-OPAQUE.pdf, p.138.-140. 9 Feb 2020

At precisely the same time, rumours of a real US Gov. investigation of the potential origins — which the NASEM initiative had never been — reached a journalist at the NY Times. On 9 Feb, that journalist, on the recommendation of Andrew C. Weber (a former Pentagon assistant secretary for nuclear,

chemical and biological defence programs under President Barack Obama), sent some questions to LeDuc about the potential involvement of US intel into the investigation.

In his reply, sent in the evening that day (9 Feb, 6:46 pm EST), LeDuc mentions that he had spoken with intel people and had told them that an accidental release was **highly unlikely**, despite having expressed a different opinion with Hamburg six hours earlier.

From: LeDuc, James W. [/O=EXCHANGELABS/OU=EXCHANGE ADMINISTRATIVE GROUP (FYDIBOHF23SPDLT)/CN=RECIPIENTS/CN=937DF08E29C4439E88A04BABFFB162AD-JWLEDUC]
Sent: 2/9/2020 6:45:06 PM
To: Broad, William [wjb@nytimes.com]
Subject: Re: Accidental release?

Bill

Thanks for the note. Suffice to say that I have spoken to folks in the intel community about the possibility that the epidemic is the result of a lab release and my position has been that I think it's highly unlikely that this is either nefarious or accidental release. These concerns continue however and I'm in the process of working with my colleagues in Wuhan to systematically assess all possible events that might suggest a lab association. I do not yet have a plan developed to share with them but hope to later today. In the meantime I continue to believe that this is a natural emergence of a new coronavirus give the similarities to SARS, the novel sequence of the virus, my knowledge and respect for Dr Shi, the world-class coronavirologist in Wuhan, and my knowledge of the labs and the safety procedures they have in place. Nonetheless I hope to convince them to conduct a very thorough assessment and to make the finding public ASAP. Happy to chat if you like. My cell is (b)(6) Probably best not to have my name associated with the story quite yet.

Thanks. Jim

Sent from my iPhone

On Feb 9, 2020, at 11:02 AM, Broad, William <wjb@nytimes.com> wrote:

WARNING: This email originated from outside of UTMB's email system. Do not click links or open attachments unless you recognize the sender and know the content is safe.

Jim: Andy Weber suggested you would be good to talk to. We keep hearing that the USG is investigating whether the China virus was an accidental release, possibly as early as November, from the Wuhan Institute of Virology, and are trying to figure out if that's true. We know something of the unclassified USG focus from an exchange of letters between OSTP and NAS (attached). But we're in the dark on whether USG intel side is doing a deeper investigation in parallel. We'd be most grateful for any guidance and would of course be happy to leave your name out of it. Bill

William J. Broad

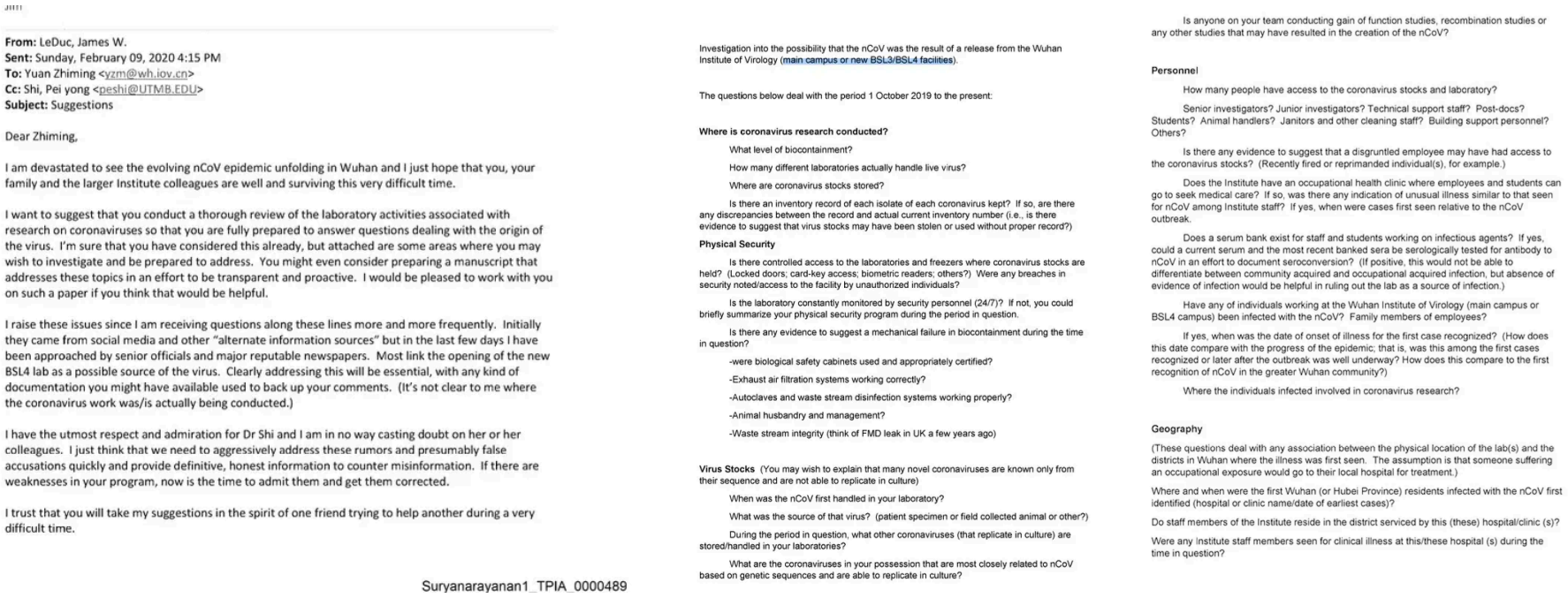
ED-20–01981-F-Dec-2022-Production-OPAQUE.pdf, p.142, 9 Feb 2020

8.6 LeDuc sends some pointed questions to Yuan Zhiming of the WIV (9 Feb)

Between these two replies he sent back on 9 Feb, to Hamburg and the NY Times journalist, LeDuc sent a fairly long email to Yuan Zhiming, director of the WIV BSL-4 lab, where he asked him some very pointed questions about his laboratory biosafety record and encouraged him to fess up to any shortcoming.

> *I just think that we need to aggressively address these rumours and **presumably** false accusations quickly and provide definitive, honest information to counter misinformation. **If there are weaknesses in your program, now is the time to admit it and get them corrected.***

His email, sent at 4:15 pm EST, was again in stark contrast with his advice to intel that he would the day before, where he wrote that he thought that an "**accidental release was highly unlikely**". Here it was just '**presumably**' false.



From: LeDuc, James W.
Sent: Sunday, February 09, 2020 4:15 PM
To: Yuan Zhiming <yzm@wh.iov.cn>
Cc: Shi, Pei yong <peshi@UTMB.EDU>
Subject: Suggestions

Dear Zhiming,

I am devastated to see the evolving nCoV epidemic unfolding in Wuhan and I just hope that you, your family and the larger Institute colleagues are well and surviving this very difficult time.

I want to suggest that you conduct a thorough review of the laboratory activities associated with research on coronaviruses so that you are fully prepared to answer questions dealing with the origin of the virus. I'm sure that you have considered this already, but attached are some areas where you may wish to investigate and be prepared to address. You might even consider preparing a manuscript that addresses these topics in an effort to be transparent and proactive. I would be pleased to work with you on such a paper if you think that would be helpful.

I raise these issues since I am receiving questions along these lines more and more frequently. Initially they came from social media and other "alternate information sources" but in the last few days I have been approached by senior officials and major reputable newspapers. Most link the opening of the new BSL4 lab as a possible source of the virus. Clearly addressing this will be essential, with any kind of documentation you might have available used to back up your comments. (It's not clear to me where the coronavirus work was/is actually being conducted.)

I have the utmost respect and admiration for Dr Shi and I am in no way casting doubt on her or her colleagues. I just think that we need to aggressively address these rumors and presumably false accusations quickly and provide definitive, honest information to counter misinformation. If there are weaknesses in your program, now is the time to admit them and get them corrected.

I trust that you will take my suggestions in the spirit of one friend trying to help another during a very difficult time.

Investigation into the possibility that the nCoV was the result of a release from the Wuhan Institute of Virology (main campus or new BSL3/BSL4 facilities).

The questions below deal with the period 1 October 2019 to the present:

Where is coronavirus research conducted?

What level of biocontainment?

How many different laboratories actually handle live virus?

Where are coronavirus stocks stored?

Is there an inventory record of each isolate of each coronavirus kept? If so, are there any discrepancies between the record and actual current inventory number (i.e., is there evidence to suggest that virus stocks may have been stolen or used without proper record?)

Physical Security

Is there controlled access to the laboratories and freezers where coronavirus stocks are held? (Locked doors; card-key access; biometric readers; others?) Were any breaches in security noted/access to the facility by unauthorized individuals?

Is the laboratory constantly monitored by security personnel (24/7)? If not, you could briefly summarize your physical security program during the period in question.

Is there any evidence to suggest a mechanical failure in biocontainment during the time in question?

-were biological safety cabinets used and appropriately certified?

-Exhaust air filtration systems working correctly?

-Autoclaves and waste stream disinfection systems working properly?

-Animal husbandry and management?

-Waste stream integrity (think of FMD leak in UK a few years ago)

Virus Stocks (You may wish to explain that many novel coronaviruses are known only from their sequence and are not able to replicate in culture)

When was the nCoV first handled in your laboratory?

What was the source of that virus? (patient specimen or field collected animal or other?)

During the period in question, what other coronaviruses (that replicate in culture) are stored/handled in your laboratories?

What are the coronaviruses in your possession that are most closely related to nCoV based on genetic sequences and are able to replicate in culture?

Is anyone on your team conducting gain of function studies, recombination studies or any other studies that may have resulted in the creation of the nCoV?

Personnel

How many people have access to the coronavirus stocks and laboratory?

Senior investigators? Junior investigators? Technical support staff? Post-docs? Students? Animal handlers? Janitors and other cleaning staff? Building support personnel? Others?

Is there any evidence to suggest that a disgruntled employee may have had access to the coronavirus stocks? (Recently fired or reprimanded individual(s), for example.)

Does the Institute have an occupational health clinic where employees and students can go to seek medical care? If so, was there any indication of unusual illness similar to that seen for nCoV among Institute staff? If yes, when were cases first seen relative to the nCoV outbreak.

Does a serum bank exist for staff and students working on infectious agents? If yes, could a current serum and the most recent banked sera be serologically tested for antibody to nCoV in an effort to document seroconversion? (If positive, this would not be able to differentiate between community acquired and occupational acquired infection, but absence of evidence of infection would be helpful in ruling out the lab as a source of infection.)

Have any of individuals working at the Wuhan Institute of Virology (main campus or BSL4 campus) been infected with the nCoV? Family members of employees?

If yes, when was the date of onset of illness for the first case recognized? (How does this date compare with the progress of the epidemic; that is, was this among the first cases recognized or later after the outbreak was well underway? How does this compare to the first recognition of nCoV in the greater Wuhan community?)

Where the individuals infected involved in coronavirus research?

Geography

(These questions deal with any association between the physical location of the lab(s) and the districts in Wuhan where the illness was first seen. The assumption is that someone suffering an occupational exposure would go to their local hospital for treatment.)

Where and when were the first Wuhan (or Hubei Province) residents infected with the nCoV first identified (hospital or clinic name/date of earliest cases)?

Do staff members of the Institute reside in the district serviced by this (these) hospital/clinic (s)?

Were any Institute staff members seen for clinical illness at this/these hospital (s) during the time in question?

Suryanarayanan1_TPIA_0000489

UTMB-LeDuc-batch-1.pdf, p.806–809, 9 Feb 2020

8.7 Track II hits a wall with the isolates request (10 Feb)

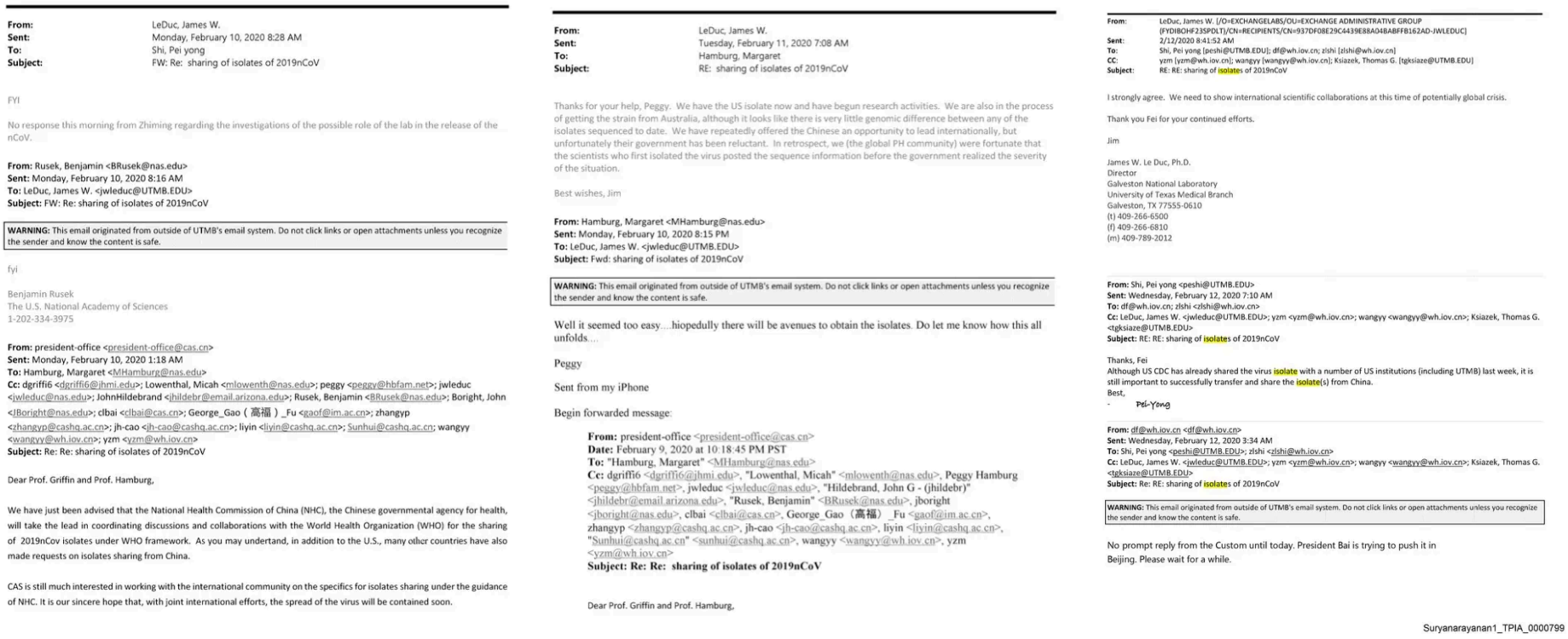
On 10 Feb, Chunli Bai, the President of the Chinese Academy of Sciences (CAS), emailed Hamburg and the other NASEM members who had worked in trying to get isolates from China. His email confirmed the dark clouds already detected in the answer of 29 Jan from Fei Deng to Pei Yong (see 8.2.c): the National Health Commission — a political body, directly under the State Council — had just informed him that it would take the lead on the question of sharing isolates.

What it meant is that all negotiations were now redirected to official government channels (Track I). The double-pronged Track II efforts, involving NASEM, the Chinese CDC and CAS, had just failed. So much for China's '*new openness to health information sharing*' that LeDuc had recently enthused about. This was a serious disappointment, since both CAS and the Chinese CDC had shown themselves willing to start sharing isolates with LeDuc's UTMB, after the first NASEM request they had received. On 3 Feb, Chunli Bai himself had even sent a rather positive response to Griffin (NAS) and Hamburg (NAM) (see 8.2.d).

The 10 Feb was the very day after LeDuc sent his letter to Yuan Zhiming, enquiring about biosafety at the WIV, four days after Fauci was quoted by ABC saying that the US would have the means to find out if the virus was a lab product, and height days after Fauci had used Chen Zhu as back-channel to try to make the Chinese government receptive to a level-handed origin investigation under the aegis of the WHO. It could not have gone worse.

CAS further explained that it was trying to intervene with the Chinese central government to clear export. Still, by late February, as the outbreak turned into a pandemic, isolates were available from many countries, but not

from China. Nevertheless, getting isolates from early Wuhan patients when NASEM was asking for them, may have allowed to better understand the virus likely future evolutionary paths at the time.



From: LeDuc, James W.
Sent: Monday, February 10, 2020 8:28 AM
To: Shi, Pei yong
Subject: FW: Re: sharing of isolates of 2019nCoV

FYI

No response this morning from Zhiming regarding the investigations of the possible role of the lab in the release of the nCoV.

From: Rusek, Benjamin <BRusek@nas.edu>
Sent: Monday, February 10, 2020 8:16 AM
To: LeDuc, James W. <jwleduc@UTMB.EDU>
Subject: FW: Re: sharing of isolates of 2019nCoV

WARNING: This email originated from outside of UTMB's email system. Do not click links or open attachments unless you recognize the sender and know the content is safe.

fyi

Benjamin Rusek
The U.S. National Academy of Sciences
1-202-334-3975

From: president-office <president-office@cas.cn>
Sent: Monday, February 10, 2020 1:18 AM
To: Hamburg, Margaret <MHamburg@nas.edu>
Cc: dgriffi6 <dgriffi6@jhmi.edu>; Lowenthal, Micah <mlowenth@nas.edu>; peggy <peggy@hbfam.net>; jwleduc <jwleduc@nas.edu>; JohnHildebrand <jhildebr@email.arizona.edu>; Rusek, Benjamin <BRusek@nas.edu>; Boright, John <JBoright@nas.edu>; clbai <clbai@cas.cn>; George_Gao（高福）_Fu <gaof@im.ac.cn>; zhangyp <zhangyp@cashq.ac.cn>; jh-cao <jh-cao@cashq.ac.cn>; liyin <liyin@cashq.ac.cn>; Sunhui@cashq.ac.cn; wangyy <wangyy@wh.iov.cn>; yzm <yzm@wh.iov.cn>
Subject: Re: Re: sharing of isolates of 2019nCoV

Dear Prof. Griffin and Prof. Hamburg,

We have just been advised that the National Health Commission of China (NHC), the Chinese governmental agency for health, will take the lead in coordinating discussions and collaborations with the World Health Organization (WHO) for the sharing of 2019nCov isolates under WHO framework. As you may undertand, in addition to the U.S., many other countries have also made requests on isolates sharing from China.

CAS is still much interested in working with the international community on the specifics for isolates sharing under the guidance of NHC. It is our sincere hope that, with joint international efforts, the spread of the virus will be contained soon.

From: LeDuc, James W.
Sent: Tuesday, February 11, 2020 7:08 AM
To: Hamburg, Margaret
Subject: RE: sharing of isolates of 2019nCoV

Thanks for your help, Peggy. We have the US isolate now and have begun research activities. We are also in the process of getting the strain from Australia, although it looks like there is very little genomic difference between any of the isolates sequenced to date. We have repeatedly offered the Chinese an opportunity to lead internationally, but unfortunately their government has been reluctant. In retrospect, we (the global PH community) were fortunate that the scientists who first isolated the virus posted the sequence information before the government realized the severity of the situation.

Best wishes, Jim

From: Hamburg, Margaret <MHamburg@nas.edu>
Sent: Monday, February 10, 2020 8:15 PM
To: LeDuc, James W. <jwleduc@UTMB.EDU>
Subject: Fwd: sharing of isolates of 2019nCoV

WARNING: This email originated from outside of UTMB's email system. Do not click links or open attachments unless you recognize the sender and know the content is safe.

Well it seemed too easy....hiopedully there will be avenues to obtain the isolates. Do let me know how this all unfolds....

Peggy

Sent from my iPhone

Begin forwarded message:

From: president-office <president-office@cas.cn>
Date: February 9, 2020 at 10:18:45 PM PST
To: "Hamburg, Margaret" <MHamburg@nas.edu>
Cc: dgriffi6 <dgriffi6@jhmi.edu>, "Lowenthal, Micah" <mlowenth@nas.edu>, Peggy Hamburg <peggy@hbfam.net>, jwleduc <jwleduc@nas.edu>, "Hildebrand, John G - (jhildebr)" <jhildebr@email.arizona.edu>, "Rusek, Benjamin" <BRusek@nas.edu>, jboright <jboright@nas.edu>, clbai <clbai@cas.cn>, George_Gao（高福）_Fu <gaof@im.ac.cn>, zhangyp <zhangyp@cashq.ac.cn>, jh-cao <jh-cao@cashq.ac.cn>, liyin <liyin@cashq.ac.cn>, "Sunhui@cashq.ac.cn" <sunhui@cashq.ac.cn>, wangyy <wangyy@wh.iov.cn>, yzm <yzm@wh.iov.cn>
Subject: Re: Re: sharing of isolates of 2019nCoV

Dear Prof. Griffin and Prof. Hamburg,

From: LeDuc, James W. [/O=EXCHANGELABS/OU=EXCHANGE ADMINISTRATIVE GROUP (FYDIBOHF23SPDLT)/CN=RECIPIENTS/CN=937DF08E29C4439E88A04BABFFB162AD-JWLEDUC]
Sent: 2/12/2020 8:41:52 AM
To: Shi, Pei yong [peshi@UTMB.EDU]; df@wh.iov.cn; zlshi [zlshi@wh.iov.cn]
CC: yzm [yzm@wh.iov.cn]; wangyy [wangyy@wh.iov.cn]; Ksiazek, Thomas G. [tgksiaze@UTMB.EDU]
Subject: RE: RE: sharing of isolates of 2019nCoV

I strongly agree. We need to show international scientific collaborations at this time of potentially global crisis.

Thank you Fei for your continued efforts.

Jim

James W. Le Duc, Ph.D.
Director
Galveston National Laboratory
University of Texas Medical Branch
Galveston, TX 77555-0610
(t) 409-266-6500
(f) 409-266-6810
(m) 409-789-2012

From: Shi, Pei yong <peshi@UTMB.EDU>
Sent: Wednesday, February 12, 2020 7:10 AM
To: df@wh.iov.cn; zlshi <zlshi@wh.iov.cn>
Cc: LeDuc, James W. <jwleduc@UTMB.EDU>; yzm <yzm@wh.iov.cn>; wangyy <wangyy@wh.iov.cn>; Ksiazek, Thomas G. <tgksiaze@UTMB.EDU>
Subject: RE: RE: sharing of isolates of 2019nCoV

Thanks, Fei
Although US CDC has already shared the virus isolate with a number of US institutions (including UTMB) last week, it is still important to successfully transfer and share the isolate(s) from China.
Best,
- Pei-Yong

From: df@wh.iov.cn <df@wh.iov.cn>
Sent: Wednesday, February 12, 2020 3:34 AM
To: Shi, Pei yong <peshi@UTMB.EDU>; zlshi <zlshi@wh.iov.cn>
Cc: LeDuc, James W. <jwleduc@UTMB.EDU>; yzm <yzm@wh.iov.cn>; wangyy <wangyy@wh.iov.cn>; Ksiazek, Thomas G. <tgksiaze@UTMB.EDU>
Subject: Re: RE: sharing of isolates of 2019nCoV

WARNING: This email originated from outside of UTMB's email system. Do not click links or open attachments unless you recognize the sender and know the content is safe.

No prompt reply from the Custom until today. President Bai is trying to push it in Beijing. Please wait for a while.

Suryanarayanan1_TPIA_0000799

ED-20–01981-F-Dec-2022-Production-OPAQUE.pdf, p.61 and 209, 10–11 Feb 2020, and UTMB-LeDuc-batch-1.pdf, p.799, 12 Feb

Sideline: Track II for the rest of us

Track II diplomacy sounds nice in principle, but here, in a very public arena, the OSTP/NASEM *pas-de-deux* ended up in a complete fiasco, with both the US public and the Chinese government getting the opposite message to what was intended.

- The drafters of the NASEM answer to OSTP went off course, by trying to give it superficial substance as an origin-statement, the kind that would normally be expected from NASEM. It all had to be expunged.
- The US public, starting with the ABC reporter, had predictable problems understanding that the purpose of the exercise was not an investigation of the origins per se, and that its apparent conclusions could be held in two short generic sentences.
- The Chinese government likely totally ignored the true message about the need to share the Wuhan isolates, when Fauci made his clumsy comment to the ABC reporter.

8.8 All roads now lead to the WHO (11–13 Feb)

In his email of 10 Feb, Chunli Bai, the president of CAS, also made the point that the sharing of isolates would have to happen via the WHO, under a WHO

framework. In a rather humiliating sentence, Chunli Bai added that many other countries, not just the US, had also asked for isolates.

> *Chunli Bai to Hamburg:*
> *The National Health Commission of China (NHC), [..] will take the lead in coordinating discussions and collaborations* ***with the World Health Organization (WHO) for the sharing of 2019nCov isolates under WHO framework. As you may understand, in addition to the U.S., many other countries have also made requests on isolates sharing from China.***

LeDuc and Hamburg had a few exchanges about this disappointing development. As to the WHO avenue, LeDuc replied privately to Hamburg that he was not exactly impressed by the ability of the WHO to obtain such isolates (11 Feb):

> *I have been engaged with the WHO discussions on sharing virus isolates and frankly have not been impressed. Maybe I was part of the wrong group.*

On 12 Feb, Hamburg put on a brave face and sent a sweet-sour message back to Chunli Bai:

> *[..] We are disappointed of course, but understand the value of sharing within the context of the WHO framework. [..]*

Exactly at the same time, the US side was also already getting very worried about not being represented on the coming WHO mission to China; after a week of negotiations led by the WHO, the prospects were not particularly encouraging (see 14.7).

The implications of the Chinese response to these two concerns should be clear: all roads lead to the WHO, and possibly to nowhere.

From: LeDuc, James W.
Sent: Tuesday, February 11, 2020 11:20 AM
To: Hamburg, Margaret
Subject: RE: sharing of isolates of 2019nCoV

I was not part of the NASEM meeting on viral origin—too bad but I suspect it was quickly called together and drew on experts inside/near the beltway.

Great idea about sharing isolates as a routine step in outbreak response. This will be a difficult sell but worth the effort. I have been engaged with the WHO discussions on sharing virus isolates and frankly have not been impressed. Maybe I was part of the wrong group.

Thanks, Jim

From: Hamburg, Margaret <MHamburg@nas.edu>
Sent: Tuesday, February 11, 2020 11:12 AM
To: LeDuc, James W. <jwleduc@UTMB.EDU>
Subject: Re: sharing of isolates of 2019nCoV

WARNING: This email originated from outside of UTMB's email system. Do not click links or open attachments unless you recognize the sender and know the content is safe.

Jim,
So glad that you are in the midst of all of this. I had hoped that we had made more progress in terms of more collaboration and transparency with our Chinese colleagues, but certainly some progress....

Where you part of the recent NASEM meeting on viral origin requested by OSTP. I think I understood that you were...you certainly should have been. I was wondering if you found it useful.

Wondering if in this modern era we should try to develop a framework for actually making it"routine" in a serious outbreak context that samples get shared and an expert scientific group to examine questions of origin and evolution.

Would welcome your thoughts.

Stay in touch.
Peggy

Sent from my iPhone

On Feb 11, 2020, at 5:08 AM, LeDuc, James W. <jwleduc@utmb.edu> wrote:

Thanks for your help, Peggy. We have the US isolate now and have begun research activities. We are also in the process of getting the strain from Australia, although it looks like there is very little genomic

1

From: Rusek, Benjamin <BRusek@nas.edu>
Sent: Thursday, February 13, 2020 2:35 PM
To: LeDuc, James W.
Cc: Dave Franz (davidrfranz@gmail.com)
Subject: FW: sharing of isolates of 2019nCoV

WARNING: This email originated from outside of UTMB's email system. Do not click links or open attachments unless you recognize the sender and know the content is safe.

FYI
Benjamin Rusek
The U.S. National Academy of Sciences
1-202-334-3975

From: Peggy Hamburg <peggy@hbfam.net>
Sent: Wednesday, February 12, 2020 9:09 PM
To: president-office <president-office@cas.cn>
Cc: Hamburg, Margaret <MHamburg@nas.edu>; dgriffi6 <dgriffi6@jhmi.edu>; Lowenthal, Micah <mlowenth@nas.edu>; jwleduc <jwleduc@nas.edu>; JohnHildebrand <jhildebr@email.arizona.edu>; Rusek, Benjamin <BRusek@nas.edu>; Boright, John <JBoright@nas.edu>; clbai <clbai@cas.cn>; George_Gao（高福）_Fu <gaof@im.ac.cn>; zhangyp <zhangyp@cashq.ac.cn>; jh-cao <jh-cao@cashq.ac.cn>; liyin <liyin@cashq.ac.cn>; Sunhui@cashq.ac.cn; wangyy <wangyy@wh.iov.cn>; yzm <yzm@wh.iov.cn>
Subject: Re: sharing of isolates of 2019nCoV

Dear President Bai,

Thank you very much for following up with this additional information. We are disappointed, of course, but understand the value of sharing within the context of the WHO framework. Such viral isolate sharing is a crucial aspect of the international collaboration needed to combat this devastating outbreak, as well as to address future threats.

We look forward to working with you on efforts to contain this virus and continue to offer our assistance as needed.

With best regards,

Diane E. Griffin, Vice President, NAS
Margaret Hamburg, Foreign Secretary, NAM

Sent from my iPhone

On Feb 9, 2020, at 10:18 PM, president-office <president-office@cas.cn> wrote:

Dear Prof. Griffin and Prof. Hamburg,

1

ED-20–01981-F-Dec-2022-Production-OPAQUE.pdf, p.223. 11 Feb 2020 and ED-20–01981-March-2022-Production_Redacted_Upload.pdf, p.362, 13 Feb 2020

Insight: NASEM real position on the origin questions

For a glimpse of the internal position of NASEM on the origin question, we may look at an email sent by Lowenthal in May 2021, to Hamburg, LeDuc, Franz, Rusek and Griffin. That email was sent in reply to LeDuc, who was discussing the work done at the time by Congress and the Lancet Covid Commission (which had just contacted him) [footnote 18]:

> *As for something public, NASEM would likely stay away from this, but if they did weigh in then my guess is that they would probably agree with Baric and Relman who now say publicly that "more investigation is still needed to determine the origin of the pandemic. Theories of accidental release from a lab and zoonotic spillover both remain viable."*

From: Lowenthal, Micah <mlowenth@nas.edu>
Sent: Tuesday, May 18, 2021 11:46 PM
To: LeDuc, James W. <jwleduc@UTMB.EDU>; Peggy Hamburg <peggy@hbfam.net>
Cc: Dave Franz (davidrfranz@gmail.com) <davidrfranz@gmail.com>; Rusek, Benjamin <BRusek@nas.edu>; Diane Griffin (dgriffi6@jhmi.edu) <dgriffi6@jhmi.edu>; Raymond Jeanloz <jeanloz@berkeley.edu>
Subject: RE: COVID-19 origins

Hi Jim,

Can you clarify what goal you have in mind? Is it to have NASEM provide what we learned from the meetings to teams that are investigating the origins? Doing that would be out of the ordinary, but we could talk about whether there are conditions under which we could do it.

As for something public, NASEM would likely stay away from this, but if they did weigh in then my guess is that they would probably agree with Baric and Relman who now say publicly that "more investigation is still needed to determine the origin of the pandemic. Theories of accidental release from a lab and zoonotic spillover both remain viable."

We generally do not publicize or publish out of the dialogues. We have, in some cases, conducted projects that are joint workshops or studies intended and agreed from the beginning to have a public product.

We have provided and can provide again summaries of all our discussions to people within the U.S. government. As Peggy noted, the people and institutions involved did not agree to make the contents of the discussions public as an outcome when we developed agendas and held the discussions.

As for what we have made public…we were concerned last year that the lack of any public description of the China bio dialogue was creating problems, suggesting that we were not forthright about our work in the context of a witch hunt and leaving UTMB and you in particular in an awkward position, having nothing on our website to which you could refer reporters. We developed a description of the engagements, we posted that description on the web here https://www.nationalacademies.org/our-work/us-china-bilateral-dialogue-on-the-challenges-of-emerging-infections-laboratory-safety-and-global-health-security, and we gave short briefings to OSTP and the S&T Advisor to the Secretary of State, so that they would not be surprised.

I think that it is important to revisit our practices and policies when a departure can provide a valuable public service and wouldn't compromise our other goals and values. Let's think through our goals, the value that achieving them would provide, how we can achieve those goals, and whether the actions needed are compatible with our principles.

Thanks.

Micah

UTMB-LeDuc-batch-1.pdf, p.2825, 18 May 2021

9. Writing a confidential report while waiting for the WHO

So far, we have reviewed the context for the Farrar call, the various constraints and objectives of its key players (including British intel, the US administration, NASEM and the WHO), the decisions and discussions that immediately followed that call, without forgetting the ongoing need for access and data that was still being sought via appeasement and quiet Track II diplomacy at the time.

In this section, we shall focus on the redaction process of the report, which was initially not meant to be public, but would end up being released online and then published, after some modifications, as 'Proximal Origin'. To do so, we first need to go back to the immediate aftermath of the Farrar call.

9.1 Andersen starts writing a report (1 Feb)

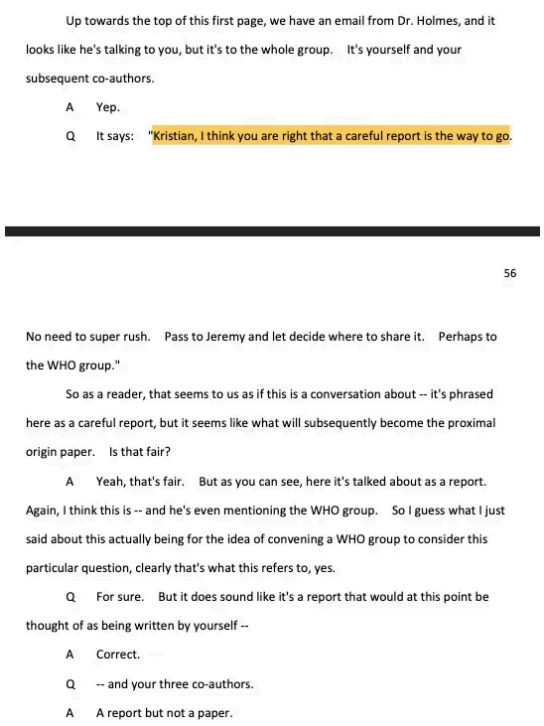
Immediately after the call, Andersen started writing down a more detailed analysis, in the form of a report. That report was not something as detailed as a paper and not yet meant for publication, the idea being to put together a confidential summary of virus features that were worth investigating, without pre-empting a conclusion on the possible origins of the virus, based on its sequence. Then that report/summary could be handed over to the WHO, once the WHO had agreed to it and set up a group to onboard the origin question.

A first version of the report (v1) was available on 1 Feb at 7:40 pm EST. That 1-page version produced by Andersen stated that two elements in the sequence were potentially problematic:

- The BamHI restriction sites at the end of the spike protein; a potential marker of genetic engineering.
- The Furin Cleavage Site with its O-linked glycans; the more problematic of the two features.

Targeted gene editing origin was discarded, based on the weak assumption that nobody would have been *a priori* aware of such an unusual target (however effective that target may be *a posteriori*). This left 'tissue culture passage' as the dominant non-natural hypothesis. [footnote 26]

Late in the evening, Andersen circulated his arguments on the Slack channel for Holmes and Rambaut to review. From that point, the Slack channel became the main venue for ad hoc discussions and to circulate draft versions between the initial kernel of three co-authors.



Up towards the top of this first page, we have an email from Dr. Holmes, and it looks like he's talking to you, but it's to the whole group. It's yourself and your subsequent co-authors.

A Yep.

Q It says: "Kristian, I think you are right that a careful report is the way to go.

56

No need to super rush. Pass to Jeremy and let decide where to share it. Perhaps to the WHO group."

So as a reader, that seems to us as if this is a conversation about -- it's phrased here as a careful report, but it seems like what will subsequently become the proximal origin paper. Is that fair?

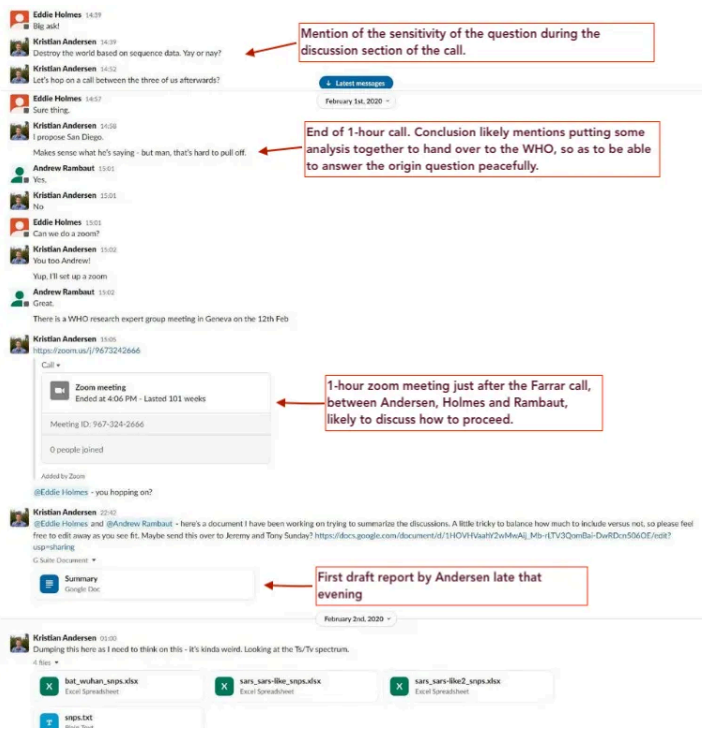
A Yeah, that's fair. But as you can see, here it's talked about as a report. Again, I think this is -- and he's even mentioning the WHO group. So I guess what I just said about this actually being for the idea of convening a WHO group to consider this particular question, clearly that's what this refers to, yes.

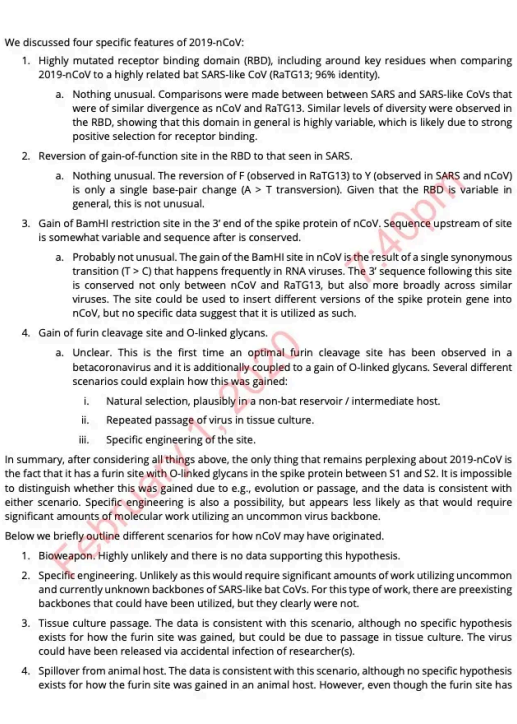
Q For sure. But it does sound like it's a report that would at this point be thought of as being written by yourself --

A Correct.

Q -- and your three co-authors.

A A report but not a paper.





We discussed four specific features of 2019-nCoV:

1. Highly mutated receptor binding domain (RBD), including around key residues when comparing 2019-nCoV to a highly related bat SARS-like CoV (RaTG13; 96% identity).
   a. Nothing unusual. Comparisons were made between between SARS and SARS-like CoVs that were of similar divergence as nCoV and RaTG13. Similar levels of diversity were observed in the RBD, showing that this domain in general is highly variable, which is likely due to strong positive selection for receptor binding.
2. Reversion of gain-of-function site in the RBD to that seen in SARS.
   a. Nothing unusual. The reversion of F (observed in RaTG13) to Y (observed in SARS and nCoV) is only a single base-pair change (A > T transversion). Given that the RBD is variable in general, this is not unusual.
3. Gain of BamHI restriction site in the 3' end of the spike protein of nCoV. Sequence upstream of site is somewhat variable and sequence after is conserved.
   a. Probably not unusual. The gain of the BamHI site in nCoV is the result of a single synonymous transition (T > C) that happens frequently in RNA viruses. The 3' sequence following this site is conserved not only between nCoV and RaTG13, but also more broadly across similar viruses. The site could be used to insert different versions of the spike protein gene into nCoV, but no specific data suggest that it is utilized as such.
4. Gain of furin cleavage site and O-linked glycans.
   a. Unclear. This is the first time an optimal furin cleavage site has been observed in a betacoronavirus and it is additionally coupled to a gain of O-linked glycans. Several different scenarios could explain how this was gained:
      i. Natural selection, plausibly in a non-bat reservoir / intermediate host.
      ii. Repeated passage of virus in tissue culture.
      iii. Specific engineering of the site.

In summary, after considering all things above, the only thing that remains perplexing about 2019-nCoV is the fact that it has a furin site with O-linked glycans in the spike protein between S1 and S2. It is impossible to distinguish whether this was gained due to e.g., evolution or passage, and the data is consistent with either scenario. Specific engineering is also a possibility, but appears less likely as that would require significant amounts of molecular work utilizing an uncommon virus backbone.

Below we briefly outline different scenarios for how nCoV may have originated.

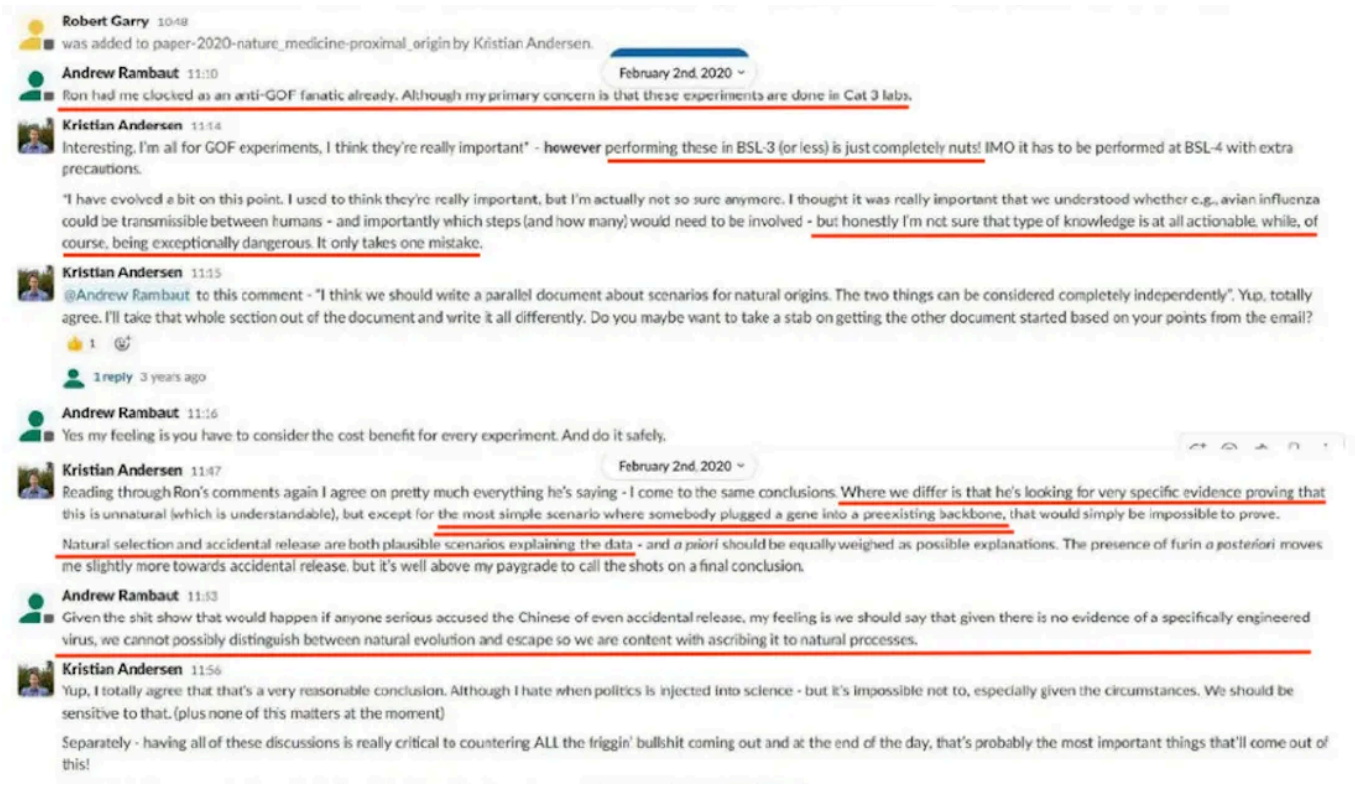
1. Bioweapon. Highly unlikely and there is no data supporting this hypothesis.
2. Specific engineering. Unlikely as this would require significant amounts of work utilizing uncommon and currently unknown backbones of SARS-like bat CoVs. For this type of work, there are preexisting backbones that could have been utilized, but they clearly were not.
3. Tissue culture passage. The data is consistent with this scenario, although no specific hypothesis exists for how the furin site was gained, but could be due to passage in tissue culture. The virus could have been released via accidental infection of researcher(s).
4. Spillover from animal host. The data is consistent with this scenario, although no specific hypothesis exists for how the furin site was gained in an animal host. However, even though the furin site has

2023.06.16-Andersen-Transcript.pdf, p.55–56, Proximal_Origin_Slack.pdf, p.4, 1 Feb 2020, and DrSSCP-Drafts-of-Proximal-Origin.pdf, p.1, 1 Feb 2020 7:40pm EST

9.2 Garry is added — conversations run free (2 Feb)

Garry (who first showed up in the Farrar call) was added at that time (2 Feb) to the PO drafting Slack, and the next day to the Google draft of the report.

This was a fertile time, coming just after the call, when the co-authors did hold some of the most (unredacted) insightful conversations, without being constrained by the wrath of Fouchier and Drosten, about the risk of GoF, the absurd burden of proof that Fouchier would like to impose, the invalidity of the arguments that a well-known backbone would have been used, the geopolitical situation and the safe angle that they may need to give to their report (see 5.6.b).



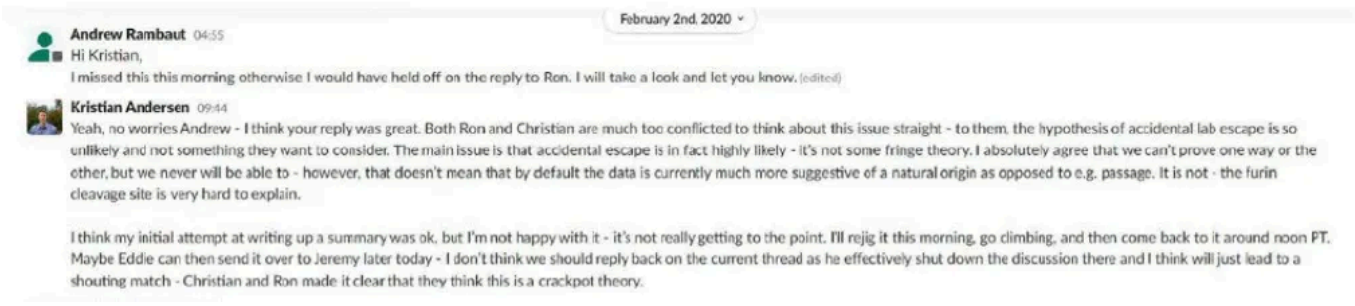
Proximal_Origin_Slack.pdf, p.5, 2 Feb 2020

9.3 Some are excluded (2 Feb)

The 'bounce-off' group (Fouchier, Drosten, Koopmans) was notably excluded from not only the Slack, but also the drafting process on Google doc. After reading Fouchier's reply to Farrar's summary (see 5.6.b), Andersen explained that it considered futile to send the notes he just shared on Slack (the very first draft of PO) to the entire group, **as Fouchier and Drosten would likely just get into a shouting match with them**.

In particular, Andersen noted that:

> *The main issue is that accidental escape is in fact highly likely — it's not some fringe theory. I absolutely agree that we can't prove one way or the other, but we never will be able to — however, that does not mean that by default the data is currently more suggestive of a natural origin as opposed to e.g. passage. It is not — the furin cleavage site is very hard to explain.*



Proximal_Origin_Slack.pdf, p.5, 2 Feb 2020

[**Note:** from that point on, as the exact times become critical for the understanding of the unfolding events, we will give the time of most highlighted messages, preferably in UK time. Times were reliably resolved as per footnote ❖]

10. Drafts are shared with Fauci and Collins (4 Feb)

10.1 Fist sent: Rough draft (4 Feb, morning UK)

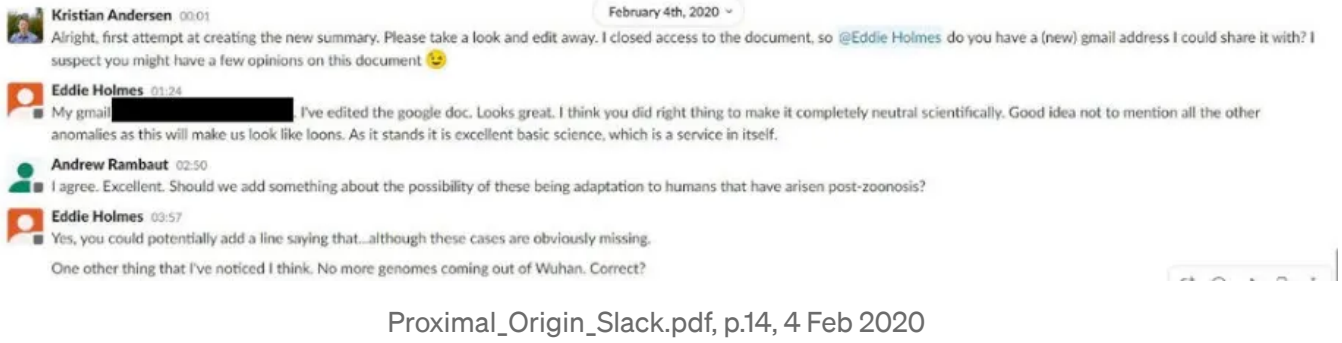
a. A rough first draft is prepared by Andersen and Holmes (~5 am UK)
On Tuesday 4 Feb, at 4:59 am UK (3 Feb, 9:39 pm in California), Andersen had just finished editing a revised summary report [see Appendix A for detailed versions].

Two minutes later, at 5:01 am UK, Andersen announced on Slack that he had a first draft of the new summary report ready for review. At 6:24 am UK (5:24 pm Sydney), Holmes wrote on Slack that he had done his edits.

b. The rough draft is shared with Farrar, then Fauci and Collins (~7 am UK)
Holmes then converted the Google doc to a Windows file ('Summary.docx'), and sent it to Farrar at 6:33 am UK, without letting his co-authors know (see version v4 of Appendix A).

Note that by that time it would be night for Garry (1:33 am NY), so **Garry's edits would anyway have had to wait.** Most likely, Holmes still wanted to share the draft with Farrar, but did not want to upset Garry by revealing that he had shared it without waiting.

Soon after, 4 Feb 7:01 am UK, Farrar forwarded the rough draft to Fauci and Collins (after converting it to PDF: 'Summary.pdf'). As it was the middle of the night for them, he would have to wait a few hours for their feedbacks. That version was rather rough, and some wildlife data needed to be reviewed, so Farrar took care to add that some updates should follow. He also added that he would push the WHO again that day.



Proximal_Origin_Slack.pdf, p.14, 4 Feb 2020

From: Jeremy Farrar
Sent: Tue, 4 Feb 2020 07:01:16 +0000
To: Fauci, Anthony (NIH/NIAID) [E];Collins, Francis (NIH/OD) [E]
Subject: FW: Prevalence of infection and stage of the epidemic in Wuhan
Attachments: Summary.docx

Please treat in confidence – a very rough first draft from Eddie and team – they will send on the edited, cleaner version later.

Pushing WHO again today

From: Edward Holmes (b) (6)
Date: Tuesday, 4 February 2020 at 06:33
To: Jeremy Farrar (b) (6)
Subject: Re: Prevalence of infection and stage of the epidemic in Wuhan

Here's our summary so far. Will be edited further.

It's fundamental science and completely neutral as written. Did not mention other anomalies as this will make us look like loons. As it stands it is excellent basic science I think, which is a service in itself.

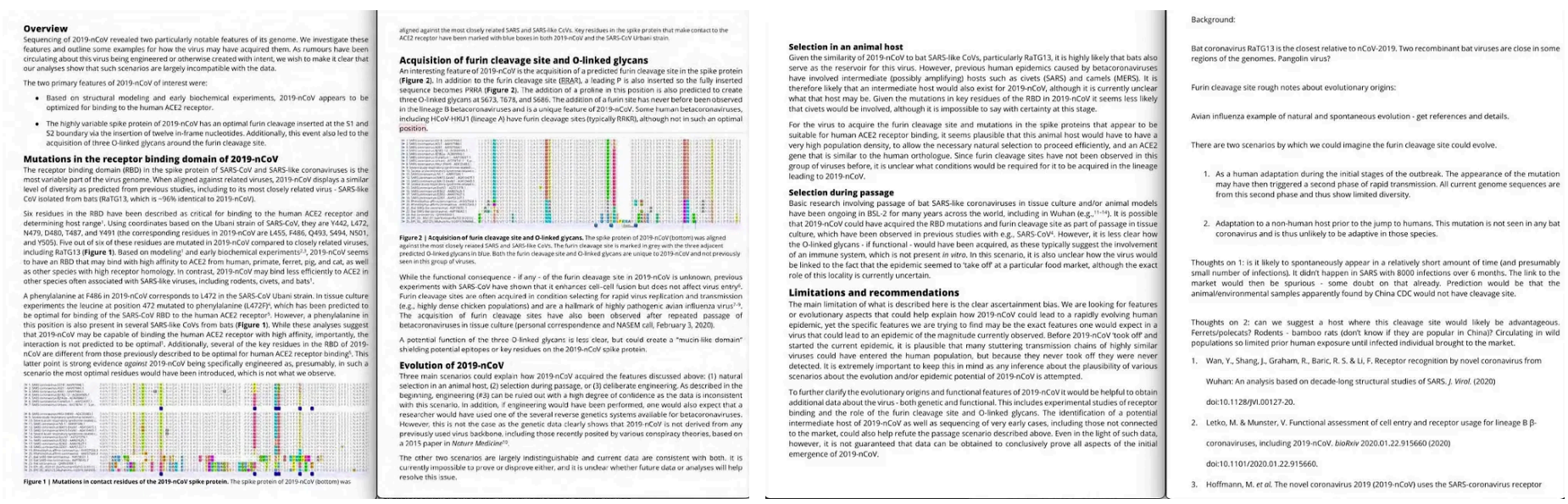
Will finish as soon as we can.

PROFESSOR EDWARD C. HOLMES FAA FRS
ARC Australian Laureate Fellow

farrar-fauci-comms.pdf, p.97, 4 Feb, 7:01 am UK

c. Analysis of the rough draft

The draft at that stage was four pages long, the last page being rather sketchy.



Overview

Sequencing of 2019-nCoV revealed two particularly notable features of its genome. We investigate these features and outline some examples for how the virus may have acquired them. As rumours have been circulating about this virus being engineered or otherwise created with intent, we wish to make it clear that our analyses show that such scenarios are largely incompatible with the data.

The two primary features of 2019-nCoV of interest were:

- Based on structural modeling and early biochemical experiments, 2019-nCoV appears to be optimized for binding to the human ACE2 receptor.
- The highly variable spike protein of 2019-nCoV has an optimal furin cleavage inserted at the S1 and S2 boundary via the insertion of twelve in-frame nucleotides. Additionally, this event also led to the acquisition of three O-linked glycans around the furin cleavage site.

Mutations in the receptor binding domain of 2019-nCoV

The receptor binding domain (RBD) in the spike protein of SARS-CoV and SARS-like coronaviruses is the most variable part of the virus genome. When aligned against related viruses, 2019-nCoV displays a similar level of diversity as predicted from previous studies, including to its most closely related virus - SARS-like CoV isolated from bats (RaTG13, which is ~96% identical to 2019-nCoV).

Six residues in the RBD have been described as critical for binding to the human ACE2 receptor and determining host range[1]. Using coordinates based on the Ubani strain of SARS-CoV, they are Y442, L472, N479, D480, T487, and Y491 (the corresponding residues in 2019-nCoV are L455, F486, Q493, S494, N501, and Y505). Five out of six of these residues are mutated in 2019-nCoV compared to closely related viruses, including RaTG13 (**Figure 1**). Based on modeling[1] and early biochemical experiments[2-5], 2019-nCoV seems to have an RBD that may bind with high affinity to ACE2 from human, primate, ferret, pig, and cat, as well as other species with high receptor homology. In contrast, 2019-nCoV may bind less efficiently to ACE2 in other species often associated with SARS-like viruses, including rodents, civets, and bats[1].

A phenylalanine at F486 in 2019-nCoV corresponds to L472 in the SARS-CoV Ubani strain. In tissue culture experiments the leucine at position 472 mutated to phenylalanine (L472F)[6], which has been predicted to be optimal for binding of the SARS-CoV RBD to the human ACE2 receptor[5]. However, a phenylalanine in this position is also present in several SARS-like CoVs from bats (**Figure 1**). While these analyses suggest that 2019-nCoV may be capable of binding the human ACE2 receptor with high affinity, importantly, the interaction is not predicted to be optimal[5]. Additionally, several of the key residues in the RBD of 2019-nCoV are different from those previously described to be optimal for human ACE2 receptor binding[5]. This latter point is strong evidence *against* 2019-nCoV being specifically engineered as, presumably, in such a scenario the most optimal residues would have been introduced, which is not what we observe.

Figure 1 | Mutations in contact residues of the 2019-nCoV spike protein. The spike protein of 2019-nCoV (bottom) was

aligned against the most closely related SARS and SARS-like CoVs. Key residues in the spike protein that make contact to the ACE2 receptor have been marked with blue boxes in both 2019-nCoV and the SARS-CoV Urbani strain.

Acquisition of furin cleavage site and O-linked glycans

An interesting feature of 2019-nCoV is the acquisition of a predicted furin cleavage site in the spike protein (**Figure 2**). In addition to the furin cleavage site (RRAR), a leading P is also inserted so the fully inserted sequence becomes PRRA (**Figure 2**). The addition of a proline in this position is also predicted to create three O-linked glycans at S673, T678, and S686. The addition of a furin site has never before been observed in the lineage B betacoronaviruses and is a unique feature of 2019-nCoV. Some human betacoronaviruses, including HCoV-HKU1 (lineage A) have furin cleavage sites (typically RRKR), although not in such an optimal position.

Figure 2 | Acquisition of furin cleavage site and O-linked glycans. The spike protein of 2019-nCoV (bottom) was aligned against the most closely related SARS and SARS-like CoVs. The furin cleavage site is marked in grey with the three adjacent predicted O-linked glycans in blue. Both the furin cleavage site and O-linked glycans are unique to 2019-nCoV and not previously seen in this group of viruses.

While the functional consequence - if any - of the furin cleavage site in 2019-nCoV is unknown, previous experiments with SARS-CoV have shown that it enhances cell–cell fusion but does not affect virus entry[6]. Furin cleavage sites are often acquired in condition selecting for rapid virus replication and transmission (*e.g.*, highly dense chicken populations) and are a hallmark of highly pathogenic avian influenza virus[7-9]. The acquisition of furin cleavage sites have also been observed after repeated passage of betacoronaviruses in tissue culture (personal correspondence and NASEM call, February 3, 2020).

A potential function of the three O-linked glycans is less clear, but could create a "mucin-like domain" shielding potential epitopes or key residues on the 2019-nCoV spike protein.

Evolution of 2019-nCoV

Three main scenarios could explain how 2019-nCoV acquired the features discussed above: (1) natural selection in an animal host, (2) selection during passage, or (3) deliberate engineering. As described in the beginning, engineering (#3) can be ruled out with a high degree of confidence as the data is inconsistent with this scenario. In addition, if engineering would have been performed, one would also expect that a researcher would have used one of the several reverse genetics systems available for betacoronaviruses. However, this is not the case as the genetic data clearly shows that 2019-nCoV is not derived from any previously used virus backbone, including those recently posited by various conspiracy theories, based on a 2015 paper in *Nature Medicine*[10].

The other two scenarios are largely indistinguishable and current data are consistent with both. It is currently impossible to prove or disprove either, and it is unclear whether future data or analyses will help resolve this issue.

Selection in an animal host

Given the similarity of 2019-nCoV to bat SARS-like CoVs, particularly RaTG13, it is highly likely that bats also serve as the reservoir for this virus. However, previous human epidemics caused by betacoronaviruses have involved intermediate (possibly amplifying) hosts such as civets (SARS) and camels (MERS). It is therefore likely that an intermediate host would also exist for 2019-nCoV, although it is currently unclear what that host may be. Given the mutations in key residues of the RBD in 2019-nCoV it seems less likely that civets would be involved, although it is impossible to say with certainty at this stage.

For the virus to acquire the furin cleavage site and mutations in the spike proteins that appear to be suitable for human ACE2 receptor binding, it seems plausible that this animal host would have to have a very high population density, to allow the necessary natural selection to proceed efficiently, and an ACE2 gene that is similar to the human orthologue. Since furin cleavage sites have not been observed in this group of viruses before, it is unclear what conditions would be required for it to be acquired in the lineage leading to 2019-nCoV.

Selection during passage

Basic research involving passage of bat SARS-like coronaviruses in tissue culture and/or animal models have been ongoing in BSL-2 for many years across the world, including in Wuhan (e.g.,[11-14]). It is possible that 2019-nCoV could have acquired the RBD mutations and furin cleavage site as part of passage in tissue culture, which have been observed in previous studies with e.g., SARS-CoV[4]. However, it is less clear how the O-linked glycans - if functional - would have been acquired, as these typically suggest the involvement of an immune system, which is not present *in vitro*. In this scenario, it is also unclear how the virus would be linked to the fact that the epidemic seemed to 'take off' at a particular food market, although the exact role of this locality is currently uncertain.

Limitations and recommendations

The main limitation of what is described here is the clear ascertainment bias. We are looking for features or evolutionary aspects that could help explain how 2019-nCoV could lead to a rapidly evolving human epidemic, yet the specific features we are trying to find may be the exact features one would expect in a virus that could lead to an epidemic of the magnitude currently observed. Before 2019-nCoV 'took off' and started the current epidemic, it is plausible that many stuttering transmission chains of highly similar viruses could have entered the human population, but because they never took off they were never detected. It is extremely important to keep this in mind as any inference about the plausibility of various scenarios about the evolution and/or epidemic potential of 2019-nCoV is attempted.

To further clarify the evolutionary origins and functional features of 2019-nCoV it would be helpful to obtain additional data about the virus - both genetic and functional. This includes experimental studies of receptor binding and the role of the furin cleavage site and O-linked glycans. The identification of a potential intermediate host of 2019-nCoV as well as sequencing of very early cases, including those not connected to the market, could also help refute the passage scenario described above. Even in the light of such data, however, it is not guaranteed that data can be obtained to conclusively prove all aspects of the initial emergence of 2019-nCoV.

Background:

Bat coronavirus RaTG13 is the closest relative to nCoV-2019. Two recombinant bat viruses are close in some regions of the genomes. Pangolin virus?

Furin cleavage site rough notes about evolutionary origins:

Avian influenza example of natural and spontaneous evolution - get references and details.

There are two scenarios by which we could imagine the furin cleavage site could evolve.

1. As a human adaptation during the initial stages of the outbreak. The appearance of the mutation may have then triggered a second phase of rapid transmission. All current genome sequences are from this second phase and thus show limited diversity.

2. Adaptation to a non-human host prior to the jump to humans. This mutation is not seen in any bat coronavirus and is thus unlikely to be adaptive in those species.

Thoughts on 1: is it likely to spontaneously appear in a relatively short amount of time (and presumably small number of infections). It didn't happen in SARS with 8000 infections over 6 months. The link to the market would then be spurious - some doubt on that already. Prediction would be that the animal/environmental samples apparently found by China CDC would not have cleavage site.

Thoughts on 2: can we suggest a host where this cleavage site would likely be advantageous. Ferrets/polecats? Rodents - bamboo rats (don't know if they are popular in China)? Circulating in wild populations so limited prior human exposure until infected individual brought to the market.

1. Wan, Y., Shang, J., Graham, R., Baric, R. S. & Li, F. Receptor recognition by novel coronavirus from Wuhan: An analysis based on decade-long structural studies of SARS. *J. Virol.* (2020) doi:10.1128/JVI.00127-20.
2. Letko, M. & Munster, V. Functional assessment of cell entry and receptor usage for lineage B β-coronaviruses, including 2019-nCoV. *bioRxiv* 2020.01.22.915660 (2020) doi:10.1101/2020.01.22.915660.
3. Hoffmann, M. *et al.* The novel coronavirus 2019 (2019-nCoV) uses the SARS-coronavirus receptor

farrar-fauci-comms.pdf, p.98–101, 4 Feb 2020, **first** version shared with Farrar and Collins

c.1. Denial of deliberate engineering

The draft started by setting the context:

> *As rumours have been circulating about this virus being engineered or otherwise created with intent, we wish to make it clear that pour analyses show that such scenarios are largely incompatible with the data.*

What is meant by '**engineered or otherwise created with intent**' is targeted gene creation/editing, i.e.: working directly with the genotype to try to achieve a phenotype. This does not include passaging, which creates selection pressure for a phenotype, without precise control of the genotype.

c.2. Three 'evolutionary origin' hypotheses

Just like the very first note written done by Andersen on 1 Feb, the report

addressed head-on the questions of adaptation to human ACE2 and the presence of a Furin Cleavage Site, and covered the same three 'evolutionary origin' hypotheses (from a molecular perspective):

1. natural selection in an animal host
2. deliberate sequence engineering
3. selection during passage. [footnote 30]

The draft went on by ruling out deliberate engineering. It did not rule out a passaged virus. Nor, unfortunately, did it consider a mix of (2) and (3): deliberate sequence engineering (such as the introduction of an FCS) followed by passaging [footnote 29].

c3. Not the same as three 'origin' hypotheses

One should remember that the draft was about '**evolutionary origins**'. This was not the same as unqualified '**origins**'. This distinction and its consequences are not artificial, they were discussed by the co-authors themselves. In particular, the draft had nothing to say about a lab-escape of a natural virus. By definition, there would be no signature at all that could point to a lab escape in the genome of a natural virus. Only the epidemiology, and particularly early cases tracing, could help ascertain that hypothesis.

But the report purely focussed on what was available at the time: the virus sequence, only released a few weeks ago. There were no reliable epidemiological data, so absolutely no way to start addressing the origin problem from an epidemiological perspective.

c.4. Passaging is a valid evolutionary origin hypothesis

The key sentence may well be:

> *The other two scenarios [2 & 3] are largely indistinguishable and current data are **consistent with both**. It is currently impossible to prove or disprove either, and it is unclear whether future data or analyses will help resolve this issue.*

That sentence is remarkably similar to the quote by Richard Ebright published in Science just a few days before, on 31 Jan:

> *Ebright tells ScienceInsider that the 2019-nCoV data are "consistent with entry into the human population as either a natural accident or a laboratory accident."*

c.5. The need for more data

The last paragraph of the draft report highlighted the needs for more data, including the '***sequencing of very early cases, including those not connected to the market'.*** A fairly similar sentence was included in the NASEM answer to OSTP, that would be published two days later (6 Feb), for which Andersen

was also consulted as an expert: *'Samples collected as early as possible in the outbreak in Wuhan [..] would be particularly valuable'* (see 8.3.f).

d. Farrar discusses the rough draft with Collins and Fauci (4 Feb, 11:08 am to 8:26 pm UK)

The local time on the US eastern coast (EST) was 2:01 am when Farrar sent the first rough draft. The first person to answer was Collins, at 5:58 am EST, or 10:58 am UK.

As can be expected, the conversation that followed focussed on the shift from concerns about deliberate sequence engineering, that had been discussed and dismissed during the call, to concerns about passaging.

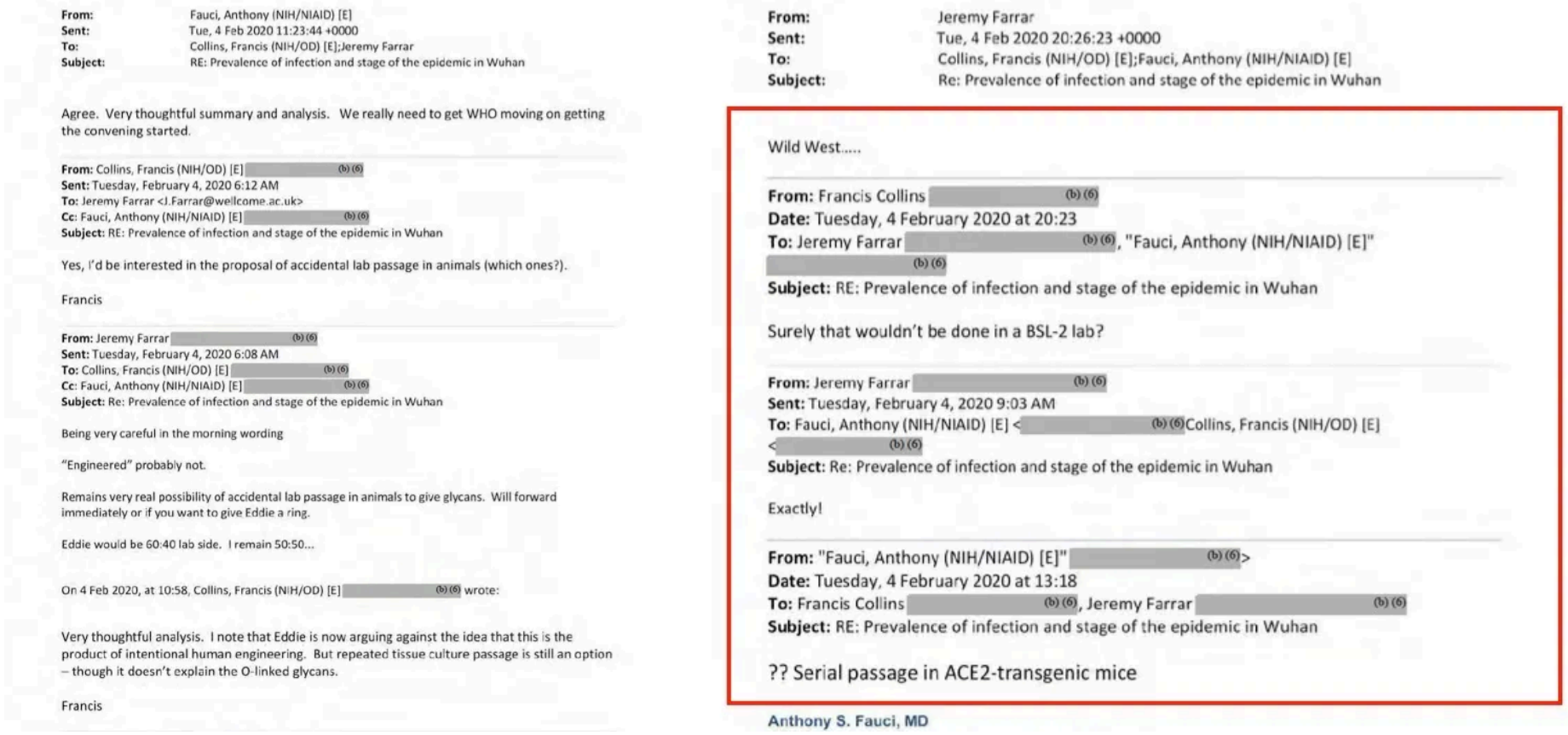
> ***Collins (4 Feb, 10:58 am UK):***
> *Very thoughtful analysis. I note that Eddie is now arguing against the idea that this is the product of international human engineering. But repeated tissue culture passage is still an option — though it doesn't explain the O-linked glycans.*

In his immediate answer to Collins, Farrar gave the latest probabilities: **60:40 lab origin for Holmes, still 50:50 for himself.**

Given the still prevalent lab hypothesis, Fauci added that the WHO needed to 'get the convening started' (11:23 am UK). The conversation then culminated with the mention of the '**Wild West**' of passaging in human-adapted mice at BSL-2 in Wuhan labs.

> ***Farrar (4 Feb, 11:08 am UK):***
> *"Engineered" probably not.*
> *Remains very real possibility of accidental lab passage in animal..*
>
> *Eddie would be 60:40 lab side. I remain 50:50..*
>
> ***Collins (11:12 am UK):***
> *Yes, 'd be interested in the proposal of accidental lab passage in animals (which ones?).*
>
> ***Fauci (11:23 am UK):***
> *Agree. Very thoughtful summary and analysis. We really need to get WHO moving on getting the convening started.*
>
> ***Fauci (1:18 pm UK):***
> *?? Serial passage in ACE2-transgenic mice*
>
> ***Farrar (2:03 pm UK):***
> *Exactly!*

> ***Collins (8:23 pm UK):***
> *Surely that wouldn't be done in a BSL-2 lab?*
>
> ***Farrar (8:26 pm UK):***
> *Wild West…..*



From: Fauci, Anthony (NIH/NIAID) [E]
Sent: Tue, 4 Feb 2020 11:23:44 +0000
To: Collins, Francis (NIH/OD) [E];Jeremy Farrar
Subject: RE: Prevalence of infection and stage of the epidemic in Wuhan

Agree. Very thoughtful summary and analysis. We really need to get WHO moving on getting the convening started.

From: Collins, Francis (NIH/OD) [E] (b) (6)
Sent: Tuesday, February 4, 2020 6:12 AM
To: Jeremy Farrar <J.Farrar@wellcome.ac.uk>
Cc: Fauci, Anthony (NIH/NIAID) [E] (b) (6)
Subject: RE: Prevalence of infection and stage of the epidemic in Wuhan

Yes, I'd be interested in the proposal of accidental lab passage in animals (which ones?).

Francis

From: Jeremy Farrar (b) (6)
Sent: Tuesday, February 4, 2020 6:08 AM
To: Collins, Francis (NIH/OD) [E] (b) (6)
Cc: Fauci, Anthony (NIH/NIAID) [E] (b) (6)
Subject: Re: Prevalence of infection and stage of the epidemic in Wuhan

Being very careful in the morning wording

"Engineered" probably not.

Remains very real possibility of accidental lab passage in animals to give glycans. Will forward immediately or if you want to give Eddie a ring.

Eddie would be 60:40 lab side. I remain 50:50…

On 4 Feb 2020, at 10:58, Collins, Francis (NIH/OD) [E] (b) (6) wrote:

Very thoughtful analysis. I note that Eddie is now arguing against the idea that this is the product of intentional human engineering. But repeated tissue culture passage is still an option – though it doesn't explain the O-linked glycans.

Francis

From: Jeremy Farrar
Sent: Tue, 4 Feb 2020 20:26:23 +0000
To: Collins, Francis (NIH/OD) [E];Fauci, Anthony (NIH/NIAID) [E]
Subject: Re: Prevalence of infection and stage of the epidemic in Wuhan

Wild West…..

From: Francis Collins (b) (6)
Date: Tuesday, 4 February 2020 at 20:23
To: Jeremy Farrar (b) (6), "Fauci, Anthony (NIH/NIAID) [E]" (b) (6)
Subject: RE: Prevalence of infection and stage of the epidemic in Wuhan

Surely that wouldn't be done in a BSL-2 lab?

From: Jeremy Farrar (b) (6)
Sent: Tuesday, February 4, 2020 9:03 AM
To: Fauci, Anthony (NIH/NIAID) [E] <(b) (6)Collins, Francis (NIH/OD) [E] <(b) (6)
Subject: Re: Prevalence of infection and stage of the epidemic in Wuhan

Exactly!

From: "Fauci, Anthony (NIH/NIAID) [E]" (b) (6)>
Date: Tuesday, 4 February 2020 at 13:18
To: Francis Collins (b) (6), Jeremy Farrar (b) (6)
Subject: RE: Prevalence of infection and stage of the epidemic in Wuhan

?? Serial passage in ACE2-transgenic mice

Anthony S. Fauci, MD

farrar-fauci-comms.pdf, p.95 and 92, 4 Feb 2020

Insight: Regulations, Common Sense and the Wild West

Baric gave a good overview of the patchy national and international regulatory oversight in his House interview, when discussing the commonality of coronavirus culturing work at BSL-2 in China. The below extract is worth reading carefully:

> *So biosafety regulations in the United States are very clear,* ***but they're heavily focused on known human pathogens****. So when you move into animal pathogens, pathogens that are in animals, where you don't really know the threat level, to some extent, that becomes a decision between the investigator and the local IBC,* ***which may or may not talk to federal authorities about whether this is appropriate or not****. So, for example, when we started working with zoonotic coronaviruses, our underlying hypothesis was that there are strains that exist in nature. They may be rare, but they could — they could potentially infect human cells. And* ***if that's your hypothesis****, then you do it under BSL-3.*
>
> *The Chinese came to a different — their biosafety regulations are different. But, again, when you ask me about specific regulations,* ***as the Chinese would say to me, Ralph Baric doesn't determine the biosafety levels in this country****, in China, right?*

> *So it's just different. So we were at a higher level containment in the United States. And then anyone who would ask me for these viruses,* ***I would insist that it be done at a higher level containment. So I kind of set the standard in the United States.***
>
> *source: source: Baric-TI-Transcript.pdf, extracts from p.20–21*

Baric then touched at the NASEM-CISAC efforts to try to drive some common sense into Chinese practices (a Track II initiative under the Defense Threats Reduction Agency [footnote 24]):

> *So National Academy of Sciences in the United States and the National Academy of Sciences in China held three joint meetings, one in Beijing, one in Harbin, and one in Galveston Island, about biosafety and biosecurity.*
> *So in the context of that,* ***there were discussions about biosafety and trying to harmonize*** *— in essence, trying to harmonize and to teach each other's group about standard practices and that kind of thing. But it wasn't more like there was a small group sessions, where we talked about biosafety. It was more of the science that we were doing* ***and the levels that it was done at.***
>
> *source: source: Baric-TI-Transcript.pdf, extracts from p.26–27*

Last, contrarily to Baric and the CISAC efforts, Daszak was quite enthusiastic of seeing some of that work done at BSL-2 in by the WIV, for convenience purpose. This resulted in a very direct dressing-down by Baric [footnotes 37, 73], showing that Baric did indeed try to enforce — as far as he could — some common sense, domestically and abroad.

10.2 Second sent: Revised draft (4 Feb, evening UK)

a. Garry does his edits (4 Feb, 6:31 pm UK)
At 1:31 pm EST, 6:31 pm UK, on 4 Feb, Garry announced that he had finished his edits. Effectively, that was the first time the three main authors, Andersen, Holmes and Garry, all had the time to do their edits and checks to the Google Doc draft created by Andersen about 14 hours ago (see 10.1).

For clarity, neither Garry nor Andersen were aware that Holmes had forwarded the draft hours ago to Farrar without waiting for Garry to do his edits (as it was then the middle of the night for him), and that Farrar had already shared it and discussed it with Fauci and Collins.

In his email, Garry described the draft as a **'cogent explanation of why concerns have been raised'**, but was careful to also point out that he

anticipated some pushback on the passaging origin hypothesis (6:31 pm UK). He could not have known at the time how right he would be proven.

Andersen agreed very much with that position, and explained that it was time for the draft to **'go up the chain',** based on the understanding that it was not '**meant for public consumption**' (4 Feb, 8:03 pm UK). He also added that working for publication '**would require much more careful crafting and attention to specific wording of key concepts**' than the summary report they had just produced, and that there was no reason to attempt to do so in any case.

b. The passaging option still requires a proper discussion via the WHO
Some important details about the WHO offload emerged in that conversation:

> ***Garry (4 Feb, 6:31 pm UK):***
> *If there is a natural explanation for CoV, it needs to be found. [.]*
> *Some, perhaps more than a few, will not like it still since it allows that the nCoV may have arisen during cell culture passage in a lab [their labs].*

The '[**their labs**]' seems to be a direct reference to the Chinese side, which would suppose that the Chinese side would be invited to comment on that document. This interpretation is reinforced by another one of Garry's emails:

> ***Garry (4 Feb, 7:14 pm UK):***
> *Another caveat is that I think there is plenty of room for additional discussion amongst the experts. Jeremy's idea (or was it Tony's) of a **face-to-face** under the auspic[es] of WHO **still** makes sense to me.*

It is worth noting the inflections in Garry's replies above: '***still*** makes sense to me', and then '*Some, perhaps more than a few, will not like it **still**, since …*"
These 'still' refers to a point that was crucial in the mind of the co-authors at the time. What Garry meant was that:

- The lab-product hypothesis was STILL very much alive, through the passaging option, even if the bioweapon hypothesis had been dismissed (rightly or wrongly) during the Farrar's call, after discussions with the bounce-off group of Fouchier, Koopmans and Drosten.
- So, the draft report would STILL be likely to be controversial for the Chinese side,
- Hence, the best way to handle this likely controversial summary/report of specific features would STILL be to confront the Chinese side in the neutral, collaborative, space offered by the WHO.

- Such a gathering would also be a good occasion to get additional inputs from experts.

On 5 Feb 2020, at 7:14 am, Garry, Robert F [redacted] wrote:

Another caveat is that I think there is plenty of room for additional discussion amongst the experts. Jeremy's idea (or was it Tony's) of a face-to-face under the auspicious of WHO still makes sense to me.

From: Edward Holmes [redacted]
Date: Tuesday, February 4, 2020 at 2:10 PM
To: Kristian Andersen [redacted]
Cc: Robert Garry [redacted], "rambaut [redacted]
Subject: Re: Summary - Invitation to edit

External Sender. Be aware of links, attachments and requests.

Works for me. Should I quickly check with Jeremy to see if he is happy for it to be circulated to the wider group?

Great job.

Professor Edward C. Holmes FAA FRS
The University of Sydney

On 5 Feb 2020, at 7:03 am, Kristian G. Andersen [redacted] wrote:

Did a final pass and I think it looks great.

Unless others have further comments, I'd say this is ready to go up the chain. Importantly, my assumption is that this will not be a document that is meant for public consumption, as that would require much more careful crafting and attention to specific wording of key concepts in the document (not really a task I think we could/should take on - that would be way, way more work).

K

On Tue, Feb 4, 2020 at 11:31 AM Garry, Robert F [redacted] wrote:

Gentlemen – I believe that the document is getting very clean.

Only a few minor points to address [or not] from my view.

I believe it is a cogent explanation why concerns were raised.

If there is a natural explanation for CoV, it needs to be found. A lot of unobserved transmission in animals/humans AND as yet unsampled Bat CoV variants (with whole or partial furin sites) must exist.

Some, perhaps more than a few, will not like it still since it allows that the nCoV may have arisen during cell culture passage in a lab (their labs).

Thanks for the great science…

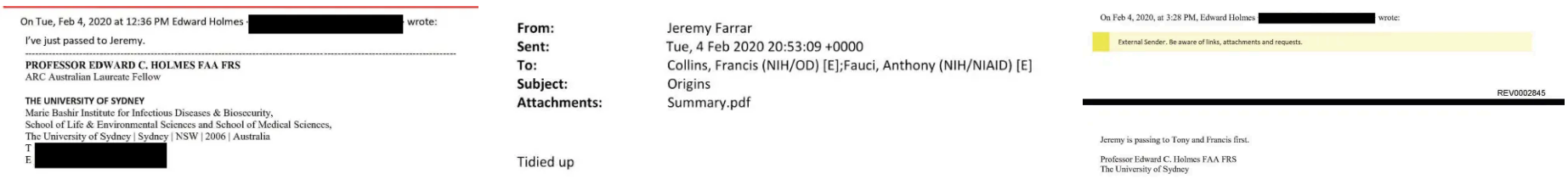
b

Proximal_Origin_Emails.pdf, p.26, 4 Feb 2020, around 7 to 8 pm UK

c. Revised draft shared with Fauci and Collins (4 Feb, ~9 pm UK)
Holmes reviewed the latest draft and next proposed to check with Farrar if it should be shared with the '**wider group**', meaning the full call group (4 Feb, 8:10 pm UK).

At 8:36 pm UK, Holmes announced by email that he had just forwarded the revised draft over to Farrar. Farrar quickly forwarded it to Fauci and Collins (8:53 pm UK)

Holmes then informed his co-drafters on Slack that '**Jeremy is passing to Tony and Francis first**' (9:38 pm UK). The 'first' may seem intriguing, but Holmes most likely meant that he would send it to the WHO after Fauci and Collins had a chance to review it first. In fact, **Holmes was lying by omission**, since he had already shared the previous draft earlier on, without waiting for Garry's edits.



On Tue, Feb 4, 2020 at 12:36 PM Edward Holmes [redacted] wrote:

I've just passed to Jeremy.

PROFESSOR EDWARD C. HOLMES FAA FRS
ARC Australian Laureate Fellow

THE UNIVERSITY OF SYDNEY
Marie Bashir Institute for Infectious Diseases & Biosecurity,
School of Life & Environmental Sciences and School of Medical Sciences,
The University of Sydney | Sydney | NSW | 2006 | Australia
T
E [redacted]

From: Jeremy Farrar
Sent: Tue, 4 Feb 2020 20:53:09 +0000
To: Collins, Francis (NIH/OD) [E];Fauci, Anthony (NIH/NIAID) [E]
Subject: Origins
Attachments: Summary.pdf

Tidied up

On Feb 4, 2020, at 3:28 PM, Edward Holmes [redacted] wrote:

External Sender. Be aware of links, attachments and requests.

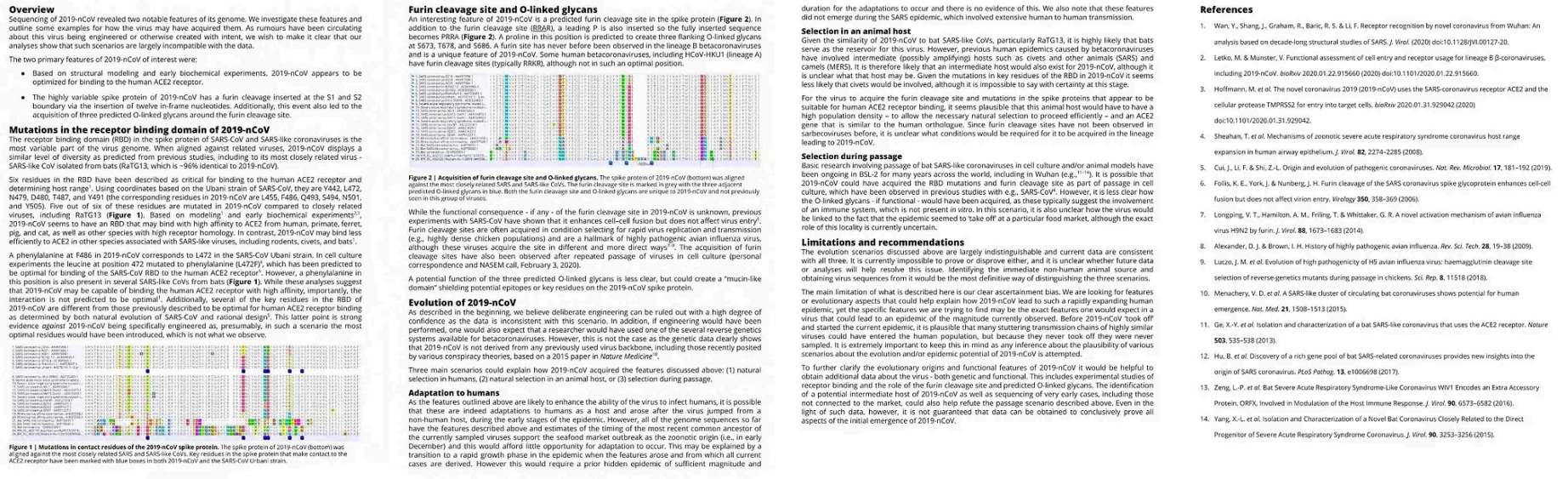
REV0002845

Jeremy is passing to Tony and Francis first.

Professor Edward C. Holmes FAA FRS
The University of Sydney

Proximal_Origin_Emails.pdf, p.25, 4 Feb 2020, 8:36 pm UK, farrar-fauci-comms.pdf, p. 87, 4 Feb 2020, 8:53 pm UK and Proximal_Origin_Emails.pdf, p.25, 4 Feb 2020, 9:38 pm UK,

That revised draft (v5a) is very similar to the first one, mostly tidied up, especially the very end that had been left rather schematic in the first version. It has the same weighing of the 3 hypotheses. 'Limitations and recommendations' is a bit more specific about the full plausibility of all three scenarios.



Overview
Sequencing of 2019-nCoV revealed two notable features of its genome. We investigate these features and outline some examples for how the virus may have acquired them. As rumours have been circulating about this virus being engineered or otherwise created with intent, we wish to make it clear that our analyses show that such scenarios are largely incompatible with the data.

The two primary features of 2019-nCoV of interest were:

- Based on structural modeling and early biochemical experiments, 2019-nCoV appears to be optimized for binding to the human ACE2 receptor.
- The highly variable spike protein of 2019-nCoV has a furin cleavage inserted at the S1 and S2 boundary via the insertion of twelve in-frame nucleotides. Additionally, this event also led to the acquisition of three predicted O-linked glycans around the furin cleavage site.

Mutations in the receptor binding domain of 2019-nCoV
The receptor binding domain (RBD) in the spike protein of SARS-CoV and SARS-like coronaviruses is the most variable part of the virus genome. When aligned against related viruses, 2019-nCoV displays a similar level of diversity as predicted from previous studies, including to its most closely related virus - SARS-like CoV isolated from bats (RaTG13, which is ~96% identical to 2019-nCoV).

Six residues in the RBD have been described as critical for binding to the human ACE2 receptor and determining host range[1]. Using coordinates based on the Ubani strain of SARS-CoV, they are Y442, L472, N479, D480, T487, and Y491 (the corresponding residues in 2019-nCoV are L455, F486, Q493, S494, N501, and Y505). Five out of six of these residues are mutated in 2019-nCoV compared to closely related viruses, including RaTG13 (**Figure 1**). Based on modeling[1] and early biochemical experiments[2], 2019-nCoV seems to have an RBD that may bind with high affinity to ACE2 from human, primate, ferret, pig, and cat, as well as other species with high receptor homology. In contrast, 2019-nCoV may bind less efficiently to ACE2 in other species associated with SARS-like viruses, including rodents, civets, and bats[1].

A phenylalanine at F486 in 2019-nCoV corresponds to L472 in the SARS-CoV Ubani strain. In cell culture experiments the leucine at position 472 mutated to phenylalanine (L472F)[2], which has been predicted to be optimal for binding of the SARS-CoV RBD to the human ACE2 receptor[3]. However, a phenylalanine in this position is also present in several SARS-like CoVs from bats (**Figure 1**). While these analyses suggest that 2019-nCoV may be capable of binding the human ACE2 receptor with high affinity, importantly, the interaction is not predicted to be optimal[1]. Additionally, several of the key residues in the RBD of 2019-nCoV are different from those previously described to be optimal for human ACE2 receptor binding as determined by both natural evolution of SARS-CoV and rational design[3]. This latter point is strong evidence *against* 2019-nCoV being specifically engineered as, presumably, in such a scenario the most optimal residues would have been introduced, which is not what we observe.

Figure 1 | Mutations in contact residues of the 2019-nCoV spike protein. The spike protein of 2019-nCoV (bottom) was aligned against the most closely related SARS and SARS-like CoVs. Key residues in the spike protein that make contact to the ACE2 receptor have been marked with blue boxes in both 2019-nCoV and the SARS-CoV Urbani strain.

Furin cleavage site and O-linked glycans
An interesting feature of 2019-nCoV is a predicted furin cleavage site in the spike protein (**Figure 2**). In addition to the furin cleavage site (RRAR), a leading P is also inserted so the fully inserted sequence becomes PRRA (**Figure 2**). A proline in this position is predicted to create three flanking O-linked glycans at S673, T678, and S686. A furin site has never before been observed in the lineage B betacoronaviruses and is a unique feature of 2019-nCoV. Some human betacoronaviruses, including HCoV-HKU1 (lineage A) have furin cleavage sites (typically RRKR), although not in such an optimal position.

Figure 2 | Acquisition of furin cleavage site and O-linked glycans. The spike protein of 2019-nCoV (bottom) was aligned against the most closely related SARS and SARS-like CoVs. The furin cleavage site is marked in grey with the three adjacent predicted O-linked glycans in blue. Both the furin cleavage site and O-linked glycans are unique to 2019-nCoV and not previously seen in this group of viruses.

While the functional consequence - if any - of the furin cleavage site in 2019-nCoV is unknown, previous experiments with SARS-CoV have shown that it enhances cell–cell fusion but does not affect virus entry[2]. Furin cleavage sites are often acquired in condition selecting for rapid virus replication and transmission (e.g., highly dense chicken populations) and are a hallmark of highly pathogenic avian influenza virus, although these viruses acquire the site in different and more direct ways[7–9]. The acquisition of furin cleavage sites have also been observed after repeated passage of viruses in cell culture (personal correspondence and NASEM call, February 3, 2020).

A potential function of the three predicted O-linked glycans is less clear, but could create a "mucin-like domain" shielding potential epitopes or key residues on the 2019-nCoV spike protein.

Evolution of 2019-nCoV
As described in the beginning, we believe deliberate engineering can be ruled out with a high degree of confidence as the data is inconsistent with this scenario. In addition, if engineering would have been performed, one would also expect that a researcher would have used one of the several reverse genetics systems available for betacoronaviruses. However, this is not the case as the genetic data clearly shows that 2019-nCoV is not derived from any previously used virus backbone, including those recently posited by various conspiracy theories, based on a 2015 paper in *Nature Medicine*[10].

Three main scenarios could explain how 2019-nCoV acquired the features discussed above: (1) natural selection in humans, (2) natural selection in an animal host, or (3) selection during passage.

Adaptation to humans
As the features outlined above are likely to enhance the ability of the virus to infect humans, it is possible that these are indeed adaptations to humans as a host and arose after the virus jumped from a non-human host, during the early stages of the epidemic. However, all of the genome sequences so far have the features described above and estimates of the timing of the most recent common ancestor of the currently sampled viruses support the seafood market outbreak as the zoonotic origin (i.e., in early December) and this would afford little opportunity for adaptation to occur. This may be explained by a transition to a rapid growth phase in the epidemic when the features arose and from which all current cases are derived. However this would require a prior hidden epidemic of sufficient magnitude and duration for the adaptations to occur and there is no evidence of this. We also note that these features did not emerge during the SARS epidemic, which involved extensive human to human transmission.

Selection in an animal host
Given the similarity of 2019-nCoV to bat SARS-like CoVs, particularly RaTG13, it is highly likely that bats serve as the reservoir for this virus. However, previous human epidemics caused by betacoronaviruses have involved intermediate (possibly amplifying) hosts such as civets and other animals (SARS) and camels (MERS). It is therefore likely that an intermediate host would also exist for 2019-nCoV, although it is unclear what that host may be. Given the mutations in key residues of the RBD in 2019-nCoV it seems less likely that civets would be involved, although it is impossible to say with certainty at this stage.

For the virus to acquire the furin cleavage site and mutations in the spike proteins that appear to be suitable for human ACE2 receptor binding, it seems plausible that this animal host would have to have a high population density – to allow the necessary natural selection to proceed efficiently – and an ACE2 gene that is similar to the human orthologue. Since furin cleavage sites have not been observed in sarbecoviruses before, it is unclear what conditions would be required for it to be acquired in the lineage leading to 2019-nCoV.

Selection during passage
Basic research involving passage of bat SARS-like coronaviruses in cell culture and/or animal models have been ongoing in BSL-2 for many years across the world, including in Wuhan (e.g.,[11–14]). It is possible that 2019-nCoV could have acquired the RBD mutations and furin cleavage site as part of passage in cell culture, which have been observed in previous studies with e.g., SARS-CoV[4]. However, it is less clear how the O-linked glycans - if functional - would have been acquired, as these typically suggest the involvement of an immune system, which is not present *in vitro*. In this scenario, it is also unclear how the virus would be linked to the fact that the epidemic seemed to 'take off' at a particular food market, although the exact role of this locality is currently uncertain.

Limitations and recommendations
The evolution scenarios discussed above are largely indistinguishable and current data are consistent with all three. It is currently impossible to prove or disprove either, and it is unclear whether future data or analyses will help resolve this issue. Identifying the immediate non-human animal source and obtaining virus sequences from it would be the most definitive way of distinguishing the three scenarios.

The main limitation of what is described here is our clear ascertainment bias. We are looking for features or evolutionary aspects that could help explain how 2019-nCoV lead to such a rapidly expanding human epidemic, yet the specific features we are trying to find may be the exact features one would expect in a virus that could lead to an epidemic of the magnitude currently observed. Before 2019-nCoV 'took off' and started the current epidemic, it is plausible that many stuttering transmission chains of highly similar viruses could have entered the human population, but because they never took off they were never sampled. It is extremely important to keep this in mind as any inference about the plausibility of various scenarios about the evolution and/or epidemic potential of 2019-nCoV is attempted.

To further clarify the evolutionary origins and functional features of 2019-nCoV it would be helpful to obtain additional data about the virus - both genetic and functional. This includes experimental studies of receptor binding and the role of the furin cleavage site and predicted O-linked glycans. The identification of a potential intermediate host of 2019-nCoV as well as sequencing of very early cases, including those not connected to the market, could also help refute the passage scenario described above. Even in the light of such data, however, it is not guaranteed that data can be obtained to conclusively prove all aspects of the initial emergence of 2019-nCoV.

References

1. Wan, Y., Shang, J., Graham, R., Baric, R. S. & Li, F. Receptor recognition by novel coronavirus from Wuhan: An analysis based on decade-long structural studies of SARS. *J. Virol.* (2020) doi:10.1128/JVI.00127-20.
2. Letko, M. & Munster, V. Functional assessment of cell entry and receptor usage for lineage B β-coronaviruses, including 2019-nCoV. *bioRxiv* 2020.01.22.915660 (2020) doi:10.1101/2020.01.22.915660.
3. Hoffmann, M. *et al.* The novel coronavirus 2019 (2019-nCoV) uses the SARS-coronavirus receptor ACE2 and the cellular protease TMPRSS2 for entry into target cells. *bioRxiv* 2020.01.31.929042 (2020) doi:10.1101/2020.01.31.929042.
4. Sheahan, T. *et al.* Mechanisms of zoonotic severe acute respiratory syndrome coronavirus host range expansion in human airway epithelium. *J. Virol.* **82**, 2274–2285 (2008).
5. Cui, J., Li, F. & Shi, Z.-L. Origin and evolution of pathogenic coronaviruses. *Nat. Rev. Microbiol.* **17**, 181–192 (2019).
6. Follis, K. E., York, J. & Nunberg, J. H. Furin cleavage of the SARS coronavirus spike glycoprotein enhances cell-cell fusion but does not affect virion entry. *Virology* **350**, 358–369 (2006).
7. Longping, V. T., Hamilton, A. M., Friling, T. & Whittaker, G. R. A novel activation mechanism of avian influenza virus H9N2 by furin. *J. Virol.* **88**, 1673–1683 (2014).
8. Alexander, D. J. & Brown, I. H. History of highly pathogenic avian influenza. *Rev. Sci. Tech.* **28**, 19–38 (2009).
9. Luczo, J. M. *et al.* Evolution of high pathogenicity of H5 avian influenza virus: haemagglutinin cleavage site selection of reverse-genetics mutants during passage in chickens. *Sci. Rep.* **8**, 11518 (2018).
10. Menachery, V. D. *et al.* A SARS-like cluster of circulating bat coronaviruses shows potential for human emergence. *Nat. Med.* **21**, 1508–1513 (2015).
11. Ge, X.-Y. *et al.* Isolation and characterization of a bat SARS-like coronavirus that uses the ACE2 receptor. *Nature* **503**, 535–538 (2013).
12. Hu, B. *et al.* Discovery of a rich gene pool of bat SARS-related coronaviruses provides new insights into the origin of SARS coronavirus. *PLoS Pathog.* **13**, e1006698 (2017).
13. Zeng, L.-P. *et al.* Bat Severe Acute Respiratory Syndrome-Like Coronavirus WIV1 Encodes an Extra Accessory Protein, ORFX, Involved in Modulation of the Host Immune Response. *J. Virol.* **90**, 6573–6582 (2016).
14. Yang, X.-L. *et al.* Isolation and Characterization of a Novel Bat Coronavirus Closely Related to the Direct Progenitor of Severe Acute Respiratory Syndrome Coronavirus. *J. Virol.* **90**, 3253–3256 (2015).

farrar-fauci-comms.pdf, p.88–91, 4 Feb 2020, **second** version shared with Farrar and Collins

Key Insight: When it started going all wrong

▶️ An attribution dilemma:

The drop of the deliberate engineering option (whether justified or not), and the new focus on passaging, generated a major burden of proof dilemma: deliberate engineering could have left key signs, 'smoking guns', but passaging is nothing else than accelerated evolution. Thus, passaging in human-adapted animals could be easily deniable, unless one had access to the lab records and/or to the key databases.

▶️ Saying things without saying it

Already on 2 Feb, one key exchange jumps out (see 9.2), that summarises perfectly that dilemma:

> ***Rambaut (4 Feb, 4:53 pm UK):***
> *Given the shit show that would happen if anyone serious accused the Chinese of even accidental release,* ***my feeling is we should say that given there is no evidence of a specifically engineered virus, we cannot possibly distinguish***

> ***between natural evolution and escape, so we are content with ascribing it to natural processes'***
>
> ***Andersen (4 Feb, 4:56 pm UK):***
> *'Yup, I totally agree. That's a very reasonable conclusion,* ***although I hate when politics is injected into science, but it's impossible not to, especially given the circumstances.*** *We should be sensitive to that.'*

In other words, the very day after the Farrar call, the report/summary drafters agreed that a non-natural origin was likely, if not more likely, especially due to the Furin Cleavage Site. However, because of the political and diplomatic dimension, they noted that they would rather give the benefit of the doubt to a natural origin.

What this precisely means is that the report drafters were trying to see how they could reconcile two imperatives, a scientific one and a political one, for their part of the process, before handing this over to the WHO:

- **Scientific:** Write down a report of noteworthy features, that an investigation of the origin under a proper WHO framework should look into.
- **Political:** Don't directly point to a possible non-natural origin of a feature, if there is an alternative natural origin for it. Instead, give the benefit of the doubt to a natural origin, while still noting and discussing the feature.

▶️ **Let the WHO deal with it**
A face-to-face with Chinese scientists, hopefully in a non-conflictual environment if under the auspice of the WHO, was, **in theory at least**, *even more necessary* now that the origin question had shifted to passaging, because of the absence of a smoking-gun with that option.

▶️ **Or not deal with it …**
But the absence of a smoking gun (compared to deliberate engineering) also meant that the Chinese side could just deny anything.

Hence, as soon as the engineering option had been discarded, the WHO, and Farrar with it, were much more likely to end up privileging the **practical perspective**, which was to put cooperation and access first, a key for saving lives in the medium terms. The alternative would have meant getting caught in a conflict with the Chinese side that offered few upsides — if any at all — and that would draw the WHO into a nasty political and diplomatic crisis on top of the existing health one.

▶️ **A schizophrenic context:**
We can see how Andersen, Rambaut, Holmes and Farrar were at that time, trying to reasonably push back against the telling down from Fouchier and

Drosten that they had faced during the Farrar call. They correctly sensed that passaging was still a key issue, and worked further on that possibility. But at the same time, they could see how difficult it would be to go down that path.

For Farrar, the game had changed. The core-group had been largely put together by Farrar, in order to obtain a confidential external evaluation by selected subject-matter experts of some of the virus genetic features, for the benefit of MI5 (which likely received it). Soon after, the initiative was recycled into support to the WHO for access, on which the US could piggyback; a new initiative that instead required publishing a suitable origin report.

It would not be incorrect to say that Farrar hijacked the efforts of Andersen, who was relatively thorough and comfortable as long as the report was meant to be non-public, to get the public statement he needed. Farrar was helped in this by Holmes, who, from the get-go, was always interested in publishing yet another paper, not a confidential report that added nothing to his prestige.

10.3 Farrar updates Fauci and Collins about the glycans (5 Feb, 6:57 am UK)

On 5 Feb, 6:57 am UK, Farrar sent a message back to Fauci and Collins to discuss a technical issue (the glycans) that had been raised, that could have affected the plausibility of passaging. In his message, he reiterated that the FCS could well be produced by passaging. [footnote 48]

He also pointed again to the importance of being able to see some experiment data from Fouchier, referring to undisclosed results from Fouchier's lab, likely mentioned during the Farrar call, that demonstrated the successful forced evolution of an FCS via passaging. With his answer, Farrar attached the latest version of the report (still v5a, already sent the previous evening).

Variation on the theme in HKU1, a virus that probably does have intense transmission infecting millions of people each year. Here the insert is three Serine residues, which pushes this site to a mucin-like patch (there are already a couple of prolines and the SSS is a turn as well)

Funny thing – not on the attachments, but those strains of MHV and HKU-1 that have o-linked glycans and the furin site ALSO have a larger patch - sometimes very large patch - of predicted o-linked glycans at the top of the prefusion form. When you see the pattern repeat itself in different viruses you start to believe it.

From: Robert Garry [redacted]
Date: Tuesday, February 4, 2020 at 5:56 PM
To: Kristian Andersen [redacted], Edward Holmes [redacted]
Cc: "rambaut [redacted]
Subject: Re: Summary - Invitation to edit

Kristian that's correct about everything he said for the P residue. It's what's shifted me to thinking that the insert of the furin site is the result of cell culture passage [or less likely intense transmission in a nonbat host]. Really need to see the data from Ron about generating the furim cleavage site on in vitro passage. Really!

CoV come with or without a furin site. CoV without a furin site are said to be non-cleaved and rely on endosomal proteases like cathepsin for entry. However if you infect a virus like SARS in culture in the presense of exogenous protease like trypsin its 100X more effective at entering because the spike gets cleaved and it can enter at the cell surface.

You have to infect flu viruses (the ones without the multibasic cleavage site) in the presence of trypsin, and include trypsin in the overlay if you want to get virus spread aka plaques.

This also contributes to the pathogenicity of - well - highly pathogenic flu virus – different tissues have different proteases and are able to "activate" flu to different extents - if the flu v has a furin cleavage site it has a lot more choices and canmore easil go systemic.

This is an excellent review on CoV fusion – deals with all the complexities:
https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3397359/

Bottom line – I think that if you put selection pressure on a Cov without a furin cleavage site in cell culture you could well generate a furin cleavage site after a number of passages (but let's see the data Ron!). It will infect a lot better if it can effectively fuse at the cell surface and doesn't have to rely on endosomal cleavage and receptor mediated endocytosis..

From: Kristian Andersen [redacted]
Date: Tuesday, February 4, 2020 at 5:08 PM
To: Edward Holmes [redacted]
Cc: Robert Garry [redacted], "rambaut [redacted]
Subject: Re: Summary - Invitation to edit

External Sender. Be aware of links, attachments and requests.

Outside my expertise, but I don't necessarily think that passage in animals would add the glycans. It's more that the glycans could suggest some sort of immune system as the glycans often work to 'shield' epitopes. So if the acquisition of glycans is adaptive, that would be suggestive of an immune system.

We didn't write this in the report, but the residues on which the glycans (S, T, and S) are all conserved in the bat virus - it's the addition of the P that makes it a specific glycan site though (not conserved in the bat, hence not predicted to be O-glycans). It's entirely possible that the 'P' works as a flexible residue for the furin cleavage site and by proxy creates the (predicted) O-linked glycans.

I'll let Bob weigh in as well - definitely not my area of expertise.

K

On Tue, Feb 4, 2020 at 2:59 PM Edward Holmes [redacted] wrote:

Agreed. Timing is perfect.

Bob - a question from Jeremy:

"Quick question though - why could passage in animals in lab work add the glycans?"

Any thoughts?

Eddie

PROFESSOR EDWARD C. HOLMES FAA FRS
ARC Australian Laureate Fellow

THE UNIVERSITY OF SYDNEY
Marie Bashir Institute for Infectious Diseases & Biosecurity,
School of Life & Environmental Sciences and School of Medical Sciences,
The University of Sydney | Sydney | NSW | 2006 | Australia
T: [redacted]
E [redacted]

On 5 Feb 2020, at 9:53 am, Garry, Robert F [redacted] wrote:

Ironically the prevailing theory now in the underbelly if the internet is that the us or other enemy engineered this bio weapon and released it on China

If the public health aspects of this were not bad enough the political fallout would be.

Good to have cogent science against the bio weapon scenario which is why I favor getting who involved in the "controversy"

Accidental release is a scenario many will not be comfortable with but it would be irresponsible to dismiss the possibility out of hand.

Sent from my iPhone

On Feb 4, 2020, at 3:28 PM, Edward Holmes [redacted] wrote:

External Sender. Be aware of links, attachments and requests.

Jeremy is passing to Tony and Francis first.

Professor Edward C. Holmes FAA FRS
The University of Sydney

From: Jeremy Farrar
Sent: Wed, 5 Feb 2020 06:57:54 +0000
To: Collins, Francis (NIH/OD) [E];Fauci, Anthony (NIH/NIAID) [E];Edward Holmes
Subject: FW: Origins
Attachments: Summary.pdf

Tony and Francis

The revised draft from Eddie, copied here.

I asked Eddie about the addition of the glycans and where these could be added accidentally by passage in lab animals or of course in the wild – the reply from Bob and Kristian

?Kristian that's correct about everything he said for the P residue. It's what's shifted me to thinking that the insert of the furin site is the result of cell culture passage [or less likely intense transmission in a nonbat host]. Really need to see the data from Ron about generating the furim cleavage site on in vitro passage. Really!

CoV come with or without a furin site. CoV without a furin site are said to be non-cleaved and rely on endosomal proteases like cathepsin for entry. However if you infect a virus like SARS in culture in the presense of exogenous protease like trypsin its 100X more effective at entering because the spike gets cleaved and it can enter at the cell surface.

You have to infect flu viruses (the ones without the multibasic cleavage site) in the presence of trypsin, and include trypsin in the overlay if you want to get virus spread aka plaques.

This also contributes to the pathogenicity of - well - highly pathogenic flu virus – different tissues have different proteases and are able to "activate" flu to different extents - if the flu v has a furin cleavage site it has a lot more choices and canmore easil go systemic.

This is an excellent review on CoV fusion – deals with all the complexities:
https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3397359/

Bottom line – I think that if you put selection pressure on a Cov without a furin cleavage site in cell culture you could well generate a furin cleavage site after a number of passages (but let's see the data Ron!). It will infect a lot better if it can effectively fuse at the cell surface and doesn't have to rely on endosomal cleavage and receptor mediated endocytosis..?

Proximal_Origin_Emails.pdf, p.30–31, 4 Feb 2020 9.59 to 10:56 pm UK, and farrar-fauci-comms.pdf, p.82–86, 5 Feb 2020,6:57 am UK

10.4 Pangolins data starts coming in (5 Feb, ~10 am UK)

The 5 February was a data crunch time: new pangolin data were popping up, received by Holmes via his good contact Tommy Lam, who was working at the time on a pangolin paper (‘Identifying SARS-CoV-2-related coronaviruses in Malayan pangolins’).

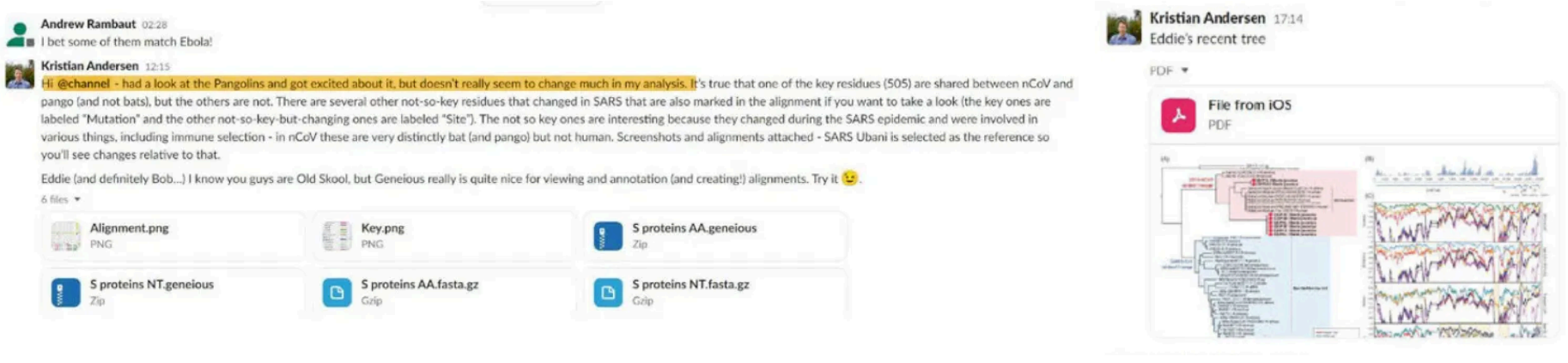
Lam sent him the main figures, of a paper that was already very well advanced, which the co-authors then discussed. After some initial excitement, Andersen and Garry concluded that the new pangolin sequences, while interesting, were globally too distant to be a progenitor of SARS-CoV-2, and locally not close enough to lead to one via recombination.

On 5 Feb 2020, at 10:31, Edward Holmes [redacted] wrote:

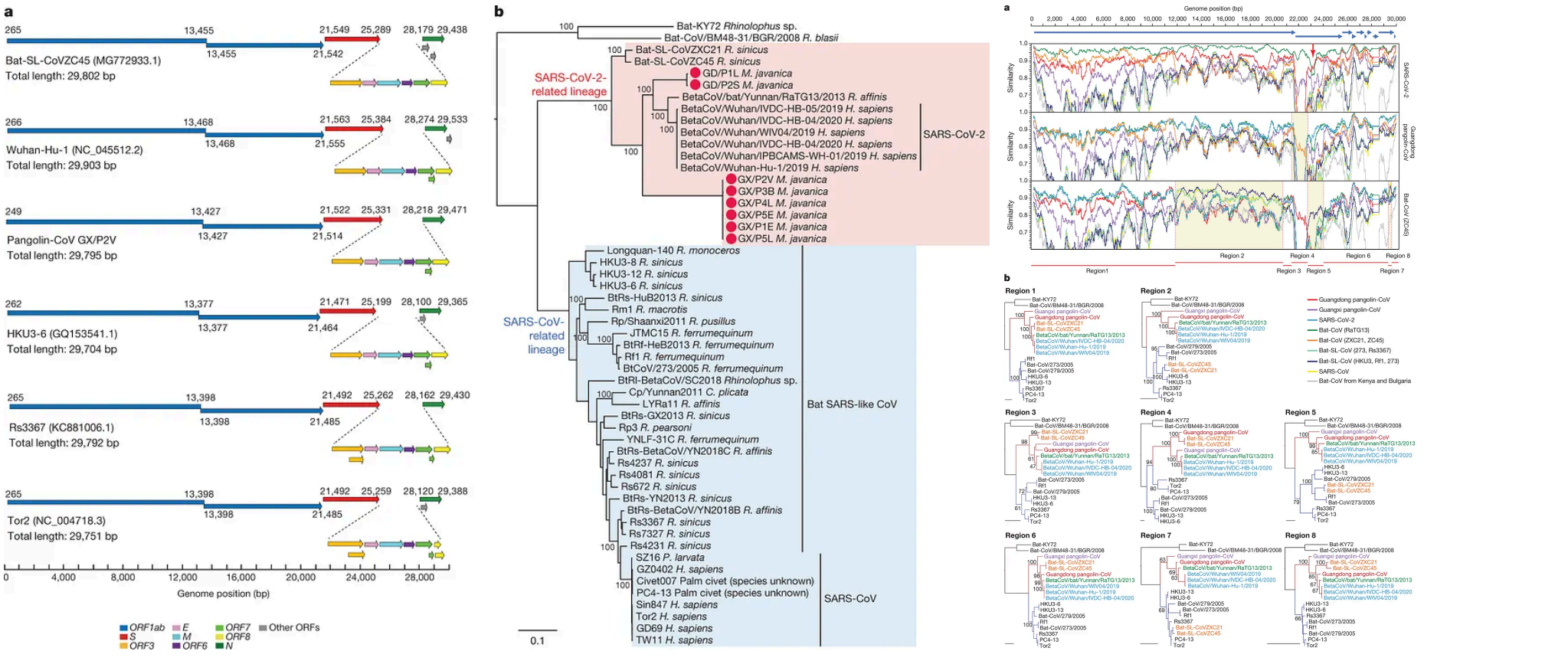
I've asked Tommy to check for synonymous changes. He's writing a paper. Only got the figure this afternoon.

Professor Edward C. Holmes FAA FRS
The University of Sydney

Proximal_Origin_Emails.pdf, p. 44, 4 Feb 2020, 10:31 am UK



Proximal_Origin_Slack.pdf, p.14, 5 Feb 2020, 5:15 pm UK, and p.16, 10:14 pm UK



Actual figures from Lam et al.

Hence, that new pangolin data (based on the figures from Lam's draft paper) did not change much their evaluation of the passaging option. For Andersen, on the contrary, the revealed presence of these sequences in a lab could actually raise the possibility of undisclosed experimentations with pangolin sequences having taken place. Also, their quick exploration of the glycans features (a question asked by Collins) did not alter either their view.

So, while the co-authors started considering updating their report to mention the pangolins, the report's conclusions were not in question. What was clear, though, was that things could change quickly with yet more data.

> ***Garry (10:52 am UK)***
> *Perhaps say we are adding new information? See whether he wants to hold off. I suspect Bethesda will be sending it round already?*
>
> *I think we need to add a section about the pangolin and possibly something about whether the glycan sites are evidence of selection by an immune system?*
>
> *[source: Proximal_Origin_Emails.pdf, p.43]*

On 5 Feb 2020, at 9:04 pm, Edward Holmes [redacted] wrote:

I think we might have dropped the ball with this pangolin virus. I ignored it when I saw it didn't have the furin cleavage site. Should now check all the key sites.

Cheers,

Eddie

PROFESSOR EDWARD C. HOLMES FAA FRS
ARC Australian Laureate Fellow

THE UNIVERSITY OF SYDNEY
Marie Bashir Institute for Infectious Diseases & Biosecurity,
School of Life & Environmental Sciences and School of Medical Sciences,
The University of Sydney | Sydney | NSW | 2006 | Australia
T [redacted]
E [redacted]

On 5 Feb 2020, at 8:44 pm, Andrew Rambaut [redacted] wrote:

I think we need to keep this document live and update it as necessary. Give it a date and version number.

Andrew

On 5 Feb 2020, at 9:52 pm, Andrew Rambaut [redacted] wrote:

Perhaps say we are adding new information? See whether he wants to hold off. I suspect Bethesda will be sending it round already?

I think we need to add a section about the pangolin and possibly something about whether the glycan sites are evidence of selection by an immune system?

A.

On 5 Feb 2020, at 10:47, Edward Holmes [redacted] wrote:

The animals are from Guangdong and Guangxi. Seized by customs. Need those Hubei pangolins.

Should I tell Jeremy to hold on sending the summary out to the group while we investigate more or does that really matter? He did say that more wildlife needed to be studied. He's sent it to the Bethesda boys.

Best wishes,

Eddie

Proximal_Origin_Emails.pdf, p.28, and p.43, 5 Feb 2020, 10::52 am UK

Garry next pitched in and reiterated the importance of carefully considering all data first. He contrasted the approach used here with the way Drosten signed a correspondence in the NEJM on asymptomatic transmission (5 Feb, 11:24 am UK). That rushed study was published on 30 Jan and was quickly picked up by Fauci in some public statement, but unfortunately, it had to be walked back. The whole story about that embarrassing mistake involving Drosten was related only two days previously (3 Feb) in a Science article.

On 5 Feb 2020, at 9:52 pm, Andrew Rambaut [redacted] wrote:

Perhaps say we are adding new information? See whether he wants to hold off. I suspect Bethesda will be sending it round already?

I think we need to add a section about the pangolin and possibly something about whether the glycan sites are evidence of selection by an immune system?

A.

On 5 Feb 2020, at 10:47, Edward Holmes [redacted] wrote:

The animals are from Guangdong and Guangxi. Seized by customs. Need those Hubei pangolins.

Should I tell Jeremy to hold on sending the summary out to the group while we investigate more or does that really matter? He did say that more wildlife needed to be studied. He's sent it to the Bethesda boys.

Best wishes,

Eddie

On Wed, Feb 5, 2020 at 6:24 AM Garry, Robert F [redacted] wrote:

Worth pointing out - if the crackpot charge comes re cell culture hypothesis - that we are discussing this in private amongst experts.

Clearly and I think correctly our approach has been different than say the flawed nejm paper -see science feb3 - about asymptomatic infection - Drosten was on the rushed out paper Tony got tripped up. Public error and pretty important. IMO they should retract the paper to send clear message.

Sent from my iPhone

On Feb 5, 2020, at 5:18 AM, Edward Holmes [redacted] wrote:

REV0001897

External Sender. Be aware of links, attachments and requests.

The pangolin virus looks like it might fall in roughly the same place on the tree as those new bat virus trees I put on Slack. Don't have the seqs of those yet.

Professor Edward C. Holmes FAA FRS
The University of Sydney

Proximal_Origin_Emails.pdf, p.42–43, 5 Feb 2020

Sideline: Holmes and Lam et al.

Holmes later explained in a NY Times article that he joined the Lam *et al.* team (of HK and Beijing scientists) a few days after the Farrar call, to help them write their paper.

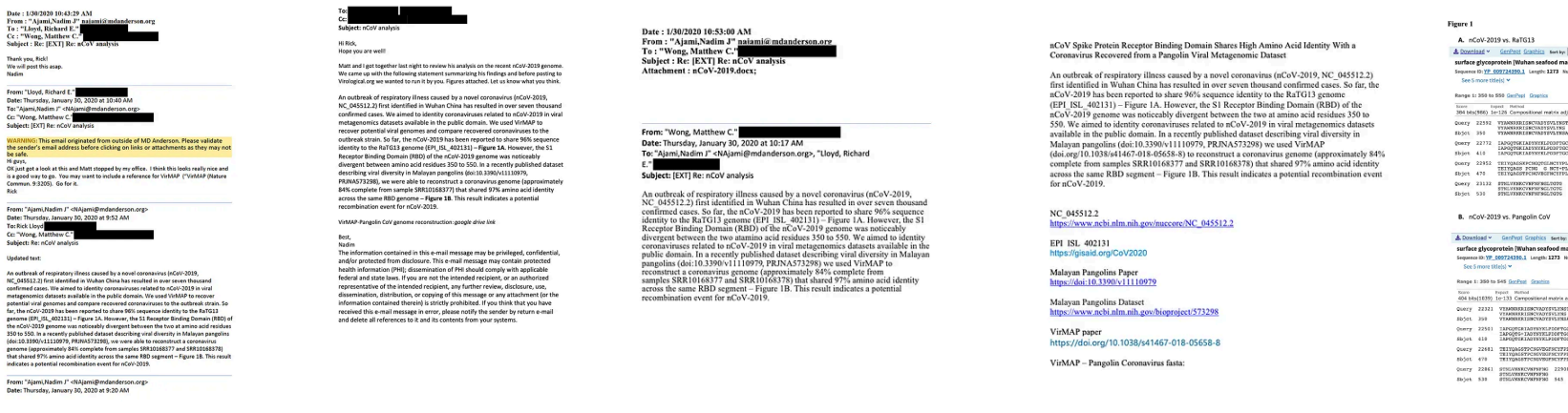
Effectively, on 5 Feb, Holmes was being drawn into a last-minute analysis of the pangolin sequences for Tommy Lam. The figures sent by Tommy Lam show a very well advanced paper — they are unchanged in the final version. Soon after, on 7 Feb, Lam *et al.* ('Identifying SARS-CoV-2-related coronaviruses in Malayan pangolins') was sent to Nature in some initial version. It was then sent to bioRxiv on 13 Feb and released as a preprint there on 18 Feb. One may conclude that Holmes provided essentially some review of the analysis and some recommendations for the text at the very end of the drafting process.

Insight: Old Pangolins in new clothes

The pangolin as animal host theory did not elicit much of an interest among the co-authors until Holmes started getting the draft of Lam *et al.* via his

friend, Tommy Lam, on **5 Feb**.

However, already on **31 Jan**, Andersen had noticed a post by an account called ‘torptube’, posted on Rambaut’s site virological.org earlier that day. Behind the then anonymous torp(edo)tube, was Matthew Wong, a bioinformatician in Joseph Petrosino’s lab at Baylor College of Medicine. Wong had picked up some RBD similarity with SARS-CoV-2 in the dataset uploaded on GISAID as part of a little-noticed paper published in Oct 2019, Liu *et al*.



Details of the review of Wong’s analysis in his lab

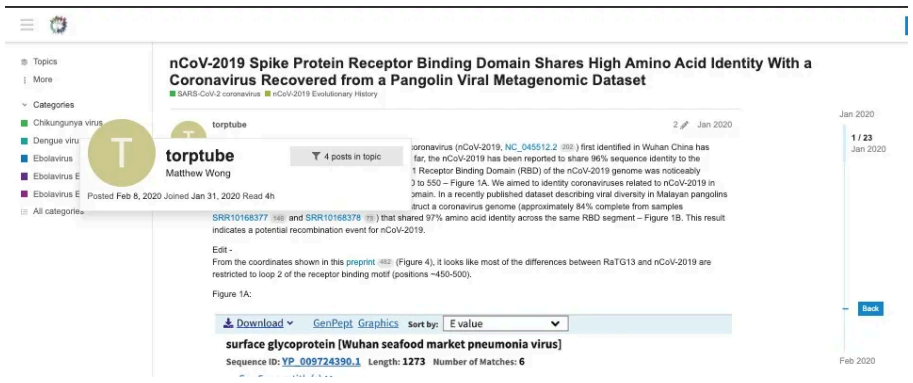
Wong had never before, and since then, posted on any other subject on virological.org. His last post was on 8 Feb 2020, just after the SCAU news, when he divulged his actual name. The events were later described in a NY Times article published soon after the SCAU news.

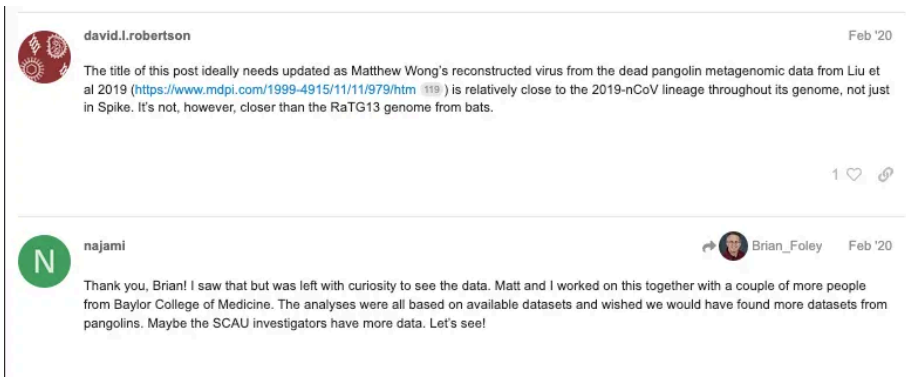
We don’t know if Andersen mentioned Wong’s post during the Farrar’s call on 1 Feb. What is sure is that there a very brief note, ‘Pangolin virus?’, in the second version of the call summary, written on **2 Feb**. Also, starting with the following version on 3 Feb, all versions would include the need for an animal host, where the FCS could have evolved, to have a very high population density if such evolution was to be possible (the alternatives being evolution in human after a zoonotic jump, or passaging). That implicitly rejected the evolution of the FCS in the solitary pangolins. Hence, the high RBD homology in the pangolin sequences would have been considered as a distant side-show in the evolution of SARS-CoV-2. There was accordingly no direct mention at all of any pangolin in the P.O. drafts until **7 Feb** (v6.a).

As Alina Chan and Shing Hei Zhan would later show in a critical review, Liu *et al.* (published in October 2019 and eventually noticed by Wong and Andersen), Lam *et al.* (which excited so much interest through Holmes) and Kangpeng Xiao *et al.* (which made worldwide news with its false 99% claims), all actually used the very same batch of pangolins, even if they confusingly used different names for the samples.

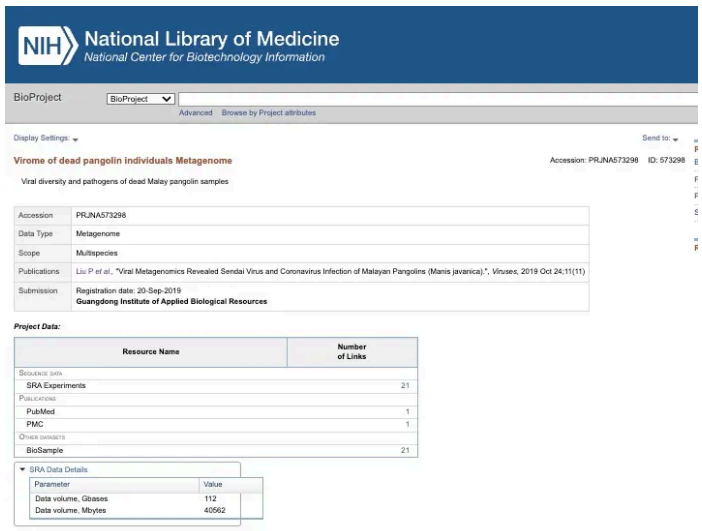
Holmes would later get rather angry at Chan, when Shen *et al.*, yet another take at these ‘new old’ pangolin sequences, for which he was a co-author alongside Kangpeng Xiao (of SCAU), was not accepted by Nature

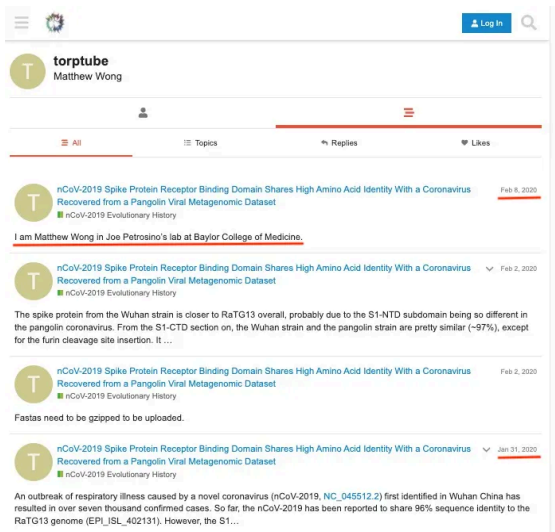
Microbiology, following the questions raised by Chan about data provenance and quality in her critical review.

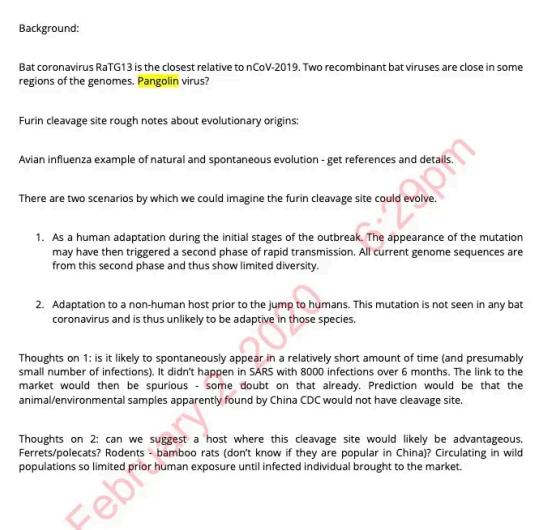




https://virological.org/t/ncov-2019-spike-protein-receptor-binding-domain-shares-high-amino-acid-identity-with-a-coronavirus-recovered-from-a-pangolin-viral-metagenomic-dataset/362,







https://www.ncbi.nlm.nih.gov/bioproject/PRJNA573298/, SSCP-Drafts-of-Proximal-Origin.pdf, p. 7

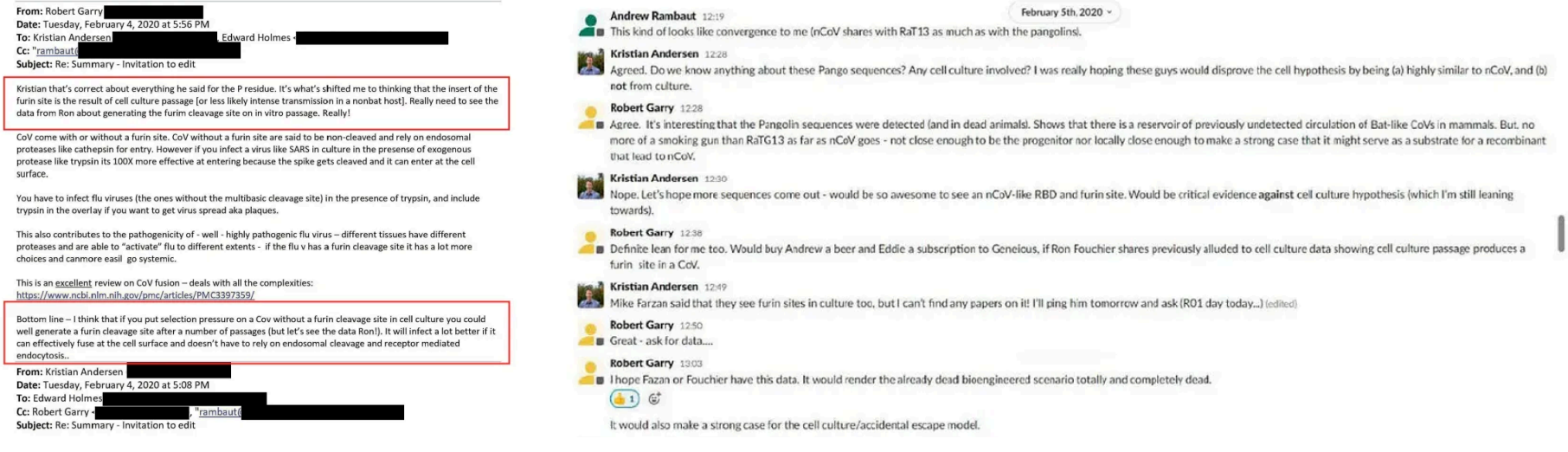
10.5 The importance of Fouchier's and Farzan's data (5 Feb, afternoon)

While they were discussing the passaging option, left unaffected by the 'new old' pangolin data in whatever fashion it was being re-analysed, the co-authors discussed on Slack and by email how Farzan and Fouchier had previously alluded on having, or knowing, of experiment data '*showing cell culture passage produces a furin site in a CoV [coronavirus]*'.

Being able to check that data was indeed becoming essential, if one started suspecting some possible passaging of (undisclosed) pangolin viruses in a lab, as the co-authors did.

> ***Garry (5 Feb, 5:38 pm UK):***
> *Would buy Andrew a beer and Eddie a subscription to Geneious [note: a leading sequence analysis software], if Ron Fouchier* ***shares previously alluded to cell culture data showing cell culture passage produces a furin site in a CoV****.*
>
> ***Andersen (5 Feb, 5:49 pm UK):***
> ***Mike Farzan said that they see furin sites in culture too,*** *but can't find any papers on it! I'll ping him tomorrow and ask (RO1 day today…).*

> ***Garry (5 Feb, 6:03 pm UK):***
> *I hope Fazan or Fouchier have this data. It would render the already dead bioengineered scenario totally and completely dead.*
> ***It would also make a strong case for the cell culture/accidental escape model.***



From: Robert Garry [redacted]
Date: Tuesday, February 4, 2020 at 5:56 PM
To: Kristian Andersen [redacted], Edward Holmes [redacted]
Cc: "rambaut@[redacted]
Subject: Re: Summary - Invitation to edit

Kristian that's correct about everything he said for the P residue. It's what's shifted me to thinking that the insert of the furin site is the result of cell culture passage [or less likely intense transmission in a nonbat host]. Really need to see the data from Ron about generating the furim cleavage site on in vitro passage. Really!

CoV come with or without a furin site. CoV without a furin site are said to be non-cleaved and rely on endosomal proteases like cathepsin for entry. However if you infect a virus like SARS in culture in the presense of exogenous protease like trypsin its 100X more effective at entering because the spike gets cleaved and it can enter at the cell surface.

You have to infect flu viruses (the ones without the multibasic cleavage site) in the presence of trypsin, and include trypsin in the overlay if you want to get virus spread aka plaques.

This also contributes to the pathogenicity of - well - highly pathogenic flu virus – different tissues have different proteases and are able to "activate" flu to different extents - if the flu v has a furin cleavage site it has a lot more choices and canmore easil go systemic.

This is an excellent review on CoV fusion – deals with all the complexities:
https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3397359/

Bottom line – I think that if you put selection pressure on a Cov without a furin cleavage site in cell culture you could well generate a furin cleavage site after a number of passages (but let's see the data Ron!). It will infect a lot better if it can effectively fuse at the cell surface and doesn't have to rely on endosomal cleavage and receptor mediated endocytosis..

From: Kristian Andersen [redacted]
Date: Tuesday, February 4, 2020 at 5:08 PM
To: Edward Holmes [redacted]
Cc: Robert Garry [redacted], "rambaut@[redacted]
Subject: Re: Summary - Invitation to edit

February 5th, 2020

Andrew Rambaut 12:19
This kind of looks like convergence to me (nCoV shares with RaT13 as much as with the pangolins).

Kristian Andersen 12:28
Agreed. Do we know anything about these Pango sequences? Any cell culture involved? I was really hoping these guys would disprove the cell hypothesis by being (a) highly similar to nCoV, and (b) not from culture.

Robert Garry 12:28
Agree. It's interesting that the Pangolin sequences were detected (and in dead animals). Shows that there is a reservoir of previously undetected circulation of Bat-like CoVs in mammals. But, no more of a smoking gun than RaTG13 as far as nCoV goes - not close enough to be the progenitor nor locally close enough to make a strong case that it might serve as a substrate for a recombinant that lead to nCoV.

Kristian Andersen 12:30
Nope. Let's hope more sequences come out - would be so awesome to see an nCoV-like RBD and furin site. Would be critical evidence **against** cell culture hypothesis (which I'm still leaning towards).

Robert Garry 12:38
Definite lean for me too. Would buy Andrew a beer and Eddie a subscription to Geneious, if Ron Fouchier shares previously alluded to cell culture data showing cell culture passage produces a furin site in a CoV.

Kristian Andersen 12:49
Mike Farzan said that they see furin sites in culture too, but I can't find any papers on it! I'll ping him tomorrow and ask (R01 day today...) (edited)

Robert Garry 12:50
Great - ask for data....

Robert Garry 13:03
I hope Fazan or Fouchier have this data. It would render the already dead bioengineered scenario totally and completely dead.
1

It would also make a strong case for the cell culture/accidental escape model.

Proximal_Origin_Emails.pdf, p.23, 4 Feb 2020 and Proximal_Origin_Slack.pdf, p.15, 5 Feb 2020

Sideline: On the passaging data

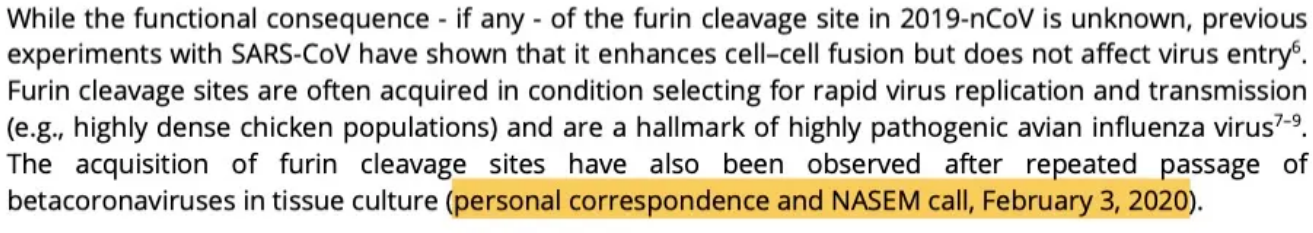
One should not assume that Fouchier and Farzan necessarily meant that they had data hinting at the possible acquisition of an FCS in passaging experiments that they had done, since it is not impossible that they were referring to experiments published by other teams. This said, the wording of one of Garry's Slack messages may point towards actual experiments by Farzan and Fouchier:

> ***Garry (7 Feb 1:24 am):***
> *Makes me think the cell culture passage scenarios possible/probably assuming this has in fact been* ***observed before by*** *Farzan and Fouchier.*

In the end, the P.O. drafts and the final published in Nature Medicine mentioned **fairly old** papers, not authored by Farzan or Fouchier, when discussing the passaging scenario.

- Version v4 of the P.O. draft (see Appendix A), written on 3 Feb, mentioned personal correspondence of the co-authors and the NASEM call of that very day as sources for knowledge of these experiments.
  (To be exhaustive, that data was also most likely mentioned during the Farrar call — but the existence of that call was not meant to be made public. The NASEM call itself was public.)
- Intermediate versions (such a v12) had references to a 30-year-old paper Li S.Q., Orlich M & Rott R (1990), and Shengqing, Y. *et al.* from 2002 in addition to Ito, T. *et al.* from 2001.

- The final version published in Nature Medicine has a single reference to a nearly **20-year-old paper** by a Japanese team instead (Ito, T. *et al.* (2001), with Kawaoka as co-author).



While the functional consequence - if any - of the furin cleavage site in 2019-nCoV is unknown, previous experiments with SARS-CoV have shown that it enhances cell–cell fusion but does not affect virus entry[6]. Furin cleavage sites are often acquired in condition selecting for rapid virus replication and transmission (e.g., highly dense chicken populations) and are a hallmark of highly pathogenic avian influenza virus[7-9]. The acquisition of furin cleavage sites have also been observed after repeated passage of betacoronaviruses in tissue culture (personal correspondence and NASEM call, February 3, 2020).

v4 of P.O. draft — 3 Feb 2020

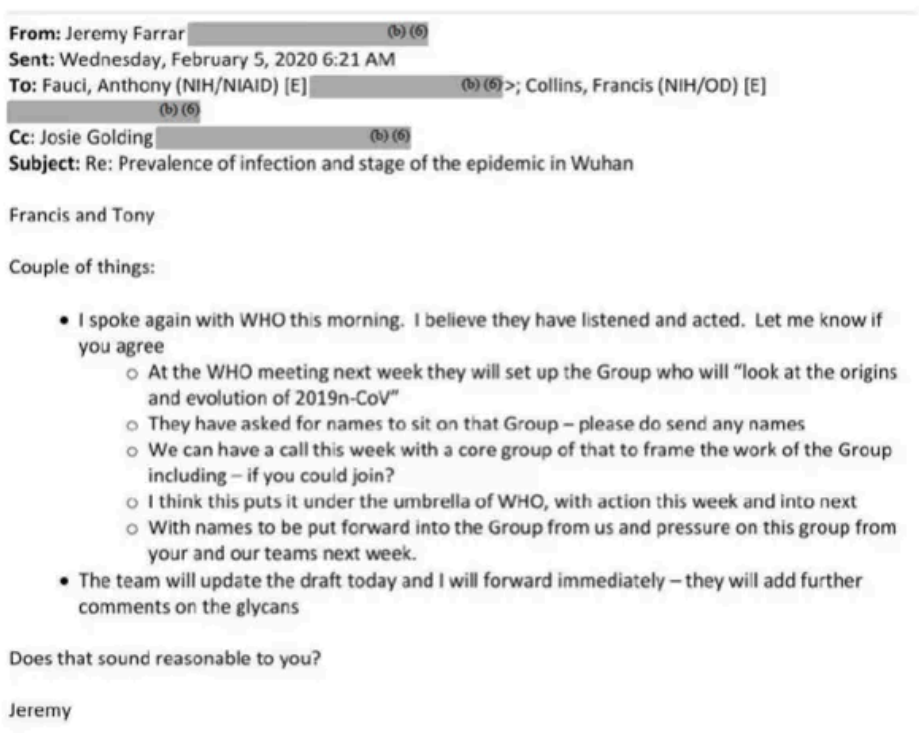
11. WHO agrees to get involved (5–7 Feb)

11.1 WHO willing to onboard the 'evolutionary origin' question (5 Feb)

During a call with the WHO, on 5 Feb morning (UK time), Farrar got the green light to set up a group at a main WHO meeting the following week (11–12 Feb), that will "**look at the origins and evolution of 2019n-CoV**".

It is likely that the group itself was meant to be the occasion of the 'face-to-face' discussion between the Chinese and International side (see 10.2, Key Insight).

Farrar penned an email the same day to Fauci and Collins, to keep them updated and discuss possible names for that group, that the WHO now needed. Collins replied by suggesting the people on the Farrar call, plus some new names.



From: Jeremy Farrar (b) (6)
Sent: Wednesday, February 5, 2020 6:21 AM
To: Fauci, Anthony (NIH/NIAID) [E] (b) (6); Collins, Francis (NIH/OD) [E] (b) (6)
Cc: Josie Golding (b) (6)
Subject: Re: Prevalence of infection and stage of the epidemic in Wuhan

Francis and Tony

Couple of things:

- I spoke again with WHO this morning. I believe they have listened and acted. Let me know if you agree
  - At the WHO meeting next week they will set up the Group who will "look at the origins and evolution of 2019n-CoV"
  - They have asked for names to sit on that Group – please do send any names
  - We can have a call this week with a core group of that to frame the work of the Group including – if you could join?
  - I think this puts it under the umbrella of WHO, with action this week and into next
  - With names to be put forward into the Group from us and pressure on this group from your and our teams next week.
- The team will update the draft today and I will forward immediately – they will add further comments on the glycans

Does that sound reasonable to you?

Jeremy

From: Fauci, Anthony (NIH/NIAID) [E]
Sent: Wednesday, February 5, 2020 5:25 PM
To: Jeremy Farrar (b) (6) Collins, Francis (NIH/OD) [E] (b) (6)
Cc: Josie Golding (b) (6) Tabak, Lawrence (NIH/OD) [E] (b) (6)
Subject: RE: Prevalence of infection and stage of the epidemic in Wuhan

Jeremy:
Thanks for the note. Looks like things are moving along with WHO. I will list below a number of names for potential members of the working group to examine the evolutionary origin of the 2019-nCoV in addition to the individuals who were on the call with us last Saturday:

Harold Varmus – Weill Cornell Medical Center – New York City
Feng Zhang – MIT (CRISPR expert)
Joseph DeRisi - Chan Zuckerberg (CZ) BioHub (he's paying close attention to the Wuhan strain vs other bat viruses and the SARS virus)
Don Ganem - University of California at San Francisco (UCSF) and the CZ BioHub (knows more about hepadnaviruses but an outstanding clinical and basic virologist)
John Coffin - Tufts and National Cancer Institute, NIH (worked out the confusion over the alleged Chronic Fatigue Syndrome retrovirus that proved to be a xenotropic MLV)
Eugene Koonin - NCBI/National Libreary of Medicine, NIH; https://www.ncbi.nlm.nih.gov/research/groups/koonin/)
Wayne Hendrickson – Columbia University and the New York Structural Biology Center
Gary Nabel – Sanofi (Boston)

Best regards,
Tony

farrar-fauci-comms.pdf, p.74, 5 Feb 2020 and p.73, 6 Feb 2020

11.2 Koopmans and WHO recommend publication (4–5 Feb)

On 4 Feb (evening UK), Farrar had a call with Koopmans, who was part of the 'bounce-off' group and was regularly working with the WHO (most recently part of the process that led to the decision to eventually declare a PHEIC on 30 Jan). Farrar discussed the situation with Koopmans, then the following day, 5 Feb, Farrar had a call with the WHO in the morning as we just saw (see 11.1).

During the two calls, both Koopmans and the WHO wondered whether **going directly for publication** could be the best option, in order to stop the spread of conspiracies about the origin.

11.3 Farrar sounds out the co-drafters about publication (6 Feb)

On 6 Feb, 7:36 am UK, Holmes shared Farrar's question about going for publication and asked his co-authors for feedback ('Thoughts?'). This was a potential massive change of plan, except maybe for Holmes, who had always eyed publication:

> *From Jeremy.*
> *[..]*
> *Talking with Marion last night and with the WHO meeting next week….both wondering whether actually publishing this sooner, but ruthlessly on the science….is worthwhile to put that flag down…"*
> *[Proximal_Origin_Emails.pdf, p.35, 5 Feb 2020 (7:36 am UK)*

11.4 Farrar asks for some toning down of the manuscript (6 Feb)

At the same time, Holmes relayed a demand from Jeremy to change the tone of the current draft:

> *From Jeremy.*
> *"Do you think in the report….possible to dampen down further the 'conspiracy' idea and make totally neutral?*
> *[..]*
>
> *[Proximal_Origin_Emails.pdf, p.35, 5 Feb 2020 (7:36 am UK)*

What constituted a conspiracy, per Jeremy's request, may not have been very consistent for all those involved in the discussions. But at least for Farrar and the WHO at that time and in that context, it would be the bioweapon stories, the HIV insert stories, and the direct sequence editing hypothesis. **This would not include passaging, which was not yet in the open anyway**. As already seen, while the draft promptly dropped the mention of 'conspiracy theories' in the opening section on 7 Feb (v6.a), it kept the possibility of passaging untouched.

It is not clear whether Farrar took the initiative of asking for that dampening-down, or if this was in caused by some initial feedback received from the WHO. In the later case, we could imagine how, when Farrar talked to them

on 5 Feb (after maybe forwarding them the current draft), the WHO could have been more than happy to try to get something published if the tone was right, based on the fact that the bioweapon/deliberate engineering option had been rejected by Farrar's experts.

Message

From: Edward Holmes [redacted]
Sent: 2/6/2020 2:36:30 AM
To: Kristian G. Andersen [redacted]
CC: Garry, Robert F [redacted]; Andrew Rambaut [redacted]
Subject: Re: Summary - Invitation to edit

From Jeremy.

"Do you think in the report....possible to dampen down further the 'conspiracy' idea and make totally neutral?

Talking with Marion last night and with the WHO meeting next week....both wondering whether actually publishing this sooner, but ruthlessly on the science....is worthwhile to put that flag down..."

Thoughts?

PROFESSOR EDWARD C. HOLMES FAA FRS
ARC Australian Laureate Fellow

Proximal_Origin_Emails.pdf, p.35, **5 Feb 2020** (7:36 am UK)

Sideline: A possible misunderstanding

On 4 Feb, Koopmans would most likely not have seen yet the draft of the report written by Andersen, Rambaut, Garry and Holmes, when she suggested to Farrar to go for publication.

- We know that on 4 Feb, Farrar's plan was to send the existing draft to Fauci and Collins first, get their feedback, then — and only after that — loop the WHO in (see 10.1).
- The email records point to Farrar only sharing the latest draft with the full call-group for the first time on 8 Feb (see 13.1).

Hence, Koopmans could well have been unaware that passaging was left wide open in that draft, when she was suggesting to go for publication.

As to the WHO, Farrar had that call with them on 5 Feb, during which he sounded the WHO about taking on the origin question. Since Farrar already had positive feedback on the draft from Fauci and Collins by that time, he may have had time to circulate it with some people at the WHO before the call.

If so, whether they paid attention to the passaging option, or even noticed it, or discussed it in the call with Farrar, is not clear. In any case, they would be trusting Farrar and his efforts to help them get access, while dealing with their own urgent priorities.

11.5 The co-drafters discuss Farrar's proposal (7 Feb, early morning UK)

After Holmes shared Farrar's question about going to publication *('Thoughts?)*, the co-drafters gathered their feedback on that idea. **In an essential quick exchange on Slack,** we see them trying to balance following the evidence against the high risk of some backlash:

▶️ **Garry** explained that the pangolin sequences to this day are simply too evolutionary distant, and further pointed to the importance of getting access to experimental data mentioned by Farzan and Fouchier about the possibility of evolving an FCS in coronavirus via passaging:

> ***Garry (7 Feb, 1:24 am):***
> *I'm thinking mostly about the PRRA to generate the furin site. Relatively easy to drop 12 bases in.*
> *The proline is the hang-up — why add that? Makes me think the cell culture passage scenarios possible/probably assuming this has in fact been observed before by Farzan and Fouchier.*

▶️ **Rambaut** agreed with that analysis:

> ***Rambaut (7 Feb, 1:34 am UK):***
> *I am quite convinced it has been put there by evolution (whether natural selection or artificial [i.e.: passaging]).*

▶️ **Andersen** then noted that the public controversy would likely amplify over the following two weeks, but that without more early cases data and an intermediate host, the question should simply be left open:

> ***Andersen (7 Feb, 1:37 am UK):***
> *Agreed — this'll amplify over the next couple of weeks. I just wish there was a way to conclusively say one or the other, but without that intermediate host or very earlier cases. there's just no telling IMO. Which all means it's back to opinions - and honestly, for this type of question I don't think opinions are helpful — unless they have some damn strong science behind them.*

▶️ **Garry** then offered a fairly good summary of the situation, reiterating the real possibility of lab escape of a passaged virus, if supported by the passaging experiments mentioned by Farzan and Fouchier:

> ***Garry (7 Feb, 1:40 am UK):***
> *Three hypotheses here.*
> *1. not likely a bat virus right into a human — could have happen[ed] long ago but not so likely.*
> *2. Wet market — ok maybe an intermediate host. I think pangolin viruses*

> *sequences **still too far afield** but could be part of an animal circulation that generated the virus.*
> *3. **lab passage I'm open to and can't discount** — that just because |don't know the data and few others do. **Either furin sites have been generated or they haven't** [in passaging experiments, as mentioned per Farzan and Fouchier]. **If they have I'm suspicious of lab escape**, but not conclusive evidence. If furin sites have not been generated in cell culture passive, then were looking at either a long circulation or a very intense circulation in either humans or animals.*
>
> *There are obviously other possibilities including lab passage combined with some ill considered GOF research.*

▶️ Next, **Holmes**, the one in contact with Jeremy, linked it all back to Jeremy's motives. In doing so he mentioned how he was careful to send a toned down version that day, so as not to attract too much attention to the passaging option (the version sent is v6.a of Appendix A, with the mention of 'conspiracy theories' toned down as requested by Farrar on 6 Feb, see 11.4).

> ***Holmes (7 Feb, 1:51 am UK):***
> *'Yes, it's going to blow. Hence why Jeremy wants us thinking about putting something out. Hence the toned down version I just sent him'*

▶️ Garry and Holmes next pondered on the gambit that the WHO seemed to be willing to take.

> ***Garry (7 Feb, 1:51 am UK):***
> *'The public space is not the place to discuss this, which WHO should be aware of [realising that in itself will pour gas on the fire'],*
>
> ***Holmes** (7 Feb, 1:51 am UK)*
> *'I agree Bob. Very tricky."*

▶️ Not only could the mention of possible passaging create a backlash for the WHO, by adding oil to the fire, but it could also affect very negatively the reputation of the co-authors if they stuck their neck out too much. **Rambaut** gave the example of Adrian Gibbs, who had previously wrongly called a lab escape and had been pilloried for it:

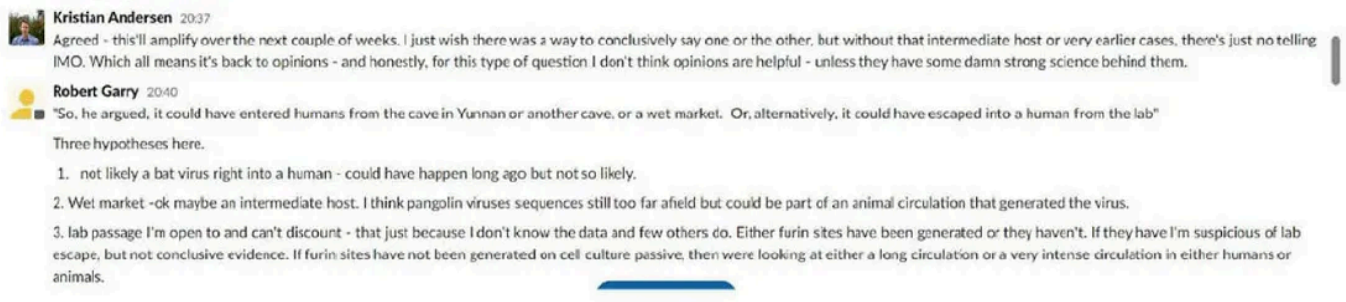
> ***Rambaut (7 Feb, 6:03 am UK):***
> *Remember when during the swine flu outbreak Adrian Gibbs suggested it was a lab escape? Caused a huge **shit show**.*

That latest comment by Rambaut was essentially a repeat of what he had said on 2 Feb. The concern had been on his mind for a while:

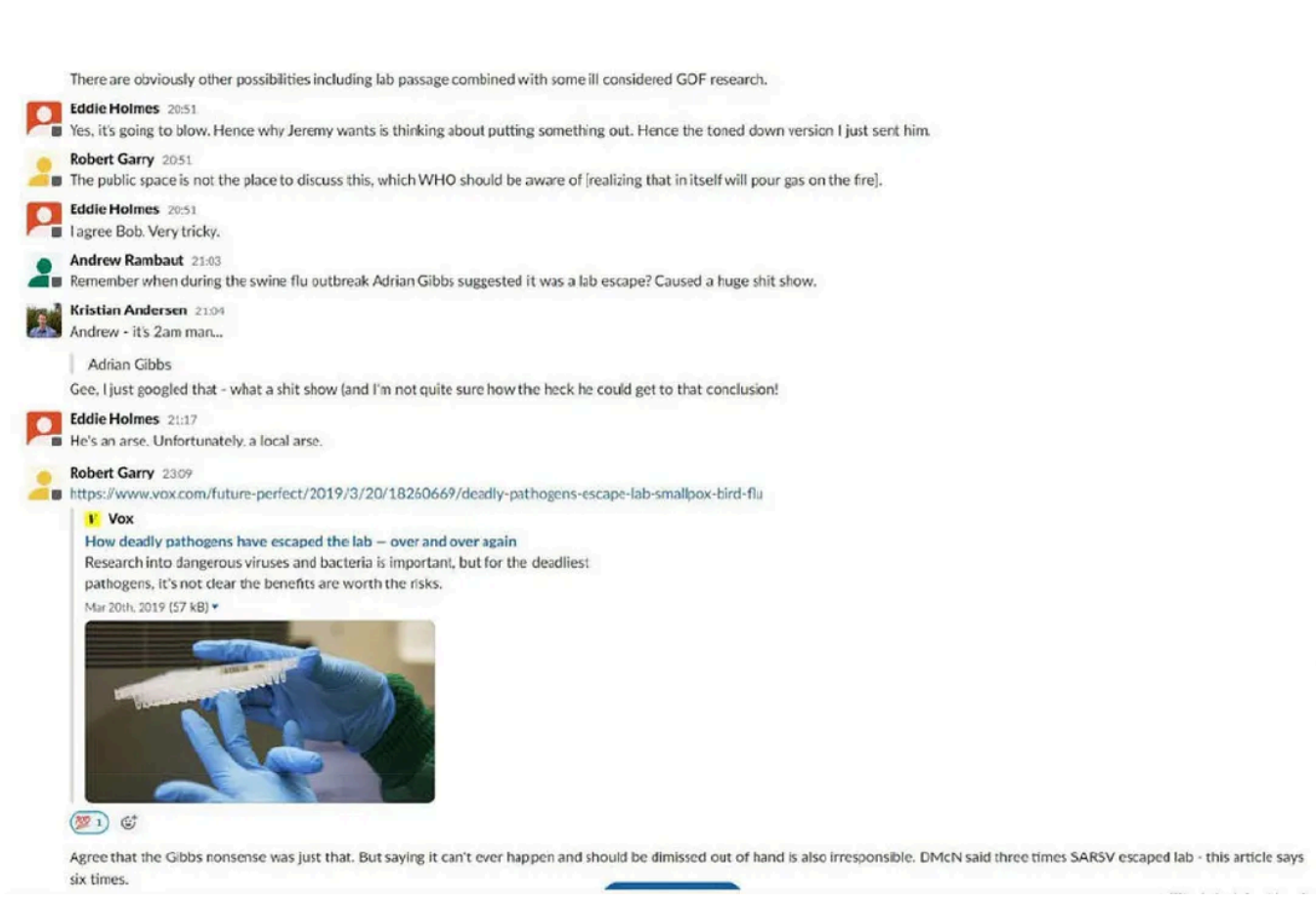
> ***Rambaut (2 Feb, 4:53 pm UK):***
> *Given the **shit show** that would happen if anyone serious accused the Chinese of even accidental release, my feeling is we should say that given there is no evidence of a specifically engineered virus, we cannot possibly distinguish between natural evolution and escape so we are content with ascribing it to natural processes.*

In essence, that conversation illustrates perfectly the dilemma the co-authors faced at that stage; Andersen, Garry and Rambaut still strongly believed in the possibility of passaging, but were under pressure from multiple fronts to play it down:

1. Farrar, acting for the WHO primarily, but also in agreement with Fauci and Collins, was under pressure to make the report public to clear the way for access to China,
2. Holmes, Farrar's main source of scientific truth, had his very own reasons to go for publication (yet another paper, among similar ones he was working on), and to encourage Farrar to do so.
3. Under the present toxic situation, Andersen, Garry and Rambaut knew only too well that sharing their reasonable doubts about passaging in a publication could be career-ending. **What they could write in a confidential scientific analysis to be shared with the WHO, would now have to be rewritten in the most defensive fashion if it was to be public.**



Proximal_Origin_Slack.pdf, p.19, 6 Feb 2020



Proximal_Origin_Slack.pdf, p.20, 6 Feb 2020

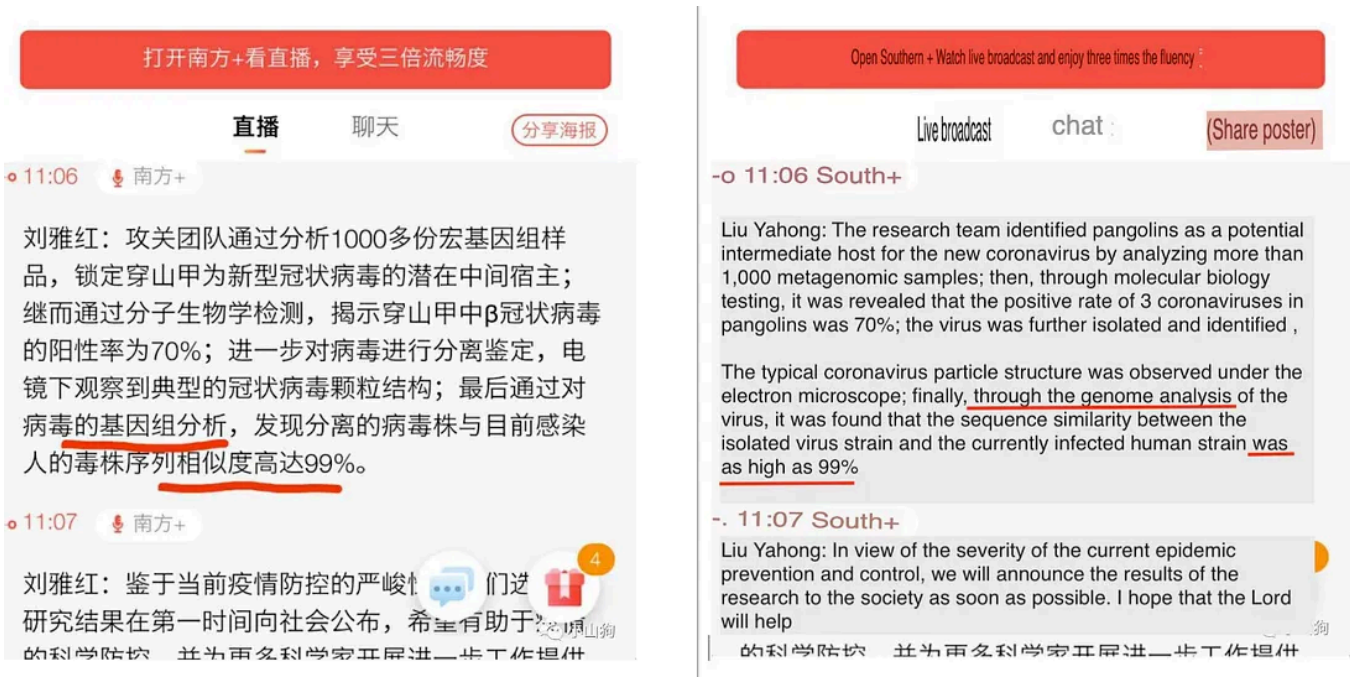
. . .

12. ‘Pangomania’ and a fateful decision (7 Feb)

12.1 The false Pangolin 99% homology news (7 Feb, 3 am UK)

a. The SCAU press conference (7 Feb, 3 am UK, 11 am in Shenzhen)
On 7 Feb, in a press conference hastily organised for 11 am in Shenzhen, the South China Agricultural University (SCAU) announced that it had found some pangolin coronavirus with a ‘99%’ homology to SARS-CoV–2. This would unleash a ‘**pangomania**’ in the words of Holmes.

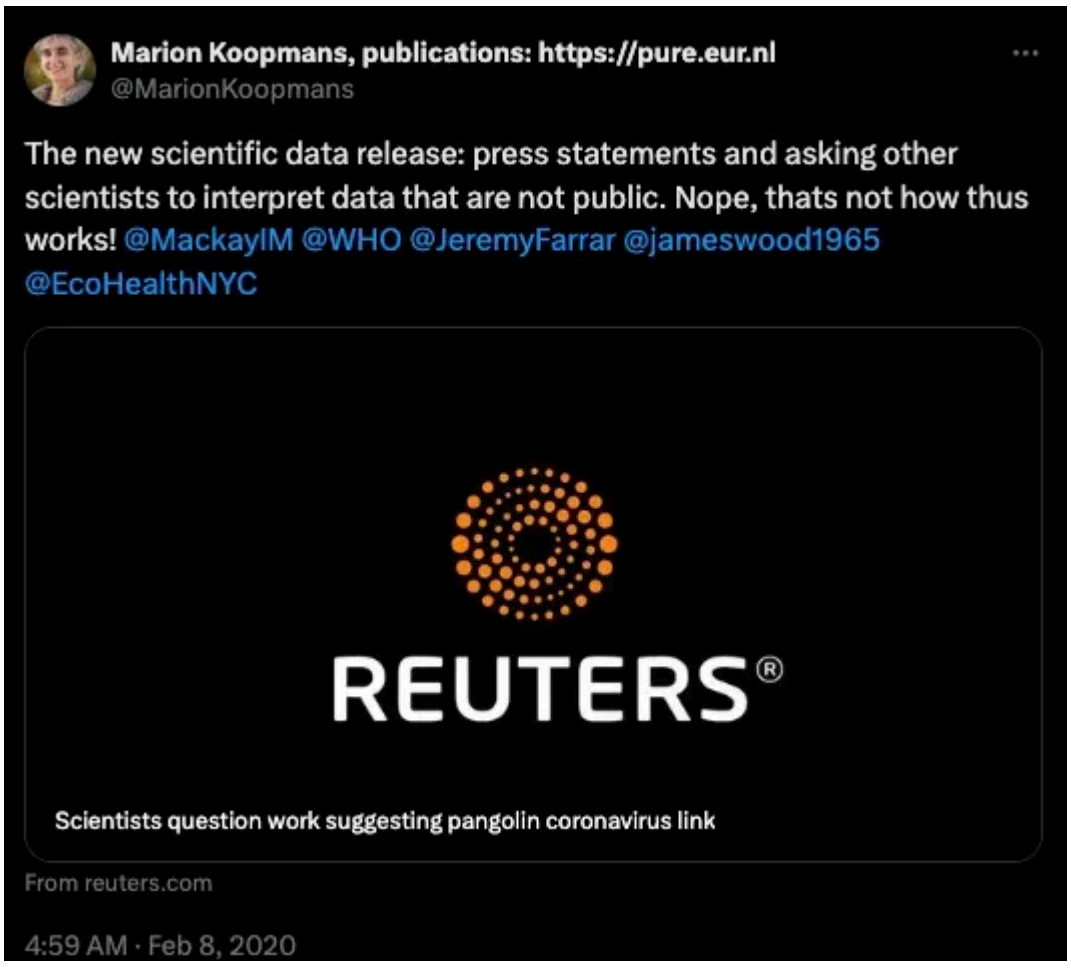
For all practical purposes, the (false) claim of 99% homology would have meant that **an essential missing link had been found.**



Official Weibo broadcast of SCAU press conference, 7 Feb, 11:06 am Shenzhen, Source: Little Mountain Dog’s blog.

b. Immediate Reservations (7 Feb)
The non-publication of the pangolin sequences at the time of the SCAU press conference on 7 Feb immediately raised issues, with some commentators noting that the press conference, in the absence of the hard data, was not the correct way of introducing findings. For instance, on the same day Reuters published an article that asked pointed questions about the supposed significance of the sequences, particularly when the data had not yet been made public.

Many of the scientists who were on the Farrar call group did also express their reservations, either privately on Slack and emails, or publicly. Koopmans went public when she tweeted her strong reservations at 4.53 pm UK on 7 Feb. In that tweet, which referred to the just published Reuters article, Koopmans pointed out that scientists should not draw conclusions until the sequences were published.



Reuters article updated on 8 Feb, published on 7 Feb, 4:53 pm UK

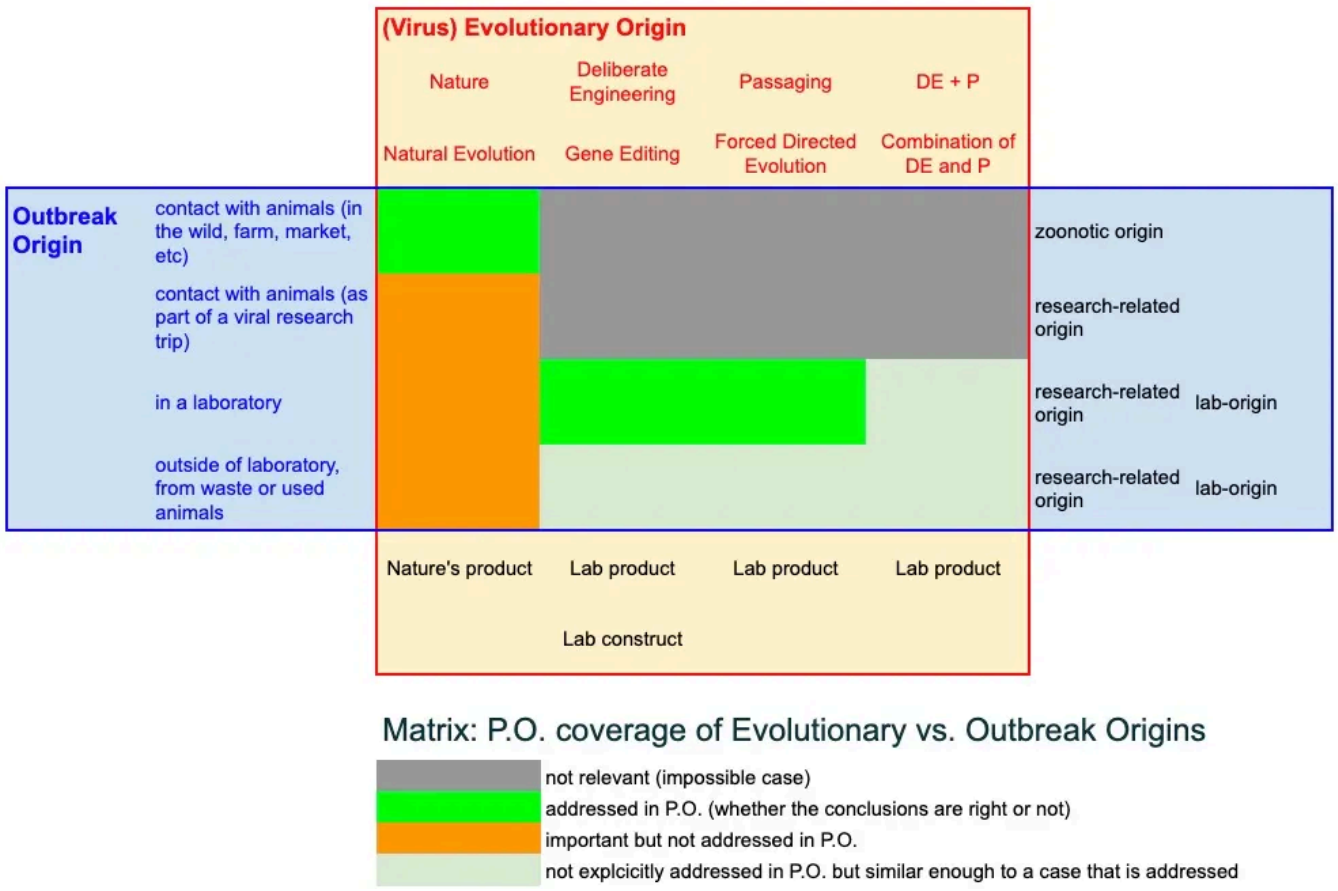
12.2 Farrar shares the latest draft with Fauci and Collins — Semantics get slippery (7 Feb, 6:09 am UK)

On 7 Feb, 6:09 am UK, Farrar shared the latest draft with Fauci and Collins 'Summary.Feb7.pdf', email subject: 'Revised Draft').

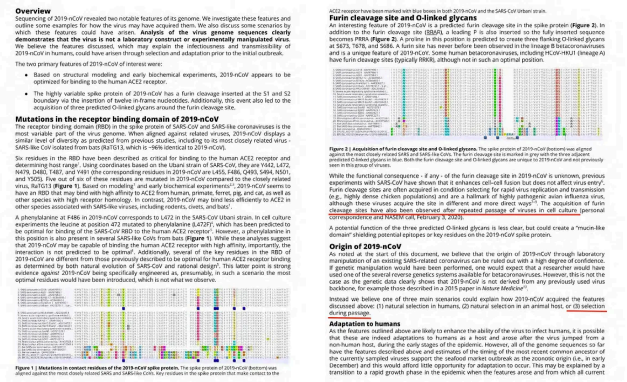
One change compared to the previous draft may seem at first innocuous, but was in fact rather portent: the '**Evolution of 2019-nCov**' section was renamed '**Origin of 2019-nCov**' (see v6a in Appendix A for the details of the evolution of the wording).

This could, and would soon, induce a confusion: the whole idea of the report was to write about the possible evolutionary origins of the virus based on the sequence, to answer doubts raised precisely by the sequence (RBD, restriction sites and FCS). But the sequence itself could not address a laboratory escape of a natural virus, since that scenario is a natural **evolutionary origin** of the virus combined with a **laboratory-origin** of the outbreak. And the situation is even worse for a sampling trip infection, a form of **research-related origin** (but not of lab-origin) of the outbreak combined again with a natural evolutionary origin of the vir**us.**

Hence, 'Origin of 2019-nCov' is misleading; the section was actually a discussion of the 'Evolutionary Origin of 2019-nCov'.



Matrix: P.O. coverage of Evolutionary vs. Outbreak Origins (source: author)



farrar-fauci-comms.pdf, p.66–71, 5 Feb 2020, third version shared with Farrar and Collins

12.3 Farrar updates Fauci and Collins about the SCAU news (7 Feb, 6.21 am UK)

Bouncing back on his latest email. Farrar shared with Fauci and Collins the just breaking (false) news about *'pangolin viruses that are 99% similar'*.

> *Plus two updates:*
> *- Reports coming out overnight that Chinese group have pangolin viruses that are 99% similar. This could be crucially important finding and if true could be the 'missing link' and explain a natural evolutionary link.*
> *- [..]*

Farrar sent that email at 6.21 am UK, only 12 minutes after sending the revised draft (hence the *'Plus'*), and less than 3 hours after the end of the SCAU conference. So Farrar most likely learnt of the SCAU conference soon after waking up, quite possibly from Holmes, when checking the latest items in his mailbox.

Since 6:21 am UK is 1:21 am EST, Collins only saw Farrar's email about 6 hours later, when he, in turn, checked his email in the morning. In his reply at 7:17 am EST (12:17 pm UK), Collins asked the right question about the presence or not of the FCS. Farrar answered about 6 hours later (6:54 pm UK) that he was chasing that FCS question.

From: Jeremy Farrar
Sent: Fri, 7 Feb 2020 18:54:20 +0000
To: Collins, Francis (NIH/OD) [E];Fauci, Anthony (NIH/NIAID) [E]
Cc: Tabak, Lawrence (NIH/OD) [E]
Subject: Re: Revised draft

I do not think so yet but ...chasing!

From: Francis Collins (b) (6)
Date: Friday, 7 February 2020 at 12:17
To: Jeremy Farrar (b) (6)>, "Fauci, Anthony (NIH/NIAID) [E]" (b) (6)
Cc: "Tabak, Lawrence (NIH/OD) [E]" (b) (6)
Subject: RE: Revised draft

Has the actual sequence of the pangolin coronavirus isolate been released? That will be VERY interesting. Does it have the furin cleavage site?

Francis

From: Jeremy Farrar (b) (6)
Sent: Friday, February 7, 2020 1:21 AM
To: Collins, Francis (NIH/OD) [E] (b) (6); Fauci, Anthony (NIH/NIAID) [E] (b) (6)
Subject: Re: Revised draft

Plus two updates

- Reports coming out overnight that Chinese group have pangolin viruses that are 99% similar. This would be a crucially important finding and if true could be the 'missing link' and explain a natural evolutionary link
- I have asked the Human Cell Atlas people to look at expression of the n-CoV in the nose/throat/respiratory tract/deep lunch tissue and if affected by age, smoking, pollution, gender, other.

farrar-fauci-comms.pdf, p.64, 7 Feb 2020 (US time)

12.4 Farrar aims for publication with Holmes (7 Feb, 6:26 am UK)

a. A timely email exchange:

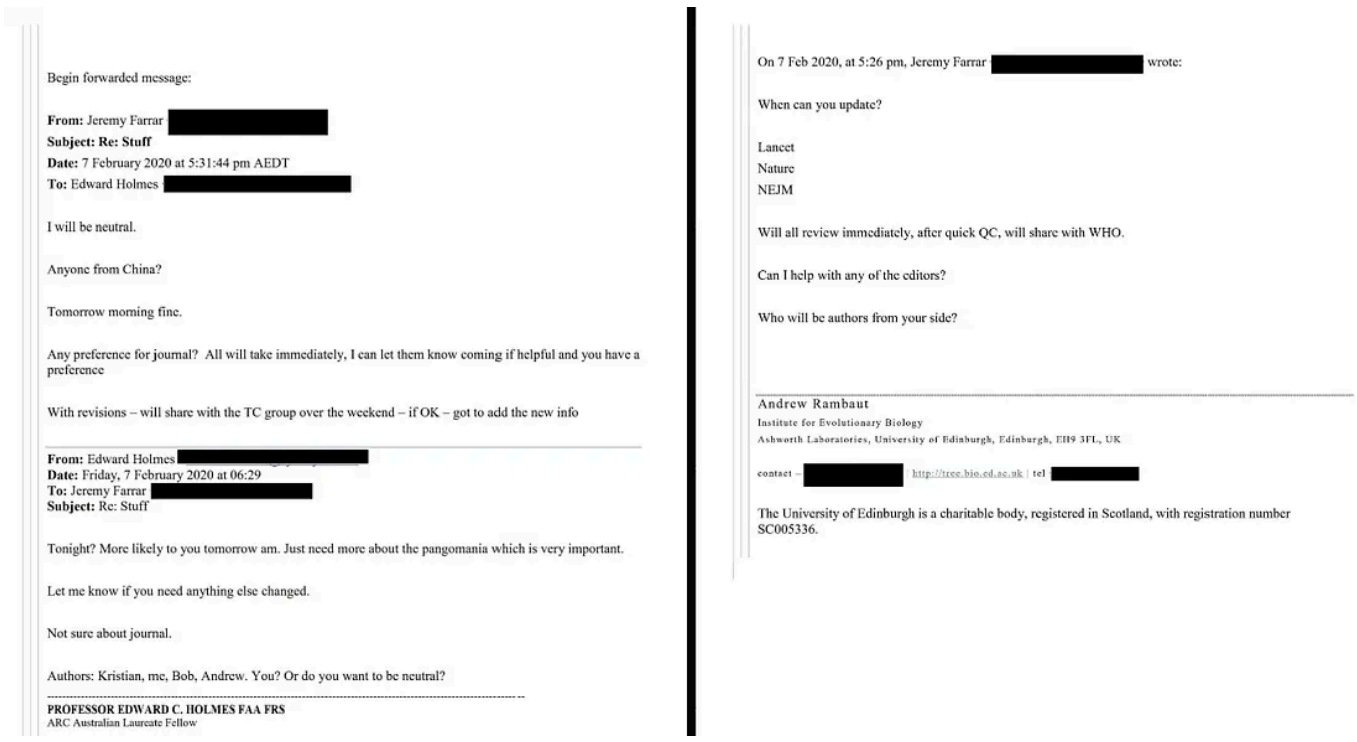
Only 5 minutes after sending that email to Fauci and Collins, and following the suggestions from WHO and Koopmans a few days earlier, Farrar explained to Holmes that he would like to go for publication and started planning for it.

In that short email exchange with Holmes that lasted **a few minutes** (Friday 7 Feb, 6:26 am UK, 5:26 pm Sydney), Farrar planned the immediate next stages for the publication of the draft report:

1. Agree on the acknowledged authors for the report.
   - This must exclude Farrar, who wants to remain ostensibly 'neutral'.
   - Farrar also suggested that the authors should include someone from China *('Anyone from China?'),* thus demonstrating an intended positioning of the report as an act of scientific cooperation between the WHO, the West and China.
2. Report to be updated to keep it up to date with the sudden interest in pangolins (*'Just need more about the pangomania which is very important'*, in Holmes' words).
3. Contact Lancet, Nature and NEJM once the report is updated. According to Farrar, these publications would consider the manuscript at the drop of a hat, most likely due to Farrar's privileged relations with them and to the essential role of such publication in the WHO game-plan.

4. After a quick Quality Check (QC), **Farrar to share the updated version with the WHO.**

5. Farrar to check with the 'TC group' over the weekend (meaning either the Tony Call group, or — more likely — the TeleConference group), i.e. the 1 Feb call group in any case.



Begin forwarded message:

From: Jeremy Farrar [redacted]
Subject: Re: Stuff
Date: 7 February 2020 at 5:31:44 pm AEDT
To: Edward Holmes [redacted]

I will be neutral.

Anyone from China?

Tomorrow morning fine.

Any preference for journal? All will take immediately, I can let them know coming if helpful and you have a preference

With revisions – will share with the TC group over the weekend – if OK – got to add the new info

---

From: Edward Holmes [redacted]
Date: Friday, 7 February 2020 at 06:29
To: Jeremy Farrar [redacted]
Subject: Re: Stuff

Tonight? More likely to you tomorrow am. Just need more about the pangomania which is very important.

Let me know if you need anything else changed.

Not sure about journal.

Authors: Kristian, me, Bob, Andrew. You? Or do you want to be neutral?

PROFESSOR EDWARD C. HOLMES FAA FRS
ARC Australian Laureate Fellow

On 7 Feb 2020, at 5:26 pm, Jeremy Farrar [redacted] wrote:

When can you update?

Lancet
Nature
NEJM

Will all review immediately, after quick QC, will share with WHO.

Can I help with any of the editors?

Who will be authors from your side?

Andrew Rambaut
Institute for Evolutionary Biology
Ashworth Laboratories, University of Edinburgh, Edinburgh, EH9 3FL, UK

contact – [redacted] http://tree.bio.ed.ac.uk | tel [redacted]

The University of Edinburgh is a charitable body, registered in Scotland, with registration number SC005336.

Proximal_Origin_Emails.pdf, p.49–50, 7 Feb 2020

For context, one must keep in mind that the Farrar-Holmes connection is the privileged one here: Farrar reached out to Holmes, who calls him his (intel) 'handler', and effectively takes care of managing the interface with the drafting team, while also bringing important working relations with Chinese scientists. It is thus logical that the two of them reviewed the strategy and decided to try to go for publication.

b. What we are missing:
Very importantly, we note that in his quick exchange with Holmes, Farrar did not discuss his decision to go for publication, while Holmes never questioned it, nor asked for any details.
That email of Farrar at 6:26 am (UK), just 5 minutes after he shared the SCAU news with Fauci and Collins, is just planning, not decision-making. The decision would have already been discussed between Farrar and Holmes.

So, most likely, we are missing a phone call or an email between Holmes and Farrar, sometimes before Farrar shared the SCAU news with Fauci and Collins. There, Holmes would have likely briefed Farrar about the (falsely) expected significance of the SCAU pangolin sequences. They would have together decided to go for publication. Then Farrar would have given some head-up to Fauci/Collins on the SCAU news themselves, and would have started planning the publication with Holmes.

Deep Dive: Farrar's and Holmes' motivations

a. Passaging was still possible:
One must keep in mind that neither Farrar nor Holmes took the decision to go for publication based on some expectation of finding the PRRA FCS in the pangolin sequence, or something close enough to it, which would allow them to reject *both* passaging and deliberate engineering (in contrast to the then situation where they had concluded that only deliberate engineering was very unlikely).

Indeed, as already seen, the rumours that Farrar was trying to address, via an authoritative published scientific statement, were about deliberate engineering, and not about passaging; passaging itself was simply not yet part of the public discussions.

In any case, Holmes likely suspected that the chance of finding such an FCS there was not that high, as this would have been a very logical feature for SCAU to report during its press conference.

So both Farrar and Holmes had other reasons to go for publication, and then jumped on the occasion (if not the distraction) to do so, offered by the pangolin homology news cycles.

b. Farrar's motivation: WHO access
Farrar was clearly concerned that the accusations of bioweapons and/or intentional human engineering would only increase if unchecked. This, in turn, would make it very difficult for the WHO to get access to China, something that NASEM itself was trying to ensure, as China had redirected the US administration to the WHO as the required platform of interaction.

As per Holmes' own words: "***Yes, it's going to blow. Hence why Jeremy wants us thinking about putting something out***" (see 11.5).

A scientific report that argued that intentional design was very unlikely, could potentially close out these rumours, provided that it received some high-profile treatment in Lancet, Nature or NEJM, and benefited from some push via the WHO and like-thinking influencers.

All of this would hopefully demonstrate the commitment of the WHO not to 'fail' China, if China let it in.

It would also look like a nice act of scientific solidarity: WHO for the international side, China with the SCAU team and Tommy Lam's team (which also included Holmes), all working in unison to resolutely advance our understanding of this virus and combat bioweapon rumours. The message would be beautiful, China would appreciate, and Holmes would forever be

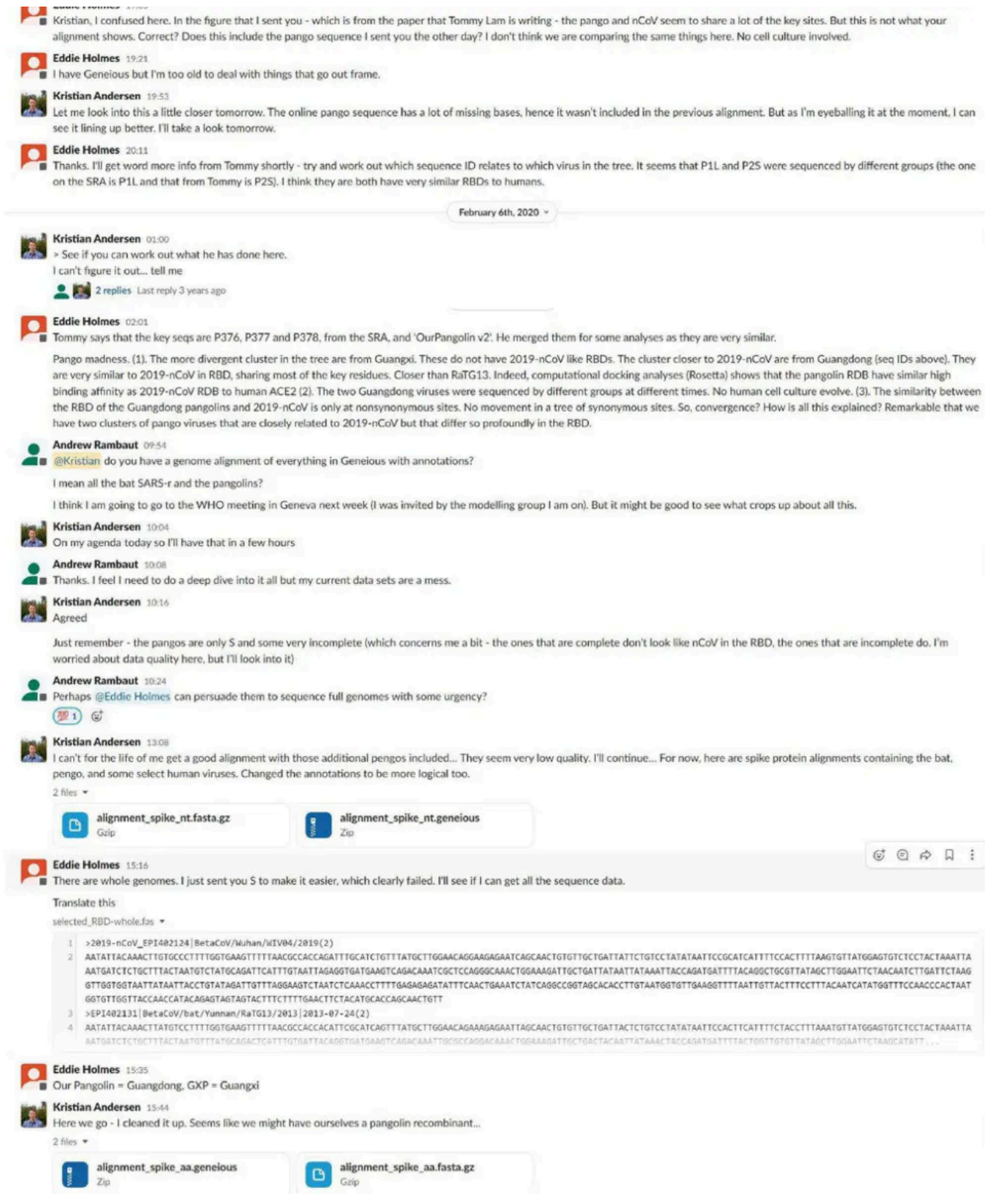
remembered as the distinguished scientist bridging between the West and China in that hour of dire need.

It looked perfect on paper, at least for those with very little understanding of the reality of China.

c. Holmes' motivation: himself
Holmes' motivation was straightforward:

- **First,** we already know that Holmes had straight from the beginning always been more keen on publication, something that is a general trait of him. [footnote 92]
- **Secondly,** Holmes had joined the Lam *et al.* team shortly after the Farrar call, to help them polish the writing of 'Identifying SARS-CoV-2-related coronaviruses in Malayan pangolins'.

This is confirmed in the Slack conversations, where on 5 and 6 Feb, Holmes and Andersen discussed at length sequences sent over by Tommy Lam:



Proximal_Origin_Slack.pdf, p.16, 5–6 Feb 2020

A first version of Lam *et al.* was sent to Nature on precisely 7 Feb (before being sent to bioRxiv on 13 Feb and released there as a preprint on 18 Feb).

According to an article in the NY Times, discussing Holmes collaboration on the Lam *et al.* paper, the pangolin sequences that Holmes was able to look at when he joined Lam *et al.* (which he also discussed on Slack on 5–6 Feb) did convince Holmes that the virus was not intentionally engineered ('a product of genetic engineering').

> *Finding such a distinct biological signature in a virus from a wild animal strengthened Dr. Holmes's confidence that SARS-CoV-2 was not the product of genetic engineering. "Suddenly what looks odd is clearly natural," Dr. Holmes said*
>
> *source: Carl Zimmer's article in the NY Times*

In short, on 7 Feb, when SCAU came up with the (false) news of 99% homology, Holmes
(1) found itself in a publishing competition with SCAU, and
(2) had a pangolin paper (Lam *et al.*) sent the same day to Nature for publication.

Holmes then proceeded to convince Farrar that the SCAU pangolin news required some publication of the draft report from the group (if not a co-publication with SCAU), if they were to **remain relevant.** It would also require some polishing to bring it more into the 'pangomania', as Holmes called it.

That convincing would not have been difficult at all. Given the urgent WHO-NASEM needs for access, Farrar was more than happy to be convinced to go for publication, and had indeed sounded Holmes about that the previous day (6 Feb, see 11.3). So when Holmes, whom he trusted above all, came back with a firm 'go' on Feb 7, Farrar was not going to hesitate a second.

12.5 The co-authors discuss Jeremy's plan (7 Feb)

a. Holmes shares Farrar's decision with his co-drafters (7 Feb, 9:36 am UK)
About 3 hours after Farrar had his quick email exchange with Holmes where he stated his decision to go for publication, Holmes shared Farrar's decision with his co-drafters and he asked them for their feedback.

Holmes also stated that he was expecting to see all the key mutations towards SARS-CoV-2 in the SCAU pangolin viruses, and also wanted to double-check if that would fully rule out the passaging option. While doing so, he sounded rather uncritical of the 99% homology claims.

When reading Holmes' email, one cannot escape the thought that Holmes was at least slightly over-doing the SCAU claims with Farrar as to their significance for passaging, while being more prudent with his co-authors, knowing that it would be a harder sale with them.

> ***Holmes (7 Feb, 9:36 am UK, emails):***
> *Jeremy wants us to publish our report somewhere. Thoughts?*
> *I'll need to update the pangolin stuff again. Not proven of course, but it makes complete sense. We don't know what the amino acid sequences of these pangolin viruses that* ***99% similar*** *to 2019-nCoV will look like, but there must be de****cent chance they have all the key mutations****. But,* ***does this swing it completely away from the passage idea****?*
> *Things are changing so fast it is hard not be redundant.*

b. The importance of checking the pangolin sequences for an FCS (9:46 am UK)

Ten minutes later, at **9:46 am UK,** Rambaut (based in the UK) answered Holmes' email by suggesting to first ask the SCAU team via personal communication ('pers-comm'), whether their pangolin sequence had some FCS:

> ***Rambaut (7 Feb, 9:46 am UK, emails):***
> *Can we at least get a pers-comm as to whether it has the insertion or not?*

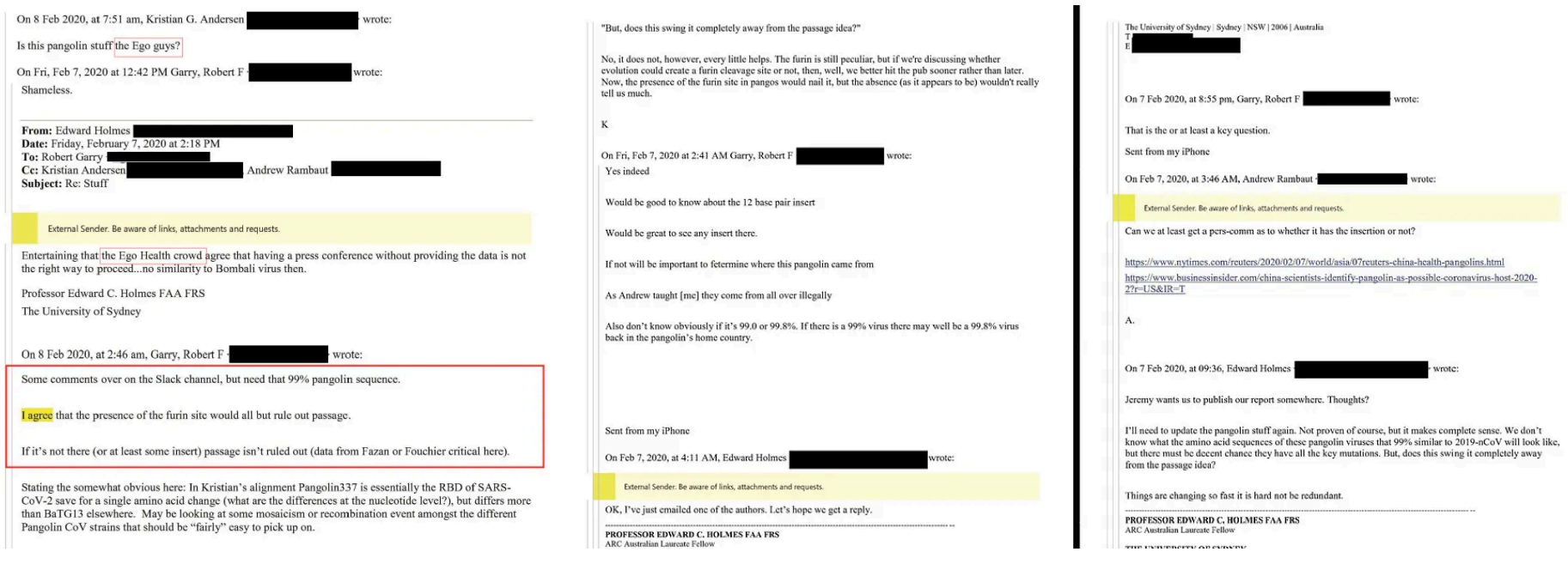
Garry (based in Florida) concurred soon after (**9:55 am UK**) with:

> ***Rambaut (7 Feb, 9:55 am UK, emails):***
> *'That is the or at least a key question'.*

c. Holmes asks the SCAU team about the FCS (10:11 am UK)

Accordingly, within half-an-hour, Holmes sent a short email to the SCAU team (~10:11 am UK, ~6:11 pm Sydney), asking them if their sequence had an FCS insertion (specifically PRRA) in the S protein. He then informed his co-authors that he had done so.

> ***Holmes (7 Feb, 10:11 am UK, emails):***
> *OK, I've just emailed one of the authors. Let's hope we get a reply.*



On 8 Feb 2020, at 7:51 am, Kristian G. Andersen [redacted] wrote:

Is this pangolin stuff the Ego guys?

On Fri, Feb 7, 2020 at 12:42 PM Garry, Robert F [redacted] wrote:

Shameless.

From: Edward Holmes [redacted]
Date: Friday, February 7, 2020 at 2:18 PM
To: Robert Garry [redacted]
Cc: Kristian Andersen [redacted] Andrew Rambaut [redacted]
Subject: Re: Stuff

External Sender. Be aware of links, attachments and requests.

Entertaining that the Ego Health crowd agree that having a press conference without providing the data is not the right way to proceed...no similarity to Bombali virus then.

Professor Edward C. Holmes FAA FRS
The University of Sydney

On 8 Feb 2020, at 2:46 am, Garry, Robert F [redacted] wrote:

Some comments over on the Slack channel, but need that 99% pangolin sequence.

I agree that the presence of the furin site would all but rule out passage.

If it's not there (or at least some insert) passage isn't ruled out (data from Fazan or Fouchier critical here).

Stating the somewhat obvious here: In Kristian's alignment Pangolin337 is essentially the RBD of SARS-CoV-2 save for a single amino acid change (what are the differences at the nucleotide level?), but differs more than BaTG13 elsewhere. May be looking at some mosaicism or recombination event amongst the different Pangolin CoV strains that should be "fairly" easy to pick up on.

"But, does this swing it completely away from the passage idea?"

No, it does not, however, every little helps. The furin is still peculiar, but if we're discussing whether evolution could create a furin cleavage site or not, then, well, we better hit the pub sooner rather than later. Now, the presence of the furin site in pangos would nail it, but the absence (as it appears to be) wouldn't really tell us much.

K

On Fri, Feb 7, 2020 at 2:41 AM Garry, Robert F [redacted] wrote:

Yes indeed

Would be good to know about the 12 base pair insert

Would be great to see any insert there.

If not will be important to fetermine where this pangolin came from

As Andrew taught [me] they come from all over illegally

Also don't know obviously if it's 99.0 or 99.8%. If there is a 99% virus there may well be a 99.8% virus back in the pangolin's home country.

Sent from my iPhone

On Feb 7, 2020, at 4:11 AM, Edward Holmes [redacted] wrote:

External Sender. Be aware of links, attachments and requests.

OK, I've just emailed one of the authors. Let's hope we get a reply.

PROFESSOR EDWARD C. HOLMES FAA FRS
ARC Australian Laureate Fellow

The University of Sydney | Sydney | NSW | 2006 | Australia
T
E [redacted]

On 7 Feb 2020, at 8:55 pm, Garry, Robert F [redacted] wrote:

That is the or at least a key question.

Sent from my iPhone

On Feb 7, 2020, at 3:46 AM, Andrew Rambaut [redacted] wrote:

External Sender. Be aware of links, attachments and requests.

Can we at least get a pers-comm as to whether it has the insertion or not?

https://www.nytimes.com/reuters/2020/02/07/world/asia/07reuters-china-health-pangolins.html
https://www.businessinsider.com/china-scientists-identify-pangolin-as-possible-coronavirus-host-2020-2?r=US&IR=T

A.

On 7 Feb 2020, at 09:36, Edward Holmes [redacted] wrote:

Jeremy wants us to publish our report somewhere. Thoughts?

I'll need to update the pangolin stuff again. Not proven of course, but it makes complete sense. We don't know what the amino acid sequences of these pangolin viruses that 99% similar to 2019-nCoV will look like, but there must be decent chance they have all the key mutations. But, does this swing it completely away from the passage idea?

Things are changing so fast it is hard not be redundant.

PROFESSOR EDWARD C. HOLMES FAA FRS
ARC Australian Laureate Fellow

Proximal_Origin_Emails.pdf, p.46–48, 7 Feb 2020 (US time)

d. Andersen gets up and joins in (7 Feb, 3:29 pm UK):

Once Andersen got up (3:29 UK, 8:29 am in California), he replied to the earlier question by Holmes about the significance of the presence or absence of an FCS in the yet-to-be-released SCAU pangolin sequence:

> ***Andersen (3:29 pm UK, emails):***
> *"But, does this swing it completely away from the passage idea?"*
> *No, it does not, however, every little helps. The furin is still peculiar, but* ***if we're discussing whether evolution could create a furin cleavage site or not****, then, well, we better hit the pub sooner rather than later.*
> ***Now, the presence of the furin site in pangos would nail it, but the absence (as it appears to be) wouldn't really tell us much.***

e. Garry and Rambaut concur with Andersen (7 Feb, 3.46 pm to 4:20 pm UK):

Switching to Slack, Garry and Rambaut explained that they would rather want to see the SCAU pangolin sequences before going for publication

> ***Garry (7 Feb, 3:46 pm, Slack):***
> *"Jeremy wants us to publish our report somewhere. Thoughts?"*
> *I think it's really important to get the pangolin sequence first (1 assume they haven't shared the FASTA file yet).*
> *The implications of a 99% similarity and a 99.8% similarity are pretty profound and at least would dramatically alter the discussion. pretty profoundly different*

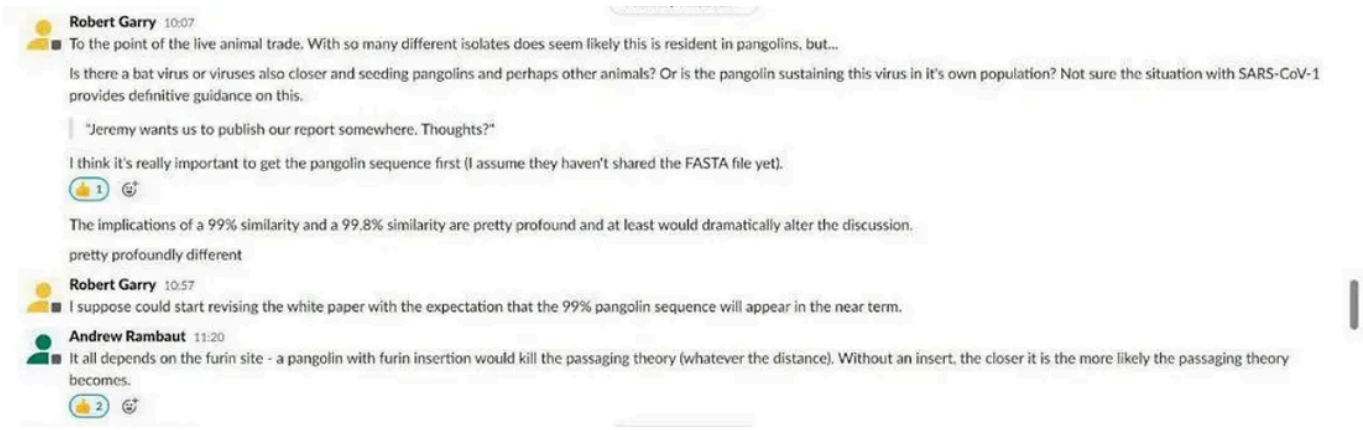
At exactly the same time, Garry wrote a similar message in the parallel email thread, referring to his just posted Slack message, and further noted again the importance to get the experimental data from either Farzan or Fouchier that showed that they could develop an FCS in a coronavirus by passaging.

> ***Garry (7 Feb, 3:46 pm, emails):***
> *Some comments over on the Slack channel, but need that 99% pangolin sequence.*

> *I agree that the presence of the furin site would all but rule out passage.*
> *If it's not there (or at least some insert) passage isn't ruled out (data from Farzan or Fouchier critical here).*

Rambaut then showed his agreement with Andersen and Garry, but went further by arguing that a SCAU pangolin sequence very close to SARS-CoV-2, but without an FCS, would actually flip things around and make passaging more likely.

> ***Rambaut (7 Feb, 4:20 pm UK, Slack)***
> *It all depends on the furin site — a pangolin with furin insertion would kill the passaging theory (whatever the distance). Without an insert, the closer it is the more likely the passaging theory becomes.*

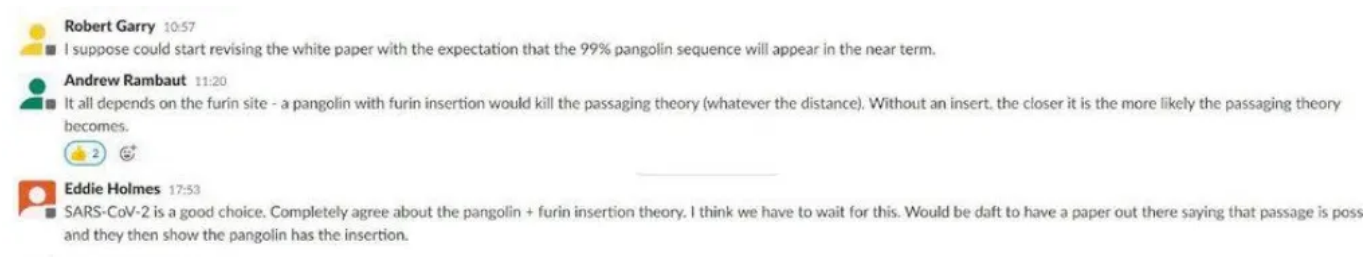


Proximal_Origin_Slack.pdf, p.46, 7 Feb 2020 (UK)

f. Holmes gets up and seems to agree (7 Feb, 10:53 pm UK)

About 7 hours later, after starting a new day, Holmes was back online at 9:53 am Sydney on 8 Feb (10:53 pm UK on 7 Feb) when he posted a message on Slack, where he agreed that it would be better to wait for the SCAU sequence:

> ***Holmes (7 Feb, 10:53 pm UK, Slack):***
> *'Completely agree about the pangolin + furin insertion theory. I think we have to wait for this. Would be daft to have a paper out there saying that passage is possible and they [SCAU] then show that the pangolin has the [PRRAR] insertion'.*



Proximal_Origin_Slack.pdf, p.21, 7 Feb 2020 (UK)

Summary: Could the SCAU sequences kill the passaging option?

The presence of not of an FCS in these SCAU pangolin sequences that had not yet been shared, was logically a key point of interest for Farrar and the report drafters. Andersen, Garry and Rambaut quickly realised that:

Having the exact PRRAR FCS of SARS-CoV-2, or something very close to it, in the SCAU sequences, however close these were to SARS-CoV-2, would make **natural-evolution much more likely**.

If, instead, there was no FCS, or only an intermediate FCS of some sort in the SCAU sequences, then:

- As long as the SCAU sequences were not too close to ARS-CoV-2, the probability of a passaging origin would be **essentially unchanged**.
- However, if these sequences were very close to SARS-CoV-2, then **passaging would be more likely**, since it would mean some close relative of SARS-CoV-2 was present in Chinese labs for a while, where it was quite possibly passaged, resulting in the acquisition of the FCS.

13. Sharing the decision with the call group and some pushback (8 Feb)

13.1 Farrar forwards the latest report to the full call group and discusses going for publication (8 Feb, 9:45 am UK)

The report/summary was reworked quickly by Andersen, Rambaut and Garry, to wave in more references to the latest pangolin sequences, but still without closing the door to passaging. Then, as he had said he would do the day before, on Saturday 8 Feb **at 9:45 am UK**, for the first time since 2 Feb, Farrar sent an email back to the entire group of attendees of the 1 Feb call, with the latest draft ('Summary.Feb7.pdf', version v6a of Appendix A).

For the sake of clarity, while this was the first draft sent to the call group since the short call summary on 2 Feb, Farrar had meanwhile already sent at least three drafts to Fauci and Collins. So the people who saw it for the first time were Fouchier, Koopmans, Drosten, Vallance, Ferguson and Golding.

In his email to the call group, Farrar asked for any comment on whether this should be published, and added that he was hoping to get the SCAU pangolin sequences before publication.

From: Jeremy Farrar [redacted]
Sent: 08 February 2020 09:45
To: Edward Holmes [redacted]; kga1978 [redacted] Andrew Rambaut [redacted]; rfgarry [redacted]
Cc: r.fouchier [redacted]; P.Vallance1 [redacted]; collinsf [redacted] afauci [redacted] Josie Golding [redacted] m.koopmans [redacted] christian.drosten [redacted]; Mike Ferguson [redacted]
Subject: FW: 2019 N-CoV

APOLOGIES WITH ALL CORRECT EMAILS

Kristen, Andrew, Bob, Eddie have reworked the summary and it is attached here.

We are pushing to get the sequence data from the reports on the pangolins, but do not have currently, clearly that is very important to incorporate.

Interested in your views

REV0000827

- Is this reasonably balanced given the data?
- Is there anything anyone disagrees with?
- Is there anything more in relation to what would seem to be the two possibilities
  - Nature, Intermediate host, evolution and passage
- Future data you may have
- Advice on whether KA, AR, RG and EH should publish this.

These and other thoughts welcome in confidence.

Proximal_Origin_Emails.pdf, p.57–58, 8 Feb 2020, 9:45 UK

13.2 Fouchier pushes back on any lab origin (8 Feb, 10.36 am UK)

With a remarkable celerity, at 10:36 am UK (11:36 am Rotterdam), Fouchier was the first to provide some feedback to Farrar [footnote 49]. In his email, which also included an annotated version of the report ('Summary.Feb7 **RF**.pdf'), Fouchier strongly pushed back on mentioning passaging in anything that would go to publication, by arguing that:

- Passaging is just another form of a laboratory manipulation, meaning manipulation of some existing virus.
- So, all that matters in the end is that, as far as one can tell, nobody in China has worked on any close relative of SARS-CoV-2, be it the (supposedly) 99% similar pangolin virus, or RaTG13 (~96%) — basically the only two (supposedly) 'close' relative of SARS-CoV-2 at the time.
- Based on the absence of evidence of such work, one should reject any idea that SARS-CoV-2 was a lab-product, be it through engineering or passaging.

Fouchier was clearly attempting to repeat his success in previously dismissing the deliberate engineering option during the Farrar call on 1 Feb, by trying to 'grandfather' the passaging option in it. The arguments he deployed were unfortunately contrived and extremely weak:

- Andersen, Rambaut and Garry had agreed during the Farrar call that there was no sign of deliberate engineering (as far as they could tell at the time), based on Fouchier feedback at the time, largely because it would have been extremely difficult to come up with such a successful high infectious variation of an existing low risk SARS-related **purely based on direct genetic manipulations.**
- However, that argument does not apply at all to passaging, which can discover its own (forced) evolutionary path towards a given target.
- So effectively, Andersen, Rambaut and Garry had no reason to drop the passaging option, especially given **the very limited knowledge one had of what was being worked on in Chinese labs.**

Likely knowing how poor his logic was, Fouchier immediately followed with two arguments that had nothing to do with science and all to do with preserving the status quo:

- Allowing for the possibility of a lab-escape of a virus enhanced via passaging would set a precedent, that would automatically mean the same questioning for future outbreaks, and even some possible revisiting of past outbreaks.
- Opening the debate on the possible risks of passaging experiments would likely start some new biosafety discussions *'that would unnecessarily obstruct future attempts of virus culturing for research and diagnostic purposes'*.

Message

From: R.A.M. Fouchier [redacted]
Sent: 2/8/2020 11:36:30 AM
To: Jeremy Farrar [redacted] Edward Holmes [redacted]; kga1978@[redacted]; Andrew Rambaut [redacted] rfgarry [redacted]
CC: P.Vallance1 [redacted]; collinsf[redacted] afauci [redacted]; Josie Golding [redacted] M.P.G.Koopmans [redacted]; christian.drosten[redacted]; Mike Ferguson [redacted]
Subject: Re: 2019 N-CoV
Attachments: Summary.Feb7 RF.pdf

I am not in favor of publishing as is. I fail to see how the last of the three discussed scenarios (passaging) does not fall under the category of "laboratory manipulation". There is no evidence that might hint to this scenario and hence it should be put aside just like the engineering option. As far as I am aware, no laboratory has worked on passaging the pangolin-origin virus, the bat-CoV RaTG13, or another closely related virus or had access to it prior to the outbreak. That nCoV-2019 could originate from a SARS-like virus in Chinese labs can also be excluded. This information could be added after reference 10 in the manuscript, to provide further argument.
If we assume passaging as a possible scenario here, we must assume it is also plausible for all outbreaks from the past, present and future. This manuscript would be much stronger if it focused on the likelihood of the first 2 scenarios as compared to intentional or accidental release. That would also limit the chance of new biosafety discussions that would unnecessarily obstruct future attempts of virus culturing for research and diagnostic purposes for any (emerging/zoonotic) virus.

I made some additional comments in the attached pdf, also in line with Andrew's comments.

With kind regards,
Ron

Proximal_Origin_Emails.pdf, p.55, 8 Feb 2020, 10:36 am UK

Sideline: Fancy Science

Quite frankly, the first argument offered by Fouchier is at best extremely credulous (basically saying that one knows everything about what work was done in China), while his two following ones (reproduced below) are totally shameful:

> ***If we assume passaging as a possible scenario here, we must assume it is also plausible for all outbreaks from the past,present and future.*** *This manuscript would be much stronger if it focused on the likelihood of the first 2 scenarios as compared to intentional or accidental release.*
>
> ***That would also limit the chance of new biosafety discussions that would unnecessarily obstruct future attempts of virus culturing for research and diagnostic purposes for any (emerging/zoonotic) virus.***
> Ron Fouchier, Proximal_Origin_Emails.pdf, p.55, 8 Feb 2020, 10:36 am UK

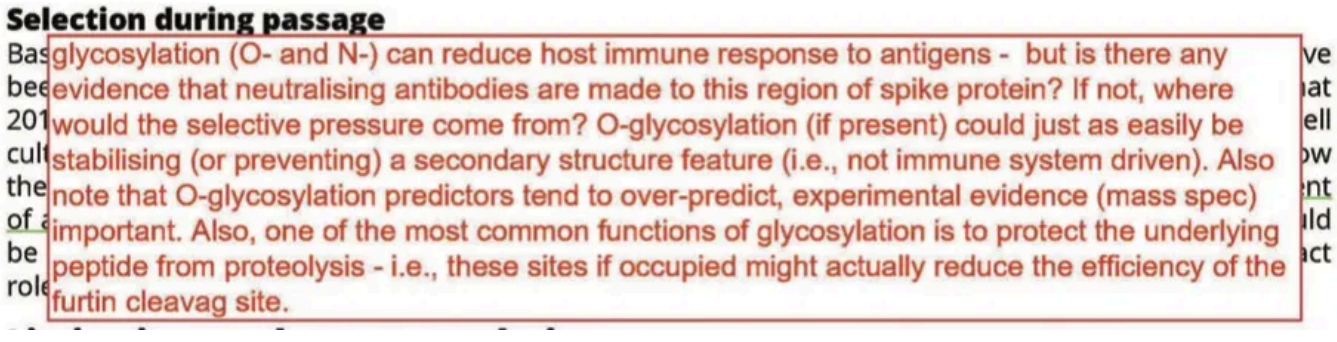
The statement by Fouchier was actually more than a preference. It ended up actively shaping the actions of the Fouchier-Koopmans-Daszak trio, in particular the torpedoing of the P.O. manuscript at *Nature*, for daring to still mention passaging (see Part 2), and the 2021 WHO mission in Wuhan.

One is thus perfectly entitled to question whether Fouchier is the right person to manage a biolaboratory doing passaging experiments, especially since he had already produced some completely fancy probabilities of a passaging accident in his own lab, which he estimated at less than one every 33 billion years, roughly 7 times the age of the earth [footnote 16]. He thus clearly has a preference for maintaining absurd estimates of passaging safety, especially when faced with a possible passaging accident.

13.3 Ferguson makes some comment and discusses the glycans (8 Feb, 12:00 pm UK)

Mike Ferguson (Wellcome Trust) was the second to provide feedback to Farrar, at 12:00 pm UK, by adding some a few comments to the draft report sent by Farrar less than two hours ago ('Summary.Feb7_**MF**.pdf').

His main comment is focussed on the glycans shield, as part of the evaluation of the possibility of a passaging origin:



Proximal_Origin_Emails.pdf, p.61, 8 Feb 2020, 12:00 pm UK

His short email also briefly addressed the glycans, which may point to Farrar having asked him separately about their significance. His reply, while rather vague, is effectively that the glycans are not incompatible with passaging. In that email, Ferguson also mentioned that Tabak (who is an expert in glycoproteins) was on the Farrar call on Saturday 1 Feb.

Message

From: Mike Ferguson [redacted]
Sent: 2/9/2020 12:00:46 PM
To: Jeremy Farrar [redacted] Edward Holmes [redacted] kga1978 [redacted]
Andrew Rambaut [redacted] rfgarry [redacted]
CC: r.fouchier [redacted] P.Vallance1 [redacted] collinsf [redacted] afauci [redacted] Josie Golding
[redacted] m.koopmans [redacted] christian.drosten [redacted]
Subject: Re: 2019 N-CoV
Attachments: Summary.Feb7_MF.pdf

Dear Jeremy et al

I have made some comments and suggestions on the pdf attached.

I am not an expert on protein O-glycosylation - however, Dr Tabak, who was on the call last weekend, is and if I were to consult anyone else on this it would be Henrik Clausen
https://icmm.ku.dk/english/research-groups/clausen-group/

However, from what I do know of general glycobiology, I am not sure one can conclude that an immune system would be required to select for O-glycosylation sites. Once an alpha-helix is disturbed by the introduction of a proline, adjacent Ser and Thr residues will be (over-)predicted to have O-glycosylation potential - hard to know the functional consequences/significance without knowing whether the potential O-sites are actually occupied.

Regards

Mike

From: Jeremy Farrar [redacted]
Sent: 08 February 2020 09:45
To: Edward Holmes [redacted] kga1978 [redacted] Andrew Rambaut [redacted] rfgarry [redacted]
Cc: r.fouchier [redacted] P.Vallance1 [redacted] collinsf [redacted] afauci [redacted] Josie Golding [redacted] m.koopmans [redacted]
christian.drosten [redacted] Mike Ferguson [redacted]
Subject: FW: 2019 N-CoV

APOLOGIES WITH ALL CORRECT EMAILS

Kristen, Andrew, Bob, Eddie have reworked the summary and it is attached here.

We are pushing to get the sequence data from the reports on the pangolins, but do not have currently, clearly that is very important to incorporate.

Interested in your views

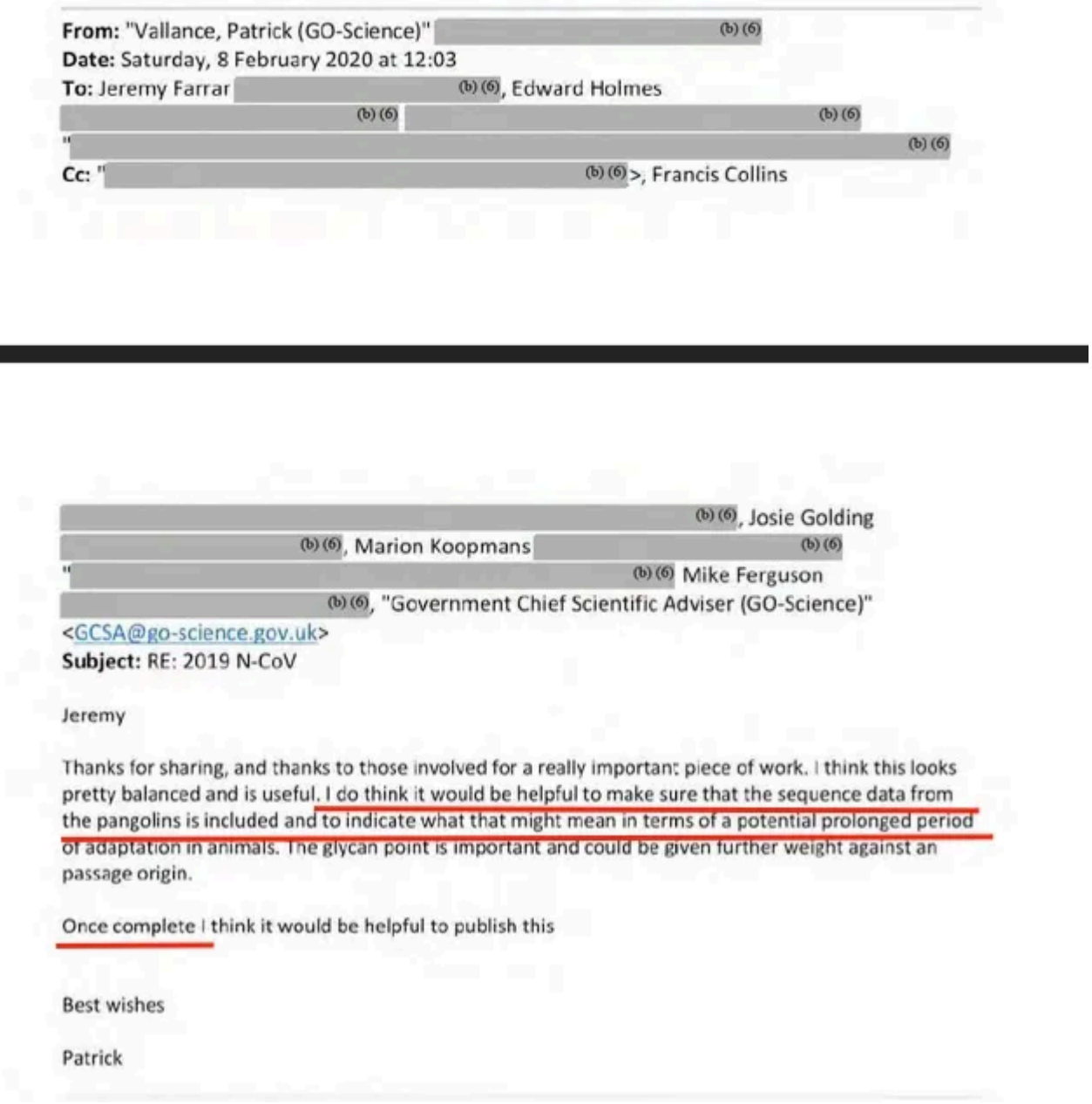
Proximal_Origin_Emails.pdf, p.56, 8 Feb 2020, 12 pm UK

13.4 Vallance supports publishing once the pangolin sequences are available (8 Feb, 12:03 pm UK)

Patrick Vallance replied at 12:03 pm UK, a few minutes after Mike Ferguson. Vallance, the UK Government Chief Scientific Adviser (GCSA) at the time, and the only extra person on the preparatory Farrar call with Fauci on 30 Jan 2020 (see 3.6).

His reply was supportive of publishing the report. It also seemed to be somewhat weighed against passaging; he discussed the importance of the SCAU sequence in terms of being able to evaluate the period of adaptation in

animals, and wanted to see the glycans being better used to give further weight against a passaging origin.



From: "Vallance, Patrick (GO-Science)" [redacted]
Date: Saturday, 8 February 2020 at 12:03
To: Jeremy Farrar [redacted], Edward Holmes
[redacted]
"[redacted]
Cc: "[redacted]>, Francis Collins

[redacted], Josie Golding
[redacted], Marion Koopmans [redacted]
"[redacted] Mike Ferguson
[redacted], "Government Chief Scientific Adviser (GO-Science)"
<GCSA@go-science.gov.uk>
Subject: RE: 2019 N-CoV

Jeremy

Thanks for sharing, and thanks to those involved for a really important piece of work. I think this looks pretty balanced and is useful. I do think it would be helpful to make sure that the sequence data from the pangolins is included and to indicate what that might mean in terms of a potential prolonged period of adaptation in animals. The glycan point is important and could be given further weight against an passage origin.

Once complete I think it would be helpful to publish this

Best wishes

Patrick

farrar-fauci-comms-full.pdf, p.62, 8 Feb 2020, 12.03 pm UK

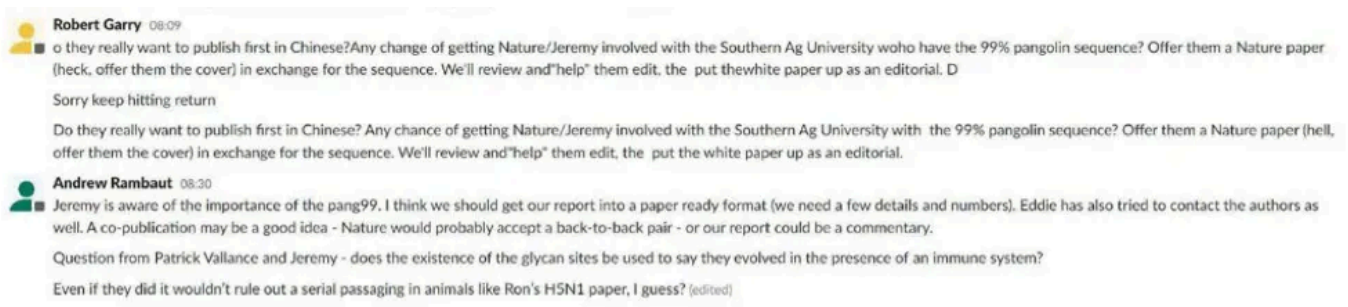
13.5 Garry and Rambaut toy with the idea of co-publishing with SCAU to speed up access to the pangolin sequences (8 Feb, 1.09 pm UK, 8.09 am EST)

One hour later, while still discussing the importance of getting hold of the SCAU sequence before publishing their analysis, Garry and Rambaut briefly toyed with the idea of getting SCAU to co-publish with them in Nature, with the help of Farrar, possibly in exchange for sharing the sequence with them so that they can finish their report.

That co-publication would have the SCAU's paper as main item, and the 'evolutionary origin' report of Andersen, Rambaut, Holmes and Garry, as either editorial or commentary:

> ***Garry (8 Feb, 1:09 pm UK):***
> *Do they really want to publish first in Chinese? Any chance of getting Nature/Jeremy involved with the Southern Ag[ricultural] University who have the 99% pangolin sequence? Offer them a Nature paper (hell, offer them the cover) in exchange for the sequence. We'll review and "help" them edit, then put the white paper up as an editorial.*

> ***Rambaut (8 Feb, 1:30 pm UK):***
> *Jeremy is aware of the importance of the pang99. I think we should get our report in a paper ready format [..]. Eddie has also tried to contact the authors as well. A co-publication may be a good idea — Nature would probably accept a back-to-back pair — or our report could be a commentary.*



Proximal_Origin_Slack.pdf, p.21, 8 Feb 2020

13.6 Focus on Passaging (8 Feb, 12:03 pm — 6:03 pm UK)

Following the circulation of the draft report to the full call group, the main technical discussions focussed largely on the possibility of the virus being a product of passaging.

▶️ First, as we saw in 13.4, Vallance thought that the glycans presence was an argument against passaging (8 Feb, 12:03 pm UK).

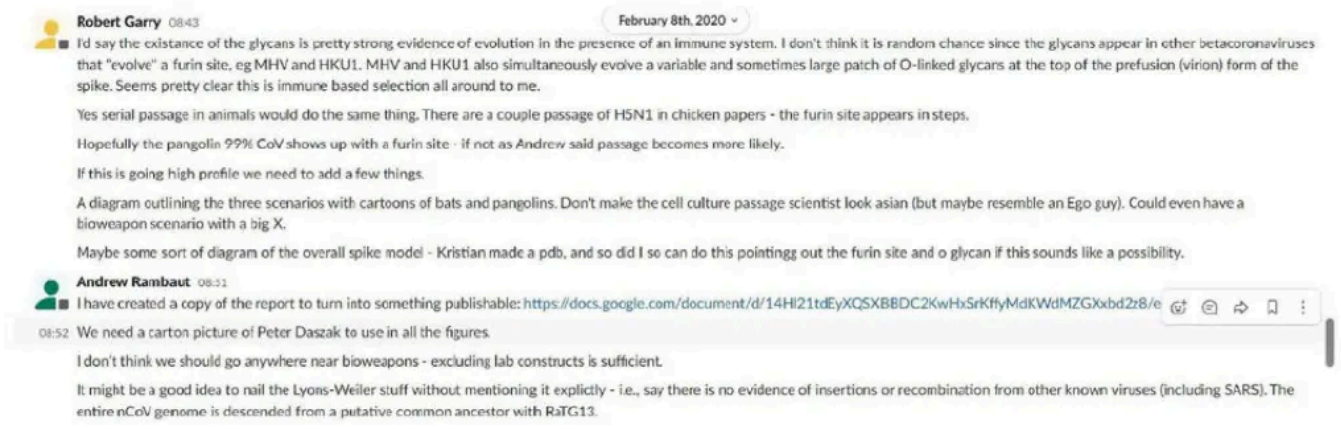
▶️ Soon after and not unrelated, Rambaut relayed to the Slack group a question from Vallance and Farrar:

> ***Rambaut (8 Feb, 1:30 pm UK)***
> *"Does the existence of the glycan sites be used to say they evolved in the presence of an immune system?*

The wording of the question was rather weird, 'Does .. be used ?' with 'Does' seemingly substituted for 'Can'. Likely someone (Farrar?) thought it better not to make it too obvious that the idea was to dig up arguments against passaging, rather than to be impartial, and thus altered the original *'**Can** the existence of the glycan sites be used to say they evolved in the presence of an immune system?'*.

▶️ Garry answered the question on Slack:

> ***Garry (8 Feb, 1:43 pm UK):***
> *I'd say the existence of the glycans is pretty strong evidence of evolution in the presence of an immune system. […] Seems pretty clear this is immune based selection all around to me.*
> *Yes serial passage in animals would do the same thing. There are a couple passage of HSN11 in chicken papers — the furin site appears in steps.*
> *Hopefully the pangolin 99% CoV shows up with a furin site — if not as Andrew said passage becomes more likely.*



Robert Garry 08:43 — February 8th, 2020

I'd say the existance of the glycans is pretty strong evidence of evolution in the presence of an immune system. I don't think it is random chance since the glycans appear in other betacoronaviruses that "evolve" a furin site, eg MHV and HKU1. MHV and HKU1 also simultaneously evolve a variable and sometimes large patch of O-linked glycans at the top of the prefusion (virion) form of the spike. Seems pretty clear this is immune based selection all around to me.

Yes serial passage in animals would do the same thing. There are a couple passage of H5N1 in chicken papers - the furin site appears in steps.

Hopefully the pangolin 99% CoV shows up with a furin site - if not as Andrew said passage becomes more likely.

If this is going high profile we need to add a few things.

A diagram outlining the three scenarios with cartoons of bats and pangolins. Don't make the cell culture passage scientist look asian (but maybe resemble an Ego guy). Could even have a bioweapon scenario with a big X.

Maybe some sort of diagram of the overall spike model - Kristian made a pdb, and so did I so can do this pointingg out the furin site and o glycan if this sounds like a possibility.

Andrew Rambaut 08:51

I have created a copy of the report to turn into something publishable: https://docs.google.com/document/d/14HI21tdEyXQSXBBDC2KwHxSrKffyMdKWdMZGXxbd2z8/e

08:52 We need a carton picture of Peter Daszak to use in all the figures.

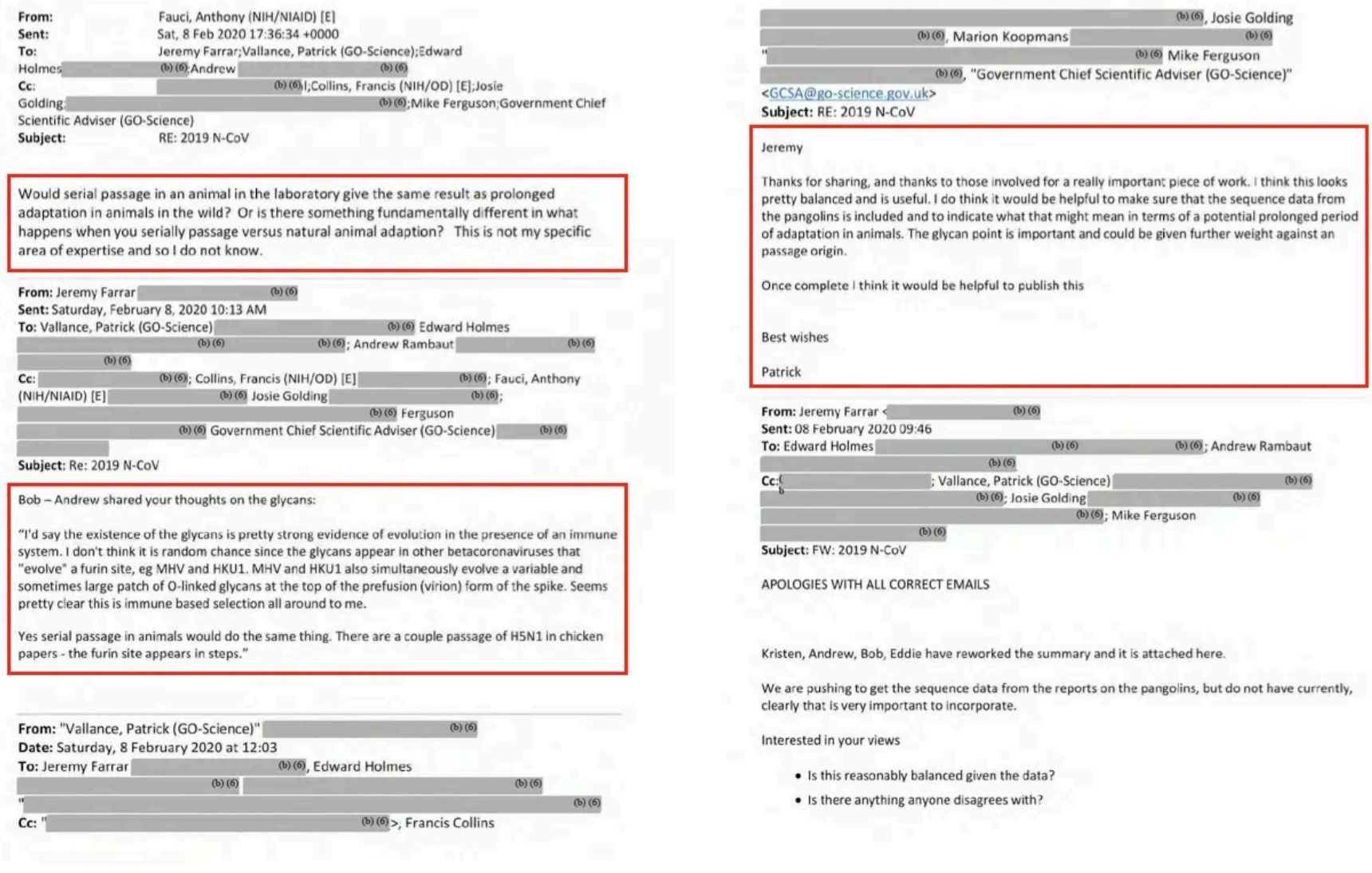
I don't think we should go anywhere near bioweapons - excluding lab constructs is sufficient.

It might be a good idea to nail the Lyons-Weiler stuff without mentioning it explictly - i.e., say there is no evidence of insertions or recombination from other known viruses (including SARS). The entire nCoV genome is descended from a putative common ancestor with RaTG13.

Proximal_Origin_Slack.pdf, p.22, 8 Feb 2020, 1:43 pm UK

▶️ Farrar then replied to Vallance's feedback (8 Feb, 3:13 pm UK) by quoting the answer given above by Garry on Slack, and forwarded to him by Rambaut, whereby Garry had argued that the glycans could happen exactly in the same way via passaging.

▶️ Next, Fauci asked for a confirmation of that general point about the impossibility of distinguishing between natural evolution or passaging in the right cells or hosts (8 Feb, 5:36 pm UK).



From: Fauci, Anthony (NIH/NIAID) [E]
Sent: Sat, 8 Feb 2020 17:36:34 +0000
To: Jeremy Farrar;Vallance, Patrick (GO-Science);Edward Holmes;Andrew [redacted];
Cc: [redacted];Collins, Francis (NIH/OD) [E];Josie Golding;[redacted];Mike Ferguson;Government Chief Scientific Adviser (GO-Science)
Subject: RE: 2019 N-CoV

Would serial passage in an animal in the laboratory give the same result as prolonged adaptation in animals in the wild? Or is there something fundamentally different in what happens when you serially passage versus natural animal adaption? This is not my specific area of expertise and so I do not know.

From: Jeremy Farrar [redacted]
Sent: Saturday, February 8, 2020 10:13 AM
To: Vallance, Patrick (GO-Science) [redacted] Edward Holmes [redacted]; Andrew Rambaut [redacted]
Cc: [redacted]; Collins, Francis (NIH/OD) [E] [redacted]; Fauci, Anthony (NIH/NIAID) [E] [redacted] Josie Golding [redacted]; [redacted] Ferguson [redacted] Government Chief Scientific Adviser (GO-Science) [redacted]
Subject: Re: 2019 N-CoV

Bob – Andrew shared your thoughts on the glycans:

"I'd say the existence of the glycans is pretty strong evidence of evolution in the presence of an immune system. I don't think it is random chance since the glycans appear in other betacoronaviruses that "evolve" a furin site, eg MHV and HKU1. MHV and HKU1 also simultaneously evolve a variable and sometimes large patch of O-linked glycans at the top of the prefusion (virion) form of the spike. Seems pretty clear this is immune based selection all around to me.

Yes serial passage in animals would do the same thing. There are a couple passage of H5N1 in chicken papers - the furin site appears in steps."

From: "Vallance, Patrick (GO-Science)" [redacted]
Date: Saturday, 8 February 2020 at 12:03
To: Jeremy Farrar [redacted], Edward Holmes [redacted]
Cc: [redacted]>, Francis Collins

[redacted], Josie Golding [redacted], Marion Koopmans [redacted] Mike Ferguson [redacted], "Government Chief Scientific Adviser (GO-Science)" <GCSA@go-science.gov.uk>
Subject: RE: 2019 N-CoV

Jeremy

Thanks for sharing, and thanks to those involved for a really important piece of work. I think this looks pretty balanced and is useful. I do think it would be helpful to make sure that the sequence data from the pangolins is included and to indicate what that might mean in terms of a potential prolonged period of adaptation in animals. The glycan point is important and could be given further weight against an passage origin.

Once complete I think it would be helpful to publish this

Best wishes

Patrick

From: Jeremy Farrar <[redacted]
Sent: 08 February 2020 09:46
To: Edward Holmes [redacted]; Andrew Rambaut [redacted]
Cc: [redacted]; Vallance, Patrick (GO-Science) [redacted]; Josie Golding [redacted]; Mike Ferguson [redacted]
Subject: FW: 2019 N-CoV

APOLOGIES WITH ALL CORRECT EMAILS

Kristen, Andrew, Bob, Eddie have reworked the summary and it is attached here.

We are pushing to get the sequence data from the reports on the pangolins, but do not have currently, clearly that is very important to incorporate.

Interested in your views

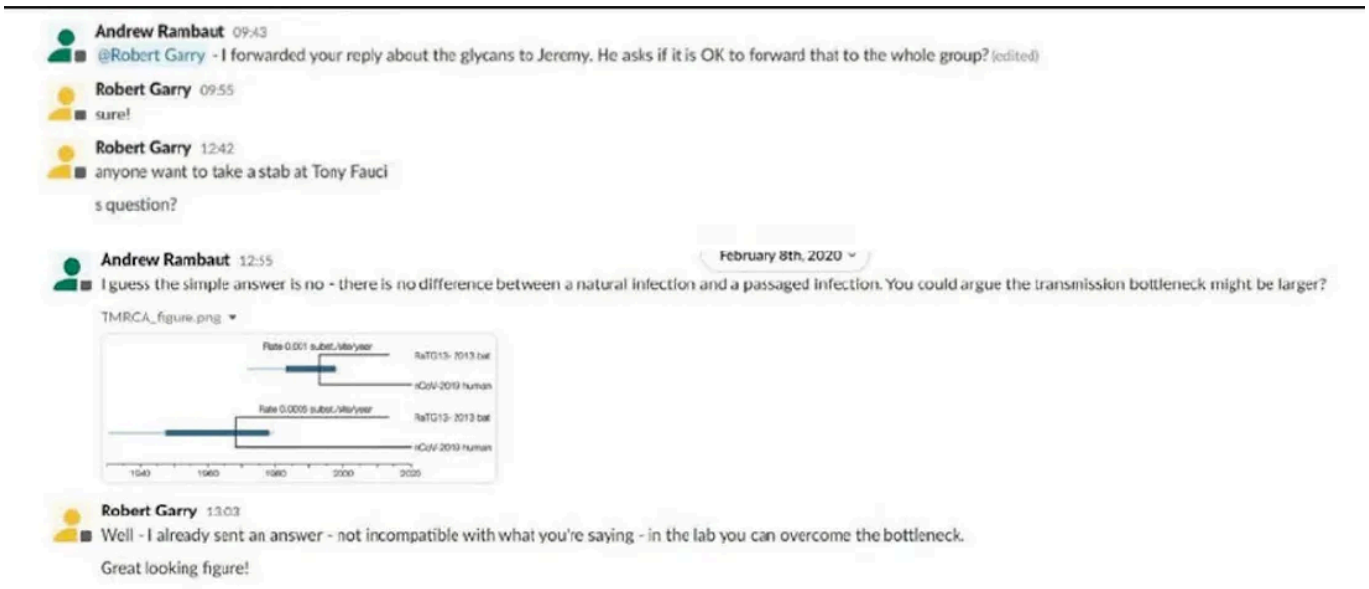
- Is this reasonably balanced given the data?
- Is there anything anyone disagrees with?

farrar-fauci-comms-full.pdf, p. 61–62 8 Feb 2020, 12:03 pm, 3:13 pm and 5:36 pm UK

▶️ Fauci's question was immediately discussed on Slack (8 Feb, 5:42 pm UK). Garry replied by email to Fauci before 6:03 pm UK (we do not have his exact email) by saying that it is indeed possible to get the same results via passaging.

> ***Garry (8 Feb, 5:42 pm UK):***
> *anyone want to take a stab at Toni Fauci's question?*

> ***Rambaut (8 Feb, 5:55 pm UK):***
> *I guess the simple answer is no — there is no difference between a natural infection and a passaged infection. You could argue the transmission bottleneck might be larger?*
>
> ***Garry (8 Feb, 6:03 pm UK):***
> *Well — I already sent an answer — not incompatible with what you're saying — in the lab you can overcome the bottleneck.*



Proximal_Origin_Slack.pdf, p.22, p.22–23, 8 Feb 2020, ~6 pm UK

13.7 Drosten wonders what the goal is (8 Feb, 7.52 pm UK)

Later that day, Drosten — who was very busy with other things that week — dropped in the email conversation for the first time since the Farrar call. His candid reaction illustrates perfectly the gap between what Andersen was trying to achieve, and what Fouchier and Drosten thought should be done.

Just as with Fouchier's (see 13.2 above), Drosten's understanding of the call discussions was that one should deny the bioweapon / HIV insert rumours and stop at that, without going in a review of the other scenario of the virus being a lab-construct, via passaging. Their reasoning was that since that passaging scenario was not being publicly discussed (i.e.: *'our own conspiracy theory'*, not the one already out there), it would make no point pouring petrol on fire by mentioning it.

> *Didn't we congregate to challenge a certain theory, and if we could, drop it? [..]*
> *Is this the case? It does not seem as if this was linked to the HIV nonsense.*
> *Who came up with this story in the beginning? Are we working on debunking our own conspiracy theory?*
>
> *source: Drosten, Proximal_Origin_Emails.pdf, p.53, 8 Feb 2020, 19:52 UK, 20:52 Germany*

Drosten later detailed his understanding of the Farrar's call group dynamic in 'Spike', his book about the pandemic.

> *In such a high-calibre group, the opportunity for a publication can arise, for example, by bringing together some expert opinions. This may also cause some clash. But what were we actually talking about?* ***Just an idea that had been put forward by two members of the group, but which others immediately recognized as baseless. That really went too far for me. It was clear to everyone that I would not participate in a publication on this basis****, neither for nor against a laboratory hypothesis.*
>
> *Then there was a lengthy email exchange between the participants in this group. The discussion focused on factual arguments and also on whether they wanted to write something. My position on this was clear, and another participant confirmed to me that they did not want to write anything. That was the end of the matter for me. Apparently part of the group did write something together with an additional author [GD: Lipkin].* ***But I didn't hear anything about it. I only found out about it when a manuscript was submitted. I was rather surprised.***
>
> *source: Drosten and Mascolo, Alles überstanden? ('Through it all?'), June 2024*

One must say that Drosten's understanding is rather limited, as it ignores:

1. the shift of emphasis from deliberate engineering to a fully possible passaging,
2. the clear confirmations sent by Farrar, to the full call group, that a publication in a journal would ensue (8 Feb and 10 Feb), and
3. the intermediate publication on virological.org on 17 Feb.

13.8 Holmes answers Drosten (8 Feb, 8.11 pm UK, 7.11 am day+1 Sydney)

A few minutes later, Holmes, who just started his day (9 Feb in Sydney), answered Drosten, explaining that the idea was not to only address some outlandish bioweapon allegations, but to also look at accidental lab escape scenarios that may appear to be more credible; basically to clear the air. Hence, the need to address the question of passaging, whether it was in the open or not.

From: Edward Holmes [redacted]
Date: Saturday, 8 February 2020 at 20:11
To: Christian Drosten [redacted]
Cc: Jeremy Farrar [redacted], "kga1978[redacted], "a.rambaut[redacted], "rfgarry[redacted], "r.fouchier[redacted], "P.Vallance1[redacted], Francis Collins <[redacted] "afauci[redacted], Josie Golding [redacted], Marion Koopmans [redacted], Mike Ferguson [redacted]
Subject: Re: [ext] 2019 N-CoV

Hi Christian,

I don't know where this story came from, but it has nothing whatsoever to do the HIV nonsense. Please don't associate this with that. This is a broader story.

Ever since this outbreak started there have suggestions that the virus escaped from the Wuhan lab, if only because of the coincidence of where the outbreak occurred and the location of the lab. I do a lot of work in China and I can you that a lot of people there believe this and believe they are being lied to. Things were made worse when Wuhan lab published the bat virus sequence - a bat sampled in a different province for which they have a large collection of samples.

I believe the aim/question here is whether we, as scientists, should try to write something balanced on the science behind this? There are arguments for and against doing this.

Personally, with the pangolin virus possessing 6/6 key sites in the receptor binding domain, I am in favour of the natural evolution theory.

Best wishes,

Eddie

PROFESSOR EDWARD C. HOLMES FAA FRS
ARC Australian Laureate Fellow

Proximal_Origin_Emails.pdf, p.52, 8 Feb 2020, 8.11 pm UK

Insight:

The two contradictory arguments against a pure lab-construct keep popping back in the origins debate, on the pure zoonotic side:

▶️ **Arg #1:** There is no known close relative in Chinese labs, so one can reject all lab constructs scenarios: (1) synthesised natural virus, (2) pure lab-design from some closely related virus(es), (3) passaging from a closely related virus (possibly on top of (2)).

👉 **Limitations:** nobody really knows what was in Chinese labs, especially since so little collected from 2016 on has been published [footnote 56].

▶️ **Arg #2:** Nobody would create such a virus via deliberate engineering. Indeed, SARS-CoV-2 does not look like an obvious deliberate engineering product, given that there would be no way to predict its phenotype from its rather complex and uncommon genotype. In other words, the argument is that a genotype-driven design could not deliver such a unique, yet rather optimal, virus.

👉 **Limitations:** The virus could be simply based on an unpublished natural virus, either isolated or synthesised from a good consensus sequence. From there some passaging (a phenotype drive), or a combination of gene editing then passaging, could have produced a functional Furin Cleavage Site adapted to human lung cells and generally optimised the virus.

In summary, Argument #1 is misplaced trust, while Argument #2 is bad science. Fouchier, well aware of the possibility of a seamless construct, chose to go for Arg #1.

Additionally, Arg #1 might have sounded plausible when RaTG13 (~96% homology) was the closest relative being discussed, but it should have been revisited when a pangolin virus with a claimed 99% homology was all of a sudden being brandished as a key exhibit for the zoonotic case. None of the co-authors seemed to notice the contradiction.

Even after the 99% claims collapsed, the simple fact that this pangolin virus was not known previously, but had been present in Chinese labs for a while, should have made it clear that there was a very real possibility that some SARS-CoV-2 related viruses, still undisclosed, were in these labs. Fouchier seemed not to have noticed that second contradiction. And while the co-authors were aware of that possibility in their private exchanges, they still used the absence of a disclosed related virus as an argument in Proximal Origin [footnote 50].

13.9 Holmes, Farrar and Rambaut confirm that they are confident enough in the pangolin claims to go for publication (8:11 pm to 9:16 pm, UK)

a. Holmes (8.11 pm UK):
On the previous day, Holmes had already expressed his conviction about the significance of the pangolin data, in his private discussions with Farrar, which led to the decision to go for publication.

In part, Holmes is likely to have been more bullish with Farrar about the supposed 99% homology of the SCAU pangolin virus, when they discussed it together on 7 Feb, than he had to be with his more careful co-authors. But on the evening of 8 Feb, in his answer to Fouchier about the reason to address an accidental lab escape, Holmes also explained that the presence of 6/6 key RBD residues, in the receptor binding domain of the Malayan pangolin virus (the one he was already analysing for Lam *et al.*), was a key factor.

> ***Holmes (5 Feb 2020, 10:04 am UK):***
> *I think we might have dropped the ball with this pangolin virus. I ignored it when I saw it didn't have the furin cleavage site. Should now check* ***all the key sites****.*
> *Cheers,*
>
> *source: Holmes to his co-authors, Proximal_Origin_Emails.pdf, p.28*
>
> ***Holmes (8 Feb 2020, 8:11 pm UK):***
> *I believe the aim/question is whether we, as scientists, should try to write*

> *something balanced on the science behind this? [..]*
>
> *Personally, with the pangolin virus possessing **6/6 key sites** in the receptor binding domain, I am in favour of the natural evolution theory.*
>
> *source: Holmes to the Farrar call group, Proximal_Origin_Emails.pdf, p.52*

b. Farrar (8.21 pm UK):
Ten minutes later, jumping on Holmes's answer to Fouchier, Farrar stated that the latest information on the pangolin virus played a key role in his decision to go for publication. The sentence is not clear: this could be a generic way of describing both the recently noticed presence of the 6 residual sites in the RBD of the Lam *et al.* Malayan pangolin virus, and/or the recent (false) 99% homology claims of SCAU.

> ***Farrar (8 Feb 2020, 8:21 pm UK)***
> *The aim of this was to bring a neutral, respected, scientific group together to look at the data and in a neutral, considered way, provide an opinion [..]*
>
> ***With the additional information on the pangolin virus, information not available 24 hours ago, I think the argument is even clearer.***
>
> *My preference is that a carefully considered piece of science, early in the public domain, will help mitigate more polarised debate. If not, that debate will increasingly happen and science will be reacting to it. Not a good position to be in.*
>
> *source: Farrar to the Farrar call group, Proximal_Origin_Emails.pdf, p.52*

c. Rambaut (9:16 pm UK):
Closing this recital, Rambaut declared himself also leaning more towards a natural origin, again due to the recent developments of the pangolin-as-host hypothesis. His position is nevertheless a bit more hesitant, since he explained that he still had some issues with passaging.

He also agreed with the reasons to go for publication (with a twist, noting the importance of avoiding some formal investigations).

> ***Rambaut (8 Feb 2020, 9:16 pm UK):***
> *I agree with Eddie, I think someone needs to lay out the science of this before it gets out of hand (and causes more formal investigations).*
>
> *I am of the view that the natural selection hypothesis is the most likely (specifically the non-bat reservoir). And as Eddie mentioned this is becoming more likely from day to day with the pangolin story.*

> *I disagree with Ron that the passaging hypothesis is evidentially equal to the engineering hypothesis. The sequence data clearly and unambiguously rules out any form of lab construct or engineering of the virus. It doesn't really have anything to say about the relative plausibility of the 3 hypotheses for selection. [note: hypothesis #3 is passaging]*
>
> *source: Rambaut to the Farrar call group, Proximal_Origin_Emails.pdf, p.51*

Rambaut next provided an excellent example of motivated reasoning:

> *I think that we need stronger arguments than an assertion that no lab has done those experiments. [..]*
>
> *Is it possible to argue that A) a passaging experiment wouldn't create the feature we see? or B) that there are logical reason why someone wouldn't do such an experiment?*
>
> *source: Rambaut to the Farrar call group, Proximal_Origin_Emails.pdf, p.51, 8 Feb 2030, 9:16 pm UK, 4:16 pm EST*

As we can see, the whole passage states the conclusion first, and then goes about lining up the arguments. It's the opposite of proper science. Additionally:

- For question (A), there is an immediate converse which is just as much worth answering: '*Can it be argued that a passaging experiment could create the features we see?*',
  The way of asking the question can indeed make marvels as to the direction of the answer; a classic *framing* issue.
- As for question (B), this is not only again a framing issues (one could just as well ask whether there are logical reason to do that passaging experiment), but it is also a case of *mirror imaging:* pretending than the other party lives in the same circumstances, has the same constraints and opportunities, and will do the same things as you.

Message

From: R.A.M. Fouchier [redacted]
Sent: 2/8/2020 2:50:00 PM
To: Andrew Rambaut [redacted]; Jeremy Farrar [redacted]
CC: Eddie Holmes [redacted]; Christian Drosten [redacted]; kga1978[redacted]; rfgarry[redacted]; p.vallance1[redacted]; collinsf[redacted]; afauci[redacted]; Josie Golding [redacted]; M.P.G. Koopmans [redacted]; Mike Ferguson [redacted]
Subject: Re: [ext] 2019 N-CoV

I do not understand Andrews argument " The sequence data clearly and unambiguously rules out any form of lab construct or engineering of the virus. ". Molecular biologists like myself can generate perfect copies of viruses without leaving a trace, eg the BamHI site. The arguments for and against passaging and engineering are the same if you ask me.
Ron

From: Andrew Rambaut [redacted]
Sent: Saturday, February 8, 2020 4:16 PM
To: Jeremy Farrar
Cc: Eddie Holmes; Christian Drosten; kga1978[redacted]; rfgarry[redacted]; r.fouchier[redacted]; p.vallance1[redacted]; collinsf[redacted]; afauci[redacted] Josie Golding; m.koopmans[redacted] Mike Ferguson
Subject: Re: [ext] 2019 N-CoV

I agree with Eddie, I think someone needs to lay out the science of this before it gets out of hand (and causes more formal investigations).

I am of the view that the natural selection hypothesis is the most likely (specifically the non-bat reservoir). And as Eddie mentioned this is becoming more likely from day to day with the pangolin story.

I disagree with Ron that the passaging hypothesis is evidentially equal to the engineering hypothesis. The sequence data clearly and unambiguously rules out any form of lab construct or engineering of the virus. It doesn't really have anything to say about the relative plausibility of the 3 hypotheses for selection.

I think we need stronger arguments than an assertion that no lab has done those experiments. We can definitely argue that it has nothing to do with RaTG13 (or SARS or any other published SARSr virus). The argument that we would need to offer this hypothesis for all other outbreaks is not a useful one in this context.

Is it possible to argue that A) a passaging experiment wouldn't create the features we see? or B) that there are logical reasons why someone wouldn't do such an experiment?

The pangolin virus that was announced in the press conference might solve this issue if it has the furin cleavage site insertion which would be all but conclusive for the natural scenario.

Andrew

On 8 Feb 2020, at 20:21, Jeremy Farrar [redacted] wrote:

The theory of the origin of the has gathered considerable momentum not in social media, but increasingly among some scientists, in main stream media, and among politicians.

The aim of this was to bring a neutral, respected, scientific group together to look at the data and in a neutral, considered way provide an opinion and we hoped to focus the discussion on the science, not on any conspiracy or other theory and to lay down a respected statement to frame whatever debate goes on – before that debate gets out of hand with potentially hugely damaging ramifications.

With the additional information on the pangolin virus, information not available even 24 hours ago, I think the argument is even clearer.

My preference is that a carefully considered piece of science, early in the public domain, will help mitigate more polarised debate. If not, that debate will increasingly happen and science will be reacting to it. Not a good position to be in.

From: Edward Holmes [redacted]
Date: Saturday, 8 February 2020 at 20:11
To: Christian Drosten [redacted]
Cc: Jeremy Farrar [redacted], "kga1978[redacted], "a.rambaut[redacted], "rfgarry[redacted], "r.fouchier[redacted], "P.Vallance1[redacted], Francis Collins <[redacted], "afauci[redacted], Josie Golding[redacted], Marion Koopmans [redacted], Mike Ferguson [redacted]
Subject: Re: [ext] 2019 N-CoV

Hi Christian,

I don't know where this story came from, but it has nothing whatsoever to do the HIV nonsense. Please don't associate this with that. This is a broader story.

Ever since this outbreak started there have suggestions that the virus escaped from the Wuhan lab, if only because of the coincidence of where the outbreak occurred and the location of the lab. I do a lot of work in China and I can you that a lot of people there believe this and believe they are being lied to. Things were made worse when Wuhan lab published the bat virus sequence - a bat sampled in a different province for which they have a large collection of samples.

I believe the aim/question here is whether we, as scientists, should try to write something balanced on the science behind this? There are arguments for and against doing this.

Personally, with the pangolin virus possessing 6/6 key sites in the receptor binding domain, I am in favour of the natural evolution theory.

Best wishes,

Eddie

PROFESSOR EDWARD C. HOLMES FAA FRS
ARC Australian Laureate Fellow

REV0000736

Proximal_Origin_Emails.pdf, p.51–52, 8 Feb 2020, 8:11 to 9:16 PM UK

d. Fast-forward:

In their review of the events with Jon Cohen, Holmes and Andersen would later cite some of these key passages to explain their position at that precise time:

5. Very shortly after the call, the pangolin data came out. This was critical, and as Eddie wrote in an email to everyone on the call on Feb 9th:

"Personally, with the pangolin virus possessing 6/6 key sites in the receptor binding domain, I am in favour of the natural evolution theory."

With Andrew Rambaut replying:

"I am of the view that the natural selection hypothesis is the most likely (specifically the non-bat reservoir). And as Eddie mentioned this is becoming more likely from day to day with the pangolin story."

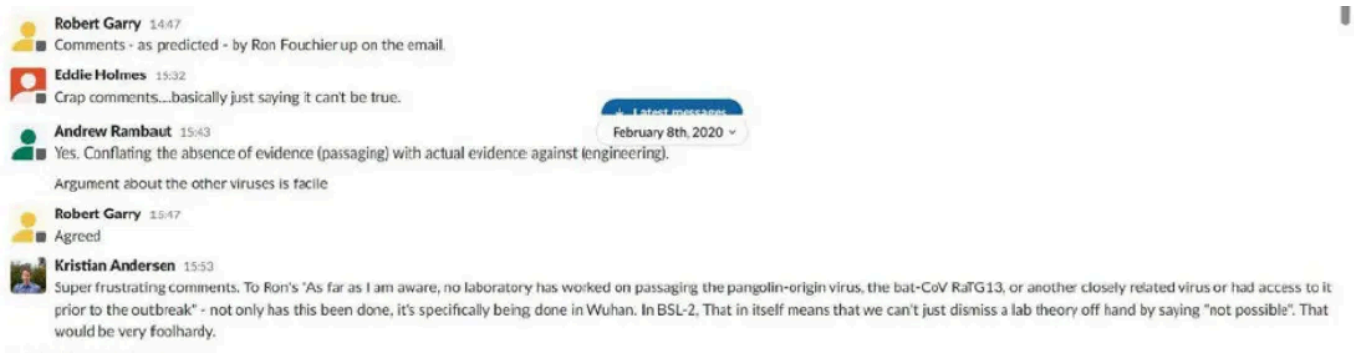
Proximal_Origin_Emails.pdf, p.150, draft email to Jon Cohen

13.10 On Slack: the co-authors discuss Fouchier (8 Feb, 7:47 pm to 8:53 pm UK)

Parallel to the discussions involving the larger call group in emails, Garry, Holmes, Rambaut and Andersen shared their impressions about Fouchier's arguments between themselves on Slack. They were less than impressed. Andersen captivated the mood, when, commenting about Fouchier assertion that '*no laboratory has worked on passaging [..] a closely related virus'*, he wrote on Slack (8:53 pm UK):

> *Not only has this been done, it's specifically being done on Wuhan. **In BSL-2**. That in itself means that we can't just dismiss a lab theory off hand by saying 'not possible'. That would be foolhardy.*

> *source: Andersen on Slack, Proximal_Origin_Slack.pdf, p.23, 8 Feb 2020, 8.53 pm UK*



Proximal_Origin_Slack.pdf, p.23, 8 Feb 2020, 8:53 pm UK

In the above, Andersen may seem to be misrepresenting Fouchier's argument, which was not whether passaging was done in Chinese labs, but instead that nobody had heard of any passaging of a **close** relative of SARS-CoV-2).

In fact, Andersen's point is correct. Indeed, absence of evidence is not evidence of absence, so Fouchier's argument was irrelevant anyway. What really matters then is precisely what Andersen highlighted: the very real risk represented by documented work in Chinese BSL-2 lab to try to make low virulence coronaviruses strains more easily adaptable to humans (on the likely basis that the low virulence coronaviruses *one started with* could be worked with at BSL-2). In these BSL-2 conditions, one just needed to be 'successful' and generate a keeper to have a good chance of causing an outbreak [footnote 37].

13.11 The Refuznik: Andersen is not keen on publishing and wants more data (8 Feb, 9:16 pm UK)

While Holmes, Farrar and Garry all felt confident enough about the relevance of the pangolin data to a zoonotic origin, and all agreed with Farrar to go for publication, Andersen threw a spanner in the works at the precise time Rambaut was agreeing by email with that plan (9:16 pm UK).

In his own response at 9:16 pm UK, Andersen reminded everybody that they first needed to see the latest pangolin data to be able to properly argue that a lab-origin was unlikely, since without that data the case was still open-ended.

In other words, Andersen stuck to the science, even if the latest version of the draft sounded much more pro-natural origin (since it was meant to be made public), and while Farrar and Holmes were confident that the existing pangolin data was by now pointing to a likely zoonotic origin, with more of it to come. In doing so, he also very wisely reminded everybody that passaging at BSL-2 conditions raises some very valid questions, that had to be answered properly.

From: "Kristian G. Andersen" [redacted]
Date: Saturday, 8 February 2020 at 21:16
To: "Drosten, Christian" [redacted]
Cc: Jeremy Farrar [redacted], Edward Holmes [redacted], "a.rambaut [redacted], "rfgarry [redacted], "r.fouchier [redacted], "P.Vallance1 [redacted], Francis Collins [redacted] "afauci [redacted], Josie Golding [redacted], Marion Koopmans [redacted], Mike Ferguson [redacted]
Subject: Re: [ext] 2019 N-CoV

A lot of good discussion here, so I just wanted to add a couple of things for context that I think are important - and why what we're considering is far from "another conspiracy theory", but rather is taking a valid scientific approach to a question that is increasingly being asked by the public, media, scientists, and politicians (e.g., I have been contacted by Science, NYT, and many other news outlets over the last couple of days about this exact question).

To Ron's question, passage of SARS-like CoVs have been ongoing for several years, and more specifically in Wuhan under BSL-2 conditions - see references 12-15 in the document for a few examples. The fact that Wuhan became the epicenter of the ongoing epidemic caused by nCoV is likely an unfortunate coincidence, but it raises questions that would be wrong to dismiss out of hand. Our main work over the last couple of weeks has been focused on trying to *disprove* any type of lab theory, but we are at a crossroad where the scientific evidence isn't conclusive enough to say that we have high confidence in any of the three main theories considered. Like Eddie - and I believe Bob, Andrew, and everybody on this email as well - I am very hopeful that the viruses from pangolins will help provide the missing pieces. For now, giving the lab theory serious consideration has been highly effective at countering many of the circulating conspiracy theories, including HIV recombinants, bioengineering, etc. - here's just one example: https://www.factcheck.org/2020/02/baseless-conspiracy-theories-claim-new-coronavirus-was-bioengineered/.

As to publishing this document in a journal, I am currently not in favor of doing so. I believe that publishing something that is open-ended could backfire at this stage. I think it's important that we try to gather additional evidence - including waiting on the pangolin virus sequences and further scrutinize the furin cleavage site and O-linked glycans - before

Proximal_Origin_Emails.pdf, p.144, 8 Feb 2020, 9:16 pm UK

Five minutes later (9:21 pm UK) on Slack, Andersen answered Rambaut, who had just asked him what his plan was, given his reluctance to go for publication:

- First, Andersen expressed the difficulty of managing this in an email chain that includes the rest of the Farrar call group. It was clear to him that the dynamic and constraints of the Farrar's call email chain were not optimal.
- Then Andersen clarified that he was not against publication per se, that he was, in fact, for drafting a proper article based on their existing report, but that it was simply too early to decide to go for publication when more data and analysis were still needed.

Proximal_Origin_Slack.pdf, p.23, 8 Feb 2020, 9:19 pm UK

13.12 The Boss: Farrar cuts the discussions short and confirms his decision to go for publication (8 Feb, 9:27 pm UK)

Ten minutes later, replying directly to Andersen's email, Farrar reiterated his main argument from only one hour ago (the pangolin sequences from Lam *et al.* and more sequences that were expected, see 13.9.c). It was a polite but firm rebuke to Andersen's objections.

He also made abundantly clear what his objective and expectations were: *'to tie this even tighter with the next iteration and make a conclusive statement which will be* ***the go to scientific statement to refer to****'.*

Message

From: Jeremy Farrar [redacted]
Sent: 2/8/2020 1:27:14 PM
To: Kristian G. Andersen [redacted] Drosten, Christian [redacted]
CC: Edward Holmes [redacted] Andrew Rambaut [redacted] rfgarry [redacted]; r.fouchier [redacted] P.Vallance1 [redacted]; collinsf[redacted] afauci[redacted] Josie Golding [redacted] m.koopmans[redacted] Mike Ferguson [redacted]
Subject: Re: [ext] 2019 N-CoV

We now have (and we will get more) the pangolin data (Eddie has) we think we can tie this up even tighter with the next iteration and make a conclusive statement which will then be the go to scientific statement to refer to.

Eddie and I have just come off a call with the National Academy of Medicine in the US – who the White House has asked to produce a report on this....

From: "Kristian G. Andersen" [redacted]
Date: Saturday, 8 February 2020 at 21:16
To: "Drosten, Christian" [redacted]
Cc: Jeremy Farrar [redacted], Edward Holmes [redacted], "a.rambaut [redacted], "rfgarry [redacted], "r.fouchier [redacted], "P.Vallance1 [redacted], Francis Collins [redacted] "afauci[redacted], Josie Golding [redacted], Marion Koopmans [redacted], Mike Ferguson [redacted]
Subject: Re: [ext] 2019 N-CoV

A lot of good discussion here, so I just wanted to add a couple of things for context that I think are important - and why what we're considering is far from "another conspiracy theory", but rather is taking a valid scientific approach to a question that is increasingly being asked by the public, media, scientists, and politicians (e.g., I have been contacted by Science, NYT, and many other news outlets over the last couple of days about this exact question).

Proximal_Origin_Emails.pdf, p.144, 8 Feb 2020, 9:27 pm UK

Exactly three minutes later (9:30 pm UK) Garry repeated that formulation on Slack. **It is thus perfectly clear that the co-authors happily went on with the editorial decisions of Farrar, who had not only decided to go for publication but was also asking for a specific tone ('conclusive') and product ('go to scientific statement to refer to').**

> *We now have (and we will get more) the pangolin data (Eddie has) we think we can tie this up even tighter with the next iteration and* ***make a conclusive statement which will then be the go to scientific statement to refer to.***

Robert Garry 16:30
We now have (and we will get more) the pangolin data (Eddie has) we think we can tie this up even tighter with the next iteration and make a conclusive statement which will then be the go to scientific statement to refer to.

Eddie and I have just come off a call with the National Academy of Medicine in the US – who the White House has asked to produce a report on this....

Moving fast - don't think we should necessarily wait on the NAM to get something out there if pango99 seq is available.

Kristian Andersen 16:40
NASEM is useless - they'll have exactly zero... Too political an organization.

Proximal_Origin_Slack.pdf, p.23, 8 Feb 2020, 9:30 pm UK

Essential Insight: Farrar's failure, or how NOT to involve domain experts

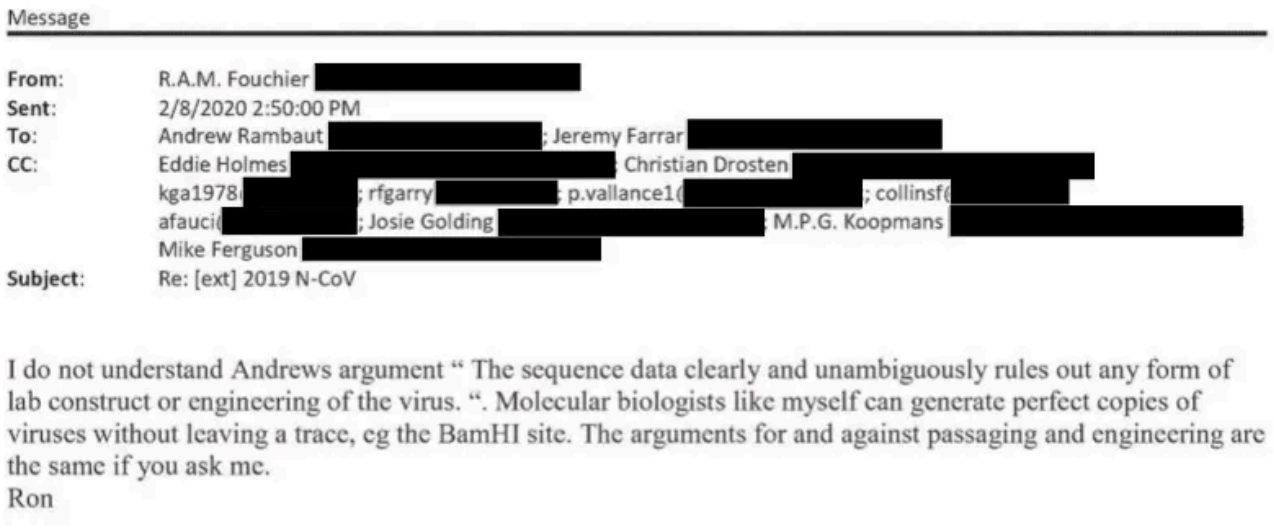
One cannot fail to notice that Farrar, Holmes, Fouchier and Andersen were too often talking past each other. This went beyond some natural

disagreement on a scientific issue, and was instead revelatory of a fractured group of scientists pursuing their own agenda and not showing their cards:

1. While obviously Holmes (who called Farrar his 'handler'), and to a certain extent Andersen (who said as much in his testimony to Congress), and from there Garry and Rambaut, were aware of the intelligence dimension that was at least partly behind the Farrar call, Fouchier, Koopmans, and Drosten would have been largely unaware of it. For them, the need to look at a possible non-natural origin must have looked not only threatening, but positively odd and a waste of everybody's time.
2. To the point, Fouchier does not seem to have shared his data showing that passaging could create an FCS. In fact, Fouchier and Koopmans would eventually feel emboldened enough to try to expunge any mention of a possible passaging, all the way down to the final publication.
3. Perhaps a better communication with that bounce-off group about the intelligence dimension of the call, and establishing some clear expectations about professionalism, would have helped. And if that was not possible, then maybe Fouchier and Koopmans (and to a lesser extent Drosten) simply should not have been invited to voice their superficial opinions. The logical understanding gap in Drosten's question *('Are we working on debunking our own conspiracy theory?),* and the way the main 'in' actors (Farrar and Holmes) rushed to try to address it, were superbly revelatory of the issue.
4. Later, Farrar does not seem to have ever explained to Andersen, Garry and Rambaut, how he had changed his focus from what was supposed to be a confidential short summary report, with some British intelligence consumers (up to the call on 1 Feb), to a summary report that would also be handled by the WHO behind closed doors (from the call to 7/8 Feb), and then to the final 'up-cycling' of that summary report into a high-profile publication that would help the WHO gain access (from 8 Feb, the day before the advance WHO team left for China). Farrar may have explained his evolving priorities better to Holmes, but we unfortunately don't have their one-to-one communications to verify it.
5. Holmes very likely did show less objectivity with Farrar as he did with his more critical co-authors, obviously because Farrar would have put direct pressure on him, whether consciously or not, and partly due to the privileged and rather easy relation he had to his 'handler' as he called him, including the fact that his one-to-one communications with Farrar were strictly private (and have remained so until today — how much of that was done on burner phones is anyone's guess).
6. Another important factor affecting Holmes, is that he was already very invested in the pangolin story with Lam *et al.,* and always very keen to have his name on yet another paper. That created a misalignment of incentives with his co-authors, but it aligned with Farrar's objectives, even if the motivations of the latter were largely distinct.

13.13 The Wicked: Fouchier tries again (8 Feb, 10:50 pm UK)

An hour and a half later (10:50 pm UK), after possibly missing Farrar's email from 9:27 pm UK, Fouchier answered the objection expressed by Rambaut at 9:16 pm UK, namely that *'The sequence data clearly and unambiguously rules out any form of lab construct or engineering of the virus'.*



Message

From: R.A.M. Fouchier [redacted]
Sent: 2/8/2020 2:50:00 PM
To: Andrew Rambaut [redacted]; Jeremy Farrar [redacted]
CC: Eddie Holmes [redacted]; Christian Drosten [redacted]
kga1978[redacted]; rfgarry [redacted]; p.vallance1[redacted]; collinsf[redacted]
afauci[redacted]; Josie Golding [redacted]; M.P.G. Koopmans [redacted]
Mike Ferguson [redacted]
Subject: Re: [ext] 2019 N-CoV

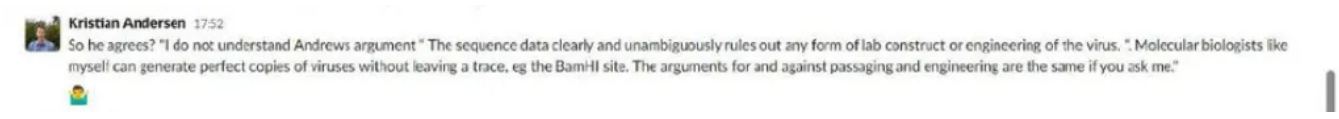
I do not understand Andrews argument " The sequence data clearly and unambiguously rules out any form of lab construct or engineering of the virus. ". Molecular biologists like myself can generate perfect copies of viruses without leaving a trace, eg the BamHI site. The arguments for and against passaging and engineering are the same if you ask me.
Ron

Proximal_Origin_Emails.pdf, p.51, 8 Feb 2020, 10:50 pm UK

This time Fouchier tried a different argument (see 13.2). He argued that:

- Yes, SARS-CoV-2 could in theory be a synthetic virus based on a good quality sequence, with possibly some point modifications.
- So 'the arguments for and again passaging and engineering are the same if you ask me'.

Only two minutes later (10:52 pm UK), Andersen correctly pointed out on Slack that Fouchier seemed to now agree that one could not reject some deliberate engineering based purely on the sequence. 🤷‍♂️



Kristian Andersen 17:52
So he agrees? "I do not understand Andrews argument " The sequence data clearly and unambiguously rules out any form of lab construct or engineering of the virus. ". Molecular biologists like myself can generate perfect copies of viruses without leaving a trace, eg the BamHI site. The arguments for and against passaging and engineering are the same if you ask me."

Proximal_Origin_Slack.pdf, p.23, 8 Feb 2020, 10:52 pm UK

While Fouchier's reasoning seems rather confusing at first, what he likely meant was that since one does not know of any closely related sequence being present at the VIW, there was no more point considering the possibility of passaging than there was of considering the possibility of an artificial introduction of the FCS in a backbone.

Obviously, that argument relies on the indefensible assumption that the WIV has published already all its sequences, since very few of the sequences harvested since 2017 have actually been published [footnote 56]. Interestingly, that indefensible part of Fouchier's argument is still to this day

repeated by scientists who should know much better, and sometime actually know better (as their FOI'd emails show), while the other part of Fouchier's argument (the fact that one can generate a perfect copy of a virus without leaving a trace) is most often ignored by the very same scientists.

13.14 The 99% pangolin claims implode (8 Feb, 11:42 pm UK)

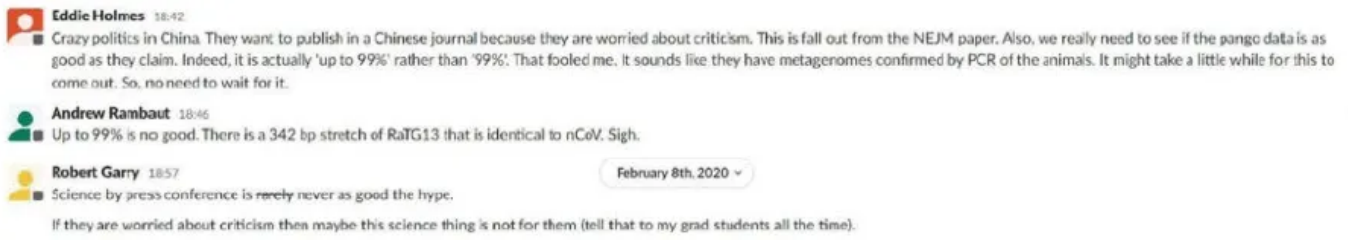
Only about two hours later, Holmes delivered some bad news on Slack, which formed a rather apt conclusion to a fateful day [footnote 51]:

1. The SCAU team wanted to publish in a Chinese journal (and not a well-known English-language one), 'because they are worried of criticism'.
   👉🏻 This might refer to the previous discussion on Slack about inviting the SCAU team to co-publish the report, in exchange for early access to their pangolin sequences (see 13.5), or it may simply be based on Holmes having asked them when their paper will be out.
2. **The 99% claims fooled him, as the real meaning was <up to> 99%'**
   👉🏻 This meant that the sequences may not be at all as decisive as claimed.

The second point is crucial. The initial enthusiasm for the promised 99% was likely part of the Farrar's decision to go to publication, at least because of the opportunity it offered of benefiting from a strong news cycle, even if the internal expectations on the SCAU sequence were not necessarily that high (in fact the 6/6 key sites of the Lam *et al.* sequences were at least as much important to Holmes).
Also, only a little more than two hours ago (see 13.12), Farrar had again ignored Andersen's warnings that one ought to wait for the data before committing to publication.

This was pure Greek tragicomedy.

> ***Holmes:*** *Crazy politics in China. They want to publish in a Chinese journal because they are worried about criticism. [..] Also, we really need to see if the pango data is as good as they claim. Indeed,* ***it is actually 'up to 99%' rather than '99%'. That fooled me.***
>
> ***Rambaut:*** *Up to 99% is no good.*
>
> ***Garry:*** *Science by press conference is never as good as the hype.*



Proximal_Origin_Slack.pdf, p.24, 8 Feb 2020, 11:42 pm UK

Key insight: A very quiet realisation

The revelation of the 99% homology being a serious misinterpretation is not discussed any further. **In particular, there is no sign in the FOI'd emails of the core-group sharing that key piece of information with the other scientists in the Farrar call group.**

Also, for sure Holmes must have discussed that piece of news with Farrar, but such discussion does not show up in the FOI's emails or in Slack. **Also, there is no mention at all of Farrar being troubled by it. Nor are the drafts of the report revised to de-emphasise the pangolins.**

So either we are missing some key emails, or — more likely — some bad news was not exactly what Farrar wanted to dwell on when doing all he could to manage access to China for the WHO, and the core-group readily understood this. **If fact, one may conclusively argue that some bad news on the SCAU side would only make it more urgent for Farrar to go for publication.**

Written by Gilles Demaneuf

261 followers · 45 following

Opinions, analyses and views expressed are purely mine and should not in any way be characterised as representing any institution.

Follow